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DIPLOMOVÁ PRÁCE

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Weilovy diferenciály

Katedra algebry

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Název práce: Weilovy diferenciály

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Abstrakt: Tato práce se zabývá tím, jak počítat lokální komponenty Weilových diferenciálů eliptického funkčního tělesa. Vzhledem k tomu, že Weilovy diferenciály tvoří vektorový prostor dimenze jedna, tak je fixován jeden konkrétní Weilův diferenciál. Pro tento diferenciál se popíše algoritmus na počítání lokálních komponent. První algoritmus funguje pro místa stupně jedna. Tento algoritmus je založen na elementárním přístupu. Definice Weilova diferenciálu není příliš zřejmá a není na první pohled jasné, k čemu je dobrá. Proto je zde popsána analogie Weilova diferenciálu s některými objekty z komplexní analýzy, jako jsou Laurentovy řady a residua. Tato analogie má lépe objasnit vlastnosti Weilova diferenciálu. Výsledkem této práce jsou další dva algoritmy, jak počítat lokální komponentu Weilova diferenciálu pomocí residuí.

Klíčová slova: algebraické funkční těleso, derivační modul, Weilův diferenciál, zúplnění, rozšíření funkčních těles

Title: Weil differentials

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Abstract: This thesis focuses upon how to calculate local components of Weil differentials of an elliptic function field. Because Weil differentials constitute a one-dimension vector space then one Weil differential is fixed. An algorithm calculating a local component is developed for the fixed one. The first algorithm computes local components of places of degree one. It is based upon elementary properties of local components. The definition of the Weil differential does not say enough about why it is defined in this way and about why it is useful. Thus there is the relationship between the Weil differential and some objects from complex analysis like the Laurent series and the residue. It provides a better understanding of properties of the Weil differential. The result of this thesis are other two algorithms calculating local components of Weil differentials. The algorithms employ the residue.

Keywords: algebraic function field, module of derivations, Weil differentials, completion, extension of function fields

Contents

Introduction

The main contribution of this thesis is a number of algorithms calculating a local component of a Weil differential of an elliptic function field and a rational function field. The algorithms for an elliptic function field are based upon three approaches. The first one uses elementary properties of local components. The second one uses the relationship between the residue and the local component. The third one uses the algorithm for a rational function field and the mapping $\mathrm{Cotr}_{E/K(x)}$. This thesis is built upon [1]. This means that the theory used in this thesis is taken from [1] and that the algorithms rely upon that theory.

Let K be a field. Throughout this thesis a coordinate ring over K determined by an irreducible polynomial $f(x_1, x_2) \in K[x_1, x_2]$ is denoted by $K[f]$. Its fractional field is denoted by $K(f)$. This notation is used for an algebraic function field determined by f.

Let F/K be an algebraic function field. Because $\dim_F \big(\Omega_F\big) = 1$ then it is possible to fix one Weil differential $\omega \in \Omega_F$ and to develop the algorithms for ω. Let $\xi \in \Omega_F$ be such that $\omega \neq \xi$. Then $\xi = u \cdot \omega$ for $u \in F$. Thus $\xi_P(v) = \big(u \cdot \omega\big)_P(v) = \omega_P(u \cdot v)$ for $v \in F$ and $P \in \mathbb{P}_F$.

Let $K(x)/K$ be a rational function field. The Weil differential $\eta \in \Omega_{K(x)}$ is defined in [1, Chapter 1.7]. It has the following properties $\eta_\infty(x^{-1}) = -1$ and $(\eta) = -2 \cdot P_\infty$. The algorithm calculating $\eta_P(r)$ for $r \in K(x)$ and an arbitrary place $P \in \mathbb{P}_{K(x)}$ is shown in Section 1.3. The needed properties of η for the algorithm is included in Section 1.3 too. The algorithm is based upon the elementary properties of local components.

Let E/K be an elliptic function field. There exists $\omega \in \mathbb{P}_E$ such that $(\omega) = 0$ and $\omega_\infty\big(\frac{y}{x}\big) = -1$. The Weil differential ω is chosen to develop an algortihm calculating $\omega_P(u)$ for $P \in \mathbb{P}_E^{(1)}$ and $u \in E$. The algorithm is based upon the elementary properties of local components. An illustration of the algorithm is in Examples 3.10 and 3.31. If $\mathrm{char}\, K \neq 2$ then the algorithm is significantly simplified. This topic is in Chapter 3.

Let $F'/K' \geq F/K$ be an finite separable extension. In [1, Chapter 3.4] and in [1, Chapter 3.5] there is a theory how to lift $\xi \in \Omega_F$ to $\xi' \in \Omega_{F'}$. It is made by the mapping $\mathrm{Cotr}_{F'/F} \colon \Omega_F \to \Omega_{F'}$. The divisor $\mathrm{Diff}(F'/F) \in \mathrm{Div}(F')$ is employed for a description of $\big(\mathrm{Cotr}_{F'/F}(\xi)\big) \in \mathrm{Div}(F')$ for $\xi \in \Omega_F$. In Section 4.2 this theory is used to describe $\mathrm{Diff}(E/K(x))$ and $\mathrm{Cotr}_{E/K(x)}(\eta)$ where E/K is an elliptic function field. The relationship between ω and $\mathrm{Cotr}_{E/K(x)}(\eta)$ is shown. It brings another algorithm calculating local components of ω and $\mathrm{Cotr}_{E/K(x)}(\eta)$ for places whose ramification indexes are two.

The relationship between the differential of completed function fields and the Weil differential described in [1, Chapter 4] is briefly recapitulated in Chapter 5. It provides a better understanding to next algorithms based upon the residue. Because in [1, Chapter 4] there is only a sketch of the proof of the theorem about the complementation of valued fields then in Section 5.2 the proof and the needed theory are finished. In Section 5.4 there are two algorithms calculating a residue. Thus because there is the relationship between the residue and the local component then the algorithms can be employed to calculate a local component of ω_P and $\mathrm{Cotr}_{E/K(x)}(\eta)$.

1. Recapitulation of Basic Knowledge

This chapter is a recapitulation of basic knowledge of algebraic geometry. It is based upon [1]. Details and proofs can be found there.

Section 1.1 is a recapitulation of a base of algebraic geometry. Most of this section is based upon [1, Chapter 1]. Proofs of fundamental theorems for this thesis are mentioned here. Lemma 1.61 is not taken from [1, Chapter 1]. It is our work.

Section 1.2 is a recapitulation of extensions of algebraic functional fields. It is based upon [1, Chapter 3]. It is in scope needed for this thesis.

Section 1.3 describes the rational function field $K(x)/K$ being the most basic algebraic function field. The set $\mathbb{P}_{K(x)}$ is studied here. The important Weil differential $\eta \in \Omega_{K(x)}$ is defined here. At the end there is an algorithm calculating η_P for $P \in \mathbb{P}_{K(x)}$.

Section 1.4 defines the elliptic function field and the Weierstrass equation. Features of the elliptic function field are described here. It is shown that an elliptic function field E/K is determined by a smooth short Weierstrass equation. There is how short Weierstrass equations looking.

1.1 Algebraic Geometry

Definition 1.1. Let F be a field. A ring $R \subsetneq F$ is a valuation ring if it has the following properties

(i) $F \neq R$;

(ii) a or a^{-1} is an element of R for all $a \in F^*$.

Definition 1.2. An integral domain R is a valuation ring if it is a valuation ring of its fraction field.

Definition 1.3. A discrete valuation of a field F is a function $\upsilon : F \to \mathbb{Z} \cup \{\infty\}$ with the following properties

(i) $\upsilon(x \cdot y) = \upsilon(x) + \upsilon(y)$ for all $x, y \in F$;

(ii) $\upsilon(x + y) \geq \min\big(\upsilon(x), \upsilon(y)\big)$ for all $x, y \in F$;

(iii) $\upsilon(x) = \infty \iff x = 0$ for all $x \in F$.

Lemma 1.4. *Let $\upsilon : F \to \mathbb{Z} \cup \{\infty\}$ be a discrete valuation of a field F. Then $\upsilon(1) = 0$, $\upsilon(-1) = 0$, $\upsilon(x) = \upsilon(-x)$ for every $x \in F$. If $\upsilon(x) \neq \upsilon(y)$ for $x, y \in F$ then $\upsilon(x + y) = \min\big(\upsilon(x), \upsilon(y)\big)$.*

Proof. The valuation $\upsilon(1) = \upsilon(1 \cdot 1) = \upsilon(1) + \upsilon(1)$. Thus $\upsilon(1) = 0$. Calculate $4 \cdot \upsilon(-1) = \upsilon\big((-1)^4\big) = \upsilon(1) = 0$. Thus $\upsilon(-1) = 0$. The valuation $\upsilon(-x) = \upsilon(-1 \cdot x) = \upsilon(-1) + \upsilon(x) = \upsilon(x) + 0$.

Suppose $\upsilon(x) > \upsilon(y)$. The inequality $\upsilon(x+y) \geq \min\big(\upsilon(x), \upsilon(y)\big)$ is satisfied by Definition 1.3. Prove the other inequality

$$\upsilon(y) = \upsilon(x+y-x) \geq \min\big(\upsilon(x+y), \upsilon(x)\big) = \upsilon(x+y).$$

The last equality is true because if $\min\big(\upsilon(x+y), \upsilon(x)\big)$ was equal to $\upsilon(x)$, $\upsilon(y)$ would be greater or equal to $\upsilon(x)$ which would be a contradiction with the assumption $\upsilon(x) > \upsilon(y)$. $\square$

Theorem 1.5. *Let F be a field and let υ be a discrete valuation of F. Then $R = \{a \in F; \upsilon(a) \geq 0\}$ is a maximal subring of F and $R^* = \{a \in F; \upsilon(a) = 0\}$. Also R is a valuation ring of F. The ring R is local.*

Definition 1.6. An integral domain R is a discrete valuation ring if there exists a discrete valuation υ of the fraction field F of R such that $R = \{a \in F; \upsilon(a) \geq 0\}$.

The ring R from Theorem 1.5 is called a discrete valuation ring of F.

Theorem 1.7. *Let R_1 and R_2 be discrete valuation rings of a field F. Denote maximal ideals of R_i by M_i for $i = 1, 2$. If $R_1 \subseteq R_2$ or $M_1 \subseteq M_2$ then $R_1 = R_2$.*

Theorem 1.8. *Let R be an integral domain that is not a field. The following statements are equivalent.*

(i) The ring R is a local ring and a principal ideal domain.

(ii) The ring R is a local and Noetherian ring whose the maximal ideal is a principal ideal.

(iii) The ring R is a discrete valuation ring.

Let these statements be satisfied. Denote the fraction field of R by F and the maximal ideal of R by M. Then there exists only one discrete valuation υ of F such that $\upsilon(x) = 1 \iff M = x \cdot R$.

Definition 1.9. The discrete valuation mentioned in Theorem 1.8 is called normalized. Every generator of the maximal ideal M mentioned in Theorem 1.8 is called a uniformizing element of M.

Definition 1.10. Let F/K be an extension of fields and let $x \in F$ be transcendental over K such that $[F : K(x)] < \infty$. The extension F/K is called an algebraic function field.

The field of constants of F/K is defined as

$$\tilde{K} := \{a \in F; a \text{ is algebraic over } K\}\ .$$

Definition 1.11. Let F/K be an algebraic function field and let R be a valuation ring of F such that $R \supsetneq K$. Then R is called a valuation ring of F/K.

Theorem 1.12. *Every valuation ring of an algebraic function field is a discrete valuation ring.*

Definition 1.13. (i) Let F/K be an algebraic function field. A place P of F/K is the maximal ideal of a valuation ring $\mathcal{O}$ of F/K.

(ii) The set of all places is denoted by $\mathbb{P}_{F/K}$ or $\mathbb{P}_F$.

Theorem 1.14. *Let F/K be an algebraic function field. If $P \in \mathbb{P}_F$ is the maximal ideal of a valuation ring $\mathcal{O}$ of F/K then P determines exactly $\mathcal{O}$. Thus $\mathcal{O}$ is denoted by $\mathcal{O}_P$.*

Lemma 1.15. *Every valuation ring of an algebraic function field contains its field of constants ($\mathcal{O}_P \supset \tilde{K}$).*

Definition 1.16. Let P be a place of F/K.

(i) There exist two fields $\mathcal{O}_P/P$ and $(K+P)/P$ such that $\mathcal{O}_P/P \geq (K+P)/P$. The degree of the place P is defined as $\deg(P) := [\mathcal{O}_P/P : (K+P)/P]$.

(ii) There exists only one normalised discrete valuation for $\mathcal{O}_P$. It is denoted by υ_P. It is usually called the valuation of the place P.

Remark 1.17. The field $(K+P)/P$ from Definition 1.16 is isomorphic to K. Thus the definition of the degree of the place P can be rewritten by the following way $\deg(P) := \deg_K(\mathcal{O}_P/P)$.

Definition 1.18. Let F/K be an algebraic function field and $i \in \mathbb{N}$. The set of places of degree i is denoted by

$$\mathbb{P}_F^{(i)} := \{P \in \mathbb{P}_F; \deg(P) = i\}.$$

Lemma 1.19. *Let F/K be an algebraic function field. A valuation υ_P is surjective for all $P \in \mathbb{P}_F$.*

Proof. Zero is mapped to infinity. Let $t \in F$ be a uniformizing element of $P \in \mathbb{P}_F$. Choose $k \in \mathbb{Z}$. Then $\upsilon_P(t^k) = k \cdot \upsilon_P(t) = k$. $\square$

Definition 1.20. Let F/K be an algebraic function field such that $K = \tilde{K}$ and K is perfect. Say that an element $z \in F$ is a separating element of F/K if $F/K(z)$ is a separable algebraic extension.

Theorem 1.21. *Let F/K be an algebraic function field such that $K = \tilde{K}$ and K is perfect. Let $z \in F$.*

(i) If $\operatorname{char} K = p > 0$ and $\upsilon_P(z) \not\equiv 0 \mod p$ for some $P \in \mathbb{P}_F$ then z is a separating element of F/K.

(ii) Let $\operatorname{char} K = p > 0$. The element $z \in F$ is a separating element of F/K if and only if $z \notin F^p$.

(iii) Let $\operatorname{char} K = 0$. If $z \in F \setminus K$ then z is a separating element of F/K.

Theorem 1.22 (Weak Approximation Theorem)**.** *Let F/K be an algebraic function field. Choose*

- *$n \in \mathbb{N}$;*
- *$x_1, \ldots, x_n \in F$;*
- *$r_1, \ldots, r_n \in \mathbb{Z}$;*

- $P_1, \dots, P_n$ *pairwise distinct places of* F/K.

Put $\upsilon_i := \upsilon_{P_i}$ *for all* $i \in \{1, \dots, n\}$. *Then there exists* $x \in F$ *such that* $\upsilon_i(x - x_i) = r_i$ *for all* $i \in \{1, \dots, n\}$.

Corollary 1.23. *Let* F/K *be an algebraic function field.*

(i) *For all* $x \in F^*$ *there exist finite sets* $\mathcal{Z}, \mathcal{P} \subset \mathbb{P}$ *with the following properties*

  (a) $\upsilon_P(x) > 0$ *for all* $P \in \mathcal{Z}$;

  (b) $\upsilon_P(x) < 0$ *for all* $P \in \mathcal{P}$;

  (c) $\upsilon_P(x) = 0$ *for all* $P \in \mathbb{P} \setminus (\mathcal{Z} \cup \mathcal{P})$.

(ii) *The set* $\mathbb{P}_F$ *is infinite.*

Convention 1.24. *Throughout this section a field of constant* K *is equal to* $\tilde{K}$.

Definition 1.25. Let F/K be an algebraic function field. The divisor group of F/K is defined as the free abelian group generated by $\mathbb{P}_F$. This group is denoted by $\mathrm{Div}(F/K)$ or $\mathrm{Div}(F)$. The elements of $\mathrm{Div}(F)$ are called the divisors of F/K. The additive notation is used for $\mathrm{Div}(F)$. In other words, the divisors are formal sums. Let $D \in \mathrm{Div}(F)$. Then

$$D = \sum_{P \in \mathbb{P}} d_P \cdot P \text{ with } d_P \in \mathbb{Z}, \text{ almost all } d_P = 0.$$

Values d_P are also denoted by $\upsilon_P(D)$ or D_P.

The support of D is defined as $\mathrm{supp}(D) := \{P \in \mathbb{P}; d_P \neq 0\}$. The divisor D can be written in the following way

$$D = \sum_{P \in \mathrm{supp}(D)} d_P \cdot P.$$

The degree of divisors is defined as a group homomorphism

$$\deg : \mathrm{Div}(F) \to \mathbb{Z};$$

$$\deg(D) = \sum_{P \in \mathrm{supp}(D)} d_P \cdot \deg(P).$$

A partial order upon $\mathrm{Div}(F)$ is defined in the following way. Let $D, E \in \mathrm{Div}(F)$. Say that $D \leq E$ if $d_P \leq e_P$ for all $P \in \mathbb{P}$.

A divisor can be split to a positive part and a negative part.

(i) The positive part of D is defined as

$$D_+ := \sum d'_P \cdot P \text{ where } d'_P := \begin{cases} d_P & \text{if } d_P \geq 0\ , \\ 0 & \text{if } d_P < 0\ . \end{cases}$$

(ii) The negative part of D is defined as

$$D_- := \sum d'_P \cdot P \text{ where } d'_P := \begin{cases} -d_P & \text{if } d_P < 0\ , \\ 0 & \text{if } d_P \geq 0\ . \end{cases}$$

We have $D = D_+ - D_-$.

The divisor D is positive (resp. negative) if $D = D_+$ (resp. $D = -D_-$) equivalently $D_- = 0$ (resp. $D_+ = 0$).

Definition 1.26. Let F/K be an algebraic function field. The principal divisor of $x \in F^*$ is defined as
$$(x) := \sum_{P \in \mathbb{P}} \upsilon_P(x) \cdot P \ .$$
It is denoted by (x).

This definition is correct because the number of places at which the valuation is not zero is finite by Corollary 1.23.

The set of all principal divisor is denoted by $\mathrm{Princ}(F/K)$ or $\mathrm{Princ}(F)$.

Lemma 1.27. *Let $x, y \in F^*$. Then*

- $(x \cdot y) = (x) + (y)$*;*
- $(x^{-1}) = -(x)$*.*

Thus $\mathrm{Princ}(F)$ *is a subgroup of* $\mathrm{Div}(F)$*.*

Lemma 1.28. *Let F/K be an algebraic function field and $P \in \mathbb{P}_F$. Then $\upsilon_P(k) = 0$ for all $k \in \tilde{K}$.*

Definition 1.29. Let F/K be an algebraic function field. Let $A \in \mathrm{Div}(F)$. The Riemann-Roch space associated to A is defined as
$$\mathcal{L}(A) := \{x \in F^*; A + (x) \geq 0\} \cup \{0\}.$$

Lemma 1.30. *A Riemann-Roch space $\mathcal{L}(A)$ is a vector space over K. Denote $l(A) := \dim_K \big(\mathcal{L}(A)\ \big)$.*

Theorem 1.31. *Let F/K be an algebraic function field. Let $x \in F \setminus \tilde{K}$. Then*
$$[F : K(x)] = \deg\big((x)_+\big) = \deg\big((x)_-\big).$$
For all $y \in F$ $\deg\big((y)\big) = 0$*.*

Definition 1.32. Let F/K be an algebraic function field.

(i) The divisor class group of F/K is defined as $\mathrm{Cls}(F) := \mathrm{Div}(F)/\mathrm{Princ}(F)$.

(ii) Picard group of F/K is defined as $\mathrm{Pic}(F) := \deg^{-1}(0)/\mathrm{Princ}(F)$.

Lemma 1.33. *The divisor class group* $\mathrm{Cls}(F) \geq \mathrm{Pic}(F)$*.*

Theorem 1.34 (Riemann's Theorem)**.** *Let F/K be an algebraic function field. There exists a constant $\gamma \in \mathbb{Z}$ such that* $\deg(A) - l(A) < \gamma$ *for all $A \in \mathrm{Div}(F)$.*

Definition 1.35. Under the assumption of Theorem 1.34, the genus of F/K is defined as the smallest γ from Theorem 1.34. The genus is usually denoted by g.

Lemma 1.36. *The genus g of an algebraic function field F/K is a non-negative integer.*

Proof. Calculate $\deg(0) - l(0) = -1$. Thus $-1 < g$ respectively $0 \leq g$. $\square$

Definition 1.37. Let F/K be an algebraic function field and let $A \in \mathrm{Div}(F)$. The index of specialization of A is defined as

$$i(A) := l(A) - \deg(A) + g - 1 \text{ where } g \text{ is the genus of } F/K\ .$$

By Lemma 1.36 $i(A) \geq 0$ for all $A \in \mathrm{Div}(F)$.

Definition 1.38. Let M, N be arbitrary sets. The set of all maps from M to N is denoted by N^M.

Definition 1.39. Let F/K be an algebraic function field. Put $\mathbb{P} := \mathbb{P}_{F/K}$. The valuations can be naturally extended to $F^{\mathbb{P}}$ in the following way

$$\upsilon_P(f) := \upsilon_P\big(f(P)\big) \text{ where } f \in F^{\mathbb{P}},\ P \in \mathbb{P}.$$

Lemma 1.40. *Under the assumption of Definition 1.39, the following statements are satisfied.*

(i) There exists an embedding of F into $F^{\mathbb{P}}$

$$\varphi : F \hookrightarrow F^{\mathbb{P}};$$

$$\varphi(x) = c_x \text{ where } c_x(P) = x \text{ for all } P \in \mathbb{P}.$$

(ii) Clearly $\upsilon_P(\varphi(x)) = \upsilon_P(c_x) = \upsilon_P(x)$ for all $P \in \mathbb{P}$ and $x \in F$.

(iii) The set of the divisors is a subset of $\mathbb{Z}^{\mathbb{P}}$.

Definition 1.41. Let F/K be an algebraic function field. Put $\mathbb{P} := \mathbb{P}_{F/K}$.

(i) The adele space of F/K is defined as

$$\mathcal{A}_F := \{f \in F^{\mathbb{P}}; f(P) \in \mathcal{O}_P \text{ for almost all } P \in \mathbb{P}\}\ .$$

The definition can be rewritten in the following way

$$\mathcal{A}_F := \{f \in F^{\mathbb{P}}; \upsilon_P(f) \geq 0 \text{ for almost all } P \in \mathbb{P}\}\ .$$

(ii) An element α of $\mathcal{A}_F$ is called an adele of F/K. An adele may be considered as an element of the direct product $\prod_{P\in\mathbb{P}} F$. Therefore the notation $\alpha = (\alpha_P)_{P\in\mathbb{P}}$ or $\alpha = (\alpha_P)$ can be used.

(iii) For $A \in \mathrm{Div}(F)$ define

$$\mathcal{A}_F(A) := \{f \in \mathcal{A}_F; \upsilon_P(f) + a_P \geq 0 \text{ for all } P \in \mathbb{P}\}\ .$$

Lemma 1.42. *Under the assumption of Definition 1.41, the following statements are satisfied.*

(i) The adele space $\mathcal{A}_F$ is a vector space over F thus also over K.

(ii) The field F is embedded into $\mathcal{A}_F$ (cf. Remark 1.40, item (i)).

(iii) The set $\mathcal{A}_F(A)$ is a vector space over K such that $\mathcal{A}_F(A) \cap F = \mathcal{L}(A)$.

Definition 1.43. Let V be a vector space and let W be a subspace of V. The set of annihilators of W in V is defined as

$$\mathrm{Ann}_V(W) := \{\varphi \in V^\star; \varphi(w) = 0 \text{ for all } w \in W\} \ .$$

The definition can be rewritten in the following way

$$\mathrm{Ann}_V(W) := \{\varphi \in V^\star; \mathrm{Ker}(\varphi) \geq W\}$$

Theorem 1.44. *Under the assumption of Definition 1.43, $\mathrm{Ann}_V(W)$ is a subspace of $V^\star$ and $\mathrm{Ann}_V(W) \simeq (V/W)^\star$.*

Definition 1.45. Let F/K be an algebraic function field.

(i) A Weil differential of F/K is defined as a K-linear mapping $\omega : \mathcal{A}_F \to K$ vanishing on $\mathcal{A}_F(A) + F$ for some $A \in \mathrm{Div}(F)$. In other words, $\omega \in \mathrm{Ann}_{\mathcal{A}_F}(\mathcal{A}_F(A) + F)$.

(ii) The vector space over K of Weil differentials vanishing on $\mathcal{A}_F(A) + K$ is denoted by
$$\Omega_F(A) := \mathrm{Ann}_{\mathcal{A}_F}(\mathcal{A}_F(A) + F) \ .$$

(iii) The vector space over K of Weil differentials is defined as

$$\Omega_F := \bigcup_{A \in \mathrm{Div}} \Omega_F(A) \ .$$

Remark 1.46. See that Ω_F is really a vector space over K. Consider $\omega_1, \omega_2 \in \Omega_F$. There exist $A_1, A_2 \in \mathrm{Div}(F)$ such that ω_i vanishes on $\mathcal{A}_F(A_i) + F$ for $i \in \{1, 2\}$. Let $A_3 \in \mathrm{Div}(F)$ such that $A_3 \leq A_1$ and $A_3 \leq A_2$. The Weil differential $\omega_3 := \omega_1 + \omega_2$ vanishes on $\mathcal{A}_F(A_3) + F$. Let $\alpha \in K$. The Weil differential $\alpha \cdot \omega_1$ vanishes on $\mathcal{A}_F(A_1) + F$. Thus Ω_F is a vector space over K.

Lemma 1.47. *Under the assumption of Definition 1.45, let $\omega \in \Omega_F$, $\alpha \in \mathcal{A}_F$ and $x \in F$. Define the Weil differential $(x \cdot \omega)(\alpha) := \omega(x \cdot \alpha)$. It is a correct definition because $F \hookrightarrow \mathcal{A}_F$ (cf. Remark 1.40, item (i)). Thus Ω_F is a vector space over F.*

Theorem 1.48. *Let F/K be an algebraic function field. The set of Weil differentials Ω_F is a subspace of $\mathcal{A}_F^\star$ over F and $dim_F(\Omega_F) = 1$.*

Definition 1.49. Let F/K be an algebraic function field. Let $0 \neq \omega \in \Omega_F$. Define the set of divisors

$$M(\omega) := \{A \in \mathrm{Div}(F); \omega \in \Omega_F(A)\} \ .$$

Remark 1.50. Under the assumption of Definition 1.49, let $A, B \in \mathrm{Div}(F)$ such that $B \leq A$ and $A \in M(\omega)$. Then $\omega \in \Omega_F(A) \subseteq \Omega_F(B)$. Thus $B \in M(\omega)$.

Lemma 1.51. *Let $0 \neq \omega \in \Omega_F$. There exists only one divisor $W \in \mathrm{Div}(F)$ such that a divisor A is an element of $M(\omega)$ if and only if $A \leq W$.*

Definition 1.52. The divisor W mentioned in Lemma 1.51 is uniquely determined by ω. Thus W is denoted by (ω). Each divisor that can be obtain this way is called canonical.

Lemma 1.53. *Let F/K be an algebraic function field.*

(i) Let $0 \neq \omega \in \Omega_F$ and $x \in F^\star$. Then $(x \cdot \omega) = (x) + (\omega)$.

(ii) Canonical divisors form a whole class $[W]$ in $\mathrm{Cls}(F)$ called the canonical class.

Lemma 1.54. *Let ω_1, ω_2 be non-zero Weil differentials. Then $(\omega_1) = (\omega_2) \iff \omega_1 = c \cdot \omega_2$ where $c \in K^\star$.*

Proof. The vector spacer Ω_F has one dimension over F. Thus there exists $c \in F^\star$ such that $\omega_1 = c \cdot \omega_2$. By Lemma 1.53 $(\omega_1) = (c \cdot \omega_2) = (c) + (\omega_2)$. The divisor (c) is zero if and only if $c \in K^\star$. $\square$

Theorem 1.55 (Riemann-Roch theorem)**.** *Let W be a canonical divisor of an algebraic function field F/K. Then for all $B \in \mathrm{Div}(F)$ we have*

$$l(W - B) = i(B) = l(B) - \deg(B) + g - 1$$

where g is genus of F/K.

Corollary 1.56. *Under the assumption of Theorem 1.55, $l(W) = g$, $\deg(W) = 2g - 2$ and $i(W) = 1$. Let $A \in \mathrm{Div}(F)$. If $\deg(A) \geq 2g - 1$ then $l(A) = \deg(A) + 1 - g$.*

Theorem 1.57. *A divisor W is canonical if and only if $\deg(W) = 2g - 2$ and $l(W) \geq g$ where g is the genus of an algebraic function field.*

Definition 1.58. Let F/K be an algebraic function field. Let $P \in \mathbb{P}_F$.

(i) Define an embedding from F into $\mathcal{A}_F$ by the following way

$$\iota_P : F \to \mathcal{A}_F;$$

$$\big(\iota_P(x)\big)(Q) = \begin{cases} x & \text{if } Q = P; \\ 0 & \text{if } Q \neq P \end{cases}$$

where $Q \in \mathrm{Div}(F)$ and $x \in \mathbb{F}$.

(ii) Let $\omega \in \Omega_F$. The local component P of ω is defined as the following mapping

$$\omega_P : F \to K;$$

$$\omega_P(x) = \omega\big(\iota_P(x)\big) \text{ where } x \in F.$$

Theorem 1.59. *Let F/K be an algebraic function field. Let $\omega \in \Omega_F$. Let $\alpha \in \mathcal{A}_F$. Then $\omega_P(\alpha_P) \neq 0$ for a finite number of places. Denote this set by $S \subset \mathbb{P}$. Then*

$$\omega(\alpha) = \sum_{P \in \mathbb{P}_F} \omega_P(\alpha_P) = \sum_{P \in S} \omega_P(\alpha_P).$$

Proof. It is trivially satisfied for $\omega = 0$. Assume that $\omega \neq 0$. Put $W := (\omega)$. Define $S_1 := \operatorname{supp}(W)$, $S_2 := \{P \in \mathbb{P}_F; \upsilon_P(\alpha_P) < 0\}$ and $S := S_1 \cup S_2$. The set S_1 is finite by Definition 1.25. The set S_2 is finite by Definition 1.41. Thus S is finite. Define $\beta = (\beta_P) \in \mathcal{A}_F$

$$\beta_P := \begin{cases} \alpha_P \text{ for } P \notin S; \\ 0 \text{ for } P \in S. \end{cases}$$

By definition of S $\beta \in \mathcal{A}_F(W)$. Thus $\omega(\beta) = 0$. The adele $\alpha = \beta + \sum_{P \in S} \iota_P(\alpha_P)$. Then

$$\omega(\alpha) = \sum_{P \in S} \omega_P(\iota_P).$$

If $P \notin S$ then $\iota_P(\alpha_P) \in \mathcal{A}_F(W)$ and thus $\omega_P(\alpha_P) = 0$. $\square$

Theorem 1.60. *Let F/K be an algebraic function field.*

(i) Let $0 \neq \omega \in \Omega_F$ and $P \in \mathbb{P}_F$. Put $W := (\omega)$. Then

  (a) $W_P = \max\{r \in \mathbb{Z};$ it holds $\omega_P(x) = 0$ for every $x \in F$ such that $\upsilon_P(x) \geq -r\}$;

  (b) $\upsilon_P(x) \geq -W_P \Rightarrow \omega_P(x) = 0$ where $x \in F$;

  (c) there exists $x \in F$ such that $\upsilon_P(x) = -W_P - 1$ and $\omega_P(x) \neq 0$.

(ii) Let $\omega, \omega' \in \Omega_F$ and $P \in \mathbb{P}_F$. Then $\omega_P = \omega'_P \iff \omega = \omega'$.

Proof. (i) If $\upsilon_P(x) \geq -W_P$ then $\iota_P(x) \in \mathcal{A}_F(W)$ thus $\omega_P(x) = 0$. Assume that $\omega_P(x) = 0$ for all $x \in F$ with the property $\upsilon_P(x) \geq -W_P - 1$. Let $\alpha = (\alpha_P)_{P \in \mathbb{P}_F} \in \mathcal{A}_F(W + P)$. The adele $\alpha = \big(\alpha - \iota_P(\alpha_P)\big) + \iota_P(\alpha_P)$ where $\big(\alpha - \iota_P(\alpha_P)\big) \in \mathcal{A}_F(W)$ and $\upsilon_P\big(\iota_P(\alpha_P)\big) \geq -W_P - 1$. Thus

$$\omega(\alpha) = \omega\big(\alpha - \iota_P(\alpha_P)\big) + \omega_P(\alpha_P) = 0.$$

Thus ω vanishes on $\mathcal{A}_F(W + P)$ which is a contradiction with the definition of W.

(ii) If $\omega = \omega'$ then $\omega_P = \omega'_P$. If $(\omega - \omega')_P = 0$ then the maximum from item *(i)* cannot be defined. Thus $\omega - \omega' = 0$.

$\square$

Lemma 1.61. *Let F/K be an algebraic function field, $0 \neq \zeta \in \Omega_F$ and $P \in \mathbb{P}_F^{(1)}$. Put $Z := (\zeta)$. Then $\zeta_P(x) \neq 0$ for every $x \in F$ such that $\upsilon_P(x) = -Z_P - 1$.*

Proof. By Theorem 1.60 there exists $y \in F$ such that

(i) $\upsilon_P(y) = -Z_P - 1$;

(ii) $\zeta_P(y) = \lambda \neq 0$.

Choose an arbitrary $y \neq x \in F$ such that $\upsilon_P(x) = -Z_P - 1$. Define $z := \frac{x}{y}$. Recall that $y \neq 0$ because $\upsilon_P(y)$ is finite. The valuation $\upsilon_P\big(\frac{x}{y}\big) = 0$. Consider z as $z = a + z' \in \mathcal{O}_P/P$ where $z' \in P$ and $a \in \mathcal{O}_P \setminus P$ because $\upsilon_P(z) = 0$. Since $P \in \mathbb{P}_F^{(1)}$ we may assume that $a \in K^\star$. The valuation $\upsilon_P(z' \cdot y) \geq -Z_P - 1 + 1 = -Z_P$. By Theorem 1.60 $\zeta_P(z' \cdot y) = 0$. The local component $\zeta_P(a \cdot y) = a \cdot \zeta_P(y) = a \cdot \lambda \neq 0$. The local component

$$\zeta_P(x) = \zeta_P(z \cdot y) = \zeta_P\big((a + z') \cdot y\big) = \zeta_P(a \cdot y) + \zeta_P(z' \cdot y) = a \cdot \lambda + 0 \neq 0.$$

$\square$

1.2 Extensions of Algebraic Function Fields

Definition 1.62. Let F'/K' and F/K be algebraic function fields.

(i) The algebraic function field F'/K' is an extension of F/K if F'/F and K'/K. It is denoted by $F'/K' \geq F/K$.

(ii) The extension $F'/K' \geq F/K$ is algebraic if F'/F is an algebraic extension.

(iii) The extension $F'/K' \geq F/K$ is finite if $[F' : F] < \infty$.

(iv) The extension $F'/K' \geq F/K$ is Galois if F'/F is a Galois extension.

Convention 1.63. *Throughout this section consider $F'/K' \geq F/K$ and put $\mathbb{P}' := \mathbb{P}_{F'/K'}$ and $\mathbb{P} := \mathbb{P}_{F/K}$.*

Lemma 1.64. *(i) If $F'/K' \geq F/K$ is algebraic then K'/K is algebraic.*

(ii) If $F'/K' \geq F/K$ is finite then $[K' : K] < \infty$.

Lemma 1.65. *Let $Q \in \mathbb{P}$ and $P' \in \mathbb{P}'$. Then*

$$\mathcal{O}_{P'} \supseteq \mathcal{O}_Q \iff P' \supseteq Q \iff \mathcal{O}_Q = \mathcal{O}_{P'} \cap F \iff Q = P' \cap F$$

and $\forall P' \in \mathbb{P}' \; \exists! Q \in \mathbb{P}$ with the mentioned properties.

Definition 1.66. Let $P \in \mathbb{P}$ and $P' \in \mathbb{P}'$ such that $P \subseteq P'$. It is denoted by $P' \mid P$. It is called that P' divides P or P' is over P or P is under P'.

Definition 1.67. Let $P \in \mathbb{P}$ and $P' \in \mathbb{P}'$ such that $P' \mid P$. The index of ramification of P' over P is defined by $e(P' \mid P) := \upsilon_{P'}(t)$ where t is a uniformizing element of P.

Remark 1.68. The ramification index $e(P' \mid P) \geq 1$ because $t \in P \subseteq P'$.

Definition 1.69. Let $P \in \mathbb{P}$ and $P' \in \mathbb{P}'$ such that $P' \mid P$. The relative degree of P' over P is defined as $f(P' \mid P) := [\mathcal{O}_{P'}/P' : (\mathcal{O}_P + P')/P']$.

Lemma 1.70. *Let $P \in \mathbb{P}$ and $P' \in \mathbb{P}'$ such that $P' \mid P$. Then* $\deg(P') \cdot [K' : K] = \deg(P) \cdot f(P' \mid P)$.

Theorem 1.71 (Fundamental Equality)**.** *Let $F'/K' \geq F/K$ be a finite extension. For all $P \in \mathbb{P}$ there exist only finitely many places $P' \in \mathbb{P}'$ such that $P' \mid P$. Denote these places $P_1, \ldots, P_n$ and put $e_i := e(P_i \mid P)$ and $f_i := f(P_i \mid P)$ for all $i = 1, \ldots, n$. Then*

$$\sum_{i=1}^{n} e_i \cdot f_i = [F' : F]$$

Definition 1.72. Let $F'/K' \geq F/K$ be a finite extension. Let $P \in \mathbb{P}$ and $P' \in \mathbb{P}'$ such that $P' \mid P$. Define two group homomorphisms on the generators of group:

(i) The Norm $N_{F'/F}$

$$N_{F'/F} : Div(F'/K') \to Div(F/K);$$

$$N_{F'/F}(P') = f(P' \mid P) \cdot P.$$

(ii) The Conorm $Con_{F'/F}$

$$Con_{F'/F} : Div(F/K) \to Div(F'/K');$$

$$Con_{F'/F}(P) = \sum_{P'|P} e(P' \mid P) \cdot P'.$$

Lemma 1.73. *Let $F''/K'' \geq F'/K' \geq F/K$ be finite extensions. Let $P \in \mathbb{P}$, $P' \in \mathbb{P}'$ and $P'' \in \mathbb{P}''$ such that $P'' \mid P'$ and $P' \mid P$. Then*

(i) $e(P'' \mid P) = e(P'' \mid P') \cdot e(P' \mid P)$;

(ii) $f(P'' \mid P) = f(P'' \mid P') \cdot f(P' \mid P)$;

(iii) $Con_{F''/F} = Con_{F''/F'} \circ Con_{F'/F}$;

(iv) $N_{F''/F} = N_{F'/F} \circ N_{F''/F'}$.

1.3 The Rational Function Field

The most basic algebraic function field is the rational function field $K(x)/K$. It is described in this section. The begin is based upon [1, Chapter 1.2] and [1, Chapter 1.7]. First, describe $\mathbb{P}_{K(x)/K}$. A place $P \in \mathbb{P}_{K(x)/K}$ is determined by an irreducible polynomial $p(x) \in K[x]$ or P is the pole. Next, show that there exists $\eta \in \Omega_{K(x)}$ with the properties $(\eta) = -2 \cdot P_\infty$ and $\eta_\infty(\frac{1}{x}) = -1$.

The other part of this section is our work. It is a tool for the calculation of $\eta_P(r)$ for an arbitrary place $P \in \mathbb{P}_{K(x)}$ and $r \in K(x)$. It is illustrated at Example 1.90. The tool is used in Section 4.2.

Lemma 1.74. *Let x be trancendental over a field K. The extension $K(x)/K$ is an algebraic function field. The genus of $K(x)/K$ is zero.*

Convention 1.75. *Throughout this section $K(x)/K$ is a rational function field and $K = \tilde{K}$.*

Theorem 1.76. *The places $\mathbb{P}_{K(x)/K}$ can be split to two cases.*

(i) A valuation ring is determined by an irreducible polynomial $p(x) \in K[x]$.

(a) The valuation ring

$$\mathcal{O}_{p(x)} = \left\{ \frac{f(x)}{g(x)}; f(x), g(x) \in K[x] \text{ and } p(x) \nmid g(x) \right\}.$$

(b) The place

$$P_{p(x)} = \left\{ \frac{f(x)}{g(x)}; f(x), g(x) \in K[x] \text{ and } p(x) \nmid g(x) \text{ and } p(x) \mid f(x) \right\}.$$

(c) Let $f(x) := \big(p(x)\big)^k \cdot \frac{a(x)}{b(x)} \in K(x)$ where $a(x), b(x) \in K[x]$ such that $p(x) \nmid a(x)$ and $p(x) \nmid b(x)$. Then $\upsilon_{p(x)}\big(f(x)\big) = k$.

(d) A uniformizing element of $P_{p(x)}$ is $p(x)$.

(e) *The degree* $\deg\big(P_{p(x)}\big) = \deg\big(p(x)\big)$.

(ii) *There exists another place in infinity. It is called the pole.*

(a) *The valuation ring*

$$\mathcal{O}_\infty = \left\{\frac{f(x)}{g(x)}; f(x), g(x) \in K[x] \text{ and } \deg\big(f(x)\big) \leq \deg\big(g(x)\big)\right\}.$$

(b) *The place*

$$P_\infty = \left\{\frac{f(x)}{g(x)}; f(x), g(x) \in K[x] \text{ and } \deg\big(f(x)\big) < \deg\big(g(x)\big)\right\}.$$

(c) *Let* $\frac{a(x)}{b(x)} \in K(x)$. *The valuation* $\upsilon_\infty\left(\frac{a(x)}{b(x)}\right) = \deg\big(b(x)\big) - \deg\big(a(x)\big)$.

(d) *A uniformizing element of* P_∞ *is* $\frac{1}{x}$.

(e) *The degree* $\deg\big(P_\infty\big) = 1$.

Convention 1.77. *Let* $\omega \in \Omega_{K(x)}$ *and* $P \in \mathbb{P}$. *Use the notation* $\omega_{p(x)} := \omega_P$ *if* P *is determined by an irreducible polynomial* $p(x)$ *and* $\omega_\infty := \omega_P$ *if* P *is the pole.*

If $P \in \mathbb{P}^{(1)}_{K(x)} \setminus \{P_\infty\}$ *then* P *is determined by* $p(x) = x - a$ *where* $a \in K$. *Let us use the following notations* $\mathcal{O}_a$, P_a, υ_a *and* ω_a *for the local component of* $\omega \in \Omega_{K(x)}$ *at* P_a.

Theorem 1.78. *There exists a unique* $\eta \in \Omega_{K(x)}$ *with the following properties*

(i) $(\eta) = -2 \cdot P_\infty$;

(ii) $\eta_\infty\big(x^{-1}\big) = -1$;

(iii) *Let* $a \in K$. *Then*

$$\eta_a\Big((x-a)^n\Big) = \begin{cases} 0 & \text{if } n \neq -1; \\ 1 & \text{if } n = -1; \end{cases}$$

$$\eta_\infty\Big((x-a)^n\Big) = \begin{cases} 0 & \text{if } n \neq -1; \\ -1 & \text{if } n = -1. \end{cases}$$

Proof. The degree $\deg(-2 \cdot P_\infty) = -2 = 2 \cdot g - 2$ and $l(-2 \cdot P_\infty) \geq 0 = g$. By Theorem 1.57 $-2 \cdot P_\infty$ is a canonical divisor.

The divisor $-2 \cdot P_\infty$ determines uniquely η up to multiplication by $c \in K^\star$ by Lemma 1.54. The local component $\eta_\infty(x^{-1}) = \lambda \neq 0$ by Lemma 1.61. Thus there exists only one $\eta \in \Omega_{K(x)}$ such that $(\eta) = -2 \cdot P_\infty$ and $\eta_\infty(x^{-1}) = -1$.

If $Q \in \mathbb{P}_{K(x)} \setminus \{P_\infty, P_\alpha\}$ then $\upsilon_Q\big((x-\alpha)^n\big) = 0$. Thus $\eta_Q\big((x-\alpha)^n\big) = 0$ by Theorem 1.60. By Definition 1.45 $\eta\big((x-\alpha)^n\big) = 0$. By Theorem 1.59 $0 = \eta_\infty\big((x-\alpha)^n\big) + \eta_\alpha\big((x-\alpha)^n\big)$.

If $n \geq 0$ then $\upsilon_\alpha\big((x-\alpha)^n\big) = n \geq 0$. Thus $\eta_\alpha\big((x-\alpha)^n\big) = 0$ by Theorem 1.60. Thus $\eta_\infty\big((x-\alpha)^n\big) = 0$.

If $n \leq -2$ then $\upsilon_\infty\big((x-\alpha)^n\big) = -n \geq 2$. Thus $\eta_\infty\big((x-\alpha)^n\big) = 0$ by Theorem 1.60. Thus $\eta_\alpha\big((x-\alpha)^n\big) = 0$.

Let $n = -1$. Then

$$\frac{1}{x} - \frac{1}{x-\alpha} = \frac{x-\alpha-x}{x\cdot(x-\alpha)} = \frac{-\alpha}{x\cdot(x-\alpha)}.$$

The valuation $\upsilon_\infty\left(\frac{-\alpha}{x\cdot(x-\alpha)}\right) = 2$. Thus

$$\eta_\infty\left(\frac{1}{x} - \frac{1}{x-\alpha}\right) = 0$$

by Theorem 1.60. We have

$$-1 = \eta_\infty\left(\frac{1}{x}\right) = \eta_\infty\left(\frac{1}{x-\alpha}\right) \text{ and } \eta_\alpha\left(\frac{1}{x-\alpha}\right) = 1.$$

□

Convention 1.79. *Throughout this section $\eta \in \Omega_{K(x)}$ is as in Theorem 1.78.*

Theorem 1.80. *Let $p(x) \in K[x]$ be irreducible and $\deg\big(p(x)\big) \geq 2$. Then $\eta_{p(x)}\Big(\big(p(x)\big)^n\Big) = 0$ for every $n \in \mathbb{Z}$.*

Proof. The valuation $\upsilon_{p(x)}\Big(\big(p(x)\big)^n\Big) = n$. By Theorem 1.60 if

$$\upsilon_{p(x)}\Big(\big(p(x)\big)^n\Big) + (\eta)_{p(x)} = n + 0 = n \geq 0$$

then $\eta_{p(x)}\Big(\big(p(x)\big)^n\Big) = 0$.

Assume that $n < 0$. If $Q \in \mathbb{P}_{K(x)} \setminus \{P_\infty, P_\alpha\}$ then $\upsilon_Q\Big(\big(p(x)\big)^n\Big) = 0$. By Theorem 1.60 $\eta_Q\Big(\big(p(x)\big)^n\Big) = 0$. The valuation $\upsilon_\infty\Big(\big(p(x)\big)^n\Big) \geq 2\cdot(-n) \geq 2$. Thus $\upsilon_\infty\Big(\big(p(x)\big)^n\Big) + (\eta)_\infty \geq 0$. By Theorem 1.60 $\eta_\infty\Big(\big(p(x)\big)^n\Big) = 0$. By Theorem 1.59 and Definition 1.45

$$\eta_{p(x)}\Big(\big(p(x)\big)^n\Big) = -\sum_{P\in\mathbb{P}\setminus\{P_{p(x)}\}} \eta_P\Big(\big(p(x)\big)^n\Big) = 0.$$

□

Lemma 1.81. *Let $a(x) \in K[x]$. Then $\eta_\infty\big(a(x)\big) = 0$.*

Proof. Put $N := (\eta)$. Let $P \in \mathbb{P}_{K(x)} \setminus \{P_\infty\}$. The valuation $\upsilon_P\big(a(x)\big) \geq 0 = -N_P$. By Theorem 1.60 $\eta_P\big(a(x)\big) = 0$. By Definition 1.45 and Theorem 1.59

$$\eta_\infty\big(a(x)\big) = -\sum_{P\in\mathbb{P}\setminus\{P_\infty\}} \eta_P\big(a(x)\big) = 0.$$

□

Theorem 1.82. *Let $u = \frac{a(x)}{b(x)} \in K(x)$ where $\gcd\big(a(x), b(x)\big) = 1$. The following statements are satisfied.*

(i) If $\deg\big(a(x)\big) + 1 < \deg\big(b(x)\big)$ *then* $\eta_\infty\Big(\frac{a(x)}{b(x)}\Big) = 0$.

(ii) If $\deg\big(a(x)\big) + 1 = \deg\big(b(x)\big)$ *then* $\eta_\infty\Big(\frac{a(x)}{b(x)}\Big) = -\frac{\mathrm{lc}\big(a(x)\big)}{\mathrm{lc}\big(b(x)\big)}$.

(iii) If $\deg\big(a(x)\big)+1 > \deg\big(b(x)\big)$ *then* $a(x) = k(x)\cdot b(x)+r(x)$ *where* $k(x), r(x) \in K[x]$ *and* $\deg\big(r(x)\big) < \deg\big(b(x)\big)$. *Then* $\eta_\infty\Big(\frac{a(x)}{b(x)}\Big) = \eta_\infty\Big(\frac{r(x)}{b(x)}\Big)$.

Proof. The valuation $\upsilon_\infty(u) = \deg\big(b(x)\big) - \deg\big(a(x)\big)$. Put $N := (\eta)$.

(i) The valuation $\upsilon_\infty(u) > \deg\big(a(x)\big) + 1 - \deg\big(a(x)\big) = 1$. Thus $\upsilon_\infty(u) \geq 2 = -N_\infty$. By Theorem 1.60 $\eta_\infty(u) = 0$.

(ii) Firstly, assume that $a(x)$ and $b(x)$ are monic. We have $\frac{a(x)}{b(x)} - \frac{1}{x} = \frac{a(x)\cdot x - b(x)}{b(x)\cdot x}$. The valuation

$$\upsilon_\infty\Big(\frac{a(x)\cdot x - b(x)}{b(x)\cdot x}\Big) \geq \deg\big(b(x)\big) + 1 - \deg\big(b(x)\big) + 1 = 2 = N_\infty.$$

By Theorem 1.60 $\eta_\infty\big(\frac{a(x)}{b(x)} - \frac{1}{x}\big) = 0$. Thus $\eta_\infty\big(\frac{a(x)}{b(x)}\big) = \eta_\infty\big(\frac{1}{x}\big) = -1$.

If $a(x)$ and $b(x)$ are not monic then define $\tilde{a}(x) := \frac{a(x)}{\mathrm{lc}(a(x))}$ and $\tilde{b}(x) := \frac{b(x)}{\mathrm{lc}(b(x))}$. Calculate

$$\eta_\infty\Big(\frac{a(x)}{b(x)}\Big) = \frac{\mathrm{lc}\big(a(x)\big)}{\mathrm{lc}\big(b(x)\big)} \cdot \eta_\infty\Big(\frac{\tilde{a}(x)}{\tilde{b}(x)}\Big) = -\frac{\mathrm{lc}\big(a(x)\big)}{\mathrm{lc}\big(b(x)\big)}.$$

(iii) By Lemma 1.81

$$\eta_\infty\Big(\frac{a(x)}{b(x)}\Big) = \eta_\infty\Big(\frac{k(x)\cdot b(x) + r(x)}{b(x)}\Big) = \eta_\infty\big(k(x)\big)+\eta_\infty\Big(\frac{r(x)}{b(x)}\Big) = \eta_\infty\Big(\frac{r(x)}{b(x)}\Big).$$

Three cases can happen.

(a) If $\deg\big(r(x)\big) + 1 = \deg\big(b(x)\big)$ then $\eta_\infty\Big(\frac{a(x)}{b(x)}\Big) = -\frac{\mathrm{lc}\big(r(x)\big)}{\mathrm{lc}\big(b(x)\big)}$.

(b) If $\deg\big(r(x)\big) + 1 < \deg\big(b(x)\big)$ then $\eta_\infty\Big(\frac{a(x)}{b(x)}\Big) = 0$.

(c) If $r(x) = 0$ then $\eta_\infty\Big(\frac{a(x)}{b(x)}\Big) = 0$.

$\square$

Remark 1.83. Theorem 1.82 brings how to calculate η_∞. Next, focus to develop an algorithm calculating η_P for $P \in \Omega_{K(x)} \setminus \{P_\infty\}$.

Lemma 1.84. *Let* $p(x) \in K[x]$ *be irreducible. Put* $n := \deg\big(p(x)\big)$. *Put* $u := \frac{x^j}{p^i(x)}$ *for* $j = 0, \ldots, n-1$ *and* $i \in \mathbb{N}$. *Then* $\eta_{p(x)}(u) = 1$ *if* $j = n-1$ *and* $i = 1$. *Otherwise* $\eta_{p(x)}(u) = 0$.

Proof. Put $N := (\eta)$. If $P \in \mathbb{P}_{K(x)} \setminus \{P_\infty, P_{p(x)}\}$ then $\upsilon_P(u) \geq 0 = -N_P$. By Theorem 1.60 $\eta_P(u) = 0$. By Definition 1.45 and Theorem 1.59 $\eta_{p(x)}(u) = -\eta_\infty(u)$. Compare degrees $\deg(x^j) \leq n-1 < n \leq n \cdot i = \deg\big(p^i(x)\big)$. By Theorem 1.82 if $j = n-1$ and $i = 1$ then $\eta_\infty(u) = -1$ and $\eta_{p(x)}(u) = 1$. Otherwise $\eta_\infty(u) = 0$ and $\eta_{p(x)}(u) = 0$. □

Theorem 1.85. *Let $u \in K(x)$. Decompose*

$$u = \gamma \cdot p_1^{\alpha_1}(x) \cdot \ldots \cdot p_n^{\alpha_n}(x)$$

where $p_i(x) \in K[x]$ are irreducible and $\alpha_i \in \mathbb{Z}$. Define

$$S_{K(x),u} := \{P_{p_i(x)}; \alpha_i < 0\} \cup \{P_\infty\}.$$

If $\eta_P(u) \neq 0$ then $P \in S_{K(x),u}$.

Proof. Prove if $P \notin S_{K(x),u}$ then $\eta_P(u) = 0$. Put $N := (\eta)$. Assume that $P \notin S_{K(x),u}$. Then $\upsilon_P(u) \geq 0 = -N_P$. By Theorem 1.60 $\eta_P(u) = 0$. □

Remark 1.86. Theorem 1.85 provides a necessary condition where η_P is non-zero. Next Algorithm calculates an exact value of these local components.

Algorithm 1.87. ***The input*** *of the algorithm are*

(i) $P_{p(x)} \in \mathbb{P}_{K(x)}$ *where $p(x) \in K[x]$ is irreducible;*

(ii) $u = \frac{a(x)}{b(x)} \in K(x)$ *such that* $\gcd\big(a(x), b(x)\big) = 1$.

The output *of the algorithm is $\eta_{p(x)}(u)$.*
The algorithm:

1. *Decompose $b(x) = p^n(x) \cdot \tilde{b}(x)$ where $p(x) \nmid \tilde{b}(x)$.*

2. *If $n = 0$ then $\eta_{p(x)}(u) = 0$ and finish the algorithm.*

3. *Use* $\gcd\big(p^n(x), \tilde{b}(x)\big) = 1$ *and find Bézout coefficients $u(x), v(x) \in K[x]$ to be satisfied*
$$1 = u(x) \cdot \tilde{b}(x) + v(x) \cdot p^n(x).$$

4. *Define and write out*
$$\tilde{a}(x) := a(x) \cdot u(x) = \sum_{i=0}^{A} a_i(x) \cdot p^i(x) \qquad (1.1)$$
where $A := \left\lfloor \frac{\deg\left(\tilde{a}(x)\right)}{\deg\left(p(x)\right)} \right\rfloor$ *and $a_i(x) \in K[x]$ such that* $\deg\big(a_i(x)\big) < \deg\big(p(x)\big)$.

5. *Denote the coefficients of $a_i(x)$ by $a_{i,j}$. Put $m := \deg\big(p(x)\big)$. The local component $\eta_{p(x)}(u) = a_{n-1,m-1}$.*

Theorem 1.88. *Algorithm 1.87 is correct.*

Proof. Keep the assumption of Algorithm 1.87. Recall that $\gcd\big(a(x), b(x)\big) = 1$. If $n = 0$ from the decomposition $b(x) = p^n(x) \cdot \tilde{b}(x)$ then by Theorem 1.85 $\eta_{p(x)}(u) = 0$.

Assume that $n \geq 1$. The equation $1 = u(x) \cdot \tilde{b}(x) + v(x) \cdot p^n(x)$ provides

$$\frac{1}{b(x)} = \frac{u(x)}{p^n(x)} + \frac{v(x)}{\tilde{b}(x)}. \tag{1.2}$$

By Equations (1.1) and (1.2)

$$\frac{a(x)}{b(x)} = \frac{u(x) \cdot a(x)}{p^n(x)} + \frac{v(x) \cdot a(x)}{\tilde{b}(x)} = \sum_{i=0}^{A} a_i(x) \cdot p^{i-n}(x) + \frac{v(x) \cdot a(x)}{\tilde{b}(x)}.$$

Put $A_i := \deg\big(a_i(x)\big)$. Use K-linearity

$$\eta_{p(x)}\Big(\frac{a(x)}{b(x)}\Big) = \sum_{i=0}^{A} \sum_{j=0}^{A_i} a_{i,j} \cdot \eta_{p(x)}\Big(\frac{x^j}{p^{n-i}(x)}\Big) + \eta_{p(x)}\Big(\frac{v(x) \cdot a(x)}{\tilde{b}(x)}\Big).$$

Put $N := (\eta)$. Recall that $p(x) \nmid \tilde{b}(x)$. Thus $\upsilon_{p(x)}\big(\tilde{b}(x)\big) = 0$. Calculate

$$\upsilon_{p(x)}\Big(\frac{v(x) \cdot a(x)}{\tilde{b}(x)}\Big) = \upsilon_{p(x)}\big(v(x)\big) + \upsilon_{p(x)}\big(a(x)\big) - \upsilon_{p(x)}\big(\tilde{b}(x)\big) \geq 0 = -N_P.$$

By Theorem 1.60 $\eta_{p(x)}\Big(\frac{v(x) \cdot a(x)}{\tilde{b}(x)}\Big) = 0$.

If $n - i \leq 0$ then $\frac{x^j}{p^{n-i}(x)} \in K[x]$. Thus $\upsilon_{p(x)}\Big(\frac{x^j}{p^{n-i}(x)}\Big) \geq 0 = -N_P$. By Theorem 1.60 $\eta_{p(x)}\Big(\frac{x^j}{p^{n-i}(x)}\Big) = 0$.

If $n - i > 0$ then $\eta_{p(x)}\Big(\frac{x^{m-1}}{p(x)}\Big) = 1$ otherwise $\eta_{p(x)}\Big(\frac{x^j}{p^i(x)}\Big) = 0$ by Lemma 1.84.

Together $\eta_{p(x)}\Big(\frac{a(x)}{b(x)}\Big) = a_{n-1,m-1}$. □

Remark 1.89. The combination of Theorems 1.82 and 1.85 and Algorithm 1.87 provides the tool for the calculation of $\eta_P(u)$ for an arbitrary $P \in \mathbb{P}_{K(x)}$ and an arbitrary $u \in K(x)$.

Example 1.90. Illustrate the tool for the calculation of local components. Consider the rational function field $\mathbb{F}_5(x)/\mathbb{F}_5$. Define

(i) $a(x) := x^2 + 3 \in \mathbb{F}_5[x]$;

(ii) $b(x) := (x^2 + 2)^2 \cdot (x + 1) = x^5 + x^4 + 4 \cdot x^3 + 4 \cdot x^2 + 4 \cdot x + 4 \in \mathbb{F}_5[x]$;

(iii) $u := \frac{a(x)}{b(x)} \in \mathbb{F}_5(x)$.

Recall that $\gcd\big(a(x), b(x)\big) = 1$. By Theorem 1.85 $\eta_P(u) \neq 0$ only for places P_∞, P_4 and P_{x^2+2}.

Because $\deg\big(a(x)\big) + 1 < \deg\big(b(x)\big)$, $\eta_\infty(u) = 0$ by Theorem 1.82.

Calculate $\eta_4(u)$ and $\eta_{x^2+2}(u)$ by Algorithm 1.87. Find Bézout coefficients $c_1(x)$ and $c_2(x)$ such that $1 = c_1(x) \cdot (x^2 + 2)^2 + c_2 \cdot (x + 1)$. The coefficient $c_1(x) = 4$ and $c_2(x) = x^3 + 4 \cdot x^2$.

(i) Calculate $\eta_{x^2+2}(u)$. Keep the notation from Algorithm 1.87, $p(x) = x^2 + 2$, $\tilde{b}(x) = x + 1$, $v(x) = 4$, $u(x) = x^3 + 4 \cdot x^2$, $n = 2$, $m = 2$ and $a(x) = x^2 + 3$. Multiply and express

$$\begin{aligned}\tilde{a}(x) = u(x) \cdot a(x) &= x^5 + 4 \cdot x^4 + 3 \cdot x^3 + 2 \cdot x^2 \\ &= p^2(x) \cdot (x + 4) + p(x) \cdot (4 \cdot x + 1) + (3 \cdot x + 2).\end{aligned}$$

Thus $\eta_{x^2+2}(u) = a_{1,1} = 4$.

(ii) Calculate $\eta_4(u)$. Keep the notation from Algorithm 1.87, $p(x) = x + 1$, $\tilde{b}(x) = x^4 + 4 \cdot x^2 + 4$, $v(x) = x^3 + 4 \cdot x^2$, $u(x) = 4$, $n = 1$, $m = 1$ and $a(x) = x^2 + 3$. Multiply and express

$$\tilde{a}(x) = u(x) \cdot a(x) = 4 \cdot x^2 + 2 = p(x) \cdot (4 \cdot x + 1) + 1.$$

Thus $\eta_4(u) = a_{0,0} = 1$.

Use Theorem 1.59 and Definition 1.45 to make the basic check

$$0 = \eta(u) = \eta_\infty(u) + \eta_4(u) + \eta_{x^2+2}(u) = 0 + 1 + 4 = 0.$$

1.4 The Elliptic Function Field

This section is a recapitulation of knowledge about the elliptic function field. The definitions of the elliptic function field and the Weierstrass equation are here. It is shown that an elliptic function field is always determined by a smooth Weierstrass equation. The K-equivalence of two Weierstrass equations is defined here. It is shown that every Weierstrass equation is K-equivalent to a short Weierstrass equation.

Convention 1.91. *Throughout this section a field of constant K is equal to $\tilde{K}$ and K is perfect.*

Definition 1.92. An algebraic function field F/K is called an elliptic function field if it satisfies

(i) the genus of F/K is one;

(ii) $\mathbb{P}_F^{(1)} \neq \emptyset$.

Definition 1.93. Let E/K be an elliptic function field. There exist $x, y \in E \setminus K$ which satisfy the following equation

$$y^2 + a_1xy + a_3y = x^3 + a_2x^3 + a_4x + a_6$$

where $a_1, a_2, a_3, a_4, a_6 \in K$. This equation is called the Weierstrass equation. We shall often work with the polynomial form of the equation

$$w(x, y) = y^2 + a_1xy + a_3y - x^3 - a_2x^3 - a_4x - a_6.$$

The equation is considered as $w(x, y) \in K[x, y]$ or $w(x_1, x_2) \in K[x_1, x_2]$.

Lemma 1.94. *A Weierstrass equation $w(x_1, x_2) \in K[x_1, x_2]$ is absolutely irreducible.*

Theorem 1.95. *Let K be a field and $f(x_1, x_2) \in K[x_1, x_2]$ be an irreducible polynomial. Define the ring $K[f] := K[x_1, x_2]/(f)$ and $K(f)$ is the fraction field of $K[f]$. Then $K(f)/K$ is an algebraic function field determined by $f(x, y)$.*

Theorem 1.96. *Let F/K be an algebraic function field such that $F = K(x, y)$ where $x, y \in F$. Let $f \in K[x_1, x_2]$ be irreducible such that $f(x, y) = 0$. Then the following statements are satisfied.*

(i) There exists a homomorphism $\sigma : K[x_1, x_2] \to K[x, y]$ such that $\sigma(x_1) = x$, $\sigma(x_2) = y$ and $\mathrm{Ker}(\sigma) = (f)$.

(ii) There exists $\varphi : K(f) \simeq F$ such that $\varphi\big(x_1 + (f)\big) = x$ and $\varphi\big(x_2 + (f)\big) = y$.

(iii) Set $d_i := deg_{x_i}(f)$ for $i = 1, 2$ then

  (a) if $d_1 \geq 1$ then $[F : K(y)] = d_1$;

  (b) if $d_2 \geq 1$ then $[F : K(x)] = d_2$.

Convention 1.97. *Under the assumption of Theorem 1.96, say F/K is determined by the polynomial f or the equation $f(x, y) = 0$.*

Theorem 1.98. *Let E/K be an elliptic function field. Then there exists a Weierstrass equation $w(x_1, x_2) \in K[x_1, x_2]$ such that E/K is determined by w. In other word, $E \simeq K(w)$ and there exist $x, y \in E$ such that $w(x, y) = 0$.*

Theorem 1.99. *Let F/K be algebraic function field determined by a Weierstrass equation $w(x, y) = 0$.*

(i) Let $Q \in \mathbb{P}_F$. Then either

  (a) $\upsilon_Q(x) \geq 0$ and $\upsilon_Q(y) \geq 0$ or

  (b) $\upsilon_Q(x) < 0$ and $\upsilon_Q(y) < 0$.

(ii) There exists only one place P such that

  (a) $\upsilon_P(x) < 0$ and $\upsilon_P(y) < 0$,

  (b) $(x)_- = 2 \cdot P$ and $(y)_- = 3 \cdot P$,

  (c) $\deg(P) = 1$.

(iii) The genus $g \in \{0, 1\}$. Then $g = 0 \iff \exists t \in F$ such that $F = K(t)$.

Definition 1.100. Let $f(x_1, x_2) \in K[x_1, x_2]$ be irreducible. Consider $f(x_1, x_2)$ as a curve.

(i) The curve f has a singularity in

$$(\alpha, \beta) \in \mathbb{V}_f \iff \Big(\frac{\partial f}{\partial x_1}(\alpha, \beta), \frac{\partial f}{\partial x_2}(\alpha, \beta)\Big) = (0, 0).$$

(ii) The curve f is smooth if it does not have any singularity. Otherwise f is singular.

(iii) The curve is smooth in K if it does not have any singularity in $\mathbb{V}_f(K)$.

Theorem 1.101. *Let F/K be algebraic function field determined by a Weierstrass equation $w(x,y)=0$. Then the following statements are equivalent.*

(i) The genus of F/K is one.

(ii) The Weierstrass equation $w(x_1,x_2)$ is smooth.

(iii) The Weierstrass equation $w(x_1,x_2)$ is smooth in K.

Theorem 1.102. *An algebraic function field is elliptic if and only if it is determined by a smooth Weierstrass equation.*

Theorem 1.103. *Let F/K be an algebraic function field determined by $f(x,y)=0$ and $P\in\mathbb{P}_F$. If $\upsilon_P(x)\geq 0$ and $\upsilon_P(y)\geq 0$ then there exist irreducible monic polynomials $p(x)\in K[x]$ and $q(y)\in K[y]$ such that*

(i) $P\cap K[x]=\big(p(x)\big)$ and $P\cap K[y]=\big(q(y)\big)$;

(ii) $\deg\big(p(x)\big)\mid\deg(P)$ and $\deg\big(q(x)\big)\mid\deg(P)$.

If $\deg(P)=1$ then $p(x)=x-\alpha$, $q(y)=y-\beta$ and $(\alpha,\beta)\in\mathbb{V}_f$. If f is not singular in $(\alpha,\beta)\in\mathbb{V}_f$ then

$$\mathcal{O}_P=\mathcal{O}_{(\alpha,\beta)}=\Big\{\frac{g}{h}\in F; g,h\in K[x,y] \text{ and } h(\alpha,\beta)\neq 0\Big\};$$

$$P=P_{(\alpha,\beta)}=\Big\{\frac{g}{h}\in F; g,h\in K[x,y] \text{ and } h(\alpha,\beta)\neq 0 \text{ and } g(\alpha,\beta)=0\Big\}.$$

Definition 1.104. Let $w_1(x_1,x_2), w_2(x_1,x_2)\in K[x_1,x_2]$ be Weierstrass equations. They are called to be K-equivalent if there exist $r,s,t\in K$ and $u\in K^\star$ such that

$$w_2\big(u^2\cdot x_1+r, u^3\cdot x_2+u^2\cdot s\cdot x_1+t\big)=u^6\cdot w_1\big(x_1,x_2\big)$$

or in other words,

$$w_2\big((x_1,x_2)\cdot A+(r,t)\big)=u^6\cdot w_1\big(x_1,x_2\big) \text{ where } A=\begin{pmatrix} u^2 & u^2\cdot s\\ 0 & u^3\end{pmatrix}.$$

Lemma 1.105. *The relation from Definition 1.104 is an equivalence.*

Theorem 1.106. *Let E/K be an elliptic function field determined by a Weierstrass equation $w_1(x,y)=0$. If a Weierstrass equation $w_2(x_1,x_2)\in K[x_1,x_2]$ is K-equivalent to $w_1(x_1,x_2)$ then $K(w_1)\simeq K(w_2)$.*

Definition 1.107. Each of the following equations is called a short Weierstrass equation

(i) $w(x_1,x_2)=x_2^2+x_1x_2-x_1^3-a_2x_1^2-a_6$ for $\operatorname{char} K=2$;

(ii) $w(x_1, x_2) = x_2^2 + a_3x_2 - x_1^3 - a_4x_1 - a_6$ for $\operatorname{char} K = 2$;

(iii) $w(x_1, x_2) = x_2^2 - x_1^3 - a_2x_1^2 - a_6$ for $\operatorname{char} K = 3$;

(iv) $w(x_1, x_2) = x_2^2 - x_1^3 - a_4x_1 - a_6$ for $\operatorname{char} K = 3$;

(v) $w(x_1, x_2) = x_2^2 - x_1^3 - a_4x_1 - a_6$ for $\operatorname{char} K \neq 2, 3$.

Theorem 1.108. *Every Weierstrass equation is K-equivalent to one of the short Weierstrass equations.*

Convention 1.109. *The short Weierstrass equations will be written*

(i) $w(x, y) = y^2 + xy - x^3 - a_2x^2 - a_6 = g_1(x, y) - f_1(x)$ *where* $g_1(x, y) := y^2 + xy$ *and* $f_1(x) := x^3 + a_2x^2 + a_6$*;*

(ii) $w(x, y) = y^2 + a_3y - x^3 - a_4x - a_6 = g_2(x, y) - f_2(x)$ *where* $g_2(x, y) := y^2 + a_3y$ *and* $f_2(x) := x^3 + a_4x + a_6$*;*

(iii) $w(x, y) = y^2 - f(x)$ *where*

  (a) $f(x) := x^3 + a_2x^2 + a_6$ *or* $f(x) := x^3 + a_4x + a_6$ *if* $\operatorname{char} K = 3$*;*

  (b) $f(x) := x^3 + a_4x + a_6$ *if* $\operatorname{char} K \neq 2, 3$.

Theorem 1.110. *Let* $w(x, y) = y^2 - f(x)$ *be a Weierstrass equation and* $\operatorname{char} K \neq 2$*. The following statements are equivalent.*

(i) The curve $w(x, y)$ is smooth.

(ii) The polynomial $f(x)$ is separable.

Corollary 1.111. *Let F/K be an algebraic function field determined by a Weierstrass equation* $w(x, y) = y^2 - f(x)$ *and* $\operatorname{char} K \neq 2$*. Then the genus $g = 1$ if and only if the polynomial $f(x)$ is separable.*

Theorem 1.112. *Let* $\operatorname{char} K = 2$.

(i) A Weierstrass equation $w(x, y) = y^2 + xy - x^3 - a_2x^2 - a_6$ *is smooth* $\iff a_6 \neq 0 \iff (0, 0) \notin \mathbb{V}_w$*. If the equation is singular then* $(0, 0) \in \mathbb{V}_w$ *is a singularity.*

(ii) A Weierstrass equation $w(x, y) = y^2 + a_3y - x^3 - a_4x - a_6$ *is smooth if and only if* $a_3 \neq 0$*. If the equation is singular then* $(\sqrt{a_4}, \sqrt{a_6}) \in \mathbb{V}_w$ *is a singularity. There exists a case such that* $(\sqrt{a_4}, \sqrt{a_6}) \in \mathbb{V}_w$ *and the equation is still smooth. It happens if* $a_6 = 0$.

Convention 1.113. *An elliptic function field E/K is always determined by a smooth short Weierstrass equation $w(x, y) = 0$ (cf. Theorems 1.102, 1.106 and 1.108). Thus E may be identified with $K(w)$.*

By Theorem 1.99 there exists only one place $P \in \mathbb{P}_E$ such that $\upsilon_P(x) < 0$ and $\upsilon_P(y) < 0$. Denote

(i) this place by P_∞;

(ii) the valuation at P_∞ by υ_∞;

(iii) the local component of $\omega \in \Omega_E$ at P_∞ by ω_∞.

Let $P \in \mathbb{P}_E^{(1)} \setminus \{P_\infty\}$. By Theorems 1.103 and 1.99 P is determined by its coordinates (α, β). Denote

(i) this place by $P_{(\alpha,\beta)}$;

(ii) the valuation at $P_{(\alpha,\beta)}$ by $\upsilon_{(\alpha,\beta)}$;

(iii) the local component of $\omega \in \Omega_E$ at $P_{(\alpha,\beta)}$ by $\omega_{(\alpha,\beta)}$.

Definition 1.114. Let F/K be an algebraic function field determined by $f(x, y) = 0$. Let $l(x, y) = a \cdot x + b \cdot y + c \in F$ where $a, b, c \in K$. The element $l(x, y)$ is called a line if $a \neq 0$ or $b \neq 0$ and $l(x, y) \neq f(x, y)$.

Lemma 1.115. *Let F/K be an algebraic function field determined by $f(x, y) = 0$. Let $P_{(\alpha,\beta)} \in \mathbb{P}_F^{(1)}$ (cf. Theorem 1.103). Let $l(x, y) \in F$ be a line. Then*

(i) l is the tangent of f at (α, β) if and only if $\upsilon_{(\alpha,\beta)}(l) \geq 2$;

(ii) l is a secant of f at (α, β) if and only if $\upsilon_{(\alpha,\beta)}(l) = 1$;

(iii) l misses (α, β) if and only if $\upsilon_{(\alpha,\beta)}(l) = 0$.

Theorem 1.116. *Let $K(w)/K$ be an elliptic function field where $w(x, y) = 0$ is a Weierstrass equation.*

(i) An element $u \in K(w)$ can be expressed as

$$u = \frac{a_1(x)}{c_1(x)} + \frac{y \cdot b_2(x)}{c_2(x)}$$

where $a_1(x), c_1(x), b_2(x), c_2(x) \in K[x]$ such that $\gcd\big(a_1(x), c_1(x)\big) = 1$ and $\gcd\big(b_2(x), c_2(x)\big) = 1$.

(ii) An element $v \in K[w] = K[x_1, x_2]/\big(w(x_1, x_2)\big)$ can be expressed as

$$v = a(x) + y \cdot b(x)$$

where $a(x), b(x) \in K[x]$.

2. Extensions of Elliptic Function Fields over Rational Function Fields

Let E/K be an elliptic function field determined by a short smooth Weierstrass equation $w(x, y) = 0$. There is a description of $E/K \geq K(x)/K$ and $E/K \geq K(y)/K$ in this chapter. Let $P \in \mathbb{P}_E^{(1)}$ and $Q \in \mathbb{P}_{K(x)}^{(1)}$ be such that $P \mid Q$. This chapter describes what it means for coordinates of P and Q. When $e(P \mid Q) = 1$ and when $e(P \mid Q) = 2$.

The main result of this chapter is Definition 2.15 that splits up $\mathbb{P}_E^{(1)} \setminus \{P_\infty\}$ to six types of a place. These types distinguishes places according to their ramification indexes to axes x and y. Theorem 2.16 brings properties of each types. The concept of the type of a place is used in next chapters.

Convention 2.1. *Throughout this chapter consider that*

(i) an elliptic function field E/K is determined by a short smooth Weierstrass equation $w(x, y) = 0$;

(ii) K is perfect and $K = \tilde{K}$.

Throughout this chapter keep the notations from Conventions 1.109, 1.113 and 1.77 and Theorem 1.76.

Lemma 2.2. *The elliptic function field E/K is a finite extension of $K(x)/K$ and $K(y)/K$.*

Proof. Clearly, $E/K(x)$ and $E/K(y)$. By Theorem 1.96 $[E : K(x)] = 2$ (resp. $[E : K(y)] = 3$). □

Lemma 2.3. *The extension $E/K(x)$ is Galois.*

(i) If $\operatorname{char} K \neq 2$ *and* $w(x, y) = y^2 - f(x)$ *then* $\operatorname{Gal}(E/K(x)) = \{\delta_1, \delta_2\}$ *where* $\delta_1 = \mathrm{id}$ *and* $\delta_2(y) = -y$.

(ii) If $\operatorname{char} K = 2$ *and* $w(x, y) = y^2 + x \cdot y - f_1(x)$ *then* $\operatorname{Gal}(E/K(x)) = \{\delta_1, \delta_2\}$ *where* $\delta_1 = \mathrm{id}$ *and* $\delta_2(y) = x + y$.

(iii) If $\operatorname{char} K = 2$ *and* $w(x, y) = y^2 + a_3 \cdot y - f_2(x)$ *then* $\operatorname{Gal}(E/K(x)) = \{\delta_1, \delta_2\}$ *where* $\delta_1 = \mathrm{id}$ *and* $\delta_2(y) = a_3 + y$.

Proof. (i) Consider $E = K(x)[y]/(y^2 - f(x))$. The roots of $y^2 - f(x)$ over E are y and $-y$. Thus $E/K(x)$ is Galois and the Galois mappings are $\delta_1 = \mathrm{id}$ and $\delta_2(y) = -y$.

(ii) Consider $E = K(x)[y]/(y^2 + x \cdot y - f_1(x))$. The roots of $y^2 + x \cdot y - f_1(x)$ over E are y and $x + y$. Thus $E/K(x)$ is Galois and the Galois mappings are $\delta_1 = \mathrm{id}$ and $\delta_2(y) = x + y$.

(iii) Consider $E = K(x)[y]/(y^2 + a_3 \cdot y - f_2(x))$. The roots of $y^2 + a_3 \cdot y - f_2(x)$ over E are y and $a_3 + y$. Thus $E/K(x)$ is Galois and the Galois mappings are $\delta_1 = \mathrm{id}$ and $\delta_2(y) = a_3 + y$.

$\square$

Lemma 2.4. *Let $Q, P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$, $P_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ and $P_\beta \in \mathbb{P}_{K(y)}^{(1)}$ where $\alpha, \beta \in K$.*

(i) Then P_α and P_β are under $P_{(\alpha,\beta)}$.

(ii) If $Q \mid P_\alpha$ then $Q = P_{(\alpha,\gamma)}$ where $\gamma \in K$. Analogically, if $Q \mid P_\beta$ then $Q = P_{(\gamma,\beta)}$.

Proof. (i) Put $P_x := P \cap K(x)$. A rational function $a(x) \in \mathcal{O}_\alpha$ is defined in α. Thus $a(x) \in \mathcal{O}_P$ and $a(x) \in \mathcal{O}_{P_x}$. Thus $\mathcal{O}_\alpha \subseteq \mathcal{O}_{P_x}$. By Theorem 1.7 $\mathcal{O}_\alpha = \mathcal{O}_{P_x}$. Prove analogically for coordinate y.

(ii) The place Q is not the pole because

$$-2 = \upsilon_\infty\big(x - \alpha\big) \neq \upsilon_Q\big(x - \alpha\big) = e\big(Q \mid P_\alpha\big) \cdot \upsilon_\alpha\big(x - \alpha\big) = e\big(Q \mid P_\alpha\big) \geq 1.$$

Because $\deg(Q) = 1$ and Q is not the pole, Q is determined by its coordinates. Let δ be coordinate x of Q. A rational function $a(x) \in \mathcal{O}_\delta$ is defined in δ. Thus $a(x) \in \mathcal{O}_Q$, $\mathcal{O}_\delta \subseteq \mathcal{O}_Q$ and $\mathcal{O}_\delta \subseteq \mathcal{O}_\alpha$. By Theorem 1.7 $\mathcal{O}_\delta = \mathcal{O}_\alpha$ and $\delta = \alpha$. Prove analogically for coordinate y.

$\square$

Lemma 2.5. *Let $P \in \mathbb{P}_E$, $P_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ and $P_\beta \in \mathbb{P}_{K(y)}^{(1)}$ where $\alpha, \beta \in K$. The valuation $\upsilon_P(x - \alpha) > 0$ if and only if $P \mid P_\alpha$. The valuation $\upsilon_P(y - \beta) > 0$ if and only if $P \mid P_\beta$.*

Proof. If $P \mid P_\alpha$ then $\upsilon_P(x - \alpha) = e(P \mid P_\alpha) \cdot \upsilon_\alpha(x - \alpha) = e(P \mid P_\alpha) > 0$.

Prove the other implication. By Theorem 1.103 there exists an irreducible polynomial $p(x) \in K[x]$ such that $P \cap K[x] = \big(p(x)\big)$. In other words, $\upsilon_P\big(a(x)\big) > 0$ for $a(x) \in K[x]$ if and only if $a(x) \in \big(p(x)\big)$. Thus $x - \alpha \in \big(p(x)\big)$. Thus $x - \alpha = c \cdot p(x)$ for $c \in K^\star$. Thus $P \cap K[x] = \big((x - \alpha)\big)$. Recall that a place $Q \in \mathbb{P}_{K(x)}$ is determined by an irreducible polynomial $q(x) \in K[x]$ if and only if $Q \cap K[x] = \big(q(x)\big)$. Put $P_x := P \cap K(x)$. Thus $P_x \cap K[x] = \big(x - \alpha\big)$. At the end, $P_x = P_\alpha$ and $P \mid P_\alpha$.

Prove analogically for coordinate y. $\square$

Lemma 2.6. *Let $P_1 := P_{(\alpha,\beta)}, P_2 := P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ where $\alpha, \beta, \gamma \in K$. Assume that $P_1 \neq P_2$. In other words, $\beta \neq \gamma$. If $\operatorname{char} K \neq 2$ then $\gamma = -\beta$. If $\operatorname{char} K = 2$ then β and γ are different roots of $g_i(\alpha, y) - f_i(\alpha)$. The places P_1, P_2 are extensions of $P_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ and $e(P_1 \mid P_\alpha) = e(P_2 \mid P_\alpha) = f(P_1 \mid P_\alpha) = f(P_2 \mid P_\alpha) = 1$.*

Proof. Because P_1 and P_2 are of degree one, their coordinates are zeros of a Weierstrass polynomial. Thus $\beta = -\gamma$ in $\operatorname{char} K \neq 2$ and β, γ are roots of $g_i(\alpha, y) - f_i(\alpha)$ in $\operatorname{char} K = 2$. By Lemma 2.4 $P_1 \mid P_\alpha$ and $P_2 \mid P_\alpha$. By Theorem 1.71 $2 = [E : K(x)] = e(P_1 \mid P_\alpha) \cdot f(P_1 \mid P_\alpha) + e(P_2 \mid P_\alpha) \cdot f(P_2 \mid P_\alpha)$. Thus $e(P_1 \mid P_\alpha) = e(P_2 \mid P_\alpha) = f(P_1 \mid P_\alpha) = f(P_2 \mid P_\alpha) = 1$. $\square$

Lemma 2.7. *Let $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ where $\alpha, \beta \in K$. Assume that there exists no other place $P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$. If $\operatorname{char} K \neq 2$ then it happens if and only if $\beta = 0$ and α is a root of $f(x)$. If $\operatorname{char} K = 2$ then it happens if and only if $w(x, y) = y^2 + xy - f_1(x)$, $\alpha = 0$ and $\beta = \sqrt{a_6}$. The place P is an extension of $P_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ and $e(P \mid P_\alpha) = 2$, $f(P \mid P_\alpha) = 1$. The place P is the only extension of P_α.*

Proof. The condition that there is no other place $P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ is equivalent that $w(\alpha, y)$ has only one root. If $\operatorname{char} K \neq 2$ then it means that $y^2 - f(\alpha)$ has only one root. It occurs if and only if $f(\alpha) = 0$ and $\beta = 0$. Look at the situation in $\operatorname{char} K = 2$. Recall that E is determined by a smooth short Weierstrass equation. Define polynomials

(i) $h_1(y) := w(\alpha, y) = g_1(\alpha, y) - f_1(\alpha) = y^2 + \alpha y - f(\alpha)$;

(ii) $h_2(y) := w(\alpha, y) = g_2(\alpha, y) - f_2(\alpha) = y^2 + a_3 y - f(\alpha)$.

Polynomials $h_1(y)$ and $h_2(y)$ have only one root if and only if the coefficient of monomial y is zero, because $h_i(y) = (y-\beta)^2 = y^2+\beta^2$. It means $a_3 = 0$ in $h_2(y)$. It is a contradiction with the smoothness of $w(x, y) = y^2 + a_3 y - x^3 - a_4 x - a_6$ by Theorem 1.112. Thus there always exists the other place $P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$. Look at h_1(y), $\alpha = 0$. Thus $h_1(y) = y^2 - f(0) = y^2 - a_6$ and $\beta = \sqrt{a_6}$.

By Lemma 2.4 $P \mid P_\alpha$.

Show that there exists no other place $Q \in \mathbb{P}_E$ such that $Q \mid P_\alpha$. Prove by a contradiction. Assume that Q exists and $\deg(Q) > 1$. Because if $\deg(Q) = 1$ then $Q = P_{(\alpha,\gamma)}$ by Lemma 2.4. It is a contradiction with the choice of P. By Theorem 1.70

$$1 < \deg(Q) = f(Q \mid P_\alpha) \cdot \deg(P_\alpha) = f(Q \mid P_\alpha).$$

By Theorem 1.71

$$2 = [E : K(x)] = e(P \mid P_\alpha) \cdot f(P \mid P_\alpha) + e(Q \mid P_\alpha) \cdot f(Q \mid P_\alpha) \geq 1 + 2 = 3.$$

It is the contradiction.

By Theorem 1.70

$$1 = \deg(P) = \deg(P_\alpha) \cdot f(P \mid P_\alpha) = f(P \mid P_\alpha).$$

By Theorem 1.71

$$2 = [E : K(x)] = e(P \mid P_\alpha) \cdot f(P \mid P_\alpha) = e(P \mid P_\alpha).$$

□

Lemma 2.8. *Let $\operatorname{char} K = 2$ and $w(x, y) = y^2 + x \cdot y - f_1(x)$. Then $P_{(0,\sqrt{a_6})} \in \mathbb{P}_E^{(1)}$ and $\upsilon_{(0,\sqrt{a_6})}(y - \sqrt{a_6}) = e(P_{(0,\sqrt{a_6})} \mid P_{\sqrt{a_6}}) = 1$ where $P_{\sqrt{a_6}} \in \mathbb{P}_{K(y)}^{(1)}$.*

Proof. Because K is perfect and $\operatorname{char} K = 2$ then $\sqrt{a_6} \in K$. Evaluate $w(0, \sqrt{a_6}) = \sqrt{a_6}^2 + 0 \cdot \sqrt{a_6} - f_1(0) = a_6 + a_6 = 0$. Thus $P_{(0,\sqrt{a_6})} \in \mathbb{P}_E^{(1)}$ by Theorem 1.103.

Calculate partial derivations of $w(x, y)$, evaluate the derivations by $(0, \sqrt{a_6})$ and find the tangent of $w(x, y)$ at point $(0, \sqrt{a_6})$. The partial derivation

$$\frac{\partial w}{\partial x}(x, y) = y - 3 \cdot x^2 - 2 \cdot a_2 \cdot x = y - x^2; \qquad \frac{\partial w}{\partial y}(x, y) = 2 \cdot y + x = x;$$
$$\frac{\partial w}{\partial x}(0, \sqrt{a_6}) = \sqrt{a_6}; \qquad \frac{\partial w}{\partial y}(0, \sqrt{a_6}) = 0.$$

The tangent of $w(x, y)$ at point $(0, \sqrt{a_6})$ is $x = 0$. Because $x = 0$ is the tangent then $\upsilon_{(0,\sqrt{a_6})}(x) = e(P_{(0,\sqrt{a_6})} \mid P_0) \geq 2$ by Lemma 1.115. By Theorem 1.71 $e(P_{(0,\sqrt{a_6})} \mid P_0)$ can be 1 or 2. Thus $e(P_{(0,\sqrt{a_6})} \mid P_0) = 2$. □

Lemma 2.9. *Let* $\operatorname{char} K = 2$ *and* $w(x, y) = g_1(x, y) - f_1(x)$. *Then* $f_1(x)$ *does not have a double root if and only if* $w(x, y)$ *is smooth.*

Proof. By Theorem 1.112 $w(x, y)$ is smooth if and only if $a_6 \neq 0$. Derive $f_1'(x) = 3 \cdot x^2 + 2 \cdot a_2 \cdot x = x^2$. Zero is the only root of $f_1'(x)$ and it is the only candidate to a double root of $f_1(x)$. We have that $f_1(0) = 0 \iff a_6 = 0$. □

Lemma 2.10. *Let* $\operatorname{char} K = 2$ *and* $w(x, y) = g_2(x, y) - f_2(x)$. *Then* $f_2(x)$ *does not have a double root if and only if* $a_6 \neq 0$.

Proof. The polynomial $f_2(x)$ with $a_6 = 0$ can be decompose $f_2(x) = x^3 + a_4 \cdot x = x \cdot (x + \sqrt{a_4}) \cdot (x + \sqrt{a_4})$. Let $f_2(x) = (x + \alpha)^2 \cdot (x + \beta) = (x^2 + \alpha^2) \cdot (x + \beta) = x^3 + \beta \cdot x^2 + \alpha^2 \cdot x + \alpha^2 \cdot \beta$. Continue by that $0 = a_2 = \beta$. Thus $a_6 = \alpha^2 \cdot \beta = 0$. □

Lemma 2.11. *Let one from the following cases be satisfied*

(i) $\operatorname{char} K \neq 2$ *and* $w(x, y) = y^2 - f(x)$;

(ii) $\operatorname{char} K = 2$ *and* $w(x, y) = g_1(x, y) - f_1(x)$;

(iii) $\operatorname{char} K = 2$ *and* $w(x, y) = g_2(x, y) - f_2(x)$ *and* $a_6 \neq 0$.

Let $P := P_{(\alpha,0)} \in \mathbb{P}_E^{(1)}$ *where* $\alpha \in K$. *Then* $P \mid P_0$, $e(P \mid P_0) = 1$ *and* $f(P \mid P_0) = 1$ *where* $P_0 \in \mathbb{P}_{K(y)}^{(1)}$ *is such that* $y \in P_0$.

Proof. By Lemma 2.4 $P \mid P_0$. By Lemma 1.70

$$1 = \deg(P) = \deg(P_0) \cdot f(P \mid P_0) = f(P \mid P_0).$$

Calculate partial derivations of $w(x, y)$, evaluate the derivations by $(\alpha, 0)$ and find the tangent at point $(\alpha, 0)$.

1. Assume that $\operatorname{char} K \neq 2$ and $w(x, y) = y^2 - f(x)$. Then

$$\frac{\partial w}{\partial x}(x, y) = -f'(x); \qquad \frac{\partial w}{\partial y}(x, y) = 2 \cdot y;$$
$$\frac{\partial w}{\partial x}(\alpha, 0) = -f'(\alpha); \qquad \frac{\partial w}{\partial y}(\alpha, 0) = 0.$$

By Theorem 1.110 $f'(\alpha) \neq 0$. The tangent at point $(\alpha, 0)$ is

$$-f'(\alpha) \cdot x + 0 \cdot y + f'(\alpha) \cdot \alpha = 0;$$
$$x - \alpha = 0.$$

Thus the line $y = 0$ is a secant at point $(\alpha, 0)$. Thus $\upsilon_P(y) = 1$ by Lemma 1.115.

2. Assume that $\operatorname{char} K = 2$ and $w(x, y) = g_1(x, y) - f_1(x)$. By Theorem 1.112 $a_6 \neq 0$ and $\alpha \neq 0$. Then

$$\frac{\partial w}{\partial x}(x, y) = y - 3 \cdot x^2 - 2 \cdot a_2 \cdot x = y - x^2; \quad \frac{\partial w}{\partial y}(x, y) = 2 \cdot y + x = x;$$
$$\frac{\partial w}{\partial x}(\alpha, 0) = -\alpha^2; \quad \frac{\partial w}{\partial y}(\alpha, 0) = \alpha.$$

The tangent at point $(\alpha, 0)$ is

$$-\alpha^2 \cdot x + \alpha \cdot y + \alpha^3 = 0;$$
$$-\alpha \cdot x + y + \alpha^2 = 0.$$

The line $y = 0$ cannot be the tangent at point $(\alpha, 0)$ because $\alpha \neq 0$. Thus the line $y = 0$ is a secant at point $(\alpha, 0)$. Thus $\upsilon_P(y) = 1$ by Lemma 1.115.

3. Assume that $\operatorname{char} K = 2$ and $w(x, y) = g_2(x, y) - f_2(x)$. Then

$$\frac{\partial w}{\partial x}(x, y) = -3 \cdot x^2 - a_4 = -x^2 - a_4; \quad \frac{\partial w}{\partial y}(x, y) = 2 \cdot y + a_3 = a_3;$$
$$\frac{\partial w}{\partial x}(\alpha, 0) = -\alpha^2 - a_4; \quad \frac{\partial w}{\partial y}(\alpha, 0) = a_3.$$

The tangent at point $(\alpha, 0)$ is

$$-(\alpha^2 + a_4) \cdot x + a_3 \cdot y + \alpha \cdot (\alpha^2 + a_4) = 0.$$

If $\alpha = \sqrt{a_4}$ then the line $y = 0$ is the tangent at point $(\alpha, 0)$. But if $(\sqrt{a_4}, 0)$ is a zero of $w(x, y)$ then $0 = 0^2 + a_3 \cdot 0 - a_4^{\frac{3}{2}} - a_4^{\frac{3}{2}} - a_6 = a_6$. It is a contradiction with the assumption. Thus the line $y = 0$ is a secant at point $(\alpha, 0)$. Thus $\upsilon_P(y) = 1$ by Lemma 1.115.

A uniformizing element of P_0 is y. By Definition 1.67 $e(P \mid P_0) = \upsilon_P(y) = 1$. □

Remark 2.12. Lemma 2.13 describes the missing cases in Lemma 2.11.

Lemma 2.13. *Consider that* $\operatorname{char} K = 2$, $w(x, y) = g_2(x, y) - f_2(x)$ *and* $a_6 = 0$. *Let* $P_{(\alpha,0)} \in \mathbb{P}_E^{(1)}$ *where* $\alpha \in K$. *Let* $P_0 \in \mathbb{P}_{K(y)}^{(1)}$ *be such that* $y \in P_0$. *Then* α *is* 0 *or* $\sqrt{a_4}$. *Put* $P_1 := P_{(\sqrt{a_4},0)}$ *and* $P_2 := P_{(0,0)}$. *Then*

(i) $P_1 \mid P_0$ *and* $P_2 \mid P_0$;

(ii) $f(P_1 \mid P_0) = 1$ *and* $f(P_2 \mid P_0) = 1$;

(iii) *if* $a_4 \neq 0$ *then* $e(P_2 \mid P_0) = 1$ *and* $e(P_1 \mid P_0) = 2$;

(iv) *if* $a_4 = 0$ *then* $P_1 = P_2$, $e(P_1 \mid P_0) = 3$ *and* $e(P_2 \mid P_0) = 3$.

Proof. By Lemma 2.4 $P_i \mid P_0$. Decompose $f_2(x) = x \cdot (x^2 + a_4) = x \cdot (x + \sqrt{a_4})^2$. If point $(\alpha, 0)$ is a zero of $w(x, y)$ then α is a root of $f_2(x)$. Thus $\alpha = 0$ or $\alpha = \sqrt{a_4}$. By Lemma 1.70 $1 = \deg(P_i) = \deg(P_0) \cdot f(P_i \mid P_0) = f(P_i \mid P_0)$. Recall that

(i) $[E : K(y)] = 3 = \deg\big((y)_-\big) = \deg\big((y)_+\big)$ (cf. Theorem 1.99);

(ii) $\frac{\partial w}{\partial x}(x, y) = -x^2 - a_4, \quad \frac{\partial w}{\partial y}(x, y) = a_3.$

The proof is split to two cases.

(i) Assume that $a_4 \neq 0$. Then $\frac{\partial w}{\partial x}(0, 0) = a_4$ and $\frac{\partial w}{\partial x}(\sqrt{a_4}, 0) = 0$. The tangent at point $(0, 0)$ is $a_4 \cdot x + a_3 \cdot y = 0$. Thus $y = 0$ is a secant at point $(0, 0)$ and $\upsilon_{P_2}(y) = 1$ by Lemma 1.115. The tangent at point $(\sqrt{a_4}, 0)$ is $a_3 \cdot y = 0$. Thus $y = 0$ is the tangent at point $(\sqrt{a_4}, 0)$ and $\upsilon_{P_1}(y) = n \geq 2$ by Lemma 1.115. A uniformizing element of P_0 is y. By Definition 1.67 $e(P_1 \mid P_0) = \upsilon_{P_1}(y) = n$ and $e(P_2 \mid P_0) = \upsilon_{P_2}(y) = 1$. Continue by that

$$\big((y)_+\big) = 1 \cdot P_2 + n \cdot P_1 + A \text{ where } A \in \mathrm{Div}(E) \text{ such that } A \geq 0;$$

$$3 = \deg\big((y)_+\big) = 1 \cdot \deg\big(P_2\big) + n \cdot \deg\big(P_1\big) + \deg(A) \geq 1 + n.$$

Thus $n \leq 2$. Thus $2 = n = e(P_1 \mid P_0)$.

(ii) Assume that $a_4 = 0$. Then $P_1 = P_2$. The line $y = 0$ is the tangent at point $(0, 0)$. Thus $\upsilon_{P_i}(y) = n \geq 2$ by Lemma 1.115. A high bound of n is 3 because

$$3 = \deg\big((y)_+\big) \geq n \cdot \deg(P_i) = n.$$

Show that there exists no other place Q such that $\upsilon_Q(y) = m > 0$. Assume that Q exists. The degree $\deg(Q) = 1$ because

$$3 = \deg\big((y)_+\big) \geq n \cdot \deg(P_2) + m \cdot \deg(Q) = n + m \cdot \deg(Q) \geq 2 + m \cdot \deg(Q);$$

$$1 \geq m \cdot \deg(Q);$$

$$m = 1 \text{ and } \deg(Q) = 1.$$

The place Q is $P_{(\beta,0)}$ where β has to be a root of $f_2(x)$ by Lemmas 2.5, 2.4 and $\upsilon_Q(y) = m = 1$. But the only root of $f_2(x)$ is 0. Thus $Q = P_i$. At the end, $n = 3 = e(P_i \mid P_0)$.

□

Lemma 2.14. *Let* $\operatorname{char} K = 2$ *and* $P_1 := P_{(\alpha,0)} \in \mathbb{P}_E^{(1)}$ *where* $\alpha \in K$. *Then there always exists* $P_2 := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ *such that* $\beta \neq 0$. *If* $w(x, y) = y^2 + xy - f_1(x)$ *then* $P_2 = P_{(\alpha,\alpha)}$ *and* $\alpha = \beta$. *If* $w(x, y) = y^2 + a_3 y - f_2(x)$ *then* $P_2 = P_{(\alpha,a_3)}$ *and* $\beta = a_3$. *The value* α *is a root of* $f_i(x)$.

Proof. It holds

(i) $g_i(\alpha, 0) - f_i(\alpha) = 0$;

(ii) $g_1(\alpha, 0) = 0^2 + \alpha \cdot 0 = 0$;

(iii) $g_2(\alpha, 0) = 0^2 + a_3 \cdot 0 = 0$.

Thus $f_i(\alpha) = 0$. Our goal is shown that there exists the other root of $g_i(\alpha, y) - f_i(\alpha)$ which is not zero.

(i) Let $w(x,y) = y^2 + xy - x^3 - a_2x^2 - a_6$. The polynomial $g_1(\alpha, y) - f_1(\alpha)$ has root zero. Thus

$$y \cdot (y - \beta) = y^2 - \beta \cdot y = y^2 + \alpha \cdot y - f_1(\alpha) = g_1(\alpha, y) - f_1(\alpha).$$

Thus $\beta = \alpha$. If $\beta = \alpha = 0$ then $0 = w(0,0) = a_6$ and $w(x,y)$ is not smooth by Theorem 1.112.

(ii) Let $w(x,y) = y^2 + a_3y - x^3 - a_4x - a_6$. The polynomial $g_2(\alpha, y) - f_2(\alpha)$ has root zero. Thus

$$y \cdot (y - \beta) = y^2 - \beta \cdot y = y^2 + a_3 \cdot y - f_2(\alpha) = g_1(\alpha, y) - f_1(\alpha).$$

Thus $\beta = a_3$. If $a_3 = 0$ then $w(x,y)$ is not smooth by Theorem 1.112.

□

Definition 2.15. Let $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$.

(i) If there exists $P \neq P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ and $\beta \neq 0$ then P is said to be of *Type 1*;

(ii) If $\operatorname{char} K \neq 2$ and $\beta = 0$ then P is said to be of *Type 2*;

(iii) If $\operatorname{char} K = 2$, $w(x,y) = g_1(x,y) - f_1(x)$, $\alpha = 0$ and $\beta = \sqrt{a_6}$ then P is said to be of *Type 3*;

(iv) If $\operatorname{char} K = 2$, $\beta = 0$ and

  (i) $w(x,y) = g_1(x,y) - f_1(x)$ or
  (ii) $w(x,y) = g_2(x,y) - f_2(x)$ and $a_6 \neq 0$ or
  (iii) $w(x,y) = g_2(x,y) - f_2(x)$, $\alpha = 0$, $a_4 \neq 0$ and $a_6 = 0$

then P is said to be of *Type 4*;

(v) If $\operatorname{char} K = 2$, $w(x,y) = g_2(x,y) - f_2(x)$, $a_4 \neq 0$, $a_6 = 0$, $\alpha = \sqrt{a_4}$ and $\beta = 0$ then P is said to be of *Type 5*;

(vi) If $\operatorname{char} K = 2$, $w(x,y) = g_2(x,y) - f_2(x)$, $a_4 = 0$, $a_6 = 0$, $\alpha = 0$ and $\beta = 0$ then P is said to be of *Type 6*.

Theorem 2.16. *Let $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$, $Q_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ and $R_\beta \in \mathbb{P}_{K(y)}^{(1)}$. Then $P_{(\alpha,\beta)} \mid Q_\alpha$ and $P_{(\alpha,\beta)} \mid R_\beta$ (cf. Lemma 2.4).*

(i) If P is of Type 1 then $e(P \mid Q_\alpha) = f(P \mid Q_\alpha) = 1$ (cf. Lemma 2.6).

(ii) If P is of Type 2 then

  (a) there exists no $\bar{P} \in \mathbb{P}_E$ such that $\bar{P} \mid Q_\alpha$ (cf. Lemma 2.7);
  (b) $e(P \mid Q_\alpha) = 2$ and $f(P \mid Q_\alpha) = 1$ (cf. Lemma 2.7);
  (c) $e(P \mid R_\beta) = 1$ (cf. Lemma 2.11).

(iii) If P is of Type 3 then

  (a) there exists no $\bar{P} \in \mathbb{P}_E$ such that $\bar{P} \mid Q_\alpha$ (cf. Lemma 2.7);

(b) $e(P \mid Q_\alpha) = 2$ and $f(P \mid Q_\alpha) = 1$ (cf. Lemma 2.7).

(iv) If P is of Type 4 then

(a) there exists $P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ such that $\gamma \neq 0$ (cf. Lemma 2.14);
(b) $e(P \mid Q_\alpha) = f(P \mid Q_\alpha) = 1$ (cf. Lemma 2.6);
(c) $e(P \mid R_\beta) = f(P \mid R_\beta) = 1$ (cf. Lemmas 2.11 and 2.13).

(v) If P is of Type 5 then

(a) there exists $P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ such that $\gamma \neq 0$ (cf. Lemma 2.14);
(b) $e(P \mid Q_\alpha) = f(P \mid Q_\alpha) = 1$ (cf. Lemma 2.6);
(c) $e(P \mid R_\beta) = 2$ and $f(P \mid R_\beta) = 1$ (cf. Lemma 2.13).

(vi) If P is of Type 6 then

(a) there exists $P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ such that $\gamma \neq 0$ (cf. Lemma 2.14);
(b) $e(P \mid Q_\alpha) = f(P \mid Q_\alpha) = 1$ (cf. Lemma 2.6);
(c) $e(P \mid R_\beta) = 3$ and $f(P \mid R_\beta) = 1$ (cf. Lemma 2.13).

Proof. It is a consequence of Lemmas 2.4, 2.6, 2.7, 2.11, 2.13 and 2.14. □

Remark 2.17. Definition 2.15 splits up $\mathbb{P}_E^{(1)} \setminus \{P_\infty\}$ to six types according to the ramification index to axe x and the ramification index to axe y if coordinate y is zero. Our effort is to characterize when $\upsilon_P(\frac{y}{(x-\alpha)^n})$ is negative. It is needed to know $e(P \mid R_\beta)$ if and only if $\beta = 0$ because $\upsilon_{P_\beta}(y) = 1$ if and only if $\beta = 0$.

These six types cover all $\mathbb{P}_E^{(1)} \setminus \{P_\infty\}$. It follows from previous lemmas.

Definition 2.18. Let $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$, $Q_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ and $R_\beta \in \mathbb{P}_{K(y)}^{(1)}$. Define

(i) $\kappa_P := 0$ if $\beta \neq 0$;

(ii) $\kappa_P := \left\lfloor \frac{e(P|R_\beta)}{e(P|Q_\alpha)} \right\rfloor$ if $\beta = 0$.

Lemma 2.19. *Let $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$, $Q_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ and $R_\beta \in \mathbb{P}_{K(y)}^{(1)}$. The value κ_P depends only on the type of P.*

(i) If P is of Type 1 then $\kappa_P = 0$;

(ii) If P is of Type 2 then $\kappa_P = \left\lfloor \frac{e(P|R_\beta)}{e(P|Q_\alpha)} \right\rfloor = \left\lfloor \frac{1}{2} \right\rfloor = 0$;

(iii) If P is of Type 3 then $\kappa_P = 0$;

(iv) If P is of Type 4 then $\kappa_P = \left\lfloor \frac{e(P|R_\beta)}{e(P|Q_\alpha)} \right\rfloor = \left\lfloor \frac{1}{1} \right\rfloor = 1$;

(v) If P is of Type 5 then $\kappa_P = \left\lfloor \frac{e(P|R_\beta)}{e(P|Q_\alpha)} \right\rfloor = \left\lfloor \frac{2}{1} \right\rfloor = 2$;

(vi) If P is of Type 6 then $\kappa_P = \left\lfloor \frac{e(P|R_\beta)}{e(P|Q_\alpha)} \right\rfloor = \left\lfloor \frac{3}{1} \right\rfloor = 3$.

Proof. It is a consequence of Theorem 2.16. ☐

Lemma 2.20. *Let* $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$. *Then*

$$n > \kappa_P \iff \upsilon_P\Big(\frac{y}{(x-\alpha)^n}\Big) < 0.$$

Proof. Let $Q_\alpha \in \mathbb{P}^{(1)}_{K(x)}$ and $R_\beta \in \mathbb{P}^{(1)}_{K(y)}$. By Lemma 2.4 $P \mid Q_\alpha$ and $P \mid R_\beta$. Calculate

$$\begin{aligned}\upsilon_P\Big(\frac{y}{(x-\alpha)^n}\Big) &= e(P \mid R_\beta)\cdot \upsilon_{R_\beta}(y) - n\cdot e(P \mid Q_\alpha)\cdot \upsilon_{Q_\alpha}(x-\alpha)\\ &= e(P \mid R_\beta)\cdot \upsilon_{R_\beta}(y) - n\cdot e(P \mid Q_\alpha).\end{aligned}$$

If $\beta \neq 0$ then $\kappa_P = 0$ and $\upsilon_{R_\beta}(y) = 0$. Thus

$$\upsilon_P\Big(\frac{y}{(x-\alpha)^n}\Big) = -n\cdot e(P \mid Q_\alpha) < 0 \iff n > 0 = \kappa_P.$$

If $\beta = 0$ then $\upsilon_{R_\beta}(y) = 1$. Thus

$$\begin{aligned}\upsilon_P\Big(\frac{y}{(x-\alpha)^n}\Big) &= e(P \mid R_\beta) - n\cdot e(P \mid Q_\alpha) < 0\\ \iff n > \frac{e(P \mid R_\beta)}{e(P \mid Q_\alpha)} &\iff n > \left\lfloor \frac{e(P \mid R_\beta)}{e(P \mid Q_\alpha)} \right\rfloor = \kappa_P.\end{aligned}$$

☐

3. How to calculate local components of Weil differential

Let E/K be an elliptic function field and $u \in E$. In this chapter there is a calculation of a local component $\omega_P(u)$ where $P \in \mathbb{P}_E^{(1)}$ and $\omega \in \Omega_E$ such that $(\omega) = 0$. First, focus on ω_∞. Theorems 3.8 and 3.6 describe exactly $\omega_\infty(u)$.

It is needed to know in which places $P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ $\omega_{(\alpha,\beta)}(u)$ can be non-zero. Thus there are necessary conditions to determine them. Next, develop algorithms to calculate $\omega_{(\alpha,\beta)}(u)$ for $P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$.

At the end there is a tweak of the calculation of $\omega_{(\alpha,\beta)}(u)$ if $\operatorname{char} K \neq 2$.

Convention 3.1. *Throughout this chapter consider that*

(i) an elliptic function field E/K is determined by a short smooth Weierstrass equation $w(x, y) = 0$;

(ii) K is perfect and $K = \tilde{K}$;

(iii) $0 = W \in \mathrm{Div}(E)$;

Throughout this chapter keep the notations from Conventions 1.109, 1.113 and 1.77 and Theorem 1.76.

Lemma 3.2. *The divisor W is a canonical divisor.*

Proof. It is a consequence of Theorem 1.57. Calculate

(i) $\deg(W) = 0 = 2g - 2$,

(ii) $l(W) = 1 \geq g$.

□

Remark 3.3. Recall the following facts.

(i) There exists $\omega \in \Omega_E$ such that $(\omega) = W$.

(ii) Let $\alpha \in \mathcal{A}_E$. By Theorem 1.59 there exists a finite subset $S \subseteq \mathbb{P}$ such that

  (a) $\omega_P(\alpha) \neq 0 \iff P \in S$;

  (b) $\omega(\alpha) = \sum_{P \in \mathbb{P}} \omega_P(\alpha) = \sum_{P \in S} \omega_P(\alpha)$.

(iii) By Definition 1.45 $\omega(u) = 0$ where $u \in E$.

(iv) By Theorem 1.116 every $u \in E$ can be expressed as $u = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x) \cdot y}{c_2(x)}$ where $a_1(x), b_2(x), c_1(x), c_2(x) \in K[x]$.

(v) By Theorem 1.99 $\upsilon_\infty(x) = -2$ and $\upsilon_\infty(y) = -3$.

(vi) By Theorem 1.60 if $\upsilon_P(u) \geq 0 = -W_P$ then $\omega_P(u) = 0$ where $u \in E$ and $P \in \mathbb{P}$.

(vii) By Theorem 1.103 every $P \in \mathbb{P}^{(1)} \setminus \{P_\infty\}$ is fully determined its affine coordinates. Thus $P = P_{(\alpha,\beta)}$ where $(\alpha, \beta) \in \mathbb{V}_w(K)$.

Theorem 3.4. *There exists only one Weil differential $\omega \in \Omega_E$ with the following properties*

(i) $(\omega) = W$;

(ii) $\omega_\infty(\frac{y}{x}) = -1$.

Proof. The divisor W determines uniquely $\omega \in \Omega_E$ up to multiplication by $c \in K^\star$ by Lemma 1.54. The local component $\omega_\infty\big(\frac{y}{x}\big) = \lambda \neq 0$ by Lemma 1.61. Thus there exist only one $\omega \in \Omega_E$ such that $(\omega) = W$ and $\omega_\infty(\frac{y}{x}) = -1$. □

Convention 3.5. *Throughout this chapter fix ω from Theorem 3.4.*

Theorem 3.6. *If $u \in K[x, y]$ or $u \in K(x)$ then $\omega_\infty(u) = 0$.*

Proof. (i) Prove the case $u \in K[x, y]$. If $P \in \mathbb{P}_E \setminus \{P_\infty\}$ then $\upsilon_P(u) \geq 0 = -W_P$. Thus $\omega_P(u) = 0$ by Theorem 1.60. By Definition 1.45 $\omega(u) = 0$. By Theorem 1.59 $\omega_\infty(u) = -\sum_{P \neq P_\infty} \omega_P(u) = 0$.

(ii) Prove the case $u \in K(x)$. Express $u = \frac{a(x)}{b(x)}$ where $a(x), b(x) \in K[x]$. If $\deg(a) \leq \deg(b)$ then $\upsilon_\infty(\frac{a}{b}) = 2\big(\deg(b) - \deg(a)\big) \geq 0$. Thus $\omega_\infty(u) = 0$ by Theorem 1.60. If $\deg(a) > \deg(b)$ then $a(x) = b(x) \cdot k(x) + r(x)$ where $k(x), r(x) \in K[x]$ and $\deg(r) < \deg(b)$. We have $\frac{a(x)}{b(x)} = k(x) + \frac{r(x)}{b(x)}$. Calculate $\omega_\infty(\frac{a}{b}) = \omega_\infty(k) + \omega_\infty(\frac{r}{b}) = 0 + 0 = 0$. By the first case $\omega_\infty(k) = 0$. Because $\deg(r) < \deg(b)$ then $\omega_\infty(\frac{r}{b}) = 0$.

□

Remark 3.7. By Theorem 1.116 every $u \in E$ can be expressed $u = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x) \cdot y}{c_2(x)}$ where $a_1(x), b_2(x), c_1(x), c_2(x) \in K[x]$ such that $\gcd(a_1, c_1) = 1$ and $\gcd(b_2, c_2) = 1$. Calculate

$$\omega_\infty(u) = \omega_\infty\Big(\frac{a_1(x)}{c_1(x)} + \frac{b_2(x) \cdot y}{c_2(x)}\Big) = \omega_\infty\Big(\frac{a_1(x)}{c_1(x)}\Big) + \omega_\infty\Big(\frac{b_2(x) \cdot y}{c_2(x)}\Big) = \omega_\infty\Big(\frac{b_2(x) \cdot y}{c_2(x)}\Big).$$

It is needed to describe how ω_∞ behaves on $\frac{b_2(x) \cdot y}{c_2(x)}$.

Theorem 3.8. *Let $u = \frac{b(x)y}{c(x)} \in E$ where $b(x), c(x) \in K[x]$ such that $\deg(b) \geq 0$, $\deg(c) \geq 0$ and $\gcd\big(b(x), c(x)\big) = 1$. Put $\beta := \deg(b) \geq 0$ and $\gamma := \deg(c) \geq 0$. The following statements are satisfied.*

(i) If $\gamma > \beta + 1$ then $\omega_\infty(u) = 0$.

(ii) If $\gamma = \beta + 1$ then $\omega_\infty(u) = -\frac{\mathrm{lc}(b)}{\mathrm{lc}(c)}$.

(iii) If $\gamma < \beta + 1$ then $b(x) = k(x) \cdot c(x) + r(x)$ where $r(x), k(x) \in K[x]$ and $\deg(r) < \gamma$. Now, $\omega_\infty(u) = \omega_\infty\big(\frac{r(x) \cdot y}{c(x)}\big)$.

Proof. Calculate $\upsilon_\infty(u) = -2 \cdot \beta - 3 + 2 \cdot \gamma = 2(\gamma - \beta - 1) - 1$.

(i) The valuation $v_\infty(u) = 2(\gamma-\beta-1)-1 > 2(\beta+1-\beta-1)-1 = -1$. Thus $v_\infty(u) \geq 0 = -W_\infty$. Thus $\omega_\infty(u) = 0$ by Theorem 1.60.

(ii) Define monic polynomials $\tilde{b}(x) := \frac{b(x)}{\mathrm{lc}(b)}$ and $\tilde{c}(x) := \frac{c(x)}{\mathrm{lc}(c)}$. Recall that $\beta = \deg(\tilde{b})$ and $\gamma = \deg(\tilde{c})$. Now,

$$\begin{aligned}
v_\infty\Big(\frac{y}{x} - \frac{\tilde{b}(x)\cdot y}{\tilde{c}(x)}\Big) &= v_\infty\Big(\frac{y\cdot(\tilde{c}(x) - x\cdot\tilde{b}(x))}{x\cdot\tilde{c}(x)}\Big)\\
&= -3 - 2\deg\Big(\tilde{c}(x) - x\cdot\tilde{b}(x)\Big) + 2 + 2\cdot\gamma\\
&\geq -1 - 2\cdot\beta + 2\cdot\gamma\\
&= -1 - 2\cdot\beta + 2\cdot(\beta+1) = 1 > -W_\infty = 0.
\end{aligned}$$

By Theorem 1.60 $\omega_\infty\big(\frac{y}{x} - \frac{\tilde{b}(x)\cdot y}{\tilde{c}(x)}\big) = 0$. Thus $\omega_\infty\big(\frac{\tilde{b}(x)\cdot y}{\tilde{c}(x)}\big) = \omega_\infty\big(\frac{y}{x}\big) = -1$. Calculate

$$\omega_\infty\Big(\frac{b(x)\cdot y}{c(x)}\Big) = \omega_\infty\Big(\frac{\mathrm{lc}(b)\cdot\tilde{b}(x)\cdot y}{\mathrm{lc}(c)\cdot\tilde{c}(x)}\Big) = \frac{\mathrm{lc}(b)}{\mathrm{lc}(c)}\cdot\omega_\infty\Big(\frac{\tilde{b}(x)\cdot y}{\tilde{c}(x)}\Big) = -\frac{\mathrm{lc}(b)}{\mathrm{lc}(c)}.$$

(iii) We have

$$\frac{b(x)\cdot y}{c(x)} = \frac{\big(k(x)\cdot c(x) + r(x)\big)\cdot y}{c(x)} = k(x)\cdot y + \frac{r(x)\cdot y}{c(x)}.$$

By Theorem 3.6 $\omega_\infty(u) = \omega_\infty\big(k(x)\cdot y\big) + \omega_\infty\big(\frac{r(x)\cdot y}{c(x)}\big) = \omega_\infty\big(\frac{r(x)\cdot y}{c(x)}\big)$. Recall that $\gamma > \deg(r)$. Three cases can happen.

(a) The first one is $\gamma > \deg(r)+1$. Then

$$\omega_\infty(u) = \omega_\infty\Big(\frac{r(x)\cdot y}{c(x)}\Big) = 0.$$

(b) The second one is $\gamma = \deg(r)+1$. Then

$$\omega_\infty(u) = \omega_\infty\Big(\frac{r(x)\cdot y}{c(x)}\Big) = -\frac{\mathrm{lc}(r)}{\mathrm{lc}(b)}.$$

(c) The third one is $r(x) = 0$. Then

$$\omega_\infty(u) = \omega_\infty\Big(\frac{0\cdot y}{c(x)}\Big) = \omega_\infty(0) = 0.$$

□

Remark 3.9. (i) If $i \leq -2$ then $\omega_\infty(y\cdot x^i) = 0$.

(ii) If $i = -1$ then $\omega_\infty(y\cdot x^i) = -1$.

(iii) If $i \geq 0$ then $\omega_\infty(y\cdot x^i) = 0$.

Proof. (i) The local component $v_\infty(y\cdot x^i) = -3-2i \geq -3+4 = 1 > -W_\infty$. Thus $\omega_\infty(y\cdot x^i) = 0$ by Theorem 1.60.

(ii) If $i = -1$, it follows from the choice of ω.

(iii) By Theorem 3.6 $\omega_\infty(y \cdot x^i) = 0$.

$\square$

Example 3.10. Illustrate using of Theorems 3.6 and 3.8. In other words, calculate ω_∞. Consider the elliptic function field $E/\mathbb{F}_5$ determined by the Weierstrass polynomial $w(x, y) = y^2 - x^3 - x - 1$. It is smooth because $4 \cdot 1^3 + 27 \cdot 1^2 \equiv 31 \not\equiv 0 \mod 5$. Work on the following examples.

(i) Put

- $a(x) := x^2 + 2$;
- $b(x) := x^2 + 3$;
- $c(x) := (x - 3)^2 \cdot (x - 2) \cdot x = x^4 + 2x^3 + x^2 + 2x$;
- $A := \frac{a(x)+y \cdot b(x)}{c(x)} \in E$.

Calculate

$$\omega_\infty(A) = \omega_\infty\left(\frac{x^2 + 2}{x^4 + 2x^3 + x^2 + 2x}\right)+\omega_\infty\left(\frac{y \cdot (x^2 + 3)}{x^4 + 2x^3 + x^2 + 2x}\right) = 0+0 = 0.$$

(ii) Put

- $a(x) := 3x^3$;
- $b(x) := 2x^3 + 1$;
- $c(x) := 4 \cdot (x - 4)^4 = 4x^4 + x^3 + 4x^2 + x + 4$;
- $B := \frac{a(x)+y \cdot b(x)}{c(x)} \in E$.

Calculate

$$\begin{aligned}\omega_\infty(B) =&\omega_\infty\left(\frac{3x^3}{4x^4 + x^3 + 4x^2 + x + 4}\right) + \omega_\infty\left(\frac{y \cdot (2x^3 + 1)}{4x^4 + x^3 + 4x^2 + x + 4}\right)\\ =&0 - \frac{2}{4} = -\frac{1}{2} = -3 = 2.\end{aligned}$$

(iii) Put

- $a(x) := x + 4$;
- $c(x) := 2 \cdot (x - 2) \cdot (x - 3) \cdot (x - 4) = 2x^3 + 2x^2 + 2x + 2$;
- $r(x) := 3x^2$;
- $k(x) := x^2 + 4$;
- $b(x) := k(x) \cdot c(x) + r(x) = 2x^5 + 2x^4 + 3x^2 + 3x + 3$;
- $C := \frac{a(x)+y \cdot b(x)}{c(x)} \in E$.

Calculate

$$\begin{aligned}
\omega_\infty(C) &= \omega_\infty\left(\frac{x+4}{2x^3+2x^2+2x+2}\right) + \omega_\infty\left(\frac{y\cdot(2x^5+2x^4+3x^2+3x+3)}{2x^3+2x^2+2x+2}\right)\\
&= \omega_\infty\left(\frac{y\cdot\Big(\big((x^2+4)\cdot(2x^3+2x^2+2x+2)\big)+(3x^2)\Big)}{2x^3+2x^2+2x+2}\right)\\
&= \omega_\infty\Big(y\cdot(x^2+4)\Big) + \omega_\infty\left(\frac{y\cdot 3x^2}{2x^3+2x^2+2x+2}\right)\\
&= 0-\frac{3}{2} = -3\cdot 3 = 2\cdot 3 = 1.
\end{aligned}$$

Remark 3.11. The calculation of $\omega_\infty(u)$ has already been known. Now, continue to develop necessary conditions when $\omega_P(u)$ can be non-zero for $u \in E$ and $P \in \mathbb{P}_E$.

Theorem 3.12. *Let* $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ *and* $P_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ *where* $\alpha,\beta \in K$. *Let* $u_1 = \frac{a_1(x)}{c_1(x)} \in E$ *and* $u_2 = \frac{b_2(x)\cdot y}{c_2(x)} \in E$ *where* $a_1(x), b_2(x), c_1(x), c_2(x) \in K[x]$ *such that* $\gcd(a_1,c_1)=1$ *and* $\gcd(b_2,c_2)=1$. *Then*

(i) $\upsilon_P(u_1) < 0 \iff c_1(\alpha)=0 \iff \upsilon_P(c_1) > 0 \iff \upsilon_\alpha(c_1) > 0$*;*

(ii) $\upsilon_P(u_2) < 0 \iff \upsilon_\alpha(c_2) > \kappa_P$*.*

Proof. By Lemma 2.4 $P \mid P_\alpha$.

(i) It follows from $\gcd(a_1,c_1)=1$.

(ii) Decompose $c_2(x) = (x-\alpha)^i\cdot\tilde{c_2}(x)$ where $\tilde{c_2}(\alpha)\neq 0$ and $i \geq 0$. Recall that $i = \upsilon_\alpha(c_2)$ and $\kappa_P \in \{0,\dots,3\}$. If $\upsilon_\alpha\big(c_2\big) = 0 \leq \kappa_P$ then

$$\upsilon_P(u_2) = \upsilon_P\big(b_2(x)\cdot y\big) - \upsilon_P\big(c_2(x)\big) = \upsilon_P(b_2(x)\cdot y) \geq 0.$$

If $\upsilon_{P_\alpha}(c_2) > 0$ then

(i) $\upsilon_P(b_2) = 0$ by $\gcd\big(b_2,c_2\big)=1$;

(ii) $i > 0$.

Calculate

$$\begin{aligned}
\upsilon_P(u_2) =& \upsilon_P\left(\frac{b_2(x)\cdot y}{(x-\alpha)^i\cdot\tilde{c_2}(x)}\right) = \upsilon_P\left(\frac{b_2(x)}{\tilde{c_2}(x)}\cdot\frac{y}{(x-\alpha)^i}\right)\\
=& \upsilon_P\big(b_2(x)\big) - \upsilon_P\big(\tilde{c_2}(x)\big) + \upsilon_P\left(\frac{y}{(x-\alpha)^i}\right) = \upsilon_P\left(\frac{y}{(x-\alpha)^i}\right).
\end{aligned}$$

In order to prove, use Lemma 2.20.

□

Remark 3.13. A number $\kappa_P = 0$ in $\operatorname{char} K \neq 2$. Thus point *(ii)* from Theorem 3.12 can be rewritten

$$\upsilon_P(u_2) < 0 \iff \upsilon_\alpha\big(c_2(x)\big) > 0 \iff c_2(\alpha) = 0.$$

Remark 3.14. By Theorem 1.60 if $\upsilon_P(x) \geq 0$ then $\omega_P(x) = 0$ for $x \in E$ and $P \in \mathbb{P}_E$. It may be used as a necessary condition when $\omega_P(x) \neq 0$. Theorem 3.12 characterises $\upsilon_P(x)$ for $P \in \mathbb{P}_E^{(1)} \setminus \{P_\infty\}$. The characterization can be used as another necessary condition when $\omega_P(x) \neq 0$.

Theorem 3.15. *Let* $u_1 = \frac{a_1(x)}{c_1(x)}, u_2 = \frac{b_2(x)\cdot y}{c_2(x)} \in E$ *where* $a_1(x), c_1(x), b_2(x), c_2(x) \in K[x]$ *such that* $\gcd(a_1, c_1) = 1$ *and* $\gcd(b_2, c_2) = 1$. *Let*

$$c_1(x) = \gamma_1 \cdot c_{1,1}^{\gamma_1}(x) \cdot \ldots \cdot c_{1,n}^{\gamma_n}(x);$$

$$c_2(x) = \gamma_2 \cdot c_{2,1}^{\delta_1}(x) \cdot \ldots \cdot c_{2,m}^{\delta_m}(x)$$

be decompositions of $c_1(x)$ *and* $c_2(x)$ *to irreducible factors in* K. *Every factor determines a place in* $K(x)$ *(cf. Theorem 1.76). Define*

$$S_{K(x)}^{(1)} := \{P_{c_{1,1}}, \ldots, P_{c_{1,n}}\} \subset \mathbb{P}_{K(x)};$$

$$S_{K(x)}^{(2)} := \{P_{c_{2,1}}, \ldots, P_{c_{2,m}}\} \subset \mathbb{P}_{K(x)}.$$

Define

$$S_E^{(i)} := \{P_E \in \mathbb{P}_E; \exists P \in S_{K(x)}^{(i)} \text{ such that } P_E | P\} \cup \{P_\infty\} \text{ for } i = 1, 2.$$

If $\omega_P(u_i) \neq 0$ *then* $P \in S_E^{(i)}$ *for* $i = 1, 2$.

Proof. Let $i = 1, 2$. Prove that if $P \notin S_E^{(i)}$ then $\omega_P(u_i) = 0$. Let $P \in \mathbb{P}_E \setminus S_E^{(i)}$. If $P \notin S_E^{(i)}$ then $\upsilon_P(c_i) = 0$. Recall that $\upsilon_P(a_1) \geq 0$ and $\upsilon_P(b_2) \geq 0$.

(i) Let $i = 1$. Calculate $\upsilon_P(u_1) = \upsilon_P(a_1) - \upsilon_P(c_1) \geq 0$. Thus $\omega_P(u_i) = 0$ by Theorem 1.60.

(ii) Let $i = 2$. Recall that $\upsilon_P(y) \geq 0$. Calculate $\upsilon_P(u_2) = \upsilon_P(b_2) + \upsilon_P(y) - \upsilon_P(c_2) \geq 0$. Thus $\omega_P(u_2) = 0$ by Theorem 1.60.

□

Remark 3.16. The necessary condition from Remark 3.14 is a more accurate version for $\mathbb{P}_E^{(1)}$ of the condition from Theorem 3.15.

The necessary conditions when $\omega_P(u)$ can be non-zero has already been developed. Now, solve how to calculate $\omega_{(\alpha,\beta)}(u)$. First, convert the calculation of $\omega_{(\alpha,\beta)}(u)$ to the calculation of $\omega_{(\alpha,\beta)}\left(\frac{1}{(x-\alpha)^i}\right)$ and $\omega_{(\alpha,\beta)}\left(\frac{y}{(x-\alpha)^i}\right)$ for $i \geq 1$. It is solved by Algorithm 3.17. Next, describe a calculation of this local components.

Algorithm 3.17. ***The inputs*** *of the algorithm are*

(i) $u = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x)\cdot y}{c_2(x)} \in E$ *where* $a_1(x), c_1(x), b_2(x), c_2(x) \in K[x]$ *such that* $\gcd(a_1, c_1) = 1$ *and* $\gcd(b_2, c_2) = 1$*;*

(ii) $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$.

Put $C_1 := \upsilon_\alpha(c_1)$ *and* $C_2 := \upsilon_\alpha(c_2)$ *where* $P_\alpha \in \mathbb{P}_{K(x)}^{(1)}$.

The outputs *of the algorithm are*

(i) *the reduction of the calculation of* $\omega_P\left(\frac{a_1(x)}{c_1(x)}\right)$ *to the calculation of* $\omega_P\left(\frac{1}{(x-\alpha)^i}\right)$ *for* $i = 1, \ldots, C_1$;

(ii) *the reduction of the calculation of* $\omega_P\left(\frac{b_2(x)\cdot y}{c_2(x)}\right)$ *to the calculation of* $\omega_P\left(\frac{y}{(x-\alpha)^i}\right)$ *for* $i = \kappa_P + 1, \ldots, C_2$.

The algorithm:

1. *Let* $i = 1, 2$.

2. *Decompose* $c_i(x) = (x-\alpha)^{C_i} \cdot \tilde{c}_i(x)$ *where* $\tilde{c}_i(\alpha) \neq 0$.

3. *Calculate* $\upsilon_\alpha(c_1) = C_1$ *and* $\upsilon_\alpha(c_2) = C_2$ *(cf. Theorem 1.76).*

4. *If* $C_1 = 0$ *then* $\omega_P\left(\frac{a_1(x)}{c_1(x)}\right) = 0$ *and stop the calculation of this branch. If* $C_2 \leq \kappa_P$ *then* $\omega_P\left(\frac{b_2(x)\cdot y}{c_2(x)}\right) = 0$ *and stop the calculation of this branch.*

5. *Use* $\gcd\left((x-\alpha)^{C_i}, \tilde{c}_i(x)\right) = 1$ *and find Bézout coefficients* $u_i(x), v_i(x) \in K[x]$ *to be satisfied*
$$1 = u_i(x) \cdot \tilde{c}_i(x) + v_i(x) \cdot (x-\alpha)^{C_i}.$$

6. *Define and express*
$$\tilde{a_1}(x) := u_1(x) \cdot a_1(x) = \sum_{i=0}^{A_t} a_{1,i} \cdot (x-\alpha)^i \text{ where } A_t := \deg(\tilde{a_1});$$
$$\tilde{b_2}(x) := u_2(x) \cdot b_2(x) = \sum_{i=0}^{B_t} b_{2,i} \cdot (x-\alpha)^i \text{ where } B_t := \deg(\tilde{b_2}).$$

7. *Calculate*
$$\omega_P(u) = \sum_{i=0}^{C_1-1} a_{1,i} \cdot \omega_P\left(\frac{1}{(x-\alpha)^{C_1-i}}\right) + \sum_{i=0}^{C_2-\kappa_P-1} b_{2,i} \cdot \omega_P\left(\frac{y}{(x-\alpha)^{C_2-i}}\right).$$

Theorem 3.18. *Algorithm 3.17 is correct.*

Proof. Keep the assumption from Algorithm 3.17. By Remark 3.14

(i) if $C_1 = 0$ then $\omega_P\left(\frac{a_1(x)}{c_1(x)}\right) = 0$;

(ii) if $C_2 \leq \kappa_P$ then $\omega_P\left(\frac{b_2(x)\cdot y}{c_2(x)}\right) = 0$.

Assume that $C_1 \geq 1$ and $C_2 \geq \kappa_P + 1 \geq 1$.

Recall that $i = 1, 2$. The equation
$$1 = u_i(x) \cdot \tilde{c}_i(x) + v_i(x) \cdot (x-\alpha)^{C_i}$$
gives
$$\frac{v_i(x)}{\tilde{c}_i(x)} + \frac{u_i(x)}{(x-\alpha)^{C_i}} = \frac{v_i(x) \cdot (x-\alpha)^{C_i} + u_i(x) \cdot \tilde{c}_i(x)}{\tilde{c}_i(x) \cdot (x-\alpha)^{C_i}} = \frac{1}{c_i(x)};$$

$$
\begin{aligned}
\frac{a_1(x)}{c_1(x)} &= \frac{v_1(x)\cdot a_1(x)}{\tilde{c_1}(x)} + \frac{u_1(x)\cdot a_1(x)}{(x-\alpha)^{C_1}};\\
\frac{b_2(x)\cdot y}{c_2(x)} &= \frac{v_2(x)\cdot b_2(x)\cdot y}{\tilde{c_2}(x)} + \frac{u_2(x)\cdot b_2(x)\cdot y}{(x-\alpha)^{C_2}}.
\end{aligned}
$$

The valuation $\upsilon_\alpha\big(\tilde{a_1}\big) = \min\big(i; a_{1,i} \neq 0\big)$ and $\upsilon_\alpha\big(\tilde{b_2}\big) = \min\big(i; b_{2,i} \neq 0\big)$. Because $C_1 > 0$, $C_2 > 0$, $\gcd\big(a_1, c_1\big) = 1$ and $\gcd\big(b_2, c_2\big) = 1$ then $\upsilon_\alpha\big(a_1\big) = 0$ and $\upsilon_\alpha\big(b_2\big) = 0$. If $\upsilon_\alpha\big(u_i\big) \geq 1$ then $(x-\alpha) \mid u_i(x)$. Recall that $C_i > 0$. Obtain in $K[x]$

$$
\begin{aligned}
&(x-\alpha) \mid \big(u_i(x)\cdot \tilde{c_i}(x) + v_i(x)\cdot (x-\alpha)^{C_i}\big);\\
&(x-\alpha) \mid 1.
\end{aligned}
$$

It is a contradiction. Thus $\upsilon_\alpha\big(u_i\big) = 0$. Thus

(i) $\upsilon_\alpha\big(\tilde{a_1}\big) = \upsilon_\alpha\big(a_1\big) + \upsilon_\alpha\big(u_1\big) = 0 = \min\big(i; a_{1,i} \neq 0\big)$;

(ii) $\upsilon_\alpha\big(\tilde{b_2}\big) = \upsilon_\alpha\big(b_2\big) + \upsilon_\alpha\big(u_2\big) = 0 = \min\big(i; b_{2,i} \neq 0\big)$.

In other words, $a_{1,0} \neq 0$ and $b_{2,0} \neq 0$.

Express

$$
u = \frac{v_1(x)\cdot a_1(x)}{\tilde{c_1}(x)} + \sum_{i=0}^{A_t} a_{1,i}\cdot (x-\alpha)^{i-C_1} + \frac{v_2(x)\cdot b_2(x)\cdot y}{\tilde{c_2}(x)} + \sum_{i=0}^{B_t} b_{2,i}\cdot y\cdot (x-\alpha)^{i-C_2}.
$$

Use K-linearity of ω_P and calculate

$$
\begin{aligned}
\omega_P(u) = \omega_P\Big(\frac{v_1(x)\cdot a_1(x)}{\tilde{c_1}(x)}\Big) &+ \sum_{i=0}^{A_t} a_{1,i}\cdot \omega_P\Big(\frac{1}{(x-\alpha)^{C_1-i}}\Big)\\
+ \omega_P\Big(\frac{v_2(x)\cdot b_2(x)\cdot y}{\tilde{c_2}(x)}\Big) &+ \sum_{i=0}^{B_t} b_{2,i}\cdot \omega_P\Big(\frac{y}{(x-\alpha)^{C_2-i}}\Big).
\end{aligned}
$$

Recall that $\upsilon_P\big(r(x,y)\big) \geq 0$ for $r(x,y) \in K[x,y]$. By Lemma 2.4 $P \mid P_\alpha$. Calculate valuations

$$
\begin{aligned}
\upsilon_P(\tilde{c_i}) &= e(P \mid P_\alpha)\cdot \upsilon_\alpha(\tilde{c_i}) = 0;\\
\upsilon_P\Big(\frac{v_1(x)\cdot a_1(x)}{\tilde{c_1}(x)}\Big) &\geq 0;\\
\upsilon_P\Big(\frac{v_2(x)\cdot a_2(x)\cdot y}{\tilde{c_2}(x)}\Big) &\geq 0;\\
\upsilon_P\Big((x-\alpha)^n\Big) &= n\cdot e(P \mid P_\alpha)\cdot \upsilon_\alpha\big((x-\alpha)\big) = n\cdot e(P \mid P_\alpha).
\end{aligned}
$$

By Theorem 1.60

$$
\begin{aligned}
\omega_P\Big(\frac{v_1(x)\cdot a_1(x)}{\tilde{c_1}(x)}\Big) &= 0;\\
\omega_P\Big(\frac{v_2(x)\cdot a_2(x)\cdot y}{\tilde{c_2}(x)}\Big) &= 0;\\
\text{if } n \geq 0 \text{ then } \omega_P\Big((x-\alpha)^n\Big) &= 0.
\end{aligned}
$$

By Lemma 2.20
$$\upsilon_P\Big(\frac{y}{(x-\alpha)^n}\Big) \geq 0 \iff n \leq \kappa_P.$$
By Theorem 1.60 if $n \leq \kappa_P$ then
$$\omega_P\Big(\frac{y}{(x-\alpha)^n}\Big) = 0.$$

It gives
$$\omega_P(u) = \sum_{i=0}^{C_1-1} a_{1,i} \cdot \omega_P\Big(\frac{1}{(x-\alpha)^{C_1-i}}\Big) + \sum_{i=0}^{C_2-\kappa_P-1} b_{2,i} \cdot \omega_P\Big(\frac{y}{(x-\alpha)^{C_2-i}}\Big).$$

At the end, it is not necessary to know $\omega_P\Big(\frac{1}{(x-\alpha)^{C_1-i}}\Big)$ or $\omega_P\Big(\frac{y}{(x-\alpha)^{C_2-i}}\Big)$ if $a_{1,i} = 0$ or $b_{2,i} = 0$. But we shall see that an algorithm to calculate $\omega_P\Big(\frac{1}{(x-\alpha)^i}\Big)$ and $\omega_P\Big(\frac{y}{(x-\alpha)^i}\Big)$ is recursive. Recall that $a_{1,0} \neq 0$ and $b_{2,0} \neq 0$. Thus the terms with coefficient zero are always calculated. □

Remark 3.19. Now, focus to develop a calculation of the terms $\omega_P\big(\frac{1}{(x-\alpha)^i}\big)$ and $\omega_P\big(\frac{y}{(x-\alpha)^i}\big)$ for $i \geq 1$ and $P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$.

Lemma 3.20. *Let $i > 0$ and $P \in \mathbb{P}_E$. If $\omega_P\big(\frac{1}{(x-\alpha)^i}\big) \neq 0$ or $\omega_P\big(\frac{y}{(x-\alpha)^i}\big) \neq 0$ then $P \mid P_\alpha$ for $P_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ or $P = P_\infty$. If $P \neq P_\infty$ then P is one from the following places.*

- *(i) There exist $P_{(\alpha,\beta)}, P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ such that $e\big(P_{(\alpha,\beta)} \mid P_\alpha\big) = f\big(P_{(\alpha,\beta)} \mid P_\alpha\big) = e\big(P_{(\alpha,\gamma)} \mid P_\alpha\big) = f\big(P_{(\alpha,\gamma)} \mid P_\alpha\big) = 1$.*
- *(ii) There exists only one $P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ such that $e\big(P_{(\alpha,\beta)} \mid P_\alpha\big) = 2$ and $f\big(P_{(\alpha,\beta)} \mid P_\alpha\big) = 1$*
- *(iii) There exists only one $P \in \mathbb{P}_E^{(2)}$. such that $e\big(P \mid P_\alpha\big) = 1$ and $f\big(P \mid P_\alpha\big) = 2$.*

Proof. That $P \mid P_\alpha$ or $P = P_\infty$ is a consequence of Theorem 3.15.

Let $P \neq P_\infty$. Describe all situation $P \mid P_\alpha$. By Theorem 1.71 there exist maximally two places P_1 and P_2 such that $P_1 \mid P_\alpha$ and $P_2 \mid P_\alpha$. If there exist both P_1 and P_2 then the situation is described by Lemma 2.6. If there exists only one P_1 such that $P_1 \mid P_\alpha$ then $\deg(P_1) = f(P_1 \mid P_\alpha) \cdot \deg(P_\alpha) = f(P_1 \mid P_\alpha)$ by Lemma 1.70. If $\deg(P_1) = 1$ then this situation is described by Lemma 2.7 and $e(P_1 \mid P_\alpha) = 2$. If $\deg(P_1) = 2$ then $e(P_1 \mid P_\alpha) = 1$. □

Theorem 3.21. *Let $P_1 := P_{(\alpha,\beta)}, P_2 := P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ where $\alpha, \beta, \gamma \in K$ such that $P_1 \neq P_2$. In other words, $\beta \neq \gamma$. Then*

- *(i) $\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big) = \frac{1}{\beta-\gamma}$;*
- *(ii) $\omega_{P_2}\Big(\frac{1}{x-\alpha}\Big) = -\frac{1}{\beta-\gamma}$;*
- *(iii) $\omega_{P_1}\Big(\frac{y}{x-\alpha}\Big) = \frac{\beta}{\beta-\gamma}$;*

(iv) $\omega_{P_2}\Big(\frac{y}{x-\alpha}\Big) = -\frac{\gamma}{\beta-\gamma}$.

Proof. By Theorem 3.6 $\omega_\infty\big(\frac{1}{x-\alpha}\big) = 0$. By Theorem 3.8 $\omega_\infty\big(\frac{y}{x-\alpha}\big) = -1$. Thus $\omega_\infty\big(\frac{y-\beta}{x-\alpha}\big) = \omega_\infty\big(\frac{y-\gamma}{x-\alpha}\big) = -1$. Use Lemma 3.20, Theorem 1.59 and facts $\omega\big(\frac{1}{x-\alpha}\big) = 0$ and $\omega\big(\frac{y}{x-\alpha}\big) = 0$ to obtain

$$\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big) + \omega_{P_2}\Big(\frac{1}{x-\alpha}\Big) = 0; \tag{3.1}$$

$$\omega_{P_1}\Big(\frac{y}{x-\alpha}\Big) + \omega_{P_2}\Big(\frac{y}{x-\alpha}\Big) = 1; \tag{3.2}$$

$$\omega_{P_1}\Big(\frac{y-\beta}{x-\alpha}\Big) + \omega_{P_2}\Big(\frac{y-\beta}{x-\alpha}\Big) = 1; \tag{3.3}$$

$$\omega_{P_1}\Big(\frac{y-\gamma}{x-\alpha}\Big) + \omega_{P_2}\Big(\frac{y-\gamma}{x-\alpha}\Big) = 1. \tag{3.4}$$

By Lemma 2.4 $P_1 \mid P_\alpha$, $P_2 \mid P_\alpha$, $P_1 \mid P_\beta$ and $P_1 \mid P_\gamma$ where $P_\alpha \in \mathbb{P}^{(1)}_{K(x)}$ and $P_\beta, P_\gamma \in \mathbb{P}^{(1)}_{K(y)}$. By Lemma 2.6 $e\big(P_1 \mid P_\alpha\big) = e\big(P_2 \mid P_\alpha\big) = 1$. Calculate valuations

$$\upsilon_{P_1}\Big(\frac{y-\beta}{x-\alpha}\Big) = e\big(P_1 \mid P_\beta\big)\cdot\upsilon_\beta\big(y-\beta\big) - e\big(P_1 \mid P_\alpha\big)\cdot\upsilon_\alpha\big(x-\alpha\big) = e\big(P_1 \mid P_\beta\big) - 1 \geq 0;$$

$$\upsilon_{P_2}\Big(\frac{y-\gamma}{x-\alpha}\Big) = e\big(P_2 \mid P_\gamma\big)\cdot\upsilon_\gamma\big(y-\gamma\big) - e\big(P_2 \mid P_\alpha\big)\cdot\upsilon_\alpha\big(x-\alpha\big) = e\big(P_2 \mid P_\gamma\big) - 1 \geq 0.$$

Thus by Theorem 1.60

$$\omega_{P_1}\Big(\frac{y-\beta}{x-\alpha}\Big) = 0;$$
$$\omega_{P_2}\Big(\frac{y-\gamma}{x-\alpha}\Big) = 0.$$

Equations (3.3) and (3.4) can be simplified

$$\omega_{P_2}\Big(\frac{y-\beta}{x-\alpha}\Big) = 1; \quad \omega_{P_2}\Big(\frac{y}{x-\alpha}\Big) = 1 + \beta\cdot\omega_{P_2}\Big(\frac{1}{x-\alpha}\Big); \tag{3.5}$$

$$\omega_{P_1}\Big(\frac{y-\gamma}{x-\alpha}\Big) = 1; \quad \omega_{P_1}\Big(\frac{y}{x-\alpha}\Big) = 1 + \gamma\cdot\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big). \tag{3.6}$$

Create a linear system from Equations (3.1), (3.2), (3.5), (3.6). Next solve it

$$\gamma\cdot\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big) + \beta\cdot\omega_{P_2}\Big(\frac{1}{x-\alpha}\Big) = -1;$$

$$\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big) + \omega_{P_2}\Big(\frac{1}{x-\alpha}\Big) = 0.$$

Recall that $\beta \neq \gamma$ and $\beta \neq 0$ in $\operatorname{char} K = 2$ by Lemma 2.14 and in $\operatorname{char} K \neq 2$ by Lemma 2.6.

$$\left(\begin{array}{cc|c} \gamma & \beta & -1 \\ 1 & 1 & 0 \end{array}\right) \simeq \left(\begin{array}{cc|c} \gamma-\beta & 0 & -1 \\ 1 & 1 & 0 \end{array}\right) \simeq \left(\begin{array}{cc|c} 1 & 0 & \frac{1}{\beta-\gamma} \\ 1 & 1 & 0 \end{array}\right) \simeq \left(\begin{array}{cc|c} 1 & 0 & \frac{1}{\beta-\gamma} \\ 0 & 1 & -\frac{1}{\beta-\gamma} \end{array}\right)$$

The results are

$$\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big) = \frac{1}{\beta-\gamma};$$
$$\omega_{P_2}\Big(\frac{1}{x-\alpha}\Big) = -\frac{1}{\beta-\gamma}.$$

Put the results into Equations (3.5) and (3.6) and obtain

$$\omega_{P_1}\Big(\frac{y}{x-\alpha}\Big) = 1+\frac{\gamma}{\beta-\gamma} = \frac{\beta}{\beta-\gamma};$$

$$\omega_{P_2}\Big(\frac{y}{x-\alpha}\Big) = 1-\frac{\beta}{\beta-\gamma} = \frac{-\gamma}{\beta-\gamma}.$$

□

Remark 3.22. If $\operatorname{char} K \neq 2$ then $\gamma = -\beta$ by Lemma 2.6. The results of Theorem 3.21 are

$$\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big) = \frac{1}{2\beta}; \qquad \omega_{P_1}\Big(\frac{y}{x-\alpha}\Big) = \frac{\beta}{2\beta} = \frac{1}{2};$$
$$\omega_{P_2}\Big(\frac{1}{x-\alpha}\Big) = -\frac{1}{2\beta}; \qquad \omega_{P_2}\Big(\frac{y}{x-\alpha}\Big) = \frac{\beta}{2\beta} = \frac{1}{2}.$$

Remark 3.23. Let $\operatorname{char} K = 2$ and P_1 be of *Type 4*. Recall that $\beta = 0$. By Lemma 2.20 and Theorem 1.60 $\omega_{P_1}\Big(\frac{y}{x-\alpha}\Big)$ should be zero. By Theorem 3.21

$$\omega_{P_1}\Big(\frac{y}{x-\alpha}\Big) = 1+\frac{\gamma}{\beta-\gamma} = 1-\frac{\gamma}{\gamma} = 0.$$

Theorem 3.24. *Let $P := P_{(\alpha,\beta)} \in \mathbb{P}^{(1)}_E$, $P_\alpha \in \mathbb{P}^{(1)}_{K(x)}$ where $\alpha, \beta \in K$ Let P be only one place such that $P \mid P_\alpha$ (cf. Lemma 2.7). Then $\omega_P\Big(\frac{1}{x-\alpha}\Big) = 0$, $\omega_P\Big(\frac{y}{x-\alpha}\Big) = 1$.*

Proof. By Theorem 3.6 $\omega_\infty\big(\frac{1}{x-\alpha}\big) = 0$. By Theorem 3.8 $\omega_\infty\big(\frac{y}{x-\alpha}\big) = -1$. Use Lemma 3.20, Theorem 1.59 and facts $\omega\big(\frac{1}{x-\alpha}\big) = 0$ and $\omega\big(\frac{y}{x-\alpha}\big) = 0$ to obtain

$$\omega_P\Big(\frac{1}{x-\alpha}\Big) = 0; \quad \omega_P\Big(\frac{y}{x-\alpha}\Big) = 1.$$

□

Remark 3.25. Algorithm 3.26 is used as a subroutine of Algorithm 3.28. The reason why the inputs of Algorithm 3.26 are $\alpha, \beta \in K$ and $n \geq 2$ comes from Algorithm 3.28.

The motivation is following. Let $(\alpha, \beta) \in \mathbb{V}_w(K)$. By Theorem 1.116 $(y-\beta)^n = a_{\beta,n}(x) + y \cdot b_{\beta,n}(x)$ where $a_{\beta,n}(x), b_{\beta,n}(x) \in K[x]$. Consider $a_{\beta,n}(x), b_{\beta,n}(x) \in K[x-\alpha]$ for a purpose of Algorithm 3.28. A recursion is used in Algorithm 3.28. Each step of the recursion needs $a_{\beta,n}(x), b_{\beta,n}(x) \in K[x-\alpha]$. Algorithm 3.26 is used for calculating $a_{\beta,n}(x), b_{\beta,n}(x) \in K[x-\alpha]$.

Algorithm 3.26. ***The inputs*** *of the algorithm are* $\alpha, \beta \in K$ *and* $n \geq 2$.

The outputs *of the algorithm are coefficients* $a_{\beta,n}^{(i)}$, $b_{\beta,n}^{(i)}$ *of the polynomials*

$$a_{\beta,n}(x) = \sum_{i=0}^{\deg\left(a_{\beta,n}\right)} a_{\beta,n}^{(i)} \cdot (x-\alpha)^i \in K[x-\alpha];$$

$$b_{\beta,n}(x) = \sum_{i=0}^{\deg\left(b_{\beta,n}\right)} b_{\beta,n}^{(i)} \cdot (x-\alpha)^i \in K[x-\alpha]$$

such that

$$\big(y-\beta\big)^n = a_{\beta,n}(x) + y \cdot b_{\beta,n}(x) \text{ in } K[w].$$

Recall that $w(x,y) = 0$ *and Theorem 1.116.*

The algorithm *is recursive. It splits according to the form of* $w(x,y)$. *Consider* $K[x-\alpha]$ *instead of* $K[x]$. *In other words, every polynomial* $p(x)$ *is considered in the variable* $x-\alpha$.

(i) Assume that $\operatorname{char} K \neq 2$ *and* $w(x,y) = y^2 - f(x)$. *Recall that* $f \in K[x-\alpha]$.

1. *Let* $n = 2$. *Then*

$$a_{\beta,2}\big(x\big) = f(x) + \beta^2; \tag{3.7}$$

$$b_{\beta,2}\big(x\big) = -2 \cdot \beta. \tag{3.8}$$

2. *Let* $a_{\beta,n}(x)$ *and* $b_{\beta,n}(x)$ *be known. Then*

$$a_{\beta,n+1}(x) = f(x) \cdot b_{\beta,n}(x) - \beta \cdot a_{\beta,n}(x); \tag{3.9}$$

$$b_{\beta,n+1}(x) = a_{\beta,n}(x) - \beta \cdot b_{\beta,n}(x). \tag{3.10}$$

(ii) Assume that $\operatorname{char} K = 2$ *and* $w(x,y) = g_1(x,y) - f_1(x)$. *Recall that* $f_1 \in K[x-\alpha]$.

1. *Let* $n = 2$. *Then*

$$a_{\beta,2}\big(x\big) = f_1(x) + \beta^2;$$

$$b_{\beta,2}\big(x\big) = -(x-\alpha) - \alpha. \tag{3.11}$$

2. *Let* $a_{\beta,n}(x)$ *and* $b_{\beta,n}(x)$ *be known. Then*

$$a_{\beta,n+1}(x) = f_1(x) \cdot b_{\beta,n}(x) - \beta \cdot a_{\beta,n}(x); \tag{3.12}$$

$$b_{\beta,n+1}(x) = a_{\beta,n}(x) - \big((x-\alpha) + \alpha + \beta\big) \cdot b_{\beta,n}(x). \tag{3.13}$$

(iii) Assume that $\operatorname{char} K = 2$ *and* $w(x,y) = g_2(x,y) - f_2(x)$. *Recall that* $f_2 \in K[x-\alpha]$.

1. *Let* $n = 2$. *Then*

$$a_{\beta,2}\big(x\big) = f_2(x) + \beta^2;$$

$$b_{\beta,2}\big(x\big) = -a_3. \tag{3.14}$$

2. Let $a_{\beta,n}(x)$ and $b_{\beta,n}(x)$ be known. Then

$$a_{\beta,n+1}(x) = f_2(x) \cdot b_{\beta,n}(x) - \beta \cdot a_{\beta,n}(x); \tag{3.15}$$

$$b_{\beta,n+1}(x) = a_{\beta,n}(x) - (a_3 + \beta) \cdot b_{\beta,n}(x). \tag{3.16}$$

Theorem 3.27. *Algorithm 3.26 is correct.*

Proof. Keep the assumption from Algorithm 3.26. Recall that $K[x - \alpha]$ is considered instead of $K[x]$.

(i) Assume that $\operatorname{char} K \neq 2$ and $w(x, y) = y^2 - f(x)$.

(a) Let $n = 2$. Then

$$\big(y - \beta\big)^2 = y^2 - 2 \cdot \beta \cdot y + \beta^2 = f(x) - 2 \cdot \beta \cdot y + \beta^2;$$

$$a_{\beta,2}\big(x\big) = f(x) + \beta^2;$$
$$b_{\beta,2}\big(x\big) = -2 \cdot \beta.$$

(b) Let $a_{\beta,n}(x)$ and $b_{\beta,n}(x)$ be known. Calculate $a_{\beta,n+1}(x)$ and $b_{\beta,n+1}(x)$

$$\begin{aligned}&\big(y - \beta\big)^{n+1} = \big(y - \beta\big) \cdot \big(a_{\beta,n}(x) + y \cdot b_{\beta,n}(x)\big)\\ =&y \cdot a_{\beta,n}(x) + y^2 \cdot b_{\beta,n}(x) - \beta \cdot a_{\beta,n}(x) - \beta \cdot y \cdot b_{\beta,n}(x)\\ =&f(x) \cdot b_{\beta,n}(x) - \beta \cdot a_{\beta,n}(x) + y\Big(a_{\beta,n}(x) - \beta \cdot b_{\beta,n}(x)\Big).\end{aligned}$$

Equations (3.9) and (3.10) are proved.

(ii) Assume that $\operatorname{char} K = 2$ and $w(x, y) = g_1(x, y) - f_1(x)$.

(a) Let $n = 2$. Then

$$\big(y - \beta\big)^2 = y^2 + \beta^2 = f_1(x) - x \cdot y + \beta^2;$$

$$a_{\beta,2}\big(x\big) = f_1(x) + \beta^2;$$
$$b_{\beta,2}\big(x\big) = -(x - \alpha) - \alpha.$$

(b) Let $a_{\beta,n}(x)$ and $b_{\beta,n}(x)$ be known. Calculate $a_{\beta,n+1}(x)$ and $b_{\beta,n+1}(x)$

$$\begin{aligned}&\big(y - \beta\big)^{n+1} = \big(y - \beta\big) \cdot \big(a_{\beta,n}(x) + y \cdot b_{\beta,n}(x)\big)\\ =&y \cdot a_{\beta,n}(x) + y^2 \cdot b_{\beta,n}(x) - \beta \cdot a_{\beta,n}(x) - \beta \cdot y \cdot b_{\beta,n}(x)\\ =&y \cdot a_{\beta,n}(x) + (f_1(x) - x \cdot y) \cdot b_{\beta,n}(x) - \beta \cdot a_{\beta,n}(x) - \beta \cdot y \cdot b_{\beta,n}(x)\\ =&f_1(x) \cdot b_{\beta,n}(x) - \beta \cdot a_{\beta,n}(x) + y\Big(a_{\beta,n}(x) - \big((x - \alpha) + \alpha + \beta\big) \cdot b_{\beta,n}(x)\Big).\end{aligned}$$

Equations (3.12) and (3.13) are proved.

(iii) Assume that $\operatorname{char} K = 2$ and $w(x, y) = g_2(x, y) - f_2(x)$.

(a) Let $n = 2$. Then

$$\big(y-\beta\big)^2 = y^2 + \beta^2 = f_1(x) - a_3 \cdot y + \beta^2;$$
$$a_{\beta,2}\big(x\big) = f_2(x) + \beta^2;$$
$$b_{\beta,2}\big(x\big) = -a_3.$$

(b) Let $a_{\beta,n}(x)$ and $b_{\beta,n}(x)$ be known. Calculate $a_{\beta,n+1}(x)$ and $b_{\beta,n+1}(x)$

$$\begin{aligned}
&\big(y-\beta\big)^{n+1} = \big(y-\beta\big)\cdot\big(a_{\beta,n}(x) + y\cdot b_{\beta,n}(x)\big)\\
=&y\cdot a_{\beta,n}(x) + y^2\cdot b_{\beta,n}(x) - \beta\cdot a_{\beta,n}(x) - \beta\cdot y\cdot b_{\beta,n}(x)\\
=&y\cdot a_{\beta,n}(x) + (f_2(x) - a_3\cdot y)\cdot b_{\beta,n}(x) - \beta\cdot a_{\beta,n}(x) - \beta\cdot y\cdot b_{\beta,n}(x)\\
=&f_2(x)\cdot b_{\beta,n}(x) - \beta\cdot a_{\beta,n}(x) + y\Big(a_{\beta,n}(x) - (a_3+\beta)\cdot b_{\beta,n}(x)\Big).
\end{aligned}$$

Equations (3.15) and (3.16) are proved.

□

Algorithm 3.28. ***The inputs*** *of the algorithm are*

(i) $n \geq 2$;

(ii) $P_1 := P_{(\alpha,\beta)}, P_2 := P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ *where* $\alpha, \beta, \gamma \in K$ *such that* $P_1 \neq P_2$. *In other words,* $\beta \neq \gamma$.

The outputs *of the algorithm are values of*

$$\omega_{P_1}\Big(\frac{1}{(x-\alpha)^n}\Big),\ \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big),\ \omega_{P_1}\Big(\frac{y}{(x-\alpha)^n}\Big),\ \omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big).$$

The algorithm *is recursive. The values of* $\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big)$, $\omega_{P_2}\Big(\frac{1}{x-\alpha}\Big)$, $\omega_{P_1}\Big(\frac{y}{x-\alpha}\Big)$, $\omega_{P_2}\Big(\frac{y}{x-\alpha}\Big)$ *can be calculated by Theorem 3.21. Let* $n \geq 2$. *Values of* $\omega_{P_1}\Big(\frac{1}{(x-\alpha)^i}\Big)$, $\omega_{P_2}\Big(\frac{1}{(x-\alpha)^i}\Big)$, $\omega_{P_1}\Big(\frac{y}{(x-\alpha)^i}\Big)$, $\omega_{P_2}\Big(\frac{y}{(x-\alpha)^i}\Big)$ *for* $i = 1, \dots n-1$ *may be assumed to be known from the previous steps of the recursion.*

1. *Calculate polynomials* $a_{\beta,n}(x), b_{\beta,n}(x), a_{\gamma,n}(x), b_{\gamma,n}(x) \in K[x-\alpha]$ *by Algorithm 3.26.*

   If $n = 2$ *then use the first step of the recursion from Algorithm 3.26. The step defines exactly* $a_{\beta,2}(x), b_{\beta,2}(x), a_{\gamma,2}(x), b_{\gamma,2}(x) \in K[x-\alpha]$.

   If $n > 2$ *then* $a_{\beta,n-1}(x), b_{\beta,n-1}(x), a_{\gamma,n-1}(x), b_{\gamma,n-1}(x) \in K[x-\alpha]$ *are known from step* $n-1$ *of this recursion. Use the other step of the recursion from Algorithm 3.26. The step calculates* $a_{\beta,n}(x), b_{\beta,n}(x), a_{\gamma,n}(x), b_{\gamma,n}(x) \in K[x-\alpha]$ *from* $a_{\beta,n-1}(x), b_{\beta,n-1}(x), a_{\gamma,n-1}(x), b_{\gamma,n-1}(x)$.

2. *Recall that* $b_{\beta,n}^{(i)}$ *is i-th coefficient of* $b_{\beta,n}(x)$. *Calculate*

$$A_{\beta,n} := b_{\beta,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\beta,n}^{(i)}\cdot\omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - \sum_{i=1}^{n-\kappa_{P_2}-1} b_{\beta,n}^{(i)}\cdot\omega_{P_2}\Big(\frac{y}{(x-\alpha)^{n-i}}\Big); \tag{3.17}$$

$$A_{\gamma,n} := b_{\gamma,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\gamma,n}^{(i)} \cdot \omega_{P_1}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - \sum_{i=1}^{n-\kappa_{P_1}-1} b_{\gamma,n}^{(i)} \cdot \omega_{P_1}\Big(\frac{y}{(x-\alpha)^{n-i}}\Big). \tag{3.18}$$

3. Obtain $\omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big)$ *and* $\omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big)$ *by solving the linear system*

$$A_{\beta,n} = a_{\beta,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) + b_{\beta,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big);$$

$$A_{\gamma,n} = -a_{\gamma,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) - b_{\gamma,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big).$$

4. The local component

$$\omega_{P_1}\Big(\frac{1}{(x-\alpha)^n}\Big) = -\omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big); \tag{3.19}$$

$$\omega_{P_1}\Big(\frac{y}{(x-\alpha)^n}\Big) = -\omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big). \tag{3.20}$$

Theorem 3.29. *Algorithm 3.28 is correct.*

Proof. Keep the assumption from Algorithm 3.28.

By Theorem 3.6 $\omega_\infty\Big(\frac{1}{(x-\alpha)^n}\Big) = 0$. Compare $\deg(1) + 1 < \deg\big((x-\alpha)^n\big)$. Thus $\omega_\infty\Big(\frac{y}{(x-\alpha)^n}\Big) = 0$ by Theorem 3.8. Use Lemma 3.20, Theorem 1.59, and facts $\omega\Big(\frac{1}{(x-\alpha)^n}\Big) = 0$ and $\omega\Big(\frac{y}{(x-\alpha)^n}\Big) = 0$ to obtain

$$\omega_{P_1}\Big(\frac{1}{(x-\alpha)^n}\Big) + \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) = 0; \tag{3.21}$$

$$\omega_{P_1}\Big(\frac{y}{(x-\alpha)^n}\Big) + \omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big) = 0. \tag{3.22}$$

The local components ω_{P_1} and ω_{P_2} can be distinguished on elements $\big(\frac{y-\beta}{x-\alpha}\big)^n$ and $\big(\frac{y-\gamma}{x-\alpha}\big)^n$. Let $P_\alpha \in \mathbb{P}^{(1)}_{K(x)}$ and $P_\beta, P_\gamma \in \mathbb{P}^{(1)}_{K(y)}$. Because

$$\begin{aligned} \upsilon_{P_1}\Big(\big(\frac{y-\beta}{x-\alpha}\big)^n\Big) &= n \cdot \Big(e\big(P_1 \mid P_\beta\big) \cdot \upsilon_\beta\big(y-\beta\big) - e\big(P_1 \mid P_\alpha\big) \cdot \upsilon_\alpha\big(x-\alpha\big)\Big) \\ &= n \cdot \Big(e\big(P_1 \mid P_\beta\big) - 1\Big) \geq 0; \end{aligned}$$

$$\begin{aligned} \upsilon_{P_1}\Big(\big(\frac{y-\gamma}{x-\alpha}\big)^n\Big) &= n \cdot \Big(e\big(P_1 \mid P_\beta\big) \cdot \upsilon_\beta\big(y-\gamma\big) - e\big(P_1 \mid P_\alpha\big) \cdot \upsilon_\alpha\big(x-\alpha\big)\Big) \\ &= n \cdot \big(0 - 1\big) = -n \end{aligned}$$

then by Theorem 1.60

$$\omega_{P_1}\Big(\big(\frac{y-\beta}{x-\alpha}\big)^n\Big) = 0; \tag{3.23}$$

$$\omega_{P_1}\Big(\big(\frac{y-\gamma}{x-\alpha}\big)^n\Big) \text{ can be non-zero.} \tag{3.24}$$

Analogically,

$$\omega_{P_2}\Big(\big(\frac{y-\beta}{x-\alpha}\big)^n\Big) \text{ can be non-zero;} \tag{3.25}$$

$$\omega_{P_2}\Big(\big(\frac{y-\gamma}{x-\alpha}\big)^n\Big) = 0. \tag{3.26}$$

Use Theorem 1.116 to express

$$\big(y-\beta\big)^n = a_{\beta,n}(x) + y \cdot b_{\beta,n}(x) \text{ where } a_{\beta,n}(x), b_{\beta,n}(x) \in K[x]; \tag{3.27}$$

$$\big(y-\gamma\big)^n = a_{\gamma,n}(x) + y \cdot b_{\gamma,n}(x) \text{ where } a_{\gamma,n}(x), b_{\gamma,n}(x) \in K[x]. \tag{3.28}$$

Express $a_{\beta,n}(x)$, $a_{\gamma,n}(x)$, $b_{\beta,n}(x)$ and $b_{\gamma,n}(x)$ as polynomials in the variable $x-\alpha$

$$a_{\beta,n}(x) = \sum_{i=0}^{\deg\left(a_{\beta,n}\right)} a_{\beta,n}^{(i)} \cdot (x-\alpha)^i.$$

Analogically, $a_{\gamma,n}(x)$, $b_{\beta,n}(x)$ and $b_{\gamma,n}(x)$.

Use Theorems 3.6 and 3.8 to deduce that

$$\begin{aligned}
\omega_\infty\Big(\frac{(y-\beta)^n}{(x-\alpha)^n}\Big) &= \omega_\infty\Big(\frac{a_{\beta,n}(x) + y \cdot b_{\beta,n}(x)}{(x-\alpha)^n}\Big) \\
&= \omega_\infty\Big(\frac{a_{\beta,n}(x)}{(x-\alpha)^n}\Big) + \omega_\infty\Big(\frac{y \cdot b_{\beta,n}(x)}{(x-\alpha)^n}\Big) \\
&= \omega_\infty\Big(\frac{y \cdot b_{\beta,n}(x)}{(x-\alpha)^n}\Big) \\
&= \sum_{i=0}^{\deg\left(b_{\beta,n}\right)} b_{\beta,n}^{(i)} \cdot \omega_\infty\Big(y \cdot (x-\alpha)^{(i-n)}\Big) \\
&= b_{\beta,n}^{(n-1)} \cdot \omega_\infty\Big(y \cdot (x-\alpha)^{-1}\Big) = -b_{\beta,n}^{(n-1)}.
\end{aligned} \tag{3.29}$$

Analogically,

$$\omega_\infty\Big(\frac{(y-\gamma)^n}{(x-\alpha)^n}\Big) = -b_{\gamma,n}^{(n-1)}. \tag{3.30}$$

Use Lemma 3.20, Theorem 1.59, Equations (3.23), (3.24), (3.25), (3.26), (3.29) and (3.30) and facts $\omega\Big(\frac{(y-\beta)^n}{(x-\alpha)^n}\Big) = 0$ and $\omega\Big(\frac{(y-\gamma)^n}{(x-\alpha)^n}\Big) = 0$ to obtain

$$\omega_{P_2}\Big(\frac{(y-\beta)^n}{(x-\alpha)^n}\Big) = b_{\beta,n}^{(n-1)};$$

$$\omega_{P_1}\Big(\frac{(y-\gamma)^n}{(x-\alpha)^n}\Big) = b_{\gamma,n}^{(n-1)}.$$

It is possible to assume that

$$\omega_{P_1}\Big(\frac{1}{(x-\alpha)^i}\Big),\ \omega_{P_2}\Big(\frac{1}{(x-\alpha)^i}\Big),\ \omega_{P_1}\Big(\frac{y}{(x-\alpha)^i}\Big),\ \omega_{P_2}\Big(\frac{y}{(x-\alpha)^i}\Big)$$

are known for $i = 1, \dots n-1$ from the recursion. Recall that the values for $i = 1$ are known from Theorem 3.21. Continue by that

$$\begin{aligned} b_{\beta,n}^{(n-1)} &= \omega_{P_2}\Big(\frac{(y-\beta)^n}{(x-\alpha)^n}\Big) = \omega_{P_2}\Big(\frac{a_{\beta,n}(x) + y \cdot b_{\beta,n}(x)}{(x-\alpha)^n}\Big) \\ &= \sum_{i=1}^{n-1} a_{\beta,n}^{(i)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) + \sum_{i=1}^{n-\kappa_{P_2}-1} b_{\beta,n}^{(i)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^{n-i}}\Big) \\ &+ a_{\beta,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) + b_{\beta,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big). \end{aligned}$$

Calculate

$$A_{\beta,n} := b_{\beta,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\beta,n}^{(i)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - \sum_{i=1}^{n-\kappa_{P_2}-1} b_{\beta,n}^{(i)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^{n-i}}\Big); \quad (3.31)$$

$$A_{\gamma,n} := b_{\gamma,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\gamma,n}^{(i)} \cdot \omega_{P_1}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - \sum_{i=1}^{n-\kappa_{P_1}-1} b_{\gamma,n}^{(i)} \cdot \omega_{P_1}\Big(\frac{y}{(x-\alpha)^{n-i}}\Big).$$

By the assumption $A_{\beta,n}$ and $A_{\gamma,n}$ are known. We have that

$$A_{\beta,n} = a_{\beta,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) + b_{\beta,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big); \quad (3.32)$$

$$A_{\gamma,n} = a_{\gamma,n}^{(0)} \cdot \omega_{P_1}\Big(\frac{1}{(x-\alpha)^n}\Big) + b_{\gamma,n}^{(0)} \cdot \omega_{P_1}\Big(\frac{y}{(x-\alpha)^n}\Big). \quad (3.33)$$

Create a linear system from Equations (3.21), (3.22), (3.32), (3.33). Next, solve it

$$A_{\beta,n} = a_{\beta,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) + b_{\beta,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big); \quad (3.34)$$

$$A_{\gamma,n} = -a_{\gamma,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) - b_{\gamma,n}^{(0)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big).$$

Prove the linear system has only one solution. Show

$$\det\begin{pmatrix} a_{\beta,n}^{(0)} & b_{\beta,n}^{(0)} \\ -a_{\gamma,n}^{(0)} & -b_{\gamma,n}^{(0)} \end{pmatrix} = -a_{\beta,n}^{(0)} \cdot b_{\gamma,n}^{(0)} + b_{\beta,n}^{(0)} \cdot a_{\gamma,n}^{(0)} \neq 0. \quad (3.35)$$

The elements of the matrix can be expressed $a_{\beta,n}^{(0)} = a_{\beta,n}(\alpha)$, $a_{\gamma,n}^{(0)} = a_{\gamma,n}(\alpha)$, $b_{\beta,n}^{(0)} = b_{\beta,n}(\alpha)$ and $b_{\gamma,n}^{(0)} = b_{\gamma,n}(\alpha)$. Evaluate Equations (3.27) and (3.28) by (α, β) and (α, γ)

$$0 = (\beta - \beta)^n = a_{\beta,n}(\alpha) + \beta \cdot b_{\beta,n}(\alpha) = a_{\beta,n}^{(0)} + \beta \cdot b_{\beta,n}^{(0)}; \quad (3.36)$$

$$0 = (\gamma - \gamma)^n = a_{\gamma,n}(\alpha) + \gamma \cdot b_{\gamma,n}(\alpha) = a_{\gamma,n}^{(0)} + \gamma \cdot b_{\gamma,n}^{(0)}. \quad (3.37)$$

Use Equations (3.36) and (3.37) to modify Equation (3.35)

$$-a_{\beta,n}^{(0)} \cdot b_{\gamma,n}^{(0)} + b_{\beta,n}^{(0)} \cdot a_{\gamma,n}^{(0)} = \beta \cdot b_{\beta,n}^{(0)} \cdot b_{\gamma,n}^{(0)} - \gamma \cdot b_{\beta,n}^{(0)} \cdot b_{\gamma,n}^{(0)} = (\beta - \gamma) \cdot b_{\beta,n}^{(0)} \cdot b_{\gamma,n}^{(0)}.$$

By the assumption $\beta - \gamma \neq 0$. Prove that $b_{\beta,n}^{(0)} \neq 0$ by an induction. The proof of $b_{\gamma,n}^{(0)} \neq 0$ is analogically. The proving is split according to the form of $w(x, y)$.

(i) Let $\operatorname{char} K \neq 2$ and $w(x,y) = y^2 - f(x)$. Because there exists the place $P_{(\alpha,\gamma)}$ then $\beta \neq 0$ by Lemma 2.7.

(a) Let $n = 2$. The coefficient $b^{(0)}_{\beta,2} = -2 \cdot \beta \neq 0$ by Equation (3.8).

(b) Let $b^{(0)}_{\beta,n} \neq 0$. Prove that $b^{(0)}_{\beta,n+1} \neq 0$. By Equations (3.10) and (3.36)

$$b^{(0)}_{\beta,n+1} = a^{(0)}_{\beta,n} - \beta \cdot b^{(0)}_{\beta,n} = -\beta \cdot b^{(0)}_{\beta,n} - \beta \cdot b^{(0)}_{\beta,n} = -2 \cdot \beta \cdot b^{(0)}_{\beta,n} \neq 0.$$

(ii) Let $\operatorname{char} K = 2$ and $w(x,y) = y^2 + x \cdot y - f_1(x)$. Because there exists the place $P_{(\alpha,\gamma)}$ then $\alpha \neq 0$ by Lemma 2.7.

(a) Let $n = 2$. The coefficient $b^{(0)}_{\beta,2} = \alpha \neq 0$ by Equation (3.11).

(b) Let $b^{(0)}_{\beta,n} \neq 0$. Prove that $b^{(0)}_{\beta,n+1} \neq 0$. By Equations (3.13) and (3.36)

$$b^{(0)}_{\beta,n+1} = a^{(0)}_{\beta,n} + (\alpha + \beta) \cdot b^{(0)}_{\beta,n} = \beta \cdot b^{(0)}_{\beta,n} + (\alpha + \beta) \cdot b^{(0)}_{\beta,n} = \alpha \cdot b^{(0)}_{\beta,n} \neq 0.$$

(iii) Let $\operatorname{char} K = 2$ and $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$. Because $w(x,y)$ is smooth then $a_3 \neq 0$.

(a) Let $n = 2$. The coefficient $b^{(0)}_{\beta,2} = a_3 \neq 0$ by Equation (3.14).

(b) Let $b^{(0)}_{\beta,n} \neq 0$. Prove that $b^{(0)}_{\beta,n+1} \neq 0$. By Equations (3.16) and (3.36)

$$b^{(0)}_{\beta,n+1} = a^{(0)}_{\beta,n} + (a_3 + \beta) \cdot b^{(0)}_{\beta,n} = \beta \cdot b^{(0)}_{\beta,n} + (a_3 + \beta) \cdot b^{(0)}_{\beta,n} = a_3 \cdot b^{(0)}_{\beta,n} \neq 0.$$

We have that $b^{(0)}_{\beta,n} \neq 0$ and $b^{(0)}_{\gamma,n} \neq 0$. Thus the determinant is not zero and the linear system has one solution and $\omega_{P_2}\left(\frac{1}{(x-\alpha)^n}\right)$, $\omega_{P_2}\left(\frac{y}{(x-\alpha)^n}\right)$ can be uniquely determined. By Equations (3.21) and (3.22) determine $\omega_{P_1}\left(\frac{1}{(x-\alpha)^n}\right)$ and $\omega_{P_1}\left(\frac{y}{(x-\alpha)^n}\right)$. $\square$

Theorem 3.30. *Let $P := P_{(\alpha,\beta)} \in \mathbb{P}^{(1)}_E$, $P_\alpha \in \mathbb{P}^{(1)}_{K(x)}$ where $\alpha, \beta \in K$ Let P be only one place such that $P \mid P_\alpha$ (cf. Lemma 2.7). If $n \geq 2$ then $\omega_P\left(\frac{1}{(x-\alpha)^n}\right) = \omega_P\left(\frac{y}{(x-\alpha)^n}\right) = 0$.*

Proof. By Theorem 3.6 $\omega_\infty\left(\frac{1}{(x-\alpha)^n}\right) = 0$. Because $\deg(1) + 1 < \deg\left((x-\alpha)^n\right)$ then $\omega_\infty\left(\frac{y}{(x-\alpha)^n}\right) = 0$ by Theorem 3.8. Use Lemma 3.20, Theorem 1.59 and $\omega\left(\frac{1}{(x-\alpha)^n}\right) = \omega\left(\frac{y}{(x-\alpha)^n}\right) = 0$ to show

$$\omega_P\left(\frac{1}{x-\alpha}\right) = \omega_P\left(\frac{y}{x-\alpha}\right) = 0.$$

$\square$

Example 3.31. Illustrate the tool from this chapter calculating $\omega_{(\alpha,\beta)}(u)$. Continue in Example 3.10. Recall that $E/\mathbb{F}_5$ is determined by $w(x,y) = y^2 - x^3 - x - 1$. The element

(i) $A = \frac{x^2+2+y\cdot\left(x^2+3\right)}{x^4+2x^3+x^2+2x} \in E$;

(ii) $B = \frac{3x^3+y\cdot\left(2x^3+1\right)}{4x^4+x^3+4x^2+x+4} \in E$;

(iii) $C = \frac{x+4+y\cdot\left(2x^5+2x^4+3x^2+3x+3\right)}{2x^3+2x^2+2x+2} \in E$;

are defined in Example 3.10. The results of Example 3.10 are $\omega_\infty\big(A\big) = 0$, $\omega_\infty\big(B\big) = 2$ and $\omega_\infty\big(C\big) = 1$.

The results of this example are values of $\omega_P(A), \omega_P(B), \omega_P(C)$ which can be non-zero. First, use Theorem 3.15 to find places where $\omega_P(A), \omega_P(B), \omega_P(C)$ can be non-zero. Next, by Algorithm 3.17 determine what local components in the forms $\omega_P\left(\frac{1}{(x-\alpha)^i}\right)$ and $\omega_P\left(\frac{y}{(x-\alpha)^i}\right)$ are necessary for the calculation of the non-zero local components. Use Theorem 3.21 and Algorithms 3.28 and 3.26 to calculate the necessary $\omega_P\left(\frac{1}{(x-\alpha)^i}\right)$ and $\omega_P\left(\frac{y}{(x-\alpha)^i}\right)$. At the end, calculate the values of the non-zero local components by Algorithm 3.17. Recall that $\kappa_P = 0$ in char $K \neq 2$ (cf. Definition 2.15 and Lemma 2.19).

(i) Work on the first example. Use Theorem 3.15 to find places where $\omega_P(A)$ can be non-zero. It holds $\gcd(a,c) = \gcd(b,c) = 1$. The denominator $c(x) = (x-3)^2 \cdot (x-2) \cdot x$ defines the following sets of places

$$S^A_{K(x)} := \{P_0, P_2, P_3\} \subset \mathbb{P}_{K(x)};$$

$$S^A_E := \{P_{(0,1)}, P_{(0,4)}, P_{(2,1)}, P_{(2,4)}, P_{(3,1)}, P_{(3,4)}, P_\infty\} \subset \mathbb{P}_E.$$

By Algorithm 3.17 it is necessary to know

$$\omega_{(0,1)}\Big(\frac{1}{x}\Big), \omega_{(0,1)}\Big(\frac{y}{x}\Big) \text{ for calculating } \omega_{(0,1)}\big(A\big);$$

$$\omega_{(0,4)}\Big(\frac{1}{x}\Big), \omega_{(0,4)}\Big(\frac{y}{x}\Big) \text{ for calculating } \omega_{(0,4)}\big(A\big);$$

$$\omega_{(2,1)}\Big(\frac{1}{x-2}\Big), \omega_{(2,1)}\Big(\frac{y}{x-2}\Big) \text{ for calculating } \omega_{(2,1)}\big(A\big);$$

$$\omega_{(2,4)}\Big(\frac{1}{x-2}\Big), \omega_{(2,4)}\Big(\frac{y}{x-2}\Big) \text{ for calculating } \omega_{(2,4)}\big(A\big);$$

$$\omega_{(3,1)}\Big(\frac{1}{(x-3)^i}\Big), \omega_{(3,1)}\Big(\frac{y}{(x-3)^i}\Big) \text{ where } i=1,2 \text{ for calculating } \omega_{(3,1)}\big(A\big);$$

$$\omega_{(3,4)}\Big(\frac{1}{(x-3)^i}\Big), \omega_{(3,4)}\Big(\frac{y}{(x-3)^i}\Big) \text{ where } i=1,2 \text{ for calculating } \omega_{(3,4)}\big(A\big).$$

(ii) Work on the second example. Use Theorem 3.15 to find places where $\omega_P(B)$ can be non-zero. It holds $\gcd(a,c) = \gcd(b,c) = 1$. The denominator $c(x) = 4 \cdot (x-4)^4$ defines the following sets of places

$$S^B_{K(x)} := \{P_4\} \subset \mathbb{P}_{K(x)};$$

$$S^B_E := \{P_{(4,2)}, P_{(4,3)}, P_\infty\} \subset \mathbb{P}_E.$$

By Algorithm 3.17 it is necessary to know

$$\omega_{(4,2)}\Big(\frac{1}{(x-4)^i}\Big), \omega_{(4,2)}\Big(\frac{y}{(x-4)^i}\Big) \text{ where } i = 1,2,3,4 \text{ for calculating } \omega_{(4,2)}\big(B\big);$$

$$\omega_{(4,3)}\Big(\frac{1}{(x-4)^i}\Big), \omega_{(4,3)}\Big(\frac{y}{(x-4)^i}\Big) \text{ where } i = 1,2,3,4 \text{ for calculating } \omega_{(4,3)}\big(B\big).$$

(iii) Work on the third example. Use Theorem 3.15 to find places where $\omega_P(A)$ can be non-zero. It holds $\gcd(a,c) = \gcd(b,c) = 1$. The denominator $c(x) = 4 \cdot (x-2) \cdot (x-3) \cdot (x-4)$ defines the following sets of places

$$S^C_{K(x)} := \{P_2, P_3, P_4\} \subset \mathbb{P}_{K(x)};$$

$$S^C_E := \{P_{(2,1)}, P_{(2,4)}, P_{(3,1)}, P_{(3,4)}, P_{(4,2)}, P_{(4,3)}, P_\infty\} \subset \mathbb{P}_E.$$

By Algorithm 3.17 it is necessary to know

$$\omega_{(2,1)}\Big(\frac{1}{x-2}\Big), \omega_{(2,1)}\Big(\frac{y}{x-2}\Big) \text{ for calculating } \omega_{(2,1)}\big(C\big);$$

$$\omega_{(2,4)}\Big(\frac{1}{x-2}\Big), \omega_{(2,4)}\Big(\frac{y}{x-2}\Big) \text{ for calculating } \omega_{(2,4)}\big(C\big);$$

$$\omega_{(3,1)}\Big(\frac{1}{x-3}\Big), \omega_{(3,1)}\Big(\frac{y}{x-3}\Big) \text{ for calculating } \omega_{(3,1)}\big(C\big);$$

$$\omega_{(3,4)}\Big(\frac{1}{x-3}\Big), \omega_{(3,4)}\Big(\frac{y}{x-3}\Big) \text{ for calculating } \omega_{(3,4)}\big(C\big);$$

$$\omega_{(4,2)}\Big(\frac{1}{x-4}\Big), \omega_{P_{(4,2)}}\Big(\frac{y}{x-4}\Big) \text{ for calculating } \omega_{(4,2)}\big(C\big);$$

$$\omega_{(4,3)}\Big(\frac{1}{x-4}\Big), \omega_{(4,3)}\Big(\frac{y}{x-4}\Big) \text{ for calculating } \omega_{(4,3)}\big(C\big).$$

Use Theorem 3.21 and calculate

$$\omega_{(0,1)}\Big(\frac{1}{x}\Big) = \omega_{(2,1)}\Big(\frac{1}{x-2}\Big) = \omega_{(3,1)}\Big(\frac{1}{x-3}\Big) = \frac{1}{1-4} = \frac{-1}{3} = -2 = 3;$$

$$\omega_{(0,4)}\Big(\frac{1}{x}\Big) = \omega_{(2,4)}\Big(\frac{1}{x-2}\Big) = \omega_{(3,4)}\Big(\frac{1}{x-3}\Big) = \frac{-1}{1-4} = \frac{1}{3} = 2;$$

$$\omega_{(0,1)}\Big(\frac{y}{x}\Big) = \omega_{(2,1)}\Big(\frac{y}{x-2}\Big) = \omega_{(3,1)}\Big(\frac{y}{x-3}\Big) = 1 + \frac{4}{1-4} = 1 - \frac{4}{3} = 1 - 8 = 3;$$

$$\omega_{(0,4)}\Big(\frac{y}{x}\Big) = \omega_{(2,4)}\Big(\frac{y}{x-2}\Big) = \omega_{(3,4)}\Big(\frac{y}{x-3}\Big) = 1 - \frac{1}{1-4} = 1 + \frac{1}{3} = 3;$$

$$\omega_{(4,2)}\Big(\frac{1}{x-4}\Big) = \frac{1}{2-3} = 4;$$

$$\omega_{(4,3)}\Big(\frac{1}{x-4}\Big) = -\frac{1}{2-3} = 1;$$

$$\omega_{(4,2)}\Big(\frac{y}{x-4}\Big) = 1 + \frac{3}{2-3} = 1 - 3 = 3;$$

$$\omega_{(4,3)}\Big(\frac{y}{x-4}\Big) = 1 - \frac{2}{2-3} = 1 + 2 = 3.$$

Use Algorithms 3.28 and 3.26 to calculate

$$\omega_{(3,1)}\Big(\frac{1}{(x-3)^2}\Big),\ \omega_{(3,1)}\Big(\frac{y}{(x-3)^2}\Big),\ \omega_{(3,4)}\Big(\frac{1}{(x-3)^2}\Big),\ \omega_{(3,4)}\Big(\frac{y}{(x-3)^2}\Big).$$

Calculate and use $y^2 = x^3 + x + 1$

$$(y-1)^2 = y^2 + 3y + 1 = x^3 + x + 2 + 3y;$$

$$(y-4)^2 = y^2 + 2y + 1 = x^3 + x + 2 + 2y.$$

Thus

$$a_{1,2}(x) = x^3 + x + 2;$$
$$b_{1,2}(x) = 3;$$
$$a_{4,2}(x) = x^3 + x + 2;$$
$$b_{4,2}(x) = 2.$$

Express $a_{1,2}(x), a_{4,2}(x)$ as polynomials in the variable $x-3$

$$a_{1,2}(x) = a_{4,2}(x) = (x-3)^3 + 4\cdot(x-3)^2 + 3\cdot(x-3) + 2.$$

Calculate

$$\begin{aligned} A_{1,2} &= b_{1,2}^{(1)} - a_{1,2}^{(1)}\cdot\omega_{P_{(3,4)}}\Big(\frac{1}{x-3}\Big) - b_{1,2}^{(1)}\cdot\omega_{P_{(3,4)}}\Big(\frac{y}{x-3}\Big) \\ &= 0 - 3\cdot 2 - 0\cdot 3 = 4; \end{aligned}$$

$$\begin{aligned} A_{4,2} &= b_{4,2}^{(1)} - a_{4,2}^{(1)}\cdot\omega_{P_{(3,1)}}\Big(\frac{1}{x-3}\Big) - b_{4,2}^{(1)}\cdot\omega_{P_{(3,1)}}\Big(\frac{y}{x-3}\Big) \\ &= 0 - 3\cdot 3 - 0\cdot 3 = 1. \end{aligned}$$

Create and solve a linear system

$$\begin{aligned} A_{1,2} &= a_{1,2}^{(0)}\cdot\omega_{(3,4)}\Big(\frac{1}{(x-3)^2}\Big) + b_{1,2}^{(0)}\cdot\omega_{(3,4)}\Big(\frac{y}{(x-3)^2}\Big); \\ A_{4,2} &= -a_{4,2}^{(0)}\cdot\omega_{(3,4)}\Big(\frac{1}{(x-3)^2}\Big) - b_{4,2}^{(0)}\cdot\omega_{(3,4)}\Big(\frac{y}{(x-3)^2}\Big); \end{aligned}$$

$$\left(\begin{array}{cc|c} 2 & 3 & 4 \\ 3 & 3 & 1 \end{array}\right) \simeq \left(\begin{array}{cc|c} 1 & 4 & 2 \\ 1 & 1 & 2 \end{array}\right) \simeq \left(\begin{array}{cc|c} 0 & 3 & 0 \\ 1 & 1 & 2 \end{array}\right) \simeq \left(\begin{array}{cc|c} 0 & 1 & 0 \\ 1 & 0 & 2 \end{array}\right).$$

It gives

$$\omega_{(3,4)}\Big(\tfrac{1}{(x-3)^2}\Big) = 2;\quad \omega_{(3,4)}\Big(\tfrac{y}{(x-3)^2}\Big) = 0.$$

By Equations (3.19) and (3.20)

$$\omega_{(3,1)}\Big(\tfrac{1}{(x-3)^2}\Big) = 3;\quad \omega_{(3,1)}\Big(\tfrac{y}{(x-3)^2}\Big) = 0.$$

Use algorithms 3.28 and 3.26 to calculate

$$\omega_{(4,2)}\Big(\frac{1}{(x-4)^i}\Big),\ \omega_{(4,2)}\Big(\frac{y}{(x-4)^i}\Big),\ \omega_{(4,3)}\Big(\frac{1}{(x-4)^i}\Big),\ \omega_{(4,3)}\Big(\frac{y}{(x-4)^i}\Big) \text{ for } i = 2, 3, 4.$$

Consider $K[x-4]$ instead of $K[x]$ during the calculation. Express $f(x)$ as a polynomial in $x-4$

$$f(x) = x^3 + x + 1 = (x-4)^3 + 2\cdot(x-4)^2 + 4\cdot(x-4) + 4.$$

Calculate and use $y^2 = x^3 + x + 1$

$$(y-2)^2 = y^2 + y + 4 = x^3 + x + 1 + y + 4 = x^3 + x + y;$$
$$(y-3)^2 = y^2 + 4y + 4 = x^3 + x + 1 + 4y + 4 = x^3 + x + 4y.$$

It gives

$$a_{2,2}(x) = x^3 + x = (x-4)^3 + 2\cdot(x-4)^2 + 4\cdot(x-4) + 3;$$
$$b_{2,2}(x) = 1;$$
$$a_{3,2}(x) = x^3 + x = (x-4)^3 + 2\cdot(x-4)^2 + 4\cdot(x-4) + 3;$$
$$b_{3,2}(x) = 4.$$

Use Equations (3.9) and (3.10) from Algorithm 3.26 to calculate recursively $a_{i,j}(x)$ and $b_{i,j}(x)$ for $i = 2,3$ and $j = 3,4$

$$a_{2,3}(x) = f(x)\cdot b_{2,2}(x) - 2\cdot a_{2,2}(x) = 4\cdot(x-4)^3 + 3\cdot(x-4)^2 + (x-4) + 3;$$
$$b_{2,3}(x) = a_{2,2}(x) - 2\cdot b_{2,2}(x) = (x-4)^3 + 2\cdot(x-4)^2 + 4\cdot(x-4) + 1;$$
$$a_{3,3}(x) = f(x)\cdot b_{3,2}(x) - 3\cdot a_{3,2}(x) = (x-4)^3 + 2\cdot(x-4)^2 + 4\cdot(x-4) + 2;$$
$$b_{3,3}(x) = a_{3,2}(x) - 3\cdot b_{3,2}(x) = (x-4)^3 + 2\cdot(x-4)^2 + 4\cdot(x-4) + 1;$$

$$a_{2,4}(x) = f(x)\cdot b_{2,3}(x) - 2\cdot a_{2,3}(x) =$$
$$(x-4)^6 + 4\cdot(x-4)^5 + 2\cdot(x-4)^4 + 3\cdot(x-4)^3 + 0\cdot(x-4)^2 + 3\cdot(x-4) + 3;$$
$$b_{2,4}(x) = a_{2,3}(x) - 2\cdot b_{2,3}(x) = 2\cdot(x-4)^3 + 4\cdot(x-4)^2 + 3\cdot(x-4) + 1;$$

$$a_{3,4}(x) = f(x)\cdot b_{3,3}(x) - 3\cdot a_{3,3}(x) =$$
$$(x-4)^6 + 4\cdot(x-4)^5 + 2\cdot(x-4)^4 + 3\cdot(x-4)^3 + 0\cdot(x-4)^2 + 3\cdot(x-4) + 3;$$
$$b_{3,4}(x) = a_{3,3}(x) - 3\cdot b_{3,3}(x) = 3\cdot(x-4)^3 + 1\cdot(x-4)^2 + 2\cdot(x-4) + 4.$$

(i) Calculate the first level of the recursion for $n = 2$

$$A_{2,2} = b_{2,2}^{(1)} - a_{2,2}^{(1)}\cdot\omega_{(4,3)}\left(\frac{1}{x-4}\right) - b_{2,2}^{(1)}\cdot\omega_{(4,3)}\left(\frac{y}{x-4}\right) = 0 - 4\cdot 1 - 0\cdot 3 = 1;$$

$$A_{3,2} = b_{3,2}^{(1)} - a_{3,2}^{(1)}\cdot\omega_{(4,2)}\left(\frac{1}{x-4}\right) - b_{3,2}^{(1)}\cdot\omega_{(4,2)}\left(\frac{y}{x-4}\right) = 0 - 4\cdot 4 - 0\cdot 3 = 4;$$

$$\left(\begin{array}{cc|c} a_{2,2}^{(0)} & b_{2,2}^{(0)} & A_{2,2} \\ -a_{3,2}^{(0)} & -b_{3,2}^{(0)} & A_{3,2} \end{array}\right) = \left(\begin{array}{cc|c} 3 & 1 & 1 \\ 2 & 1 & 4 \end{array}\right) \simeq \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 2 & 1 & 4 \end{array}\right) \simeq \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 0 \end{array}\right).$$

It gives

$$\omega_{(4,3)}\left(\tfrac{1}{(x-4)^2}\right) = 2;\quad \omega_{(4,3)}\left(\tfrac{y}{(x-4)^2}\right) = 0.$$

By Equations (3.19) and (3.20)

$$\omega_{(4,2)}\left(\tfrac{1}{(x-4)^2}\right) = 3;\quad \omega_{(4,2)}\left(\tfrac{y}{(x-4)^2}\right) = 0.$$

(ii) Calculate the second level of the recursion for $n = 3$

$$\begin{aligned} A_{2,3} &= b_{2,3}^{(2)} \\ &\quad - a_{2,3}^{(1)} \cdot \omega_{(4,3)}\Big(\frac{1}{(x-4)^2}\Big) - a_{2,3}^{(2)} \cdot \omega_{(4,3)}\Big(\frac{1}{x-4}\Big) \\ &\quad - b_{2,3}^{(1)} \cdot \omega_{(4,3)}\Big(\frac{y}{(x-4)^2}\Big) - b_{2,3}^{(2)} \cdot \omega_{(4,3)}\Big(\frac{y}{x-4}\Big) \\ &= 2 - 1 \cdot 2 - 3 \cdot 1 - 4 \cdot 0 - 2 \cdot 3 = 1; \end{aligned}$$

$$\begin{aligned} A_{3,3} &= b_{3,3}^{(2)} \\ &\quad - a_{3,3}^{(1)} \cdot \omega_{(4,2)}\Big(\frac{1}{(x-4)^2}\Big) - a_{3,3}^{(2)} \cdot \omega_{(4,2)}\Big(\frac{1}{x-4}\Big) \\ &\quad - b_{3,3}^{(1)} \cdot \omega_{(4,2)}\Big(\frac{y}{(x-4)^2}\Big) - b_{3,3}^{(2)} \cdot \omega_{(4,2)}\Big(\frac{y}{x-4}\Big) \\ &= 2 - 4 \cdot 3 - 2 \cdot 4 - 4 \cdot 0 - 2 \cdot 3 = 1; \end{aligned}$$

$$\left(\begin{array}{cc|c} a_{2,3}^{(0)} & b_{2,3}^{(0)} & A_{2,3} \\ -a_{3,3}^{(0)} & -b_{3,3}^{(0)} & A_{3,3} \end{array}\right) = \left(\begin{array}{cc|c} 3 & 1 & 1 \\ 3 & 4 & 1 \end{array}\right) \simeq \left(\begin{array}{cc|c} 3 & 1 & 1 \\ 0 & 1 & 0 \end{array}\right) \simeq \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 0 \end{array}\right).$$

It gives

$$\omega_{(4,3)}\Big(\tfrac{1}{(x-4)^3}\Big) = 2; \quad \omega_{(4,3)}\Big(\tfrac{y}{(x-3)^3}\Big) = 0.$$

By Equations (3.19) and (3.20)

$$\omega_{(4,2)}\Big(\tfrac{1}{(x-4)^3}\Big) = 3; \quad \omega_{(4,2)}\Big(\tfrac{y}{(x-4)^3}\Big) = 0.$$

(iii) Calculate the third level of the recursion for $n = 4$

$$\begin{aligned} A_{2,4} &= b_{2,4}^{(3)} \\ &\quad - a_{2,4}^{(1)} \cdot \omega_{(4,3)}\Big(\frac{1}{(x-4)^3}\Big) - a_{2,4}^{(2)} \cdot \omega_{(4,3)}\Big(\frac{1}{(x-4)^2}\Big) - a_{2,4}^{(3)} \cdot \omega_{(4,3)}\Big(\frac{1}{x-4}\Big) \\ &\quad - b_{2,4}^{(1)} \cdot \omega_{(4,3)}\Big(\frac{y}{(x-4)^3}\Big) - b_{2,4}^{(2)} \cdot \omega_{(4,3)}\Big(\frac{y}{(x-4)^2}\Big) - b_{2,4}^{(3)} \cdot \omega_{(4,3)}\Big(\frac{y}{x-4}\Big) \\ &= 2 - 3 \cdot 2 - 0 \cdot 2 - 3 \cdot 1 - 3 \cdot 0 - 4 \cdot 0 - 2 \cdot 3 \\ &= 2 - 1 - 0 - 3 - 0 - 0 - 1 = 2; \end{aligned}$$

$$\begin{aligned} A_{3,4} &= b_{3,4}^{(3)} \\ &\quad - a_{3,4}^{(1)} \cdot \omega_{P_{(4,2)}}\Big(\frac{1}{(x-4)^3}\Big) - a_{3,4}^{(2)} \cdot \omega_{P_{(4,2)}}\Big(\frac{1}{(x-4)^2}\Big) - a_{3,4}^{(3)} \cdot \omega_{P_{(4,2)}}\Big(\frac{1}{x-4}\Big) \\ &\quad - b_{3,4}^{(1)} \cdot \omega_{P_{(4,2)}}\Big(\frac{y}{(x-4)^3}\Big) - b_{3,4}^{(2)} \cdot \omega_{P_{(4,2)}}\Big(\frac{y}{(x-4)^2}\Big) - b_{3,4}^{(3)} \cdot \omega_{P_{(4,2)}}\Big(\frac{y}{x-4}\Big) \\ &= 3 - 3 \cdot 3 - 0 \cdot 3 - 3 \cdot 4 - 2 \cdot 0 - 1 \cdot 0 - 3 \cdot 3 \\ &= 3 - 4 - 0 - 2 - 0 - 0 - 4 = 3; \end{aligned}$$

$$\left(\begin{array}{cc|c} a_{2,4}^{(0)} & b_{2,4}^{(0)} & A_{2,4} \\ -a_{3,4}^{(0)} & -b_{3,4}^{(0)} & A_{3,4} \end{array}\right) = \left(\begin{array}{cc|c} 3 & 1 & 2 \\ 2 & 1 & 3 \end{array}\right) \simeq \left(\begin{array}{cc|c} 1 & 0 & 4 \\ 2 & 1 & 3 \end{array}\right) \simeq \left(\begin{array}{cc|c} 1 & 0 & 4 \\ 0 & 1 & 0 \end{array}\right).$$

It gives

$$\omega_{P_{(4,3)}}\left(\tfrac{1}{(x-4)^4}\right) = 4;\quad \omega_{P_{(4,3)}}\left(\tfrac{y}{(x-3)^4}\right) = 0.$$

By Equations (3.19) and (3.20)

$$\omega_{P_{(4,2)}}\left(\tfrac{1}{(x-4)^4}\right) = 1;\quad \omega_{P_{(4,2)}}\left(\tfrac{y}{(x-4)^4}\right) = 0.$$

Use algorithm 3.17 to calculate of the needed local components.

(i) Work on the first example.

(a) Calculate $\omega_{(0,1)}(A)$ and $\omega_{(0,4)}(A)$. Decompose $c(x) = \tilde{c}(x)\cdot x$ where $\tilde{c}(x) := (x-3)^2\cdot(x-2) = x^3+2x^2+x+2$. Find Bézout coefficients $u(x)$ and $v(x)$ such that $1 = u(x)\cdot\tilde{c}(x) + v(x)\cdot x$. The coefficient $u(x) = 3$ and $v(x) = 2x^2+4x+2$. Make $\tilde{a}(x) = u(x)\cdot a(x) = 3x^2+1$ and $\tilde{b}(x) = u(x)\cdot b(x) = 3x^2+4$. It gives

$$\omega_{(0,1)}(A) = 1\cdot\omega_{(0,1)}\left(\frac{1}{x}\right) + 4\cdot\omega_{(0,1)}\left(\frac{y}{x}\right) = 1\cdot 3 + 4\cdot 3 = 0;$$

$$\omega_{(0,4)}(A) = 1\cdot\omega_{(0,4)}\left(\frac{1}{x}\right) + 4\cdot\omega_{(0,4)}\left(\frac{y}{x}\right) = 1\cdot 2 + 4\cdot 3 = 4.$$

(b) Calculate $\omega_{(2,1)}(A)$ and $\omega_{(2,4)}(A)$. Decompose $c(x) = \tilde{c}(x)\cdot(x-2)$ where $\tilde{c}(x) := (x-3)^2\cdot x = x^3+4x^2+4x$. Find Bézout coefficients $u(x)$ and $v(x)$ such that $1 = u(x)\cdot\tilde{c}(x) + v(x)\cdot(x-2)$. The coefficient $u(x) = 3$ and $v(x) = 2x^2+2x+2$. Make $\tilde{a}(x) = u(x)\cdot a(x) = 3x^2+1$ and $\tilde{b}(x) = u(x)\cdot b(x) = 3x^2+4$. Express $\tilde{a}(x), \tilde{b}(x)$ as polynomials in the variable $x-2$

$$\tilde{a}(x) = (3x+1)\cdot(x-2)+3;$$

$$\tilde{b}(x) = (3x+1)\cdot(x-2)+1.$$

It gives

$$\omega_{(2,1)}(A) = 3\cdot\omega_{(2,1)}\left(\frac{1}{x-2}\right) + 1\cdot\omega_{(2,1)}\left(\frac{y}{x-2}\right) = 3\cdot 3 + 1\cdot 3 = 2;$$

$$\omega_{(2,4)}(A) = 3\cdot\omega_{(2,4)}\left(\frac{1}{x-2}\right) + 1\cdot\omega_{(2,4)}\left(\frac{y}{x-2}\right) = 3\cdot 2 + 1\cdot 3 = 4.$$

(c) Calculate $\omega_{(3,1)}(A)$ and $\omega_{(3,4)}(A)$. Decompose $c(x) = \tilde{c}(x)\cdot(x-3)^2$ where $\tilde{c}(x) := (x-2)\cdot x = x^2+3x$. Find Bézout coefficients $u(x)$ and $v(x)$ such that $1 = u(x)\cdot\tilde{c}(x) + v(x)\cdot(x-3)^2$. The coefficient $u(x) = 4x$ and $v(x) = x+4$. Make $\tilde{a}(x) = u(x)\cdot a(x) = 4x^3+3x$ and $\tilde{b}(x) = u(x)\cdot b(x) = 4x^3+2x$. Express $\tilde{a}(x), \tilde{b}(x)$ as polynomials in the variable $x-3$

$$\tilde{a}(x) = (4x+4)\cdot(x-3)^2 + 1\cdot(x-3)+2;$$

$$\tilde{b}(x) = (4x+4)\cdot(x-3)^2 + 0\cdot(x-3) + 4.$$

It gives

$$\begin{aligned}\omega_{(3,1)}\big(A\big) &= 1\cdot\omega_{(3,1)}\Big(\frac{1}{x-3}\Big) + 2\cdot\omega_{(3,1)}\Big(\frac{1}{(x-3)^2}\Big) + 4\cdot\omega_{(3,1)}\Big(\frac{y}{(x-3)^2}\Big)\\ &= 1\cdot 3 + 2\cdot 3 + 4\cdot 0 = 4;\end{aligned}$$

$$\begin{aligned}\omega_{(3,4)}\big(A\big) &= 1\cdot\omega_{(3,4)}\Big(\frac{1}{x-3}\Big) + 2\cdot\omega_{(3,4)}\Big(\frac{1}{(x-3)^2}\Big) + 4\cdot\omega_{(3,4)}\Big(\frac{y}{(x-3)^2}\Big)\\ &= 1\cdot 2 + 2\cdot 2 + 4\cdot 0 = 1.\end{aligned}$$

There is a recapitulation of the calculated values from the first example

$$\begin{array}{ll}\omega_{(0,1)}\big(A\big) = 0; & \omega_{(0,4)}\big(A\big) = 4;\\ \omega_{(2,1)}\big(A\big) = 2; & \omega_{(2,4)}\big(A\big) = 4;\\ \omega_{(3,1)}\big(A\big) = 4; & \omega_{(3,4)}\big(A\big) = 1;\\ \omega_{\infty}\big(A\big) = 0. & \end{array}$$

By Theorem 1.59 and Definition 1.45 $\omega\big(A\big) = 0$. Make the basic check

$$\omega\big(A\big) = \sum_{P\in S_E^A} \omega_P\big(A\big) = 0.$$

(ii) Work on the second example. Calculate $\omega_{(4,2)}\big(B\big)$ and $\omega_{(4,3)}\big(B\big)$. Decompose $c(x) = \tilde{c}(x)\cdot(x-4)^4$ where $\tilde{c}(x) := 4$. Find Bézout coefficients $u(x)$ and $v(x)$ such that $1 = u(x)\cdot\tilde{c}(x) + v(x)\cdot(x-4)^4$. The coefficient $u(x) = 4$ and $v(x) = 0$. Make $\tilde{a}(x) = u(x)\cdot a(x) = 2x^3$ and $\tilde{b}(x) = u(x)\cdot b(x) = 3x^3 + 4$. Express $\tilde{a}(x), \tilde{b}(x)$ as polynomials in the variable $x-4$

$$\tilde{a}(x) = 2\cdot(x-4)^3 + 4\cdot(x-4)^2 + 1\cdot(x-4) + 3;$$

$$\tilde{b}(x) = 3\cdot(x-4)^3 + 1\cdot(x-4)^2 + 4\cdot(x-4) + 1.$$

It gives

$$\begin{aligned}\omega_{(4,2)}\big(B\big) =& 3\cdot\omega_{(4,2)}\Big(\frac{1}{(x-4)^4}\Big) + 1\cdot\omega_{(4,2)}\Big(\frac{1}{(x-4)^3}\Big)\\ &+4\cdot\omega_{(4,2)}\Big(\frac{1}{(x-4)^2}\Big) + 2\cdot\omega_{(4,2)}\Big(\frac{1}{x-4}\Big)\\ &+1\cdot\omega_{(4,2)}\Big(\frac{y}{(x-4)^4}\Big) + 4\cdot\omega_{(4,2)}\Big(\frac{y}{(x-4)^3}\Big)\\ &+1\cdot\omega_{(4,2)}\Big(\frac{y}{(x-4)^2}\Big) + 3\cdot\omega_{(4,2)}\Big(\frac{y}{x-4}\Big)\\ =& 3\cdot 1 + 1\cdot 3 + 4\cdot 3 + 2\cdot 4 + 1\cdot 0 + 4\cdot 0 + 1\cdot 0 + 3\cdot 3\\ =& 3 + 3 + 2 + 3 + 0 + 0 + 0 + 4 = 0;\end{aligned}$$

$$
\begin{aligned}
\omega_{(4,3)}\big(B\big) =& 3\cdot\omega_{(4,3)}\Big(\frac{1}{(x-4)^4}\Big)+1\cdot\omega_{(4,3)}\Big(\frac{1}{(x-4)^3}\Big)\\
&+4\cdot\omega_{(4,3)}\Big(\frac{1}{(x-4)^2}\Big)+2\cdot\omega_{(4,3)}\Big(\frac{1}{x-4}\Big)\\
&+1\cdot\omega_{(4,3)}\Big(\frac{y}{(x-4)^4}\Big)+4\cdot\omega_{(4,3)}\Big(\frac{y}{(x-4)^3}\Big)\\
&+1\cdot\omega_{(4,3)}\Big(\frac{y}{(x-4)^2}\Big)+3\cdot\omega_{(4,3)}\Big(\frac{y}{x-4}\Big)\\
=& 3\cdot 4+1\cdot 2+4\cdot 2+2\cdot 1+1\cdot 0+4\cdot 0+1\cdot 0+3\cdot 3\\
=& 2+2+3+2+0+0+0+4=3.
\end{aligned}
$$

There is a recapitulation of the calculated values from the second example

$$\omega_{(4,2)}\big(B\big)=0;\quad \omega_{(4,3)}\big(B\big)=3;\quad \omega_{\infty}\big(B\big)=2.$$

By Theorem 1.59 and Definition 1.45 $\omega\big(A\big)=0$. Make the basic check

$$\omega\big(B\big)=\sum_{P\in S_E^B}\omega_P\big(B\big)=0.$$

(iii) Work on the third example.

(a) Calculate $\omega_{(2,1)}\big(C\big)$ and $\omega_{(2,4)}\big(C\big)$. Decompose $c(x)=\tilde{c}(x)\cdot(x-2)$ where $\tilde{c}(x):=2\cdot(x-3)\cdot(x-4)=2x^2+x+4$. Find Bézout coefficients $u(x)$ and $v(x)$ such that $1=u(x)\cdot\tilde{c}(x)+v(x)\cdot(x-2)$. The coefficient $u(x)=4$ and $v(x)=2x$. Make $\tilde{a}(x)=u(x)\cdot a(x)=4x+1$ and $\tilde{b}(x)=u(x)\cdot b(x)=3x^5+3x^4+2x^2+2x+2$. Express $\tilde{a}(x),\tilde{b}(x)$ as polynomials in the variable $x-2$

$$\tilde{a}(x)=4\cdot(x-2)+4;$$

$$\tilde{b}(x)=(3x^4+4x^3+3x^2+3x+3)\cdot(x-2)+3.$$

It gives

$$\omega_{(2,1)}\big(C\big)=4\cdot\omega_{(2,1)}\Big(\frac{1}{x-2}\Big)+3\cdot\omega_{(2,1)}\Big(\frac{y}{x-2}\Big)=4\cdot 3+3\cdot 3=1;$$

$$\omega_{(2,4)}\big(C\big)=4\cdot\omega_{(2,4)}\Big(\frac{1}{x-2}\Big)+3\cdot\omega_{(2,4)}\Big(\frac{y}{x-2}\Big)=4\cdot 2+3\cdot 3=2.$$

(b) Calculate $\omega_{(3,1)}\big(C\big)$ and $\omega_{(3,4)}\big(C\big)$. Decompose $c(x)=\tilde{c}(x)\cdot(x-3)$ where $\tilde{c}(x):=2\cdot(x-2)\cdot(x-4)=2x^2+3x+1$. Find Bézout coefficients $u(x)$ and $v(x)$ such that $1=u(x)\cdot\tilde{c}(x)+v(x)\cdot(x-3)$. The coefficient $u(x)=2$ and $v(x)=x+2$. Make $\tilde{a}(x)=u(x)\cdot a(x)=2x+3$ and $\tilde{b}(x)=u(x)\cdot b(x)=4x^5+4x^4+x^2+x+1$. Express $\tilde{a}(x),\tilde{b}(x)$ as polynomials in the variable $x-3$

$$\tilde{a}(x)=2\cdot(x-3)+4;$$

$$\tilde{b}(x)=(4x^4+x^3+3x^2+1)\cdot(x-3)+4.$$

It gives

$$\omega_{(3,1)}\big(C\big) = 4 \cdot \omega_{(3,1)}\Big(\frac{1}{x-3}\Big) + 4 \cdot \omega_{(3,1)}\Big(\frac{y}{x-3}\Big) = 4 \cdot 3 + 4 \cdot 3 = 4;$$

$$\omega_{(3,4)}\big(C\big) = 4 \cdot \omega_{(3,4)}\Big(\frac{1}{x-3}\Big) + 4 \cdot \omega_{(3,4)}\Big(\frac{y}{x-3}\Big) = 4 \cdot 2 + 4 \cdot 3 = 0.$$

(c) Calculate $\omega_{(4,2)}\big(C\big)$ and $\omega_{(4,3)}\big(C\big)$. Decompose $c(x) = \tilde{c}(x) \cdot (x-4)$ where $\tilde{c}(x) := 2 \cdot (x-2) \cdot (x-3) = 2x^2 + 2$. Find Bézout coefficients $u(x)$ and $v(x)$ such that $1 = u(x) \cdot \tilde{c}(x) + v(x) \cdot (x-4)$. The coefficient $u(x) = 4$ and $v(x) = 2x+3$. Make $\tilde{a}(x) = u(x) \cdot a(x) = 4x+1$ and $\tilde{b}(x) = u(x) \cdot b(x) = 3x^5 + 3x^4 + 2x^2 + 2x + 2$. Express $\tilde{a}(x), \tilde{b}(x)$ as polynomials in the variable $x-4$

$$\tilde{a}(x) = 4 \cdot (x-4) + 2;$$
$$\tilde{b}(x) = (3x^4 + 2x) \cdot (x-4) + 2.$$

It gives

$$\omega_{(4,2)}\big(C\big) = 2 \cdot \omega_{(4,2)}\Big(\frac{1}{x-4}\Big) + 2 \cdot \omega_{(4,2)}\Big(\frac{y}{x-4}\Big) = 2 \cdot 4 + 2 \cdot 3 = 4;$$

$$\omega_{(4,3)}\big(C\big) = 2 \cdot \omega_{(4,3)}\Big(\frac{1}{x-4}\Big) + 2 \cdot \omega_{(4,3)}\Big(\frac{y}{x-4}\Big) = 2 \cdot 1 + 2 \cdot 3 = 3.$$

There is a recapitulation of the calculated values from the third example

$$\begin{aligned} &\omega_{(2,1)}\big(C\big) = 1; \quad \omega_{(2,4)}\big(C\big) = 2;\\ &\omega_{(3,1)}\big(C\big) = 4; \quad \omega_{(3,4)}\big(C\big) = 0;\\ &\omega_{(4,2)}\big(C\big) = 4; \quad \omega_{(4,3)}\big(C\big) = 3;\\ &\omega_{\infty}\big(C\big) = 1. \end{aligned}$$

By Theorem 1.59 and Definition 1.45 $\omega\big(A\big) = 0$. Make the basic check

$$\omega\big(C\big) = \sum_{P \in S_E^C} \omega_P\big(C\big) = 0.$$

Remark 3.32. In Example 3.31 all local components of the form $\omega_{(\alpha,\beta)}\Big(\frac{y}{(x-\alpha)^n}\Big)$ are zero for $n \geq 2$. It is not a coincidence. Show that $\omega_{(\alpha,\beta)}\Big(\frac{y}{(x-\alpha)^n}\Big) = 0$ if $\operatorname{char} K \neq 2$, $w(x,y) = y^2 - f(x)$, $n \geq 2$ and there exists $P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ such that $\beta \neq \gamma$.

Lemma 3.33. *Let* $\operatorname{char} K \neq 2$*,* $w(x,y) = y^2 - f(x)$ *and* $n \geq 2$*. Let* $(\alpha,\beta), (\alpha,\gamma) \in \mathbb{V}_w(K)$ *such that* $\beta \neq \gamma$*. Define the following polynomials in* $K[x-\alpha]$

$$a_{\beta,n}(x) = \sum_{i=0}^{\deg\left(a_{\beta,n}\right)} a_{\beta,n}^{(i)} \cdot (x-\alpha)^i; \quad b_{\beta,n}(x) = \sum_{i=0}^{\deg\left(b_{\beta,n}\right)} b_{\beta,n}^{(i)} \cdot (x-\alpha)^i;$$

$$a_{\gamma,n}(x) = \sum_{i=0}^{\deg\left(a_{\gamma,n}\right)} a_{\gamma,n}^{(i)} \cdot (x-\alpha)^i; \quad b_{\gamma,n}(x) = \sum_{i=0}^{\deg\left(b_{\gamma,n}\right)} b_{\gamma,n}^{(i)} \cdot (x-\alpha)^i$$

such that

$$a_{\beta,n}(x) + y \cdot b_{\beta,n}(x) = (y-\beta)^n;$$
$$a_{\gamma,n}(x) + y \cdot b_{\gamma,n}(x) = (y-\gamma)^n \text{ (cf. Theorem 1.116).}$$

Then

(i) $a_{\beta,n}(x) = (-1)^n \cdot a_{\gamma,n}(x)$ *and* $b_{\beta,n}(x) = (-1)^{n+1} \cdot b_{\gamma,n}(x)$*;*

(ii) $a^{(0)}_{\beta,n} = (-1)^n \cdot 2^{n-1} \cdot \beta^n$ *and* $b^{(0)}_{\beta,n} = (-1)^{n-1} \cdot 2^{n-1} \cdot \beta^{n-1}$.

Proof. Recall that

1. $\beta = -\gamma$ and $\beta \neq 0$ by Lemma 2.6;
2. $f(\alpha) = \beta^2$.

(i) Use an induction.

1. Let $n = 2$. Use Equations (3.7) and (3.8)

$$a_{\beta,2}(x) = f(x) + \beta^2;$$
$$a_{\gamma,2}(x) = f(x) + \gamma^2 = f(x) + (-\beta)^2 = a_{\beta,2}(x);$$
$$b_{\beta,2}(x) = -2 \cdot \beta;$$
$$b_{\gamma,2}(x) = -2 \cdot \gamma = 2 \cdot \beta = -b_{\beta,2}(x).$$

2. Assume that the assertion is satisfied for $n-1$. Prove it for n. Use Equations (3.9) and (3.10)

$$a_{\beta,n}(x) = f(x) \cdot b_{\beta,n-1}(x) - \beta \cdot a_{\beta,n-1}(x);$$

$$\begin{aligned} a_{\gamma,n}(x) &= f(x) \cdot b_{\gamma,n-1}(x) - \gamma \cdot a_{\gamma,n-1}(x) \\ &= f(x) \cdot b_{\beta,n-1}(x) \cdot (-1)^n - \beta \cdot a_{\beta,n-1}(x) \cdot (-1)^n \\ &= (-1)^n \cdot a_{\beta,n}(x); \end{aligned}$$

$$b_{\beta,n}(x) = a_{\beta,n-1}(x) - \beta \cdot b_{\beta,n-1}(x);$$

$$\begin{aligned} b_{\gamma,n}(x) &= a_{\gamma,n-1}(x) - \gamma \cdot b_{\gamma,n-1}(x) \\ &= a_{\beta,n-1}(x) \cdot (-1)^{n-1} - \beta \cdot b_{\beta,n-1}(x) \cdot (-1)^{n+1} \\ &= (-1)^{n+1} \cdot b_{\beta,n}(x). \end{aligned}$$

(ii) Use an induction.

1. Let $n = 2$. Use Equations (3.7) and (3.8)

$$a^{(0)}_{\beta,2} = a_{\beta,2}(\alpha) = f(\alpha) + \beta^2 = 2 \cdot \beta^2;$$

$$b^{(0)}_{\beta,2} = b_{\beta,2}(\alpha) = -2 \cdot \beta.$$

2. Assume that the assertion is satisfied for $n-1$. Prove it for n. Use Equations (3.9) and (3.10)

$$\begin{aligned}
a^{(0)}_{\beta,n} &= a_{\beta,n}(\alpha)\\
&= f(\alpha)\cdot b_{\beta,n-1}(\alpha) - \beta\cdot a_{\beta,n-1}(\alpha)\\
&= \beta^2\cdot(-1)^{n-2}\cdot 2^{n-2}\cdot\beta^{n-2} - \beta\cdot(-1)^{n-1}\cdot 2^{n-2}\cdot\beta^{n-1}\\
&= (-1)^n\cdot\beta^n\cdot 2^{n-2} + (-1)^n\cdot\beta^n\cdot 2^{n-2}\\
&= (-1)^n\cdot\beta^n\cdot 2^{n-1};
\end{aligned}$$

$$\begin{aligned}
b^{(0)}_{\beta,n} &= b_{\beta,n}(\alpha)\\
&= a_{\beta,n-1}(\alpha) - \beta\cdot b_{\beta,n-1}(\alpha)\\
&= (-1)^{n-1}\cdot 2^{n-2}\cdot\beta^{n-1} - \beta\cdot(-1)^{n-2}\cdot 2^{n-2}\cdot\beta^{n-2}\\
&= (-1)^{n-1}\cdot\beta^{n-1}\cdot 2^{n-1}.
\end{aligned}$$

□

Theorem 3.34. *Let* char $K\neq 2$ *and* $w(x,y)=y^2-f(x)$. *If* $P_1 := P_{(\alpha,\beta)}, P_2 := P_{(\alpha,\gamma)}\in\mathbb{P}^{(1)}_E$ *such that* $\beta\neq\gamma$ *and* $n\geq 2$ *then* $\omega_{P_1}\Big(\frac{y}{(x-\alpha)^n}\Big)=\omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big)=0.$

Proof. Show that the solution of the linear system from step 3 of Algorithm 3.28 is $\omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big)=0$. By Equation (3.20) if $\omega_{P_2}\Big(\frac{y}{(x-\alpha)^n}\Big)=0$ then $\omega_{P_1}\Big(\frac{y}{(x-\alpha)^n}\Big)=0$. Use Cramer's rule to show

$$\det\begin{pmatrix} a^{(0)}_{\beta,n} & A_{\beta,n}\\ -a^{(0)}_{\gamma,n} & A_{\gamma,n}\end{pmatrix} = a^{(0)}_{\beta,n}\cdot A_{\gamma,n} + a^{(0)}_{\gamma,n}\cdot A_{\beta,n} = 0.$$

By Lemma 3.33 the determinant can be expressed

$$\begin{aligned}
&a^{(0)}_{\beta,n}\cdot A_{\gamma,n} + a^{(0)}_{\gamma,n}\cdot A_{\beta,n}\\
=&(-1)^n\cdot 2^{n-1}\cdot\beta^n\cdot A_{\gamma,n} + (-1)^n\cdot 2^{n-1}\cdot(-\beta)^n\cdot A_{\beta,n}\\
=&(-1)^n\cdot 2^{n-1}\cdot\beta^n\cdot\big(A_{\gamma,n}+(-1)^n\cdot A_{\beta,n}\big).
\end{aligned}$$

The term $(-1)^n\cdot 2^{n-1}\cdot\beta^n\neq 0$ by the assumption. Show that $A_{\gamma,n}=(-1)^{n+1}\cdot A_{\beta,n}$ by an induction. Recall that $\kappa_P=0$ in char $K\neq 2$.

(i) Let $n=2$. By Lemma 3.33

$$\begin{aligned}
A_{\beta,2} &= b^{(1)}_{\beta,2} - a^{(1)}_{\beta,2}\cdot\omega_{P_2}\Big(\frac{1}{(x-\alpha)^1}\Big) - b^{(1)}_{\beta,2}\cdot\omega_{P_2}\Big(\frac{y}{(x-\alpha)^1}\Big) \quad \text{(cf. Equation (3.17))}\\
&= 0 - a^{(1)}_{\beta,2}\cdot\frac{-1}{2\beta} - 0\cdot\frac{1}{2} = \frac{1}{2\beta}\cdot a^{(1)}_{\beta,2};
\end{aligned}$$

$$\begin{aligned}
A_{\gamma,2} &= b^{(1)}_{\gamma,2} - a^{(1)}_{\gamma,2}\cdot\omega_{P_1}\Big(\frac{1}{(x-\alpha)^1}\Big) - b^{(1)}_{\gamma,2}\cdot\omega_{P_1}\Big(\frac{y}{(x-\alpha)^1}\Big) \quad \text{(cf. Equation (3.18))}\\
&= 0 - a^{(1)}_{\gamma,2}\cdot\frac{1}{2\beta} - 0\cdot\frac{1}{2} = \frac{-1}{2\beta}\cdot a^{(1)}_{\gamma,2} = \frac{1}{2\beta}\cdot a^{(1)}_{\beta,2}\cdot(-1)^3.
\end{aligned}$$

(ii) Assume that $A_{\gamma,i} = (-1)^{i+1} \cdot A_{\beta,i}$ for $i = 2, \dots n-1$. Thus $\omega_{P_1}\Big(\frac{y}{(x-\alpha)^i}\Big) = 0$ and $\omega_{P_2}\Big(\frac{y}{(x-\alpha)^i}\Big) = 0$ for $i = 2, \dots n-1$. By Lemma 3.33

$$\begin{aligned}
A_{\beta,n} &= b_{\beta,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\beta,n}^{(i)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - \sum_{i=1}^{n-1} b_{\beta,n}^{(i)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^{n-i}}\Big) \\
&= b_{\beta,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\beta,n}^{(i)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - b_{\beta,n}^{(n-1)} \cdot \omega_{P_2}\Big(\frac{y}{(x-\alpha)^{1}}\Big) \\
&= \frac{1}{2} b_{\beta,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\beta,n}^{(i)} \cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big);
\end{aligned}$$

$$\begin{aligned}
A_{\gamma,n} &= b_{\gamma,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\gamma,n}^{(i)} \cdot \omega_{P_1}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - \sum_{i=1}^{n-1} b_{\gamma,n}^{(i)} \cdot \omega_{P_1}\Big(\frac{y}{(x-\alpha)^{n-i}}\Big) \\
&= b_{\gamma,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\gamma,n}^{(i)} \cdot \omega_{P_1}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - b_{\gamma,n}^{(n-1)} \cdot \omega_{P_1}\Big(\frac{y}{(x-\alpha)^{1}}\Big) \\
&= \frac{1}{2} b_{\gamma,n}^{(n-1)} - \sum_{i=1}^{n-1} a_{\gamma,n}^{(i)} \cdot \omega_{P_1}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) \\
&= \frac{1}{2} b_{\beta,n}^{(n-1)} \cdot (-1)^{n+1} - \sum_{i=1}^{n-1} a_{\beta,n}^{(i)} \cdot (-1)^{n} \cdot -\omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) \\
&= (-1)^{n+1} \cdot A_{\beta,n}.
\end{aligned}$$

□

Remark 3.35. Algorithm 3.28 can be simplified for char $K \neq 2$ by Theorem 3.34. Next algorithm is the simplified one.

Algorithm 3.36. *Let* char $K \neq 2$ *and* $w(x, y) = y^2 - f(x)$.
The inputs *of the algorithm are*

(i) $n \geq 2$*;*

(ii) $P_1 := P_{(\alpha,\beta)}, P_2 := P_{(\alpha,\gamma)} \in \mathbb{P}_E^{(1)}$ *where* $\alpha, \beta, \gamma \in K$ *such that* $P_1 \neq P_2$*. In other words,* $\beta \neq \gamma$*.*

The outputs *of the algorithm are values of* $\omega_{P_1}\Big(\frac{1}{(x-\alpha)^n}\Big)$, $\omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big)$.

The algorithm *is recursive. The values of* $\omega_{P_1}\Big(\frac{1}{x-\alpha}\Big)$ *and* $\omega_{P_2}\Big(\frac{1}{x-\alpha}\Big)$ *can be calculated by Theorem 3.21. Let* $n \geq 2$*. Values of* $\omega_{P_1}\Big(\frac{1}{(x-\alpha)^i}\Big)$, $\omega_{P_2}\Big(\frac{1}{(x-\alpha)^i}\Big)$*, for* $i = 1, \dots n-1$ *may be assumed to be known from the previous steps of the recursion.*

1. *Calculate polynomials* $a_{\beta,n}(x), b_{\beta,n}(x), a_{\gamma,n}(x), b_{\gamma,n}(x) \in K[x-\alpha]$ *by Algorithm 3.26.*

If $n = 2$ then use the first step of the recursion from Algorithm 3.26. The step defines exactly $a_{\beta,2}(x), b_{\beta,2}(x), a_{\gamma,2}(x), b_{\gamma,2}(x) \in K[x-\alpha]$.

If $n > 2$ then $a_{\beta,n-1}(x), b_{\beta,n-1}(x), a_{\gamma,n-1}(x), b_{\gamma,n-1}(x) \in K[x-\alpha]$ are known from step $n-1$ of this recursion. Use the other step of the recursion from Algorithm 3.26. The step calculates $a_{\beta,n}(x), b_{\beta,n}(x), a_{\gamma,n}(x), b_{\gamma,n}(x) \in K[x-\alpha]$ from $a_{\beta,n-1}(x), b_{\beta,n-1}(x), a_{\gamma,n-1}(x), b_{\gamma,n-1}(x)$.

2. *Recall that $b^{(i)}_{\beta,n}$ is i-th coefficient of $b_{\beta,n}(x)$. Calculate*

$$\begin{aligned}\omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) &= \frac{1}{a^{(0)}_{\beta,n}} \cdot \Big[\frac{1}{2}\cdot b^{(n-1)}_{\beta,n} - \sum_{i=1}^{n-1} a^{(i)}_{\beta,n}\cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big)\Big]\\ &= \frac{1}{(-1)^n\cdot 2^{n-1}\cdot \beta^n}\cdot\Big[\frac{1}{2}\cdot b^{(n-1)}_{\beta,n} - \sum_{i=1}^{n-1} a^{(i)}_{\beta,n}\cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big)\Big];\end{aligned}$$

$$\omega_{P_1}\Big(\frac{1}{(x-\alpha)^n}\Big) = -\omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big).$$

Theorem 3.37. *Algorithm 3.36 is correct.*

Proof. Keep the assumption from Algorithm 3.36. The thought of the proof is same as the thought of the proof of Theorem 3.29 but it uses that $\omega_{P_i}\Big(\frac{y}{(x-\alpha)^n}\Big) = 0$ from Theorem 3.34. Recall that $\kappa_P = 0$ in $\operatorname{char} K \neq 2$. Modify Equations (3.34) and (3.31)

$$\begin{aligned}A_{\beta,n} &= a^{(0)}_{\beta,n}\cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big)\\ &= (-1)^n\cdot 2^{n-1}\cdot\beta^n\cdot\omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) \quad \text{(cf. Lemma 3.33)};\end{aligned}$$

$$\begin{aligned}A_{\beta,n} &= b^{(n-1)}_{\beta,n} - \sum_{i=1}^{n-1} a^{(i)}_{\beta,n}\cdot\omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - \sum_{i=1}^{n-1} b^{(i)}_{\beta,n}\cdot\omega_{P_2}\Big(\frac{y}{(x-\alpha)^{n-i}}\Big)\\ &= b^{(n-1)}_{\beta,n} - \sum_{i=1}^{n-1} a^{(i)}_{\beta,n}\cdot\omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) - b^{(n-1)}_{\beta,n}\cdot\omega_{P_2}\Big(\frac{y}{(x-\alpha)^{1}}\Big)\\ &= \frac{1}{2}\cdot b^{(n-1)}_{\beta,n} - \sum_{i=1}^{n-1} a^{(i)}_{\beta,n}\cdot\omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big) \quad \text{(cf. Remark 3.22)}.\end{aligned}$$

Together

$$\omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big) = \frac{1}{(-1)^n\cdot 2^{n-1}\cdot \beta^n}\cdot\Big[\frac{1}{2}\cdot b^{(n-1)}_{\beta,n} - \sum_{i=1}^{n-1} a^{(i)}_{\beta,n}\cdot \omega_{P_2}\Big(\frac{1}{(x-\alpha)^{n-i}}\Big)\Big].$$

By Equation (3.21) $\omega_{P_1}\Big(\frac{1}{(x-\alpha)^n}\Big) = -\omega_{P_2}\Big(\frac{1}{(x-\alpha)^n}\Big)$. □

Lemma 3.38. *Let $\operatorname{char} K \neq 2$ and $w(x,y) = y^2 - f(x)$. Let $(\alpha,\beta), (\alpha,\gamma) \in \mathbb{V}_w(K)$ such that $\beta \neq \gamma$. Let $n \geq 2$. Then the determinant from Equation (3.35)*

$$\det\begin{pmatrix} a^{(0)}_{\beta,n} & b^{(0)}_{\beta,n}\\ -a^{(0)}_{\gamma,n} & -b^{(0)}_{\gamma,n}\end{pmatrix} = (-1)^{3n-1}\cdot 2^{2n-1}\cdot\beta^{2n-1} \neq 0.$$

Proof. Recall that $\beta \neq 0$ by Lemma 2.6. By Lemma 3.33

$$
\begin{aligned}
\det \begin{pmatrix} a^{(0)}_{\beta,n} & b^{(0)}_{\beta,n} \\ -a^{(0)}_{\gamma,n} & -b^{(0)}_{\gamma,n} \end{pmatrix} &= - a^{(0)}_{\beta,n} \cdot b^{(0)}_{\gamma,n} + b^{(0)}_{\beta,n} \cdot a^{(0)}_{\gamma,n} \\
&= - (-1)^n \cdot 2^{n-1} \cdot \beta^n \cdot (-1)^{n-1} \cdot 2^{n-1} \cdot (-1)^{n-1} \cdot \beta^{n-1} \\
&+ (-1)^{n-1} \cdot 2^{n-1} \cdot \beta^{n-1} \cdot (-1)^n \cdot 2^{n-1} \cdot (-1)^n \cdot \beta^n \\
&= (-1)^{3n-1} \cdot 2^{2n-2} \cdot \beta^{2n-1} + (-1)^{3n-1} \cdot 2^{2n-2} \cdot \beta^{2n-1} \\
&= (-1)^{3n-1} \cdot 2^{2n-1} \cdot \beta^{2n-1} \neq 0.
\end{aligned}
$$

□

4. Cotrace and Different

Let E/K be an elliptic function field. The cotrace $\mathrm{Cotr}_{E/K(x)} : \Omega_{K(x)} \to \Omega_E$ and $\mathrm{Diff}(E/K(x))$ are described in this chapter. Next, there is a relationship between $\mathrm{Cotr}_{E/K(x)}(\eta)$ and ω where ω is as in Theorem 3.4 and η is as in Theorem 1.78. At the end of this chapter it is used to calculate ω_P.

4.1 Recapitulation

It is a recapitulation of facts needed for this chapter. It is based upon [1, Chapter 3.4], [1, Chapter 3.5] and [1, Chapter 3.6]. Details and proofs can be found there. The numbering in brackets refers to [1].

Convention 4.1. *Throughout this section consider a finite separable extension of algebraic function fields $F'/K' \geq F/K$ and consider that $K = \tilde{K}$, $K' = \tilde{K}'$ and K and K' are perfect.*

Definition 4.2 (Definition 3.4.1)**.** Let $P \in \mathbb{P}_F$. Denote the integral closure of $\mathcal{O}_P$ in F' by $\mathcal{O}'_P := \mathrm{ic}_{F'}(\mathcal{O}_P)$. Then the set

$$\mathcal{C}_P := \{z \in F'; \mathrm{Tr}_{F'/F}\left(z \cdot \mathcal{O}'_P\right) \subseteq \mathcal{O}_P\}$$

is called the complementary module over $\mathcal{O}_P$.

Theorem 4.3 (Proposition 3.4.2)**.** *Let $P \in \mathbb{P}_F$.*

(i) The set $\mathcal{C}_P$ is an $\mathcal{O}'_P$-module and $\mathcal{O}'_P \subseteq \mathcal{C}_P$.

(ii) There exists an element $t \in F'$ depending on P such that $\mathcal{C}_P = t \cdot \mathcal{O}'_P$. The element t satisfies

(a) $v_{P'}(t) \leq 0$ for all $P' \mid P$;

(b) for every $t' \in F'$

$$\mathcal{C}_P = t' \cdot \mathcal{O}'_P \iff v_{P'}(t') = v_{P'}(t) \text{ for all } P' \mid P.$$

(iii) The set $\mathcal{C}_P = \mathcal{O}'_P$ for almost all $P \in \mathbb{P}_F$.

Definition 4.4 (Definition 3.4.3)**.** Let $P \in \mathbb{P}_F$ and $P' \in \mathbb{P}_{F'}$ such that $P' \mid P$. Let $\mathcal{C}_P = t \cdot \mathcal{O}'_P$ be the complementary module over $\mathcal{O}_P$. Define the different exponent of $P' \mid P$ by $d(P' \mid P) := -v_{P'}(t)$.

Define

$$\mathrm{Diff}(F'/F) := \sum_{P \in \mathbb{P}_F} \sum_{P' \mid P} d(P' \mid P) \cdot P'.$$

This divisor is called the different of F'/F.

Remark 4.5. By item *(ii)* of Theorem 4.3 $d(P' \mid P)$ is well-defined and $d(P' \mid P) \geq 0$.

By item *(iii)* of Theorem 4.3 $\mathcal{C}_P = 1 \cdot \mathcal{O}'_P$ for almost all $P \in \mathbb{P}_F$. Thus $d(P' \mid P) = 0$ for almost all $P \in \mathbb{P}_F$. Thus $\mathrm{Diff}(F'/F)$ is well-defined and $\mathrm{Diff}(F'/F) \geq 0$.

Definition 4.6 (Definition 3.4.5)**.** Put

$$\mathcal{A}_{F'/F} := \{\alpha \in \mathcal{A}_{F'}; \alpha_{P'} = \alpha_{Q'} \text{ whenever } P' \cap F = Q' \cap F\};$$

$$\mathcal{A}_{F'/F}(A) := \mathcal{A}_{F'}(A) \cap \mathcal{A}_{F'/F} \text{ where } A \in \mathrm{Div}(F').$$

The set $\mathcal{A}_{F'/F}$ is an F'-subspace of $\mathcal{A}_{F'}$.

Extend $\mathrm{Tr}_{F'/F} : F' \to F$ to the mapping $\mathrm{Tr}_{F'/F} : \mathcal{A}_{F'/F} \to \mathcal{A}_F$ by the following way

$$\big(\mathrm{Tr}_{F'/F}(\alpha)\big)_P := \mathrm{Tr}_{F'/F}(\alpha_{P'})$$

where $\alpha \in \mathcal{A}_{F'/F}$ and $P \in \mathbb{P}_F$, $P' \in \mathbb{P}_{F'}$ are such that $P' \mid P$.

Theorem 4.7 (Theorem 3.4.6)**.** *Let $\omega \in \Omega_F$. There exists a unique $\omega' \in \Omega_{F'}$ such that*

$$\mathrm{Tr}_{K'/K}\big(\omega'(\alpha)\big) = \omega\big(\mathrm{Tr}_{F'/F}(\alpha)\big) \text{ for all } \alpha \in \mathcal{A}_{F'/F}. \tag{4.1}$$

The Weil differential ω' is called the cotrace of ω in F'/F. It is denoted by $\mathrm{Cotr}_{F'/F}(\omega)$.

If $\omega \neq 0$ then

$$\big(\mathrm{Cotr}_{F'/F}(\omega)\big) = \mathrm{Con}_{F'/F}((\omega)) + \mathrm{Diff}(F'/F).$$

Corollary 4.8 (Corollary 3.4.7)**.** *Let F/K be an algebraic function field and $x \in F$ such that $F/K(x)$ is separable. Let η be the Weil differential of $K(x)/K$ mentioned in Theorem 1.78. Then*

$$\big(\mathrm{Cotr}_{F/K(x)}(\eta)\big) = -2 \cdot (x)_- + \mathrm{Diff}(F/K(x)).$$

Remark 4.9 (Remark 3.4.8). Let $P \in \mathbb{P}_F$ and $P_1, \ldots, P_n \in \mathbb{P}_{F'}$ such that $P_i \mid P$. Let $\omega \in \Omega_F$ and $\omega' = \mathrm{Cotr}_{F'/F}(\omega)$. Then

$$\mathrm{Tr}_{K'/K}\Big(\sum_{i=1}^{n} \omega'_{P_i}(y)\Big) = \omega_P\big(\mathrm{Tr}_{F'/F}(y)\big) \text{ for every } y \in F'.$$

Proof. Define

$$\alpha := \sum_{i=1}^{n} \iota_{P_i}(y) \in \mathcal{A}_{F'/F}.$$

The trace $\mathrm{Tr}_{F'/F}(\alpha) = \iota_P\big(\mathrm{Tr}_{F'/F}(y)\big)$. By Theorem 1.59 express Equation (4.1) as

$$\mathrm{Tr}_{K'/K}\Big(\sum_{i=1}^{n} \omega'_{P_i}(y)\Big) = \mathrm{Tr}_{K'/K}\big(\omega'(\alpha)\big) = \omega\big(\mathrm{Tr}_{F'/F}(\alpha)\big) = \omega_P\big(\mathrm{Tr}_{F'/F}(y)\big).$$

□

Theorem 4.10 (Theorem 3.4.13; Hurwitz Genus Formula)**.** *Let g', g be the genera of F'/K' and F/K. Then*

$$2g' - 2 = \frac{[F' : F]}{[K' : K]} \cdot (2g - 2) + \deg\Big(\mathrm{Diff}(F'/F)\Big).$$

Theorem 4.11 (Theorem 3.5.1; Dedekind's Different Theorem)**.** *Let $P \in \mathbb{P}_F$ and $P' \in \mathbb{P}_{F'}$ be such that $P' \mid P$. Then*

(i) $d(P' \mid P) \geq e(P' \mid P) - 1$*;*

(ii) $d(P' \mid P) = e(P' \mid P) - 1$ *if and only if* $\operatorname{char} K \nmid e(P' \mid P)$*.*

Theorem 4.12 (Theorem 3.6.3)**.** *Let K'/K be an algebraic field extension. Let F/K be an algebraic function field. Define $F' := FK'$. The following statements are satisfied.*

(i) The extension F'/K' is an algebraic function field.

(ii) If $P \in \mathbb{P}_F$ and $P' \in \mathbb{P}_{F'}$ are such that $P' \mid P$ then $e(P' \mid P) = 1$.

(iii) The genus of F'/K' is same as the genus of F/K.

Theorem 4.13. *Let F/K be an algebraic function field determined by an absolutely irreducible polynomial $f(x, y) = 0$. Let $K' \geq K$ be an algebraic extension. Put $F' := FK'$. Then F'/K' is determined by $f(x, y) = 0$.*

4.2 Cotrace and Different for Elliptic Curves

Let E/K be an elliptic function field. Let ω be as in Theorem 3.4 and η be as in Theorem 1.78. Use the knowledge from Section 4.1 to describe $\operatorname{Diff}(E/K(x))$. If $\operatorname{Diff}(E/K(x))$ is known then $\big(\operatorname{Cotr}_{E/K(x)}(\eta)\big)$ is known by Corollary 4.8. Then find $u \in E^\star$ such that $\omega = u \cdot \operatorname{Cotr}_{E/K(x)}(\eta)$. This relationship and Remark 4.9 convert the calculation of ω_P to the calculation of η_Q if $e(P \mid Q) = 2$.

The main idea of the description of $\operatorname{Diff}(E/K(x))$ is to find all places $P \in \mathbb{P}_E$ such that $e\big(P \mid P \cap K(x)\big) = 2$. It follows from Theorem 4.11. Lemma 2.7 describes this situation for $\mathbb{P}_E^{(1)}$. Thus consider that $K = \bar{K}$. Next, use Lemma 2.7 to find the places and move them down to K if $K \neq \bar{K}$. At the end, set the exact values of $d\big(P \mid P \cap K(x)\big)$ for the found places P.

Convention 4.14. *Throughout this section consider that*

(i) E/K is an elliptic function field determined by a short smooth Weierstrass equation $w(x, y) = 0$;

(ii) K is perfect and $K = \tilde{K}$;

(iii) $\eta \in \Omega_{K(x)}$ is as in Theorem 1.78;

(iv) $\omega \in \Omega_E$ is as in Theorem 3.4.

Throughout this chapter keep the notations from Conventions 1.109, 1.113 and 1.77 and Theorem 1.76.

Theorem 4.15. *Let $P_{\infty,E}$ be the pole in E and let $P_{\infty,K(x)}$ be the pole in $K(x)$.*

(i) The degree of $\operatorname{Diff}(E/K(x))$ *is* 4*.*

(ii) *Let* $\operatorname{char} K \neq 2$ *and* $w(x,y) = y^2 - f(x)$. *The decomposition of* $f(x) = f_1(x) \cdot \cdots \cdot f_n(x)$ *in* $K[x]$ *gives* $P_{f_1(x)}, \ldots, P_{f_n(x)} \in \mathbb{P}_{K(x)}$. *The number* $n \leq 3$ *and the all factors are pairwise distinct by smoothness of* $w(x,y) = 0$. *There exists one place in* E *over each* $P_{f_i(x)} \in \mathbb{P}_{K(x)}$. *Denote the places in* E *by* $P_{(f_i(x),0)}$. *The ramification index* $e(P_{(f_i(x),0)} \mid P_{f_i(x)}) = 2$, $d(P_{(f_i(x),0)} \mid P_{f_i(x)}) = 1$, $e(P_{\infty,E} \mid P_{\infty,K(x)}) = 2$ *and* $d(P_{\infty,E} \mid P_{\infty,K(x)}) = 1$. *There exists no other else place* $P \in \mathbb{P}_E$ *such that* $e\big(P \mid P \cap K(x)\big) = 2$ *and* $d\big(P \mid P \cap K(x)\big) > 0$. *The degree* $\deg\big(P_{(f_i(x),0)}\big) = \deg\big(f_i(x)\big)$. *The different* $\operatorname{Diff}(E/K(x)) = P_\infty + \sum_{i=1}^{n} P_{(f_i(x),0)}$ *and* $\big(\operatorname{Cotr}_{E/K(x)}(\eta)\big) = -3 \cdot P_\infty + \sum_{i=1}^{n} P_{(f_i(x),0)}$.

(iii) *Let* $\operatorname{char} K = 2$ *and* $w(x,y) = y^2 + y \cdot x - f_1(x)$. *The different exponent* $d(P_{(0,\sqrt{a_6})} \mid P_{\sqrt{a_6}}) = 2$ *and* $d(P_{\infty,E} \mid P_{\infty,K(x)}) = 2$. *There exists no other else place* $P \in \mathbb{P}_E$ *such that* $e\big(P \mid P \cap K(x)\big) = 2$ *and* $d\big(P \mid P \cap K(x)\big) > 0$. *The different* $\operatorname{Diff}(E/K(x)) = 2 \cdot P_\infty + 2 \cdot P_{(0,\sqrt{a_6})}$ *and* $\big(\operatorname{Cotr}_{E/K(x)}(\eta)\big) = -2 \cdot P_\infty + 2 \cdot P_{(0,\sqrt{a_6})}$.

(iv) *Let* $\operatorname{char} K = 2$ *and* $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$. *The different exponent* $d(P_{\infty,E} \mid P_{\infty,K(x)}) = 4$. *There exists no other else place* $P \in \mathbb{P}_E$ *such that* $e\big(P \mid P \cap K(x)\big) = 2$ *and* $d\big(P \mid P \cap K(x)\big) > 0$. *The different* $\operatorname{Diff}(E/K(x)) = 4 \cdot P_\infty$ *and* $\big(\operatorname{Cotr}_{E/K(x)}(\eta)\big) = 0$.

Proof. (i) By Theorem 4.10 $2 \cdot 1 - 2 = \frac{2}{1} \cdot (2 \cdot 0 - 2) + \deg\big(\operatorname{Diff}(E/K(x))\big)$. Thus $\deg\big(\operatorname{Diff}(E/K(x))\big) = 4$.

Prove the remaining items. Let $P \in \mathbb{P}_{K(x)}$ and $P' \in \mathbb{P}_E$ be such that $P' \mid P$. Recall that $e(P' \mid P)$ can be 1 or 2 by Theorem 1.71. By Theorem 4.11 if $e(P' \mid P) = 1$ then $d(P' \mid P) = e(P' \mid P) - 1 = 0$. Thus find places such that $e(P' \mid P) = 2$. By Theorem 1.99 $e(P_{\infty,E} \mid P_{\infty,K(x)}) = 2$. Assume that $K = \bar{K}$ in the first step. Then every place is of degree one. Use Lemma 2.7 to find the rest of places P' such that $e(P' \mid P) = 2$ and $P' \neq P_\infty$. Split up the following step according to the form of $w(x,y)$.

(ii) Because $K = \bar{K}$ then $f(x) = (x - \alpha_1) \cdot (x - \alpha_2) \cdot (x - \alpha_3)$ in $K[x]$. By Lemma 2.7 there exist only places $P_{(\alpha_i,0)} \in \mathbb{P}_E$ such that $e(P_{(\alpha_i,0)} \mid P_{\alpha_i}) = 2$ where $P_{\alpha_i} \in \mathbb{P}_{K(x)}$.

(iii) By Lemma 2.7 there exists only place $P_{(0,\sqrt{a_6})} \in \mathbb{P}_E$ such that $e(P_{(0,\sqrt{a_6})} \mid P_0) = 2$ where $P_0 \in \mathbb{P}_{K(x)}$.

(iv) By Lemma 2.7 there exists no place P such that $P \neq P_\infty$ and $e(P \mid P \cap K(x)) = 2$.

If $K \neq \bar{K}$ then put $E' := E\bar{K}$. Then $E'/\bar{K}$ is determined by $w(x,y) = 0$ by Theorem 4.13. The following assertions are satisfied

1. $E'/\bar{K} \geq E/K \geq K(x)/K$;

2. $E'/\bar{K} \geq \bar{K}(x)/\bar{K} \geq K(x)/K$;

Move the behaviour of a ramification index from $E'/\bar{K} \geq \bar{K}(x)/\bar{K}$ to $E/K \geq K(x)/K$. Let $Q \in \mathbb{P}_{K(x)}$, $Q' \in \mathbb{P}_{\bar{K}(x)}$ $P \in \mathbb{P}_E$ and $P' \in \mathbb{P}_{E'}$ be such that $Q' \mid Q$, $P' \mid Q'$, $P \mid Q$ and $P' \mid P$. By Lemma 1.73 $e(P' \mid Q') \cdot e(Q' \mid Q) = e(P' \mid Q)$ and $e(P' \mid P) \cdot e(P \mid Q) = e(P' \mid Q)$. By Theorem 4.12 $e(P' \mid P) = 1$ and

$e(Q' \mid Q) = 1$. Together $e(P' \mid Q') = e(P \mid Q)$. We shall use this idea to describe all places in E without the pole such that their ramification indexes are two. Recall that $e(P_{\infty,E} \mid P_{\infty,K(x)}) = 2$. Split up the following step according to the form of $w(x,y)$.

(ii) There exist only $\bar{P}_{\alpha_i} \in \mathbb{P}_{\bar{K}(x)}$ and $\bar{P}_{(\alpha_i,0)} \in \mathbb{P}_{E'}$ such that $e(\bar{P}_{(\alpha_i,0)} \mid \bar{P}_{\alpha_i}) = 2$. Recall that α_i are roots of $f(x)$.

If $\alpha_i \in K$ then $P_{\alpha_i} := \bar{P}_{\alpha_i} \cap K(x)$ and $P_{(\alpha_i,0)} := \bar{P}_{(\alpha_i,0)} \cap E$. The ramification index $e(P_{(\alpha_i,0)} \mid P_{\alpha_i}) = e(\bar{P}_{(\alpha_i,0)} \mid \bar{P}_{\alpha_i}) = 2$. By Theorem 1.71 $f(P_{(\alpha_i,0)} \mid P_{\alpha_i}) = 1$. The degree $\deg(P_{(\alpha_i,0)}) = 1$.

Assume that $\alpha_i \notin K$. Let $f(x) = f_1(x) \cdot \dots \cdot f_n(x)$ be the decomposition of f in $K[x]$. Assume that $f_i(x)$ are monic. There exists a factor $f_k(x)$ such that $f_k(\alpha_i) = 0$. Decompose $f_k(x) = (x - \alpha_1) \cdot \dots \cdot (x - \alpha_m)$. Prove that $\bar{P}_{\alpha_j} \cap K(x) = P_{f_k(x)}$ for $j = 1, \dots, m$ and $\bar{P}_{(\alpha_1,0)} \cap E = \dots = \bar{P}_{(\alpha_m,0)} \cap E =: P_{(f_i(x),0)}$. Let $r = \frac{a(x)}{c(x)} \in K(x)$ where $a(x), c(x) \in K[x]$ such that $\gcd(a(x), c(x)) = 1$. Let j be an arbitrary value from $1, \dots, m$. If $r \in \bar{P}_{\alpha_j}$ then $a(\alpha_j) = 0$ and $c(\alpha_j) \neq 0$. Thus $f_k(x) \mid a(x)$ and $f_k(x) \nmid c(x)$. Thus $r \in P_{f_k(x)}$. Thus $\bar{P}_{\alpha_j} \cap K(x) \subseteq P_{f_k(x)}$. By Theorem 1.7 $\bar{P}_{\alpha_j} \cap K(x) = P_{f_k(x)}$. Let $P \in \mathbb{P}_E$ be such that $P \mid P_{f_k(x)}$ and $\bar{P}_{(\alpha_j,0)} \mid P$ for an arbitrary $j = 1, \dots, m$. Thus $e(P \mid P_{f_k(x)}) = e(\bar{P}_{(\alpha_i,0)} \mid \bar{P}_{\alpha_i}) = 2$. Thus P is only one place in E over $P_{f_k(x)}$ by Theorem 1.71. Because j is arbitrary then $\bar{P}_{(\alpha_j,0)} \mid P$ for all $j = 1, \dots, m$. By Theorem 1.71 $f(P \mid P_{f_k(x)}) = 1$. By Lemma 1.70 $\deg(P) = f(P \mid P_{f_k(x)}) \cdot \deg(P_{f_k(x)}) = \deg(f_k(x))$.

At the end, every factor $f_i(x)$ determines $P_{(f_i(x),0)} \in \mathbb{P}_E$ such that

(i) $\deg(P_{(f_i(x),0)}) = \deg\big(f_i(x)\big)$;

(ii) $e(P_{(f_i(x),0)} \mid P_{f_i(x)}) = 2$;

(iii) $\deg\Big(\sum_{i=1}^n P_{(f_i(x),0)}\Big) = \deg\big(f(x)\big) = 3$.

This places are only places such that their ramification index are two and they are not the pole.

(iii) There exist $\bar{P}_0 \in \mathbb{P}_{\bar{K}(x)}$ and $\bar{P}_{(0,\sqrt{a_6})} \in \mathbb{P}_{E'}$ such that $e(\bar{P}_{(0,\sqrt{a_6})} \mid \bar{P}_0) = 2$. The squared root $\sqrt{a_6} \in K$ because K is perfect and $\operatorname{char} K = 2$. Thus $P_0 := \bar{P}_0 \cap K(x)$ and $P_{(0,\sqrt{a_6})} := \bar{P}_{(0,\sqrt{a_6})} \cap E$. By Lemma 2.4 $P_{(0,\sqrt{a_6})} \mid P_0$. The ramification index $e(P_{(0,\sqrt{a_6})} \mid P_0) = e(\bar{P}_{(0,\sqrt{a_6})} \mid \bar{P}_0) = 2$.

(iv) There exists no place $\bar{P} \in \mathbb{P}_{E'}$ such that $e(\bar{P} \mid \bar{P} \cap \bar{K}(x)) = 2$ and $\bar{P} \neq P_\infty$. Thus there exists no place $P \in \mathbb{P}_E$ such that $e(P \mid P \cap K(x)) = 2$ and $P \neq P_\infty$.

Continue to build $\operatorname{Diff}(E/K(x))$ and $\big(\operatorname{Cotr}_{E/K(x)}(\eta)\big)$. Split up the following step according to the form of $w(x,y)$.

(ii) Because $\operatorname{char} K \nmid e(P_{(f_i(x),0)} \mid P_{f_k(x)})$ then by Theorem 4.11 $d(P_{(f_i(x),0)} \mid P_{f_k(x)}) = e(P_{(f_i(x),0)} \mid P_{f_k(x)}) - 1 = 1$. Thus $\operatorname{Diff}(E/K(x)) = P_\infty + \sum_{i=1}^n P_{(f_i(x),0)}$. Check that $\deg\big(\operatorname{Diff}(E/K(x))\big) = 4$ and $\operatorname{Diff}(E/K(x)) \geq 0$. By Corollary 4.8 $\big(\operatorname{Cotr}_{E/K(x)}(\eta)\big) = -3 \cdot P_\infty + \sum_{i=1}^n P_{(f_i(x),0)}$.

(iii) The different $\mathrm{Diff}(E/K(x)) = c_1 \cdot P_{(0,\sqrt{a_6})} + c_2 \cdot P_\infty$. By Theorem 4.11

(i) $1 = e(P_{(0,\sqrt{a_6})} \mid P_0) - 1 \leq d(P_{(0,\sqrt{a_6})} \mid P_0) = c_1$;

(ii) $1 = e(P_{\infty,E} \mid P_{\infty,K(x)}) - 1 \leq d(P_{\infty,E} \mid P_{\infty,K(x)}) = c_2$.

Calculate $4 = \deg\big(\mathrm{Diff}(E/K(x))\big) = c_1 + c_2$. By Corollary 4.8

$$\big(\mathrm{Cotr}_{E/K(x)}(\eta)\big) = (c_2 - 4) \cdot P_\infty + c_1 \cdot P_{(0,\sqrt{a_6})}.$$

By Lemma 2.3 $E/K(x)$ is Galois and the Galois mappings are $\delta_1 = \mathrm{id}$ and $\delta_2(y) = x + y$. The trace $\mathrm{Tr}_{E/K(x)}$ can be calculated. By remark 4.9

$$\begin{aligned}
\mathrm{Cotr}_{E/K(x)}(\eta)_\infty\Big(\frac{y}{x^2}\Big) &= \eta_\infty\Big(\mathrm{Tr}_{E/K(x)}\,\big(\frac{y}{x^2}\big)\Big)\\
&= \eta_\infty\Big(\delta_1\big(\frac{y}{x^2}\big) + \delta_2\big(\frac{y}{x^2}\big)\Big)\\
&= \eta_\infty\Big(\frac{y}{x^2} + \frac{y+x}{x^2}\Big) = \eta_\infty\Big(\frac{1}{x}\Big) = -1 \neq 0.
\end{aligned}$$

The valuation $\upsilon_\infty\big(\frac{y}{x^2}\big) = -3 + 4 = 1$. Thus by Theorem 1.60

$$\Big(\mathrm{Cotr}_{E/K(x)}(\eta)\Big)_\infty = (c_2 - 4) < -\upsilon_\infty\Big(\frac{y}{x^2}\Big) = -1.$$

Thus $c_2 < 3$.

By remark 4.9

$$\begin{aligned}
\mathrm{Cotr}_{E/K(x)}(\eta)_{(0,\sqrt{a_6})}\Big(\frac{y-\sqrt{a_6}}{x^2}\Big) &= \eta_0\Big(\mathrm{Tr}_{E/K(x)}\,\big(\frac{y-\sqrt{a_6}}{x^2}\big)\Big)\\
&= \eta_0\Big(\delta_1\big(\frac{y-\sqrt{a_6}}{x^2}\big) + \delta_2\big(\frac{y-\sqrt{a_6}}{x^2}\big)\Big)\\
&= \eta_0\Big(\frac{y-\sqrt{a_6}}{x^2} + \frac{y+x-\sqrt{a_6}}{x^2}\Big)\\
&= \eta_0\Big(\frac{1}{x}\Big) = 1 \neq 0.
\end{aligned}$$

By Lemmas 2.7 and 2.8

$$\begin{aligned}
\upsilon_{(0,\sqrt{a_6})}\Big(\frac{y-\sqrt{a_6}}{x^2}\Big) &= e(P_{(0,\sqrt{a_6})} \mid P_{\sqrt{a_6}}) \cdot \upsilon_{\sqrt{a_6}}(y - \sqrt{a_6})\\
&\quad - 2 \cdot e(P_{(0,\sqrt{a_6})} \mid P_0) \cdot \upsilon_0(x)\\
&= 1 \cdot 1 - 2 \cdot 2 \cdot 1 = -3.
\end{aligned}$$

Thus by Theorem 1.60

$$\Big(\mathrm{Cotr}_{E/K(x)}(\eta)\Big)_{(0,\sqrt{a_6})} = c_1 < -\upsilon_{(0,\sqrt{a_6})}\Big(\frac{y-\sqrt{a_6}}{x^2}\Big) = 3.$$

Together $c_1 = 2$, $c_2 = 2$ and $\mathrm{Diff}(E/K(x)) = 2 \cdot P_\infty + 2 \cdot P_{(0,\sqrt{a_6})}$. Check that $\deg\big(\mathrm{Diff}(E/K(x))\big) = 4$ and $\mathrm{Diff}(E/K(x)) \geq 0$. By Corollary 4.8 $\big(\mathrm{Cotr}_{E/K(x)}(\eta)\big) = -2 \cdot P_\infty + 2 \cdot P_{(0,\sqrt{a_6})}$.

(iv) The pole is only place with the ramification index two. The different $\mathrm{Diff}(E/K(x)) = 4 \cdot P_\infty$ because $\deg\big(\mathrm{Diff}(E/K(x))\big) = 4$. By Corollary 4.8 $\big(\mathrm{Cotr}_{E/K(x)}(\eta)\big) = 0$.

$\square$

Theorem 4.16.

(i) If $\operatorname{char} K \neq 2$ *and* $w(x,y) = y^2 - f(x)$ *then* $\omega = \frac{1}{2\cdot y} \cdot \mathrm{Cotr}_{E/K(x)}(\eta)$.

(ii) If $\operatorname{char} K = 2$ *and* $w(x,y) = y^2 + x \cdot y - f_1(x)$ *then* $\omega = \frac{1}{x} \cdot \mathrm{Cotr}_{E/K(x)}(\eta)$.

(iii) If $\operatorname{char} K = 2$ *and* $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$ *then* $\omega = \frac{1}{a_3} \cdot \mathrm{Cotr}_{E/K(x)}(\eta)$.

Proof. By Theorem 1.48 $\omega = u \cdot \mathrm{Cotr}_{E/K(x)}(\eta)$ where $u \in E^\star$. By Lemma 1.53 $(\omega) = (u) + \Big(\mathrm{Cotr}_{E/K(x)}(\eta)\Big)$. Thus $(u) = (\omega) - \Big(\mathrm{Cotr}_{E/K(x)}(\eta)\Big) = -\Big(\mathrm{Cotr}_{E/K(x)}(\eta)\Big)$. Recall that $(\omega) = 0$.

(i) Decompose $f(x) = \sum_{i=1}^n f_i(x)$ where $n \leq 3$. By Theorem 4.15 every factor determines $P_{(f_i(x),0)} \in \mathbb{P}_E$ such that $e\big(P_{(f_i(x),0)} \mid P_{f_i(x)}\big) = 2$ where $P_{f_i(x)} \in \mathbb{P}_{K(x)}$. Find u such that $(u) = +3 \cdot P_\infty - \sum_{i=1}^n P_{(f_i(x),0)}$. Put $u := \frac{1}{y}$.

(a) The valuation $\upsilon_\infty\big(\frac{1}{y}\big) = 3$.

(b) Define $\tilde{f}_i(x) := \frac{f(x)}{f_i(x)}$. Then $\upsilon_{f_i(x)}(\tilde{f}_i(x)) = 0$. Calculate

$$\begin{aligned}
\upsilon_{(f_i(x),0)}\Big(\frac{1}{y}\Big) &= \upsilon_{(f_i(x),0)}\Big(\frac{y}{y^2}\Big) = \upsilon_{(f_i(x),0)}\Big(\frac{y}{f(x)}\Big) \\
&= \upsilon_{(f_i(x),0)}\Big(\frac{y}{f_i(x) \cdot \tilde{f}_i(x)}\Big) \\
&= \upsilon_{(f_i(x),0)}(y) - e\big(P_{(f_i(x),0)} \mid P_{f_i(x)}\big) \cdot \Big(\upsilon_{f_i(x)}(f_i(x)) + \upsilon_{f_i(x)}(\tilde{f}_i(x))\Big) \\
&= -\upsilon_{(f_i(x),0)}\Big(\frac{1}{y}\Big) - 2; \\
\upsilon_{(f_i(x),0)}\Big(\frac{1}{y}\Big) &= -1.
\end{aligned}$$

By Theorem 1.99 P_∞ is only one place where a valuation of $\frac{1}{y}$ is positive. Thus $\deg\big((\frac{1}{y})_+\big) = 3$. By Theorem 4.15

$$\deg\Big(\sum_{i=1}^n P_{(f_i(x),0)}\Big) = \sum_{i=1}^n \deg(f_i(x)) = \deg(f(x)) = 3.$$

Thus $\big((\frac{1}{y})_-\big) = \sum_{i=1}^n P_{(f_i(x),0)}$. Thus $\big(\frac{1}{y}\big) = 3 \cdot P_\infty - \sum_{i=1}^n P_{(f_i(x),0)}$.

By Lemma 2.3 $E/K(x)$ is Galois and the Galois mappings are $\delta_1 = \mathrm{id}$ and $\delta_2(y) = -y$. Thus $\mathrm{Tr}_{E/K(x)}$ can be calculated.

By Lemma 1.54 $\left(\frac{1}{y}\right) = \left(\frac{1}{c\cdot y}\right)$ for $c \in K^\star$. By Remark 4.9

$$\begin{aligned}\eta_\infty\Big(\operatorname{Tr}_{E/K(x)}\big(\frac{1}{x}\big)\Big) &= \eta_\infty\Big(\frac{2}{x}\Big) = -2\\ =&\operatorname{Cotr}_{E/K(x)}(\eta)_\infty\Big(\frac{1}{x}\Big) = \Big(\frac{c\cdot y}{c\cdot y}\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_\infty\Big(\frac{1}{x}\Big)\\ =&(c\cdot y\cdot\omega)_\infty\Big(\frac{1}{x}\Big) = (\omega)_\infty\Big(\frac{c\cdot y}{x}\Big) = -c.\end{aligned}$$

Thus $c = 2$ and $u = \frac{1}{2\cdot y}$.

(ii) Find u such that $(u) = +2\cdot P_\infty - 2\cdot P_{(0,\sqrt{a_6})}$. Put $u := \frac{1}{x}$.

(a) The valuation $\upsilon_\infty\left(\frac{1}{x}\right) = 2$.

(b) By Lemma 2.7 $e(P_{(0,\sqrt{a_6})} \mid P_0) = 2$. Thus $\upsilon_{(0,\sqrt{a_6})}\left(\frac{1}{x}\right) = -e(P_{(0,\sqrt{a_6})} \mid P_0)\cdot\upsilon_0(x) = -2$ where $P_0 \in \mathbb{P}_{K(x)}$.

By Theorem 1.99 P_∞ is only one place where a valuation of $\frac{1}{x}$ is positive. Thus $\deg\left((\frac{1}{x})_+\right) = 2$. The degree $\deg\left(2\cdot P_{(0,\sqrt{a_6})}\right) = 2$. Thus $\left((\frac{1}{x})_-\right) = 2\cdot P_{(0,\sqrt{a_6})}$. Thus $\left(\frac{1}{x}\right) = 2\cdot P_\infty - 2\cdot P_{(0,\sqrt{a_6})}$.

By Lemma 2.3 $E/K(x)$ is Galois and the Galois mappings are $\delta_1 = \mathrm{id}$ and $\delta_2(y) = x + y$. Thus $\operatorname{Tr}_{E/K(x)}$ can be calculated.

By Lemma 1.54 $\left(\frac{1}{x}\right) = \left(\frac{1}{c\cdot x}\right)$ for $c \in K^\star$. By Remark 4.9

$$\begin{aligned}\eta_\infty\Big(\operatorname{Tr}_{E/K(x)}\big(\frac{y}{x^2}\big)\Big) &= \eta_\infty\Big(\frac{y+x+y}{x^2}\Big) = \eta_\infty\Big(\frac{1}{x}\Big) = -1\\ =&\operatorname{Cotr}_{E/K(x)}(\eta)_\infty\Big(\frac{y}{x^2}\Big) = \Big(\frac{c\cdot x}{c\cdot x}\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_\infty\Big(\frac{y}{x^2}\Big)\\ =&(c\cdot x\cdot\omega)_\infty\Big(\frac{y}{x^2}\Big) = (\omega)_\infty\Big(\frac{c\cdot y}{x}\Big) = -c.\end{aligned}$$

Thus $c = 1$ and $u = \frac{1}{x}$.

(iii) Because $\Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big) = 0$, $u \in K^\star$.

By Lemma 2.3 $E/K(x)$ is Galois and the Galois mappings are $\delta_1 = \mathrm{id}$ and $\delta_2(y) = a_3 + y$. Thus $\operatorname{Tr}_{E/K(x)}$ can be calculated.

By Remark 4.9

$$\begin{aligned}\eta_\infty\Big(\operatorname{Tr}_{E/K(x)}\big(\frac{y}{x\cdot a_3}\big)\Big) &= \eta_\infty\Big(\frac{y+a_3+y}{x\cdot a_3}\Big) = \eta_\infty\Big(\frac{1}{x}\Big) = -1\\ &= \operatorname{Cotr}_{E/K(x)}(\eta)_\infty\Big(\frac{y}{x\cdot a_3}\Big).\end{aligned}$$

Oh the other hand, $\omega_\infty\Big(\frac{y}{x\cdot a_3}\Big) = -\frac{1}{a_3}$. Set $u := \frac{1}{a_3}$.

$\square$

Theorem 4.17. *Let* $u = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x)\cdot y}{c_2(x)} \in E$ *be such that* $\gcd\big(a_1(x), c_1(x)\big) = 1$ *and* $\gcd\big(b_2(x), c_2(x)\big) = 1$. *Let* ω *be as in Theorem 3.4. Let* $P \in \mathbb{P}_E$ *and* $Q \in \mathbb{P}_{K(x)}$ *be such that* $e\big(P \mid Q\big) = 2$. *Then* $\omega_P(u) = \eta_Q\Big(\frac{b_2(x)}{c_2(x)}\Big)$.

(i) If char $K \neq 2$ *and* $w(x,y) = y^2 - f(x)$ *then*

$$\Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P(u) = \eta_Q\Big(\frac{2 \cdot a_1(x)}{c_1(x)}\Big).$$

(ii) If char $K = 2$ *and* $w(x,y) = y^2 + x \cdot y - f_1(x)$ *then*

$$\Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P(u) = \eta_Q\Big(\frac{x \cdot b_2(x)}{c_2(x)}\Big).$$

(iii) If char $K = 2$ *and* $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$ *then*

$$\Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P(u) = \eta_Q\Big(\frac{a_3 \cdot b_2(x)}{c_2(x)}\Big).$$

Proof. Calculate $\operatorname{Tr}_{E/K(x)}(u)$. Split up the calculation according to the form of $w(x,y)$. By Lemma 2.3

(i) $\operatorname{Tr}_{E/K(x)}(u) = \delta_1(u) + \delta_2(u) = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x) \cdot y}{c_2(x)} + \frac{a_1(x)}{c_1(x)} - \frac{b_2(x) \cdot y}{c_2(x)} = \frac{2 \cdot a_1(x)}{c_1(x)}$;

(ii) $\operatorname{Tr}_{E/K(x)}(u) = \delta_1(u) + \delta_2(u) = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x) \cdot y}{c_2(x)} + \frac{a_1(x)}{c_1(x)} + \frac{b_2(x)}{c_2(x)} \cdot (x+y) = \frac{x \cdot b_2(x)}{c_2(x)}$;

(iii) $\operatorname{Tr}_{E/K(x)}(u) = \delta_1(u) + \delta_2(u) = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x) \cdot y}{c_2(x)} + \frac{a_1(x)}{c_1(x)} + \frac{b_2(x)}{c_2(x)} \cdot (a_3 + y) = \frac{a_3 \cdot b_2(x)}{c_2(x)}$.

Because P is only one place over Q then $\big(\operatorname{Cotr}_{E/K(x)}(\eta)\big)_P(u) = \eta_Q\big(\operatorname{Tr}_{E/K(x)}(u)\big)$ by Remark 4.9. Next, calculate $\omega_P(u)$ by Theorem 4.16. Split up the following step according to the form of $w(x,y)$. Then

(i) $\Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P(u) = \eta_Q\Big(\frac{2 \cdot a_1(x)}{c_1(x)}\Big)$;

$$\begin{aligned}\omega_P(u) &= 2y \cdot \omega_P\Big(\frac{u}{2y}\Big) = \Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P\Big(\frac{u}{2y}\Big)\\ &= \Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P\Big(\frac{a_1(x) \cdot y}{2 \cdot f(x) \cdot c_1(x)} + \frac{b_2(x)}{2 \cdot c_2(c)}\Big) = \eta_Q\Big(\frac{b_2(x)}{c_2(c)}\Big);\end{aligned}$$

(ii) $\Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P(u) = \eta_Q\Big(\frac{x \cdot b_2(x)}{c_2(x)}\Big)$;

$$\begin{aligned}\omega_P(u) &= x \cdot \omega_P\Big(\frac{u}{x}\Big) = \Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P\Big(\frac{u}{x}\Big)\\ &= \Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P\Big(\frac{a_1(x)}{c_1(x) \cdot x} + \frac{b_2(x) \cdot y}{c_2(c) \cdot x}\Big) = \eta_Q\Big(\frac{b_2(x)}{c_2(c)}\Big);\end{aligned}$$

(iii) $\Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P(u) = \eta_Q\Big(\frac{a_3 \cdot b_2(x)}{c_2(x)}\Big)$;

$$\begin{aligned}\omega_P(u) &= a_3 \cdot \omega_P\Big(\frac{u}{a_3}\Big) = \Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P\Big(\frac{u}{a_3}\Big)\\ &= \Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_P\Big(\frac{a_1(x)}{c_1(x) \cdot a_3} + \frac{b_2(x) \cdot y}{c_2(c) \cdot a_3}\Big) = \eta_Q\Big(\frac{b_2(x)}{c_2(c)}\Big).\end{aligned}$$

$\square$

Remark 4.18.

(i) Theorem 4.17 converts the calculation of a local component of ω and of $\operatorname{Cotr}_{E/K(x)}(\eta)$ to the calculation of a local component of η. Recall that Section 1.3 provides the tool calculating a local component of η.

(ii) If Theorems 4.17 and 1.82 are used to calculate ω_∞ then they give the same result as Theorem 3.8. It means that the conditions about degrees are same.

(iii) If $e(P \mid Q) = 1$ then there exists $P_2 \in \mathbb{P}_E$ such that $P_2 \mid Q$. Put $P_1 := P$. Then

$$\Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_{P_1}(u) + \Big(\operatorname{Cotr}_{E/K(x)}(\eta)\Big)_{P_2}(u) = \eta_Q\Big(\operatorname{Tr}_{E/K(x)}(u)\Big)$$

by Remark 4.9. It is not known how to split $\eta_Q\big(\operatorname{Tr}_{E/K(x)}(u)\big)$ between two local components.

5. Derivations and Weil Differentials

The definition of the Weil differential does not say enough about why it is useful and about why it is defined in this way. This chapter brings a relationship between the Weil differential and some objects from analysis such as the derivation, the differential, the residue and the Laurent series. It helps to illustrate what contribution the Weil differential has. At the end of this chapter, the theory is applied to an elliptic function field. It provides another tool calculating ω_P for ω as in Theorem 3.4.

Section 5.1 is a recapitulation of the needed facts for this chapter. The section is based upon [1, Chapter 4.1]. Details and proofs can be found there. There are definitions and a theory of the derivation and the differential of algebraic function fields.

The real number field $\mathbb{R}$ is the completion of the rational number field $\mathbb{Q}$ with respect to the ordinary absolute value. In Section 5.2 there is a theory which provides the completion of an algebraic function field F/K with respect the ordering by υ_P for $P \in \mathbb{P}_F^{(1)}$. The theory is built generally for a valued field (T, υ).

In Section 5.3 the completion of a valued field is used to (F, υ_P) where F/K is an algebraic function field and $P \in \mathbb{P}_F^{(1)}$. It provides the representation of every $z \in F$ as a power series and the definition of the residue. At the end, there is a relationship between the Weil differential and the differential. The section is based upon [1, Chapter 4.2].

There is an application of the previous sections to an elliptic function field E/K in Section 5.4. It provides two algorithms calculating $\mathrm{res}_{P,t}$ if $e\big(P \mid P \cap K(x)\big) = 1$. The relationship between Ω_E and Δ_E from Section 5.3 and the relationship between ω and $\mathrm{Cotr}_{E/K(x)}(\eta)$ from Chapter 4 give a tool how to calculate ω_P by using $\mathrm{res}_{P,t}$. Thus there is another way how to calculate ω_P.

Convention 5.1. *Throughout this chapter consider that if F/K is an algebraic function field then K is perfect and $K = \tilde{K}$. Throughout this chapter keep the notations from Conventions 1.109, 1.113 and 1.77 and Theorem 1.76.*

Remark 5.2. A separating element is used a lot in this chapter. Thus recall Definition 1.20 and Theorem 1.21.

A consequence of them is that every uniformizing element of a place is always a separating element. It is important for this chapter.

5.1 Derivations and Differentials

It is not possible to define the derivation and the differential as in analysis because there is no metric of an arbitrary algebraic function field. Thus the definitions have a formal character and they keep their basic features as in analysis. Lemma 5.10 shows this theory for an elliptic function field. The numbering in brackets refers to [1, Chapter 4.1].

Definition 5.3 (Definition 4.1.1)**.** Let F/K be an algebraic function field and M be an F-module. A mapping $\delta : F \to M$ is derivation if δ is K-linear and it satisfies the product formula

$$\delta(u \cdot v) = \delta(u) \cdot v + u \cdot \delta(v) \text{ for all } u, v \in F.$$

Lemma 5.4 (Lemma 4.1.2)**.** *Let $\delta : F \to M$ be a derivation from an algebraic function field F/K to an F-module. Then the following assertions are satisfied.*

(i) The derivation $\delta(a) = 0$ for each $a \in K$.

(ii) The derivation $\delta(z^n) = n \cdot z^{n-1} \cdot \delta(z)$ for $z \in F$ and $n \geq 0$.

(iii) If $\operatorname{char} K = p$ then $\delta(z^p) = 0$ for each $z \in F$.

(iv) The derivation $\delta\left(\frac{x}{y}\right) = \frac{\delta(x) \cdot y - x \cdot \delta(y)}{y^2}$ for $0 \neq y, x \in F$.

Proof. (i) Let $x \in F^\star$. By the product formula $\delta(a \cdot x) = \delta(a) \cdot x + a \cdot \delta(x)$. By K-linearity $\delta(a \cdot x) = a \cdot \delta(x)$. Together, $\delta(a) \cdot x = 0$. Thus $\delta(a) = 0$.

(ii) Use an induction. If $n = 0$ then $0 = \delta(1) = 0 \cdot \frac{1}{z} \cdot \delta(z) = 0$. If $n = 1$ then $\delta(z) = 1 \cdot z^0 \cdot \delta(z) = \delta(z)$.

Let $n \geq 2$ and the assertion be satisfied for $n - 1$. By the product formula and the induction assumption

$$\begin{aligned}\delta(z \cdot z^{n-1}) =& \delta(z) \cdot z^{n-1} + z \cdot \delta(z^{n-1}) = \delta(z) \cdot z^{n-1} + z \cdot (n-1) \cdot z^{n-2} \cdot \delta(z) \\ =& \delta(z) \cdot z^{n-1} + (n-1) \cdot z^{n-1} \cdot \delta(z) = n \cdot z^{n-1} \cdot \delta(z).\end{aligned}$$

(iii) By the previous item $\delta(z^p) = p \cdot z^{p-1} \cdot \delta(z) = 0$.

(iv) By the product formula

$$0 = \delta(1) = \delta(y \cdot y^{-1}) = \delta(y) \cdot y^{-1} + y \cdot \delta(y^{-1}).$$

Thus $\delta(y^{-1}) = -\frac{\delta(y)}{y^2}$. By the product formula

$$\delta(x \cdot y^{-1}) = \delta(x) \cdot y^{-1} + x \cdot \delta(y^{-1}) = \delta(x) \cdot y^{-1} - x \cdot \frac{\delta(y)}{y^2} = \frac{\delta(x) \cdot y - x \cdot \delta(y)}{y^2}.$$

□

Lemma 5.5 (Lemma 4.1.3)**.** *Let $\delta_1, \delta_2 : F \to M$ be derivations of an algebraic function field F/K to an F-module M. If $\delta_1(x) = \delta_2(x)$ where x is a separating element of F/K then $\delta_1 = \delta_2$.*

Lemma 5.6 (Proposition 4.1.4)**.** *Let F/K be an algebraic function field.*

(i) Let E/F be a finite separable extension of fields and N/E be an extension of fields. Let $\delta_0 : F \to N$ be a derivation. Then there exists a derivation $\delta : E \to N$ uniquely determined by δ_0.

(ii) Let N/F be an extension of fields and x be a separating element of F/K. Then there exists a unique derivation $\delta : F \to N$ of F/K with the property $\delta(x) = 1$.

Definition 5.7 (Definition 4.1.5). (i) Let F/K be an algebraic function field and x be a separating element of F/K. The unique derivation $\delta_x : F \to F$ of F/K with the property $\delta_x(x) = 1$ (cf. Lemma 5.6) is called the derivation with respect x.

(ii) Define

$$\mathrm{Der}_F := \{\epsilon : F \to F; \epsilon \text{ is a derivation of } F/K\}.$$

Let $\epsilon_1, \epsilon_2 \in \mathrm{Der}_F$ and $u, z \in F$. Define

$$\big(\epsilon_1 + \epsilon_2\big)(z) := \epsilon_1(z) + \epsilon_2(z) \quad \text{and} \quad \big(u \cdot \epsilon_1\big)(z) := u \cdot \epsilon_1(z).$$

The mapping $\epsilon_1 + \epsilon_2$ and $u \cdot \epsilon_1$ are obviously derivations of F/K. Thus Der_F becomes an F-module in this manner and it is called the module of derivations of F/K.

Remark 5.8. Let $\delta_x \in \mathrm{Der}_F$ and $f(x) \in K[x]$. Then $\delta_x\big(f(x)\big) = f'(x)$.

Proof. Calculate $\delta_x\big(f(x)\big) = \sum_i f_i \cdot i \cdot x^{i-1} \cdot \delta_x(x) = \sum_i f_i \cdot i \cdot x^{i-1} = f'(x)$. □

Lemma 5.9 (Lemma 4.1.6). *Let F/K be an algebraic function field and x be a separating element of F/K. The following statements are satisfied.*

(i) Let $\epsilon \in \mathrm{Der}_F$. Then $\epsilon = \epsilon(x) \cdot \delta_x$. Thus $\deg_F\big(\mathrm{Der}_F\big) = 1$.

(ii) (Chain rule) If $x \neq y \in F$ is a separating element of F/K then

$$\delta_y = \delta_y(x) \cdot \delta_x. \tag{5.1}$$

(iii) Let $t \in F$. Then

$$\delta_x(t) \neq 0 \iff t \text{ is a separating element.}$$

Theorem 5.10. *Let E/K be an elliptic function field determined by a short smooth Weierstrass equation $w(x, y) = 0$. Then*

(i) If $\mathrm{char}\, K \neq 2$ and $w(x, y) = y^2 - f(x)$ then $\delta_x(y) = \frac{f'(x)}{2y}$ and $\delta_y(x) = \frac{2y}{f'(x)}$.

(ii) If $\mathrm{char}\, K = 2$ and $w(x, y) = y^2 + x \cdot y - f_1(x)$ then $\delta_x(y) = \frac{f_1'(x) - y}{x}$ and $\delta_y(x) = \frac{x}{f_1'(x) - y}$.

(iii) If $\mathrm{char}\, K = 2$ and $w(x, y) = y^2 + a_3 \cdot y - f_2(x)$ then $\delta_x(y) = \frac{f_2'(x)}{a_3}$ and $\delta_y(x) = \frac{a_3}{f_2'(x)}$.

(iv) The derivation $\delta_x(y) = -\frac{\frac{\partial w}{\partial x}(x,y)}{\frac{\partial w}{\partial y}(x,y)}$.

Proof. (i) Calculate $\delta_x(y^2) = 2 \cdot y \cdot \delta_x(y)$ and $\delta_x(y^2) = \delta_x(f(x)) = f'(x)$. Thus $\delta_x(y) = \frac{f'(x)}{2y}$.

By Lemma 5.9

$$1 = \delta_y(y) = \delta_y(x) \cdot \delta_x(y) = \delta_y(x) \cdot \frac{f'(x)}{2y}.$$

Thus $\delta_y(x) = \frac{2y}{f'(x)}$.

(ii) Use the equation $w(x,y)=0$ to express $y=\frac{f_1(x)-y^2}{x}$. Derive

$$\begin{aligned}\delta_x(y) =&\delta_x\Big(\frac{f_1(x)-y^2}{x}\Big) = \frac{\big(f_1'(x)-2\cdot y\cdot\delta_x(y)\big)\cdot x-\big(f_1(x)-y^2\big)\cdot\delta_x(x)}{x^2}\\ =&\frac{f_1'(x)\cdot x-\big(f_1(x)-y^2\big)}{x^2}=\frac{f_1'(x)}{x}-\frac{y}{x}=\frac{f_1'(x)-y}{x}.\end{aligned}$$

By Lemma 5.9 $\delta_y(x)=\frac{x}{f_1'(x)-y}$.

(iii) Use the equation $w(x,y)=0$ to express $y=\frac{f_2(x)-y^2}{a_3}$. Derive

$$\begin{aligned}\delta_x(y) =&\delta_x\Big(\frac{f_2(x)-y^2}{a_3}\Big) = \frac{1}{a_3}\cdot\delta_x\big(f_2(x)-y^2\big)\\ =&\frac{1}{a_3}\cdot\big(f_2'(x)-2\cdot y\cdot\delta_x(y)\big)=\frac{f_2'(x)}{a_3}.\end{aligned}$$

By Lemma 5.9 $\delta_y(x)=\frac{a_3}{f_2'(x)}$.

(iv) It is a consequence of the previous items.

□

Definition 5.11 (Definition 4.1.7)**.** Let F/K be an algebraic function field.

(i) Define $Z:=\{(u,x)\in F\times F; x \text{ is separating}\}$.
Define a relation $\sim$ on Z by

$$(u,x)\sim(v,y) \iff v=u\cdot\delta_y(x). \tag{5.2}$$

The relation $\sim$ is an equivalence by Equation (5.1).

(ii) Denote the equivalence class $(u,x)\in Z$ with respect $\sim$ by $u\,dx$. Call it a differential of F/K. The denotation of the equivalence class $(1,x)\in Z$ is simplified to dx. The differentials in the form dx are called exact.
Rewrite Equation (5.2)

$$u\,dx=v\,dy \iff v=u\cdot\delta_y(x). \tag{5.3}$$

(iii) Denote the set of differentials of F/K by

$$\Delta_F:=\{u\,dx;(u,x)\in Z\}.$$

Let $u\,dx, v\,dy\in\Delta_F$. Let z be a separating element of F/K and let $t\in F$. By Equation (5.3)

$$u\,dx=\big(u\cdot\delta_z(x)\big)\,dz \quad\text{and}\quad v\,dy=\big(v\cdot\delta_z(y)\big)\,dz.$$

Define

$$u\,dx+v\,dy:=\big(u\cdot\delta_z(x)+v\cdot\delta_z(y)\big)\,dz. \tag{5.4}$$

By Equation (5.1) the definition from Equation (5.4) is independent on the choice of z.
Define

$$t\cdot(u\,dx):=(tu)\,dx\in\Delta_F. \tag{5.5}$$

The definitions from Equations (5.4) and (5.5) gives that Δ_F is an F-module.

(iv) Define the mapping

$$d : F \to \Delta_F; \tag{5.6}$$
$$t \mapsto \begin{cases} dt & \text{if } t \text{ is separating;} \\ 0 & \text{if } t \text{ is non-separating.} \end{cases}$$

Theorem 5.12 (Proposition 4.1.8)**.** *Let F/K be an algebraic function field.*

(i) Let $z \in F$ be separating of F/K. Let $\omega \in \Delta_F$. Then $dz \neq 0$ and ω can be uniquely written in the form $\omega = u\,dx$ where $u \in F$. Thus $\deg_F\left(\Delta_F\right) = 1$*.*

(ii) The mapping $d : F \to \Delta_F$ from Equation (5.6) is a derivation of F/K.

(iii) Let $t \in F$. Then $dt \neq 0 \iff t$ is separating.

(iv) Let $\delta : F \to M$ be a derivation of F/K to an F-module M. Then there exists a unique F-linear map $\mu : \Delta_F \to M$ such that $\delta = \mu \circ d$.

Remark 5.13 (Remark 4.1.9). Let F/K be an algebraic function field.

(i) The exact differentials of F/K constitute a K-subspace of Δ_F.

(ii) Let $\omega_1, 0 \neq \omega_2 \in \Delta_F$. Since $\deg_F(\Delta_F) = 1$, the quotient $\frac{\omega_1}{\omega_2} \in F$ can be defined by the following way

$$\frac{\omega_1}{\omega_2} = \frac{u \cdot \omega_2}{\omega_2} = u \iff \omega_1 = u \cdot \omega_2.$$

In case, $\omega_1 = dt$ and $\omega_2 = dx$ where $x, t \in F$ and x is separating. The quotient $\frac{dt}{dx} = \delta_x(t)$. Recall that if t is separating then it follows from Definition 5.11 and if t is not separating then $dt = 0$ and $\delta_x(t) = 0$.

By Equation (5.1) and the previous thought

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}$$

where $x, y, z \in F$ and z, x are separating.

5.2 Completion of Valued Field

This section describes the theory of the completion of a valued field (T, υ). The ideas of this section are similar to analysis. At the beginning there are definitions of the sequence, the convergent sequence, the Cauchy sequence, the series and the limit. Next, the needed theory is developed like the arithmetic of limits, a condition when a series converges. The main result of this section is Theorem 5.28 saying that is is possible to complete an arbitrary valued field.

Definition 5.14. Let T be a field and $\upsilon : T \to \mathbb{Z}_\infty$ be a surjective discrete valuation of T (cf. Definition 1.3). The pair (T, υ) is called a valued field.

Remark 5.15. The theory of valued fields is applied to (F, υ_P) where F/K is an algebraic function field and $P \in \mathbb{P}_F$. The valuation υ_P is surjective by Lemma 1.19. It is a reason why a discrete valuation in Definition 5.14 is considered surjective.

Convention 5.16. *Throughout this section a pair (T, υ) is a valued field.*

Definition 5.17. Let $(x_n)_{n\geq 0}$ be a sequence in T.

(i) The sequence $(x_n)_{n\geq 0}$ is convergent if there exists $x \in T$ which satisfies

$$\forall \epsilon \in \mathbb{R} \text{ there exists } n_0 \in \mathbb{N}_0 \text{ such that } \upsilon(x - x_n) > \epsilon \text{ whenever } n \geq n_0.$$

(ii) The sequence $(x_n)_{n\geq 0}$ is a Cauchy sequence if it has the following property

$$\forall \epsilon \in \mathbb{R} \text{ there exists } n_0 \in \mathbb{N}_0 \text{ such that } \upsilon(x_n - x_m) > \epsilon \text{ whenever } n, m \geq n_0 \ .$$

Remark 5.18. In analysis $x, y \in \mathbb{C}$ are more closely if $|x - y|$ is smaller. In our case $x, y \in T$ are more closely if $\upsilon(x - y)$ is greater. It causes by $\upsilon(0) = \infty$.

Lemma 5.19. *If a sequence $(x_n)_{n\geq 0}$ is convergent then its limit $x \in T$ is unique.*

Proof. Prove by a contradiction. Let $x, y \in T$ be limits of $(x_n)_{n\geq 0}$ such that $x \neq y$. Thus $\upsilon(x - y) =: K \in \mathbb{Z}$. Set $\epsilon := K$. By Definition 5.17 there exist $n_1, n_2 \in \mathbb{N}_0$ such that

- $\upsilon(x - x_n) > \epsilon = K$ whenever $n \geq n_1$;
- $\upsilon(y - x_n) > \epsilon = K$ whenever $n \geq n_2$.

Define $n_o := \max(n_1, n_2)$. Then

$$K = \upsilon(x - y) = \upsilon(x - x_n + x_n - y) \geq \min\big(\upsilon(x - x_n), \upsilon(y - x_n)\big) > \epsilon = K$$

whenever $n \geq n_0$. □

Lemma 5.20. *If a sequence $(x_n)_{n\geq 0}$ is convergent then it is Cauchy.*

Proof. Let $x \in T$ be a limit of $(x_n)_{n\geq 0}$. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exist $n_0 \in \mathbb{N}_0$ such that $\upsilon(x - x_n) > \epsilon$ whenever $n \geq n_0$. Then

$$\upsilon(x_n - x_m) = \upsilon(x_n - x + x - x_m) \geq \min\big(\upsilon(x - x_n), \upsilon(x - x_m)\big) > \epsilon$$

whenever $m, n \geq n_0$. Thus $(x_n)_{n\geq 0}$ is Cauchy. □

Remark 5.21. In general, a Cauchy sequence does not have to have a limit. It means that an element which could be a limit of the Cauchy sequence misses in T.

Lemma 5.22. *Let $(x_n)_{n\geq 0}$ be a Cauchy sequence. If there exists a convergent subsequence $(x_{n_k})_{k\geq 0}$ and $\lim_{k\to\infty} x_{n_k} = x$, then $\lim_{n\to\infty} x_n = x$.*

Proof. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_1 \in \mathbb{N}_0$ such that $\upsilon(x - x_{n_k}) > \epsilon$ whenever $k \geq n_1$. By Definition 5.17 there exists $n_2 \in \mathbb{N}_0$ such that $\upsilon(x_m - x_n) > \epsilon$ whenever $m, n \geq n_2$. Choose $n_k \in \mathbb{N}_0$ such that $k \geq n_1$ and $n_k \geq n_2$. Then

$$\upsilon(x - x_n) = \upsilon(x - x_{n_k} + x_{n_k} - x_n) \geq \min\big(\upsilon(x - x_{n_k}), \upsilon(x_{n_k} - x_n)\big) > \epsilon$$

whenever $n \geq n_2$. Thus $\lim_{n\to\infty} x_n = x$. □

Lemma 5.23. *Let $(x_n)_{n\geq 0}$ be a Cauchy sequence such that $\lim_{n\to\infty} x_n \neq 0$. Then there exists $n_0 \in \mathbb{N}_0$ such that $\upsilon(x_n) = K \in \mathbb{Z}$ whenever $n \geq n_0$. If $\lim_{n\to\infty} x_n = x$ then $\upsilon(x) = K$.*

Proof. Proved by a contradiction. Suppose that

$$\forall n_0 \in \mathbb{N}_0 \text{ there exist } m, n \geq n_0 \text{ such that } \upsilon(x_m) > \upsilon(x_n). \tag{5.7}$$

Make a subsequence $(x_{t_l})_{l\geq 0}$ of $(x_n)_{n\geq 0}$ by the following iterated process

1. Define the element x_{t_1}. Choose an arbitrary $\epsilon_0 \in \mathbb{R}$. For this ϵ_0 there exists $n_0 \in \mathbb{N}_0$ such that $\upsilon(x_n - x_m) > \epsilon_0$ whenever $m, n \geq n_0$ by Definition 5.17. By Assumption (5.7) there exist $k, l \geq n_0$ such that $\upsilon(x_k) > \upsilon(x_l)$. Thus $\epsilon_0 < \upsilon(x_k - x_l) = \min\big(\upsilon(x_k), \upsilon(x_l)\big) = \upsilon(x_l)$. Put $t_0 := l$ and $x_{t_1} := x_l$.

2. Define the element $x_{t_{i+1}}$. Assume that t_i and x_{t_i} are defined. Set $\epsilon_{i+1} := \upsilon(x_{t_i})$. For this ϵ_{i+1} there exists n'_{i+1} such that $\upsilon(x_n - x_m) > \epsilon_{i+1}$ whenever $m, n \geq n'_{i+1}$ by Definition 5.17. Define $n_{i+1} := \max(n'_{i+1}, t_i + 1)$. By Assumption (5.7) there exist $k, l \geq n_{i+1}$ such that $\upsilon(x_k) > \upsilon(x_l)$. Thus $\upsilon(x_{t_i}) = \epsilon_{i+1} < \upsilon(x_k - x_l) = \min\big(\upsilon(x_k), \upsilon(x_l)\big) = \upsilon(x_l)$. Set $t_{i+1} := l$ and $x_{t_{i+1}} := x_l$.

The subsequence $(x_{t_l})_{l\geq 0}$ has the property $\upsilon(x_{t_i}) < \upsilon(x_{t_{i+1}})$ for all $i \in \mathbb{N}_0$. Thus $\forall \epsilon \in \mathbb{R}$ there exists $n_0 \in \mathbb{N}_0$ such that $\upsilon(x_{t_n}) > \epsilon$ whenever $n \geq n_0$. Thus $\lim_{n\to\infty} x_{t_n} = 0$. By Lemma 5.22 the sequence $(x_n)_{n\geq 0}$ converges to 0 which is the contradiction with the property of this sequence.

Prove that $\upsilon(x) = K$ by a contradiction. Suppose that $\upsilon(x) \neq K$. Set $M := \min(\upsilon(x), K)$. Choose an arbitrary $\epsilon > M$. By the previous statement of this lemma there exists $m_0 \in \mathbb{N}_0$ such that $\upsilon(x_n) = K$ whenever $n \geq m_0$. The valuation $\upsilon(x - x_n) = \min\big(\upsilon(x), \upsilon(x_n)\big) = M < \epsilon$ whenever $n \geq m_0$. Thus there exists no $n_0 \in \mathbb{N}_0$ for ϵ according to Definition 5.17. So $\lim_{n\to\infty} x_n \neq x$ which is the contradiction. □

Theorem 5.24. *Let $(x_n)_{n\geq 0}$ and $(y_n)_{n\geq 0}$ be convergent sequences in T. Let $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$. Then the following statements are satisfied.*

(i) The limit $\lim_{n\to\infty}(x_n + y_n) = \lim_{n\to\infty}(x_n) + \lim_{n\to\infty}(y_n) = x + y$;

(ii) The limit $\lim_{n\to\infty}(x_n \cdot y_n) = \lim_{n\to\infty}(x_n) \cdot \lim_{n\to\infty}(y_n) = x \cdot y$.

Proof. Choose an arbitrary $\epsilon \in \mathbb{R}$.

(i) By Definition 5.17 there exist $n_1, n_2 \in \mathbb{N}_0$ such that $\upsilon(x - x_n) > \epsilon$ whenever $n \geq n_1$ and $\upsilon(y - y_n) > \epsilon$ whenever $n \geq n_2$. Define $n_0 := \max(n_1, n_2)$. Then

$$\upsilon(x + y - x_n - y_n) \geq \min\big(\upsilon(x - x_n), \upsilon(y - y_n)\big) > \epsilon$$

whenever $n \geq n_0$.

(ii) The proof of multiplication is split to three cases.

(a) Suppose that $\lim_{n\to\infty}(x_n) = 0$ and $\lim_{n\to\infty}(y_n) = 0$. By Definition 5.17 there exist $n_1, n_2 \in \mathbb{N}_0$ such that $\upsilon(x_n) > \frac{\epsilon}{2}$ whenever $n \geq n_1$ and $\upsilon(y_n) > \frac{\epsilon}{2}$ whenever $n \geq n_2$. Set $n_0 := \max(n_1, n_2)$. Then

$$\upsilon(x_n \cdot y_n) = \upsilon(x_n) + \upsilon(y_n) > \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

whenever $n \geq n_0$.

(b) Suppose that $\lim_{n\to\infty}(x_n) = x \neq 0$ and $\lim_{n\to\infty}(y_n) = 0$. By Lemma 5.23 there exists $n_1 \in \mathbb{N}_0$ such that $\upsilon(x_n) = K_x \in \mathbb{Z}$ whenever $n \geq n_1$. By Definition 5.17 there exists $n_2 \in \mathbb{N}_0$ such that $\upsilon(y_n) > \epsilon - K_x$ whenever $n \geq n_2$. Define $n_0 := \max(n_1, n_2)$. Then

$$\upsilon(x_n \cdot y_n) = \upsilon(x_n) + \upsilon(y_n) > K_x + \epsilon - K_x = \epsilon$$

whenever $n \geq n_0$.

(c) Suppose that $\lim_{n\to\infty}(x_n) = x \neq 0$ and $\lim_{n\to\infty}(y_n) = y \neq 0$. Define $K_y := \upsilon(y) \in \mathbb{Z}$. By Lemma 5.23 there exists $n_1 \in \mathbb{N}_0$ such that $\upsilon(x_n) = \upsilon(x) = K_x \in \mathbb{Z}$ whenever $n \geq n_1$. Define $K := \min(K_x, K_y)$. By Definition 5.17 there exist $n_2, n_3 \in \mathbb{N}_0$ such that $\upsilon(x - x_n) > \frac{\epsilon}{2} - K$ whenever $n \geq n_2$ and $\upsilon(y - y_n) > \frac{\epsilon}{2}$ whenever $n \geq n_3$. Set $n_0 := \max(n_1, n_2, n_3)$. Then

$$\begin{aligned}
&\upsilon(x \cdot y - x_n \cdot y_n) \\
=&\upsilon(x \cdot y - x_n \cdot y + x_n \cdot y - x_n \cdot y_n) \\
=&\upsilon\big(y \cdot (x - x_n) + x_n \cdot (y - y_n)\big) \\
\geq&\min\big(\upsilon(y) + \upsilon(x - x_n), \upsilon(x_n) + \upsilon(y - y_n)\big) \\
=&\min\big(K_y + \upsilon(x - x_n), K_x + \upsilon(y - y_n)\big) \\
\geq&\min\big(K + \upsilon(x - x_n), K + \upsilon(y - y_n)\big) \\
=&\min\big(\upsilon(x - x_n), \upsilon(y - y_n)\big) + K \\
>&\frac{\epsilon}{2} - K + K + \frac{\epsilon}{2} = \epsilon
\end{aligned}$$

whenever $n \geq n_0$.

□

Definition 5.25. Let $(x_n)_{n\geq 0}$ be a sequence in T. Define its partial sum $s_m := \sum_{i=0}^{m} x_i$. Say that the infinity series $\sum_{i=0}^{\infty} x_i$ is a convergent if the sequence of its partial sums $(s_m)_{m\geq 0}$ is convergent. Use the denotation $\sum_{i=0}^{\infty} x_i := \lim_{m\to\infty} s_m$.

Definition 5.26.

(i) A valued field (T, v) is called complete if every Cauchy sequence in T is convergent.

(ii) A valued field $(\hat{T}, \hat{v})$ is called a completion of a valued field (T, v) if the following properties are satisfied

(1) $\hat{T}/T$ and $\hat{v}\restriction_T = v$;

(2) $(\hat{T}, \hat{v})$ is complete;

(3) T is dense in $\hat{T}$; i.e. $\forall z \in \hat{T}$ there is a sequence $(x_n)_{n\geq 0}$ in T with the property $\lim_{n\to\infty} x_n = z$.

Theorem 5.27. *Suppose that (T, v) is complete. Let $(x_n)_{n\geq 0}$ be a sequence in T. The infinity series $\sum_{i=0}^{\infty} x_i$ is convergent $\iff$ $\lim_{n\to\infty} x_n = 0$.*

Proof. Define the partial sum $s_m := \sum_{i=0}^{m} x_i$ of $(x_n)_{n\geq 0}$. Firstly, show $\Leftarrow$. Show that $(s_m)_{m\geq 0}$ is Cauchy. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_0 \in \mathbb{N}_0$ such that $v(x_n) > \epsilon$ whenever $n \geq n_0$. Choose arbitrary $m, n \geq n_0$. Show that $v(s_m - s_n) > \epsilon$. Assume that $m \geq n$. Calculate

$$v(s_m - s_n) = v\Big(\sum_{i=0}^{m} x_i - \sum_{i=0}^{n} x_i\Big) = v\Big(\sum_{i=n+1}^{m} x_i\Big) = \min\big(v(x_i); i = n+1, \dots, m\big) > \epsilon.$$

Thus $(s_m)_{m\geq 0}$ is Cauchy. Because (T, v) is complete then $(s_m)_{m\geq 0}$ is convergent.

Show $\Rightarrow$. Let $s \in T$ be a sum of $\sum_{i=0}^{\infty} x_i$. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_0 \in \mathbb{N}_0$ such that $v(s - s_n) > \epsilon$ whenever $n \geq n_0$. Choose an arbitrary $n \geq n_0 + 1$. Calculate

$$v(x_n) = v(s - s_{n-1} - s + s_n) \geq \min\big(v(s - s_{n-1}), v(s - s_n)\big) > \epsilon.$$

Thus $\lim_{n\to\infty} x_n = 0$. $\square$

Theorem 5.28. *Let (T, v) be a valued field. There exists a completion $(\hat{T}, \hat{v})$ of (T, v). This completion is unique in the following sense. If $(\tilde{T}, \tilde{v})$ is another completion of (T, v) then there exists a T-isomorphism $f : \hat{T} \to \tilde{T}$ such that $\hat{v} = \tilde{v} \circ f$. Because $(\hat{T}, \hat{v})$ is unique up to isomorphisms, it is called the completion of (T, v).*

Proof. The proof is technical and long. Thus at the beginning there is a sketch of the proof for a better understanding. First, define a ring $R := \{(x_n)_{n\geq 0}; (x_n)_{n\geq 0}$ is a Cauchy sequence in $T\}$ and prove that it is really a ring. Next, define $I := \{(x_n)_{n\geq 0} \in R; \lim_{n\to\infty} x_n = 0\}$ and prove that it is a maximal ideal of R. Define $\hat{T} := R/I$. Because I is maximal then $\hat{T}$ is a field. The field $\hat{T} \geq T$. Define a valuation $\hat{v}$ of $\hat{T}$. Let $\hat{t} = (x_n)_{n\geq 0} + I \in \hat{T}$ where $(x_n)_{n\geq 0}$ is a Cauchy sequence in T. If $\lim_{n\to\infty} x_n = 0$ then $\hat{v}(\hat{t}) = \infty$. If $\lim_{n\to\infty} x_n \neq 0$ then there exists $K \in \mathbb{Z}$ by Lemma 5.23 and $\hat{v}(\hat{t}) = K$. Prove that $\hat{v}$ is independent on the choice of a representative of a residual class and that $\hat{v}$ is a valuation. Next, prove that $v = \hat{v}\restriction_T$. Let $(z_m)_{m\geq 0}$ be Cauchy in $\hat{T}$. For every $m \in \mathbb{N}_0$ $z_m = (x_{mn})_{n\geq 0} + I$ where $(x_{mn})_{n\geq 0}$ is a Cauchy sequence in T. Then $\lim_{m\to\infty} z_m = (x_{mm})_{m\geq 0} + I$.

Thus $(\hat{T}, \hat{v})$ is complete. Let $z = (x_n)_{n\geq 0} + I \in \hat{T}$ where $(x_n)_{n\geq 0}$ is a Cauchy sequence in T. Then $\lim_{n\to\infty} x_n = z$. Thus T is dense in $\hat{T}$. Let $(\tilde{T}, \tilde{v})$ be another completion of (T, v). Define a mapping $f : \hat{T} \to \tilde{T}$ by the following way $f(z) = \tilde{v}\text{-}\lim_{n\to\infty} x_n$ where $\tilde{v}$-lim is the limit in $\tilde{T}$. Prove that the mapping f is a T-isomorphism and that $\hat{v} = \tilde{v} \circ f$.

Now, make the proof. Define a set

$$R := \{(x_n)_{n\geq 0}; (x_n)_{n\geq 0} \text{ is a Cauchy sequence in } T\}.$$

The set R is a ring. The structure of the ring on R is defined the following way. Let $(x_n)_{n\geq 0}, (y_n)_{n\geq 0} \in R$.

(1) The zero is defined by $0 := (0)_{n\geq 0}$. This sequence is obviously Cauchy.

(2) The unit is defined by $1 := (1)_{n\geq 0}$. This sequence is obviously Cauchy.

(3) The operator minus is defined by $-(x_n)_{n\geq 0} := (-x_n)_{n\geq 0}$. The sequence $(-x_n)_{n\geq 0}$ is obviously Cauchy if $(x_n)_{n\geq 0}$ is Cauchy.

(4) The operator plus is defined by $(x_n)_{n\geq 0} + (y_n)_{n\geq 0} := (x_n + y_n)_{n\geq 0}$. Prove that $(x_n + y_n)_{n\geq 0}$ is Cauchy. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_0 \in \mathbb{N}_0$ such that $v(x_n - x_m) > \epsilon$ and $v(y_n - y_m) > \epsilon$ whenever $m, n \geq n_0$. Then $v(x_n + y_n - x_m - y_m) \geq \min\big(v(x_n - x_m), v(y_n - y_m)\big) > \epsilon$ whenever $m, n \geq n_0$.

(5) The operator multiplication is defined by $(x_n)_{n\geq 0} \cdot (y_n)_{n\geq 0} = (x_n \cdot y_n)_{n\geq 0}$. Prove that $(x_n \cdot y_n)_{n\geq 0}$ is Cauchy. This proof is split to three cases.

  (i) Suppose that $\lim_{n\to\infty} x_n = 0$ and $\lim_{n\to\infty} y_n = 0$. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_0 \in \mathbb{N}_0$ such that $v(x_n) > \frac{\epsilon}{2}$ and $v(y_n) > \frac{\epsilon}{2}$ whenever $n \geq n_0$. Then

  $$v(x_n \cdot y_n) = v(x_n) + v(y_n) > \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \text{ whenever } n \geq n_0 .$$

  Thus $(x_n \cdot y_n)_{n\geq 0}$ converges to 0. By Lemma 5.20 $(x_n \cdot y_n)_{n\geq 0}$ is Cauchy.

  (ii) Suppose that $\lim_{n\to\infty} x_n = 0$ and $\lim_{n\to\infty} y_n \neq 0$. By Lemma 5.23 there exists $n_1 \in \mathbb{N}_0$ such that $v(y_n) =: K \in \mathbb{Z}$ whenever $n \geq n_1$. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_2 \in \mathbb{N}_0$ such that $v(x_n) > \epsilon - K$ whenever $n \geq n_2$. Define $n_0 := \max(n_1, n_2)$. Then

  $$v(x_n \cdot y_n) = v(x_n) + v(y_n) > \epsilon - K + K = \epsilon \text{ whenever } n \geq n_0.$$

  Thus $(x_n \cdot y_n)_{n\geq 0}$ converges to 0. By Lemma 5.20 $(x_n \cdot y_n)_{n\geq 0}$ is Cauchy.

  (iii) Suppose that $\lim_{n\to\infty} x_n \neq 0$ and $\lim_{n\to\infty} y_n \neq 0$. By Lemma 5.23 there exist $n_1, n_2 \in \mathbb{N}_0$ such that $v(x_n) = K_x \in \mathbb{Z}$ whenever $n \geq n_1$ and $v(y_n) = K_y \in \mathbb{Z}$ whenever $n \geq n_2$. Set $K := \min(K_x, K_y)$. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exist $n_3, n_4 \in \mathbb{N}_0$ such that

$\upsilon(x_n - x_m) > \epsilon - K$ whenever $m, n \geq n_3$ and $\upsilon(y_n - y_m) > \epsilon - K$ whenever $m, n \geq n_4$. Define $n_0 := \max(n_1, n_2, n_3, n_4)$. Then

$$\begin{aligned}
&\upsilon(x_n \cdot y_n - x_m \cdot y_m) \\
=&\upsilon(x_n \cdot y_n - x_m \cdot y_n + x_m \cdot y_n - x_m \cdot y_m) \\
=&\upsilon\big(y_n \cdot (x_n - x_m) + x_m \cdot (y_n - y_m)\big) \\
\geq& \min\Big(\upsilon\big(y_n \cdot (x_n - x_m)\big), \upsilon\big(x_m \cdot (y_n - y_m)\big)\Big) \\
=& \min\Big(\upsilon(y_n) + \upsilon(x_n - x_m), \upsilon(x_m) + \upsilon(y_n - y_m)\Big) \\
=& \min\Big(K_y + \upsilon(x_n - x_m), K_x + \upsilon(y_n - y_m)\Big) \\
\geq& \min\Big(K + \upsilon(x_n - x_m), K + \upsilon(y_n - y_m)\Big) \\
=& \min\big(\upsilon(x_n - x_m), \upsilon(y_n - y_m)\big) + K \\
>&\epsilon - K + K = \epsilon
\end{aligned}$$

whenever $m, n \geq n_0$. Thus $(x_n \cdot y_n)_{n\geq 0}$ is Cauchy.

Define an ideal

$$I := \{(x_n)_{n\geq 0} \in R; \lim_{n\to\infty} x_n = 0\}.$$

Prove that I is a proper ideal. The zero is obviously in I. The unit is not obviously in I. Let $(x_n)_{n\geq 0}, (y_n)_{n\geq 0} \in I$ and $(r_n)_{n\geq 0} \in R$.

Prove that I is closed under the internal addition, $(x_n + y_n)_{n\geq 0} \in I$. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exist $n_1, n_2 \in \mathbb{N}_0$ such that $\upsilon(x_n) > \epsilon$ whenever $n \geq n_1$ and $\upsilon(y_n) > \epsilon$ whenever $n \geq n_2$. Define $n_0 := \max(n_1, n_2)$. Then $\upsilon(x_n + y_n) \geq \min\big(\upsilon(x_n), \upsilon(y_n)\big) > \epsilon$ whenever $n \geq n_0$.

Prove that I is closed under the external multiplication, $(x_n \cdot r_n)_{n\geq 0} \in I$. This proof is split to two cases.

(i) Suppose that $\lim_{n\to\infty} r_n = 0$;

(ii) Suppose that $\lim_{n\to\infty} r_n \neq 0$.

The both cases are proved above during the proving that R is closed to multiplication (cf. item 5).

Prove that I is a maximal ideal of R. Suppose an ideal J of R such that $I \subsetneqq J \subseteq R$. There exists a sequence $(a_n)_{n\geq 0} \in J \setminus I$. Thus $\lim_{n\to\infty} a_n \neq 0$. By Lemma 5.23 there exists $n_0 \in \mathbb{N}_0$ such that $\upsilon(a_n) = K \in \mathbb{Z}$ whenever $n \geq n_0$. Define a sequence $(i_n)_{n\geq 0}$.

(i) If $n < n_0$ and $a_n = 0$ then $i_n := 1$;

(ii) if $n < n_0$ and $a_n \neq 0$ then $i_n := 0$;

(iii) if $n \geq n_0$ then $i_n := 0$.

Then $(i_n)_{n\geq 0} \in I$ because $\lim_{n\to\infty} i_n = 0$ Define another sequence $(b_n)_{n\geq 0} := (i_n)_{n\geq 0} + (a_n)_{n\geq 0} \in J \setminus I$. For all $n \in \mathbb{N}_0$ $b_n \neq 0$. Thus $(b_n)_{n\geq 0} \in J$ is an invertible element of R. Thus $J = R$ and I is a maximal ideal.

Therefore the residual class ring $\hat{T} := R/I$ is a field. Define a mapping $\varrho : T \to \hat{T}$ by $\varrho(t) := (t)_{n\geq 0} + I$ where $(t)_{n\geq 0}$ is a constant sequence. The mapping ϱ is an injective homomorphism of fields. Let $t, s \in T$. Then

(i) $\varrho(t) = (t)_{n\geq 0} + I = 0 \in \hat{T} \Leftrightarrow \lim_{n\to\infty} t = 0 \Leftrightarrow t = 0$;

(ii) $\varrho(t+s) = (t+s)_{n\geq 0} + I = (t)_{n\geq 0} + (s)_{n\geq 0} + I = \varrho(t) + \varrho(s)$;

(iii) $\varrho(t\cdot s) = (t\cdot s)_{n\geq 0} + I = (t)_{n\geq 0}\cdot(s)_{n\geq 0} + I = \varrho(t)\cdot\varrho(s)$.

Thus T is embedded into $\hat{T}$ by ϱ.

Let $\hat{t} = (x_n)_{n\geq 0} + I \in \hat{T}$ where $(x_n)_{n\geq 0}$ is a Cauchy sequence in T. If $\lim_{n\to\infty} x_n \neq 0$ then there exist $K_x \in \mathbb{Z}$ and $n_0 \in \mathbb{N}_0$ such that $\upsilon(x_n) = K_x$ whenever $n \geq n_0$ by Lemma 5.23. A valuation $\hat{\upsilon}$ on $\hat{T}$ is defined by

$$\hat{\upsilon}(\hat{t}) := \begin{cases} K_x & \text{if } \lim_{n\to\infty} x_n \neq 0; \\ \infty & \text{if } \lim_{n\to\infty} x_n = 0. \end{cases} \tag{5.8}$$

Show that the definition of $\hat{\upsilon}$ is independent on the choice of a representative of a residual class. Let $(x_n)_{n\geq 0} + I = (y_n)_{n\geq 0} + I \in \hat{T}$. If $(x_n)_{n\geq 0}, (y_n)_{n\geq 0} \in I$ then $\hat{\upsilon}((x_n)_{n\geq 0} + I) = \hat{\upsilon}((x_n)_{n\geq 0} + I) = \infty$ by the definition of $\hat{\upsilon}$.

Suppose that $(x_n)_{n\geq 0} + I = (y_n)_{n\geq 0} + I \neq I$. It is satisfied

$$(x_n)_{n\geq 0} + I = (y_n)_{n\geq 0} + I \Leftrightarrow (x_n - y_n)_{n\geq 0} \in I \Leftrightarrow \lim_{n\to\infty}(x_n - y_n) = 0. \tag{5.9}$$

By Lemma 5.23 there exist $n_1, n_2 \in \mathbb{N}_0$ such that $\hat{\upsilon}\big((x_n)_{n\geq 0} + I\big) = \upsilon(x_n) = K_x \in \mathbb{Z}$ whenever $n \geq n_1$ and $\hat{\upsilon}\big((y_n)_{n\geq 0} + I\big) = \upsilon(y_n) = K_y \in \mathbb{Z}$ whenever $n \geq n_2$. Our goal is to prove that $K_x = K_y$. Prove by a contradiction. Suppose that $K_x < K_y$. Choose $\epsilon \in \mathbb{R}$ such that $\epsilon > K_y$. By the assumption from Equation (5.9) there exists $n_3 \in \mathbb{N}_0$ such that $\upsilon(x_n - y_n) > \epsilon$ whenever $n \geq n_3$. Define $n_0 := \max(n_1, n_2, n_3)$. Then

$$K_y < \epsilon < \upsilon(x_n - y_n) = \min\big(\upsilon(x_n), \upsilon(y_n)\big) = \min(K_x, K_y) = K_x$$

whenever $n \geq n_0$. It is the contradiction. Thus $K_x = K_y$.

Show that $\hat{\upsilon}$ is really a valuation. By the definition of $\hat{\upsilon}$

$$\hat{\upsilon}\big((x_n)_{n\geq 0} + I\big) = \infty \Leftrightarrow \lim_{n\to\infty} x_n = 0 \Leftrightarrow (x_n)_{n\geq 0} \in I \Leftrightarrow (x_n)_{n\geq 0} + I = 0 \in \hat{T}.$$

Let $(x_n)_{n\geq 0} + I, (y_n)_{n\geq 0} + I \in \hat{T}$ Show that $\hat{\upsilon}\big((x_n)_{n\geq 0} + I\big) + \hat{\upsilon}\big((y_n)_{n\geq 0} + I\big) = \hat{\upsilon}\big((x_n \cdot y_n)_{n\geq 0} + I\big)$.

(i) Suppose that $(x_n)_{n\geq 0} + I \neq I$ and $(y_n)_{n\geq 0} + I \neq I$. By Lemma 5.23 there exist $n_1, n_2 \in \mathbb{N}_0$ such that $\hat{\upsilon}\big((x_n)_{n\geq 0} + I\big) = \upsilon(x_n) =: K_x \in \mathbb{Z}$ whenever $n \geq n_1$ and $\hat{\upsilon}\big((y_n)_{n\geq 0} + I\big) = \upsilon(y_n) =: K_y \in \mathbb{Z}$ whenever $n \geq n_2$. Define $n_0 := \max(n_1, n_2)$. Calculate $\upsilon(x_n \cdot y_n) = \upsilon(x_n) + \upsilon(y_n) = K_x + K_y$ whenever $n \geq n_0$. Thus $\hat{\upsilon}((x_n \cdot y_n)_{n\geq 0} + I) = K_x + K_y$. On the other hand $\hat{\upsilon}\big((x_n)_{n\geq 0} + I\big) + \hat{\upsilon}\big((y_n)_{n\geq 0} + I\big) = K_x + K_y$.

(ii) Suppose that $(x_n)_{n\geq 0} + I = I$ and $(y_n)_{n\geq 0} + I \neq I$. By Lemma 5.23 there exists $n_2 \in \mathbb{N}_0$ such that $\hat{\upsilon}\big((y_n)_{n\geq 0} + I\big) = \upsilon(y_n) =: K_y \in \mathbb{Z}$ whenever $n \geq n_2$. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_1 \in \mathbb{N}_0$ such that $\upsilon(x_n) > \epsilon - K_y$ whenever $n \geq n_1$. Define $n_0 := \max(n_1, n_2)$. Calculate $\upsilon(x_n \cdot y_n) = \upsilon(x_n) + \upsilon(y_n) > \epsilon - K_y + K_y = \epsilon$ whenever $n \geq n_0$. Thus $\hat{\upsilon}\big((x_n \cdot y_n)_{n\geq 0} + I\big) = \infty$. On the other hand $\hat{\upsilon}\big((x_n)_{n\geq 0} + I\big) + \hat{\upsilon}\big((y_n)_{n\geq 0} + I\big) = \infty + K_y = \infty$.

(iii) Suppose that $(x_n)_{n\geq 0}+I=I$ and $(y_n)_{n\geq 0}+I=I$. Choose an arbitrary $\epsilon\in\mathbb{R}$. By Definition 5.17 there exists $n_1,n_2\in\mathbb{N}_0$ such that $\upsilon(x_n)>\frac{\epsilon}{2}$ whenever $n\geq n_1$ and $\upsilon(y_n)>\frac{\epsilon}{2}$ whenever $n\geq n_2$. Define $n_0:=\max(n_1,n_2)$. Calculate $\upsilon(x_n\cdot y_n)=\upsilon(x_n)+\upsilon(y_n)>\frac{\epsilon}{2}+\frac{\epsilon}{2}=\epsilon$ whenever $n\geq n_0$. Thus $\hat{\upsilon}\big((x_n\cdot y_n)_{n\geq 0}+I\big)=\infty$. On the other hand $\hat{\upsilon}\big((x_n)_{n\geq 0}+I\big)+\hat{\upsilon}\big((y_n)_{n\geq 0}+I\big)=\infty+\infty=\infty$.

Show that $\min\Big(\hat{\upsilon}\big((x_n)_{n\geq 0}+I\big),\hat{\upsilon}\big((y_n)_{n\geq 0}+I\big)\Big)\leq\hat{\upsilon}\big((x_n+y_n)_{n\geq 0}+I\big)$.

(i) Suppose that $(x_n)_{n\geq 0}+I\neq I$ and $(y_n)_{n\geq 0}+I\neq I$. By Lemma 5.23 there exist $n_1,n_2\in\mathbb{N}_0$ such that $\hat{\upsilon}\big((x_n)_{n\geq 0}+I\big)=\upsilon(x_n)=:K_x\in\mathbb{Z}$ whenever $n\geq n_1$ and $\hat{\upsilon}\big((y_n)_{n\geq 0}+I\big)=\upsilon(y_n)=:K_y\in\mathbb{Z}$ whenever $n\geq n_2$. Suppose that $K_x\geq K_y$. Define $n_0:=\max(n_1,n_2)$. Calculate $\upsilon(x_n+y_n)\geq\min\big(\upsilon(x_n),\upsilon(y_n)\big)=\min\big(K_x,K_y\big)=\min\Big(\hat{\upsilon}\big((x_n)_{n\geq 0}+I\big),\hat{\upsilon}\big((y_n)_{n\geq 0}+I\big)\Big)=K_y\in\mathbb{Z}$ whenever $n\geq n_0$. If $\lim_{n\to\infty}(x_n+y_n)=0$ then $\hat{\upsilon}\big((x_n+y_n)_{n\geq 0}+I\big)=\infty>K_y$. If $\lim_{n\to\infty}(x_n+y_n)\neq 0$ then there exist $K_z\in\mathbb{Z}$ and $n_3\in\mathbb{N}_0$ such that $\upsilon(x_n+y_n)=K_z=\hat{\upsilon}\big((x_n+y_n)_{n\geq 0}+I\big)$ whenever $n\geq n_3$. Define $N_0:=\max(n_1,n_2,n_3)$. Then $\hat{\upsilon}\big((x_n+y_n)_{n\geq 0}+I\big)=\upsilon(x_n+y_n)\geq K_y$ whenever $n\geq N_0$.

(ii) Suppose that $(x_n)_{n\geq 0}+I=I$ and $(y_n)_{n\geq 0}+I\neq I$. By Lemma 5.23 there exists $n_2\in\mathbb{N}_0$ such that $\hat{\upsilon}\big((y_n)_{n\geq 0}+I\big)=\upsilon(y_n)=:K_y\in\mathbb{Z}$ whenever $n\geq n_2$. Choose $\epsilon>K_y$. By Definition 5.17 there exists $n_1\in\mathbb{N}_0$ such that $\upsilon(x_n)>\epsilon$ whenever $n\geq n_1$. Define $n_0:=\max(n_1,n_2)$. Calculate $\upsilon(x_n+y_n)=\min\big(\upsilon(x_n),\upsilon(y_n)\big)=K_y$ whenever $n\geq n_0$. Thus $\hat{\upsilon}\big((x_n+y_n)_{n\geq 0}+I\big)=K_y$. On the other hand $\min\Big(\hat{\upsilon}\big((x_n)_{n\geq 0}+I\big),\hat{\upsilon}\big((y_n)_{n\geq 0}+I\big)\Big)=\min\big(\infty,K_y\big)=K_y$.

(iii) Suppose that $(x_n)_{n\geq 0}+I=I$ and $(y_n)_{n\geq 0}+I=I$. Choose an arbitrary $\epsilon\in\mathbb{R}$. By Definition 5.17 there exists $n_1,n_2\in\mathbb{N}_0$ such that $\upsilon(x_n)>\epsilon$ whenever $n\geq n_1$ and $\upsilon(y_n)>\epsilon$ whenever $n\geq n_2$. Define $n_0:=\max(n_1,n_2)$. Calculate $\upsilon(x_n+y_n)=\min\big(\upsilon(x_n),\upsilon(y_n)\big)>\epsilon$ whenever $n\geq n_0$. Thus $\hat{\upsilon}\big((x_n+y_n)_{n\geq 0}+I\big)=\infty$. On the other hand $\min\Big(\hat{\upsilon}\big((x_n)_{n\geq 0}+I\big),\hat{\upsilon}\big((y_n)_{n\geq 0}+I\big)\Big)=\min(\infty,\infty)=\infty$.

Prove that $\hat{\upsilon}\restriction_T$ is υ. Let $t\in T$.

(i) If $t=0$ then $\hat{\upsilon}(\varrho(0))=\hat{\upsilon}\big((0)_{n\geq 0}+I\big)=\infty=\upsilon(0)$.

(ii) If $t\neq 0$ then $\hat{\upsilon}(\varrho(t))=\hat{\upsilon}\big((t)_{n\geq 0}+I\big)=\upsilon(t)$.

Now $(\hat{T},\hat{\upsilon})$ is a valued field, $\hat{T}/T$ and $\hat{\upsilon}\restriction_T=\upsilon$.

Show that the valued field $(\hat{T},\hat{\upsilon})$ is complete. Let $(z_m)_{m\geq 0}$ be a Cauchy sequence in $\hat{T}$. For every $m\in\mathbb{N}_0$ $z_m=(x_{mn})_{n\geq 0}+I$ where $(x_{mn})_{n\geq 0}$ is a Cauchy sequence in T. Define $z:=(x_{nn})_{n\geq 0}+I\in\hat{T}$. The desired result is $\lim_{m\to\infty}z_m=z$.

The first step of the proof is shown that $(x_{nn})_{n\geq 0}$ is a Cauchy sequence in T otherwise the definition of z is not correct. Choose an arbitrary $\epsilon\in\mathbb{R}$. For every $l\in\mathbb{N}_0$ $(x_{ln})_{n\geq 0}$ is Cauchy. By Definition 5.17

$$\exists n_l\in\mathbb{N}_0 \text{ such that } \upsilon(x_{la}-x_{lb})>\epsilon \text{ whenever } a,b\geq n_l. \tag{5.10}$$

It is able to consider that

$$n_l \leq l \text{ for every } l \in \mathbb{N}_0 \tag{5.11}$$

Because if there exists $l \in \mathbb{N}_0$ such that $n_l > l$, make a subsequence $(x_{ln_k})_{k\geq 0}$ where $n_k := n_l - l + k$. By Lemma 5.29 $(x_{ln_k})_{k\geq 0} + I = (x_{ln})_{n\geq 0} + I$. The new representative satisfies that $n_l \leq l$. By Definition 5.17 there exists $m_0 \in \mathbb{N}_0$ such that $\hat{\upsilon}(z_r - z_s) > \epsilon$ whenever $r, s \geq m_0$. Choose an arbitrary $r, s \geq m_0$. Write out $z_r - z_s = (x_{rn} - x_{sn})_{n\geq 0} + I$ where $(x_{rn} - x_{sn})_{n\geq 0}$ is Cauchy in T. Write out the definition of $\hat{\upsilon}$ (cf. Equation (5.8)).

(i) If $\lim_{n\to\infty}(x_{rn} - x_{sn}) = 0$ then $\hat{\upsilon}(z_r - z_s) = \infty$ and by Definition 5.17 there exists $n_{rs} \in \mathbb{N}_0$ such that $\upsilon(x_{rk} - x_{sk}) > \epsilon$ whenever $k \geq n_{rs}$.

(ii) If $\lim_{n\to\infty}(x_{rn} - x_{sn}) \neq 0$ then by Lemma 5.23 and the definition of $\hat{\upsilon}$ there exist $n_{rs} \in \mathbb{N}_0$ and $K_{rs} \in \mathbb{Z}$ such that $\upsilon(x_{rk} - x_{sk}) = K_{rs} = \hat{\upsilon}(z_r - z_s) > \epsilon$ whenever $k \geq n_{rs}$.

Thus

$$\exists n_{rs} \in \mathbb{N}_0 \text{ such that } \upsilon(x_{rk} - x_{sk}) > \epsilon \text{ whenever } k \geq n_{rs}. \tag{5.12}$$

Define $B := \max(n_r, n_s, n_{rs})$. Choose an arbitrary $t \geq B$. The following statement are satisfied.

(i) The valuation $\upsilon(x_{rr} - x_{rt}) > \epsilon$ following from assumptions (5.11) and (5.10).

(ii) The valuation $\upsilon(x_{rt} - x_{st}) > \epsilon$ following from assumption (5.12).

(iii) The valuation $\upsilon(x_{st} - x_{ss}) > \epsilon$ following from assumptions (5.11) and (5.10).

Calculate

$$\begin{aligned}\upsilon(x_{rr} - x_{ss}) &= \upsilon(x_{rr} - x_{rt} + x_{rt} - x_{st} + x_{st} - x_{ss}) \\ &\geq \min\big(\upsilon(x_{rr} - x_{rt}), \upsilon(x_{rt} - x_{st}), \upsilon(x_{st} - x_{ss})\big) > \epsilon\end{aligned}$$

whenever $r, s \geq m_0$ and $t \geq B$. It has been proved that $(x_{nn})_{n\geq 0}$ is a Cauchy sequence in T.

The second step of the proof is shown that $\lim_{m\to\infty} z_m = z$. Choose an arbitrary $\epsilon \in \mathbb{R}$. The sequence $(x_{nn})_{n\geq 0}$ is Cauchy; $z = (x_{nn})_{n\geq 0} + I$. By Definition 5.17 there exists $n_0 \in \mathbb{N}_0$ such that $\upsilon(x_{ll} - x_{kk}) > \epsilon$ whenever $k, l \geq n_0$. Choose an arbitrary $m \geq n_0$. Prove that $\hat{\upsilon}(z - z_m) > \epsilon$. Write out $z - z_m = (x_{nn} - x_{mn})_{n\geq 0} + I$ where $(x_{nn} - x_{mn})_{n\geq 0}$ is Cauchy in T. Write out the definition of $\hat{\upsilon}$ (cf. Equation (5.8)). If $\lim_{n\to\infty}(x_{nn} - x_{mn}) = 0$ then $\hat{\upsilon}(z - z_m) = \infty > \epsilon$. It is done. If $\lim_{n\to\infty}(x_{nn} - x_{mn}) \neq 0$ then by Lemma 5.23 and the definition of $\hat{\upsilon}$ there exist $k_0 \in \mathbb{N}_0$ and $K_m \in \mathbb{Z}$ such that $\upsilon(x_{kk} - x_{mk}) = K_m = \hat{\upsilon}(z - z_m)$ whenever $k \geq k_0$. Choose $k \geq B := \max(n_0, k_0, n_m)$. Recall that n_m follows from assumption (5.10). Then

(i) $\upsilon(x_{kk} - x_{mm}) > \epsilon$;

(ii) $\upsilon(x_{mm} - x_{mk}) > \epsilon$ following from assumptions (5.11) and (5.10).

Calculate

$$\begin{aligned}\hat{\upsilon}(z - z_m) &= \upsilon(x_{kk} - x_{mk})\\ &= \upsilon(x_{kk} - x_{mm} + x_{mm} - x_{mk})\\ &\geq \min\big(\upsilon(x_{kk} - x_{mm}), \upsilon(x_{mm} - x_{mk})\big) > \epsilon\end{aligned}$$

whenever $m \geq n_0$ and $k \geq B$. It has been proved that $\lim_{m\to\infty} z_m = z$.

Show that T is dense in $\hat{T}$. Let $z = (x_n)_{n\geq 0} + I \in \hat{T}$ where $(x_n)_{n\geq 0}$ is Cauchy in T. Prove that $\lim_{n\to\infty} \varrho(x_n) = z$. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_0 \in \mathbb{N}_0$ such that $\upsilon(x_n - x_m) > \epsilon$ whenever $n, m \geq n_0$. Choose an arbitrary $k \geq n_0$. Prove that $\hat{\upsilon}\big(\varrho(x_k) - z\big) > \epsilon$. Write out $\varrho(x_k) - z = (x_k - x_n)_{n\geq 0} + I$ where $(x_k - x_n)_{n\geq 0}$ is Cauchy in T. Write out the definition of $\hat{\upsilon}$ (cf. Equation (5.8)). If $\lim_{n\to\infty}(x_k - x_n) = 0$ then $\hat{\upsilon}(\varrho(x_k) - z) = \infty > \epsilon$. It is done. If $\lim_{n\to\infty}(x_k - x_n) \neq 0$ then by Lemma 5.23 and the definition of $\hat{\upsilon}$ there exist $n_k \in \mathbb{N}_0$ and $K_k \in \mathbb{Z}$ such that $\hat{\upsilon}(\varrho(x_k) - z) = \upsilon(x_k - x_n) = K_k$ whenever $n \geq n_k$. Choose $n \geq B := \max(n_k, n_0)$. Then $\hat{\upsilon}(\varrho(x_k) - z) = \upsilon(x_k - x_n) > \epsilon$ whenever $k \geq n_0$. It has been proved that $(\hat{T}, \hat{\upsilon})$ is a completion of (T, υ) according to Definition 5.26.

The last thing to prove is the uniqueness of $(\hat{T}, \hat{\upsilon})$. Suppose that $(\tilde{T}, \tilde{\upsilon})$ is another completion of (T, υ). Denote

(i) the limit in $(\hat{T}, \hat{\upsilon})$ by $\hat{\upsilon}$- lim,

(ii) the limit in $(\tilde{T}, \tilde{\upsilon})$ by $\tilde{\upsilon}$- lim,

(iii) the limit in (T, υ) by υ- lim.

Let $z_1 = (x_n)_{n\geq 0} + I \in \hat{T}$ and $z_2 = (y_n)_{n\geq 0} + I \in \hat{T}$. Define the mapping $f : \hat{T} \to \tilde{T}$ by the following way

$$f(z_1) := \tilde{\upsilon}\text{-}\lim_{n\to\infty} x_n \ .$$

Recall that f is defined correctly because $(x_n)_{n\geq 0}$ is Cauchy in T thus it is Cauchy in $\tilde{T}$ which is complete. Thus $(x_n)_{n\geq 0}$ converges in $\tilde{T}$. Show that f is a field homomorphism by using Lemma 5.24.

$$f(z_1 + z_2) = \tilde{\upsilon}\text{-}\lim_{n\to\infty}(x_n + y_n) = \tilde{\upsilon}\text{-}\lim_{n\to\infty} x_n + \tilde{\upsilon}\text{-}\lim_{n\to\infty} y_n = f(z_1) + f(z_2);$$

$$f(z_1 \cdot z_2) = \tilde{\upsilon}\text{-}\lim_{n\to\infty}(x_n \cdot y_n) = \tilde{\upsilon}\text{-}\lim_{n\to\infty} x_n \cdot \tilde{\upsilon}\text{-}\lim_{n\to\infty} y_n = f(z_1) \cdot f(z_2).$$

The mapping f is a T-homomorphism because

$$f(t) = \tilde{\upsilon}\text{-}\lim_{n\to\infty} t = t \text{ whenever } t \in T.$$

The mapping f is an injective homomorphism because

$$f(z_1) = 0 \Leftrightarrow \tilde{\upsilon}\text{-}\lim_{n\to\infty} x_n = 0.$$

The elements $0, x_n \in T$ for all $n \in \mathbb{N}_0$ and $\upsilon = \tilde{\upsilon} \restriction_T = \hat{\upsilon} \restriction_T$. Then

$$\tilde{\upsilon}\text{-}\lim_{n\to\infty} x_n = 0 \Leftrightarrow \upsilon\text{-}\lim_{n\to\infty} x_n = 0 \Leftrightarrow z_1 = (x_n)_{n\geq 0} + I = I \Leftrightarrow z_1 = 0 \in \hat{T}.$$

Let $\tilde{z} \in \tilde{T}$. There exists a sequence $(u_n)_{n\geq 0}$ in T such that $\tilde{\upsilon}\text{-}\lim_{n\to\infty} u_n = \tilde{z}$. It follows from that T is dense in $\tilde{T}$. Recall that $(u_n)_{n\geq 0}$ converges in $\tilde{T}$ thus it is Cauchy in $\tilde{T}$. But $(u_n)_{n\geq 0}$ is a sequence in T and $\upsilon = \tilde{\upsilon} \restriction_T$. Thus $(u_n)_{n\geq 0}$ is Cauchy in T. Define $z := (u_n)_{n\geq 0} + I \in \hat{T}$. Then

$$f(z) = \tilde{\upsilon}\text{-}\lim_{n\to\infty} u_n = \tilde{z}.$$

Thus f is a surjective homomorphism. In all, f is a T-isomorphism.

Prove that $\hat{\upsilon} = \tilde{\upsilon} \circ f$. Let $\hat{t} = (x_n)_{n\geq 0} + I \in \hat{T}$. If $\hat{t} = 0$ then $f(\hat{t}) = 0$ because f is a T-isomorphism. Thus $\tilde{\upsilon}\big(f(\hat{t})\big) = \tilde{\upsilon}(0) = \infty = \hat{\upsilon}(\hat{t})$. If $\hat{t} \neq 0$ then $\upsilon\text{-}\lim_{n\to\infty} x_n \neq 0$. Thus by Lemma 5.23 there exist $n_1 \in \mathbb{N}_0$ and $\hat{K} \in \mathbb{Z}$ such that $\hat{\upsilon}(\hat{t}) = \upsilon(x_n) = \hat{K}$ whenever $n \geq n_1$. Define $\tilde{t} := f(\hat{t}) \neq 0$ and $\tilde{K} := \tilde{\upsilon}(\tilde{t}) \neq \infty$. It follows from $\hat{t} \neq 0$ and f is a T-isomorphism. Prove that $\tilde{K} = \hat{K}$ by a contradiction. Suppose that $\tilde{K} \neq \hat{K}$. Define $M := \max(\tilde{K}, \hat{K})$ and $m := \min(\tilde{K}, \hat{K})$. Because $\tilde{\upsilon} \restriction_T = \upsilon$ then $\tilde{\upsilon}(x_n) = \upsilon(x_n) = \hat{K} = \hat{\upsilon}(\hat{t})$ whenever $n \geq n_1$. By the definition of f $\tilde{\upsilon}\text{-}\lim_{n\to\infty} x_n = \tilde{t}$. Choose $\epsilon > M$. By Definition 5.17 there exist $n_2 \in \mathbb{N}_0$ such that $\tilde{\upsilon}(x_n - \tilde{t}) > \epsilon$ whenever $n \geq n_2$. Define $n_0 := \max(n_1, n_2)$. Then

$$M < \epsilon < \tilde{\upsilon}(x_n - \tilde{t}) = \min\big(\tilde{\upsilon}(x_n), \tilde{\upsilon}(\tilde{t})\big) = \min(\hat{K}, \tilde{K}) = m \text{ whenever } n \geq n_0.$$

It has been proved that $\hat{\upsilon} = \tilde{\upsilon} \circ f$. □

Lemma 5.29. *Keep the assumption of Theorem 5.28. The valued field $(\hat{T}, \hat{\upsilon})$ is as in the proof of Theorem 5.28. Recall that $\hat{T} = R/I$. Let $(x_n)_{n\geq 0} + I \in \hat{T}$ where $(x_n)_{n\geq 0}$ is Cauchy in T. A subsequence $(x_{n_k})_{k\geq 0}$ is a representative of the residual class $(x_n)_{n\geq 0} + I$. It means*

$$\begin{aligned}(x_{n_k})_{k\geq 0} + I = (x_n)_{n\geq 0} + I &\Leftrightarrow (x_{n_k} - x_k)_{k\geq 0} + I = I \\ \Leftrightarrow (x_{n_k} - x_k)_{k\geq 0} \in I &\Leftrightarrow \lim_{k\to\infty} (x_{n_k} - x_k) = 0.\end{aligned}$$

Proof. A sequence $(x_n)_{n\geq 0}$ is Cauchy. Choose an arbitrary $\epsilon \in \mathbb{R}$. By Definition 5.17 there exists $n_0 \in \mathbb{N}_0$ such that $\upsilon(x_n - x_m) > \epsilon$ whenever $m, n \geq n_0$. The sequence $(n_k)_{k\geq 0}$ is strictly increasing by the definition of a subsequence. Thus $n_k \geq k$ for every $k \in \mathbb{N}_0$. Then $\upsilon(x_{n_k} - x_k) > \epsilon$ whenever $k \geq n_0$. Thus $\lim_{k\to\infty} x_{n_k} - x_k = 0$. □

5.3 The P-adic completion

Section 5.2 provides the completion of a valued field. The completion of (F, υ_P) where F/K is an algebraic function field and $P \in \mathbb{P}_F^{(1)}$ provides the representation of every $z \in F$ as a power series. It is similar to the Laurent series from complex analysis. The definition of the residue in this section is same as in complex analysis. At the end, this theory gives a relationship between the Weil differential and the diferential. In other word, there exists an isomorphism of Δ_F onto Ω_F. The section is based upon [1, Chapter 4.2] and [1, Chapter 4.3]. The numbering in brackets refers to [1].

Definition 5.30 (Definition 4.2.5)**.** Let P be a place of an algebraic function field F/K. The completion of a valued field (F, υ_P) is called the P-adic completion of F. Denote the completion by $(\hat{F}_P, \upsilon_P)$. The denotation $\hat{F}_P$ is almost used only. The valuation υ_P is known from a context.

Convention 5.31. *Throughout this section consider that F/K is an algebraic function field, $P \in \mathbb{P}_F^{(1)}$ and $t \in F$ is a uniformizing element of P.*

Theorem 5.32 (Theorem 4.2.6)**.** *Every $z \in \hat{F}_P$ can be uniquely represented in the form*

$$z = \sum_{i=n}^{\infty} a_i \cdot t^i \text{ where } n \in \mathbb{Z} \text{ and } a_i \in K.$$

This representation is called the P-adic power series expansion of z with respect t.

If $(c_i)_{i \geq n}$ is a sequence in K and $n \in \mathbb{Z}$ then $\sum_{i=n}^{\infty} c_i \cdot t^i$ converges and $\upsilon_P\left(\sum_{i=n}^{\infty} c_i \cdot t^i\right) = \min\left(i; c_i \neq 0\right)$.

Theorem 5.33 (Proposition 4.2.7)**.** *If $z = \sum_{i=n}^{\infty} a_i \cdot t^i$ is the P-adic power series expansion of $z \in F$ with respect t then*

$$\delta_t(z) = \frac{dz}{dt} = \sum_{i=n}^{\infty} a_i \cdot i \cdot t^{i-1}.$$

Definition 5.34 (Definition 4.2.8)**.** Suppose that $z = \sum_{i=n}^{\infty} a_i \cdot t^i$ is the P-adic power series expansion of $z \in F$ with respect t. Define the residue of z with respect P and t by the K-linear map

$$\begin{aligned} &\mathrm{res}_{P,t} : F \to K; \\ &\mathrm{res}_{P,t}(z) = a_{-1}. \end{aligned}$$

Remark 5.35. If $\upsilon_P(z) \geq 0$ then $\mathrm{res}_{P,t}(z) = 0$.

Theorem 5.36. *Let $t, s \in F$ be uniformizing elements of P. Let $z \in F$. Then*

$$\mathrm{res}_{P,s}(z) = \mathrm{res}_{P,t}\left(z \cdot \frac{ds}{dt}\right).$$

Definition 5.37 (Definition 4.2.10)**.** Let $\omega \in \Delta_F$. Express $\omega = u\,dt$ where $u \in F$. Then define the residue of ω at P by $\mathrm{res}_P(\omega) := \mathrm{res}_{P,t}(u)$.

Remark 5.38. Definition 5.37 is correct. Let $s \in F$ be another uniformizing element of P. Then $\omega = u\,dt = \left(u \cdot \frac{dt}{ds}\right) ds$. By Theorem 5.36 $\mathrm{res}_{P,t}(u) = \mathrm{res}_{P,s}\left(\left(u \cdot \frac{dt}{ds}\right)\right)$. Thus the definition of $\mathrm{res}_P(\omega)$ is independent on the choice of a uniformizing element.

Definition 5.39 (Definition 4.3.1)**.** Define the mapping

$$\begin{aligned} &\delta : F \to \Omega_F; \\ &\delta(x) = \begin{cases} \mathrm{Cotr}_{F/K(x)}(\eta) & \text{if } x \text{ a separating element of } F/K(x); \\ 0 & \text{if } x \text{ a non-separating element of } F/K(x) \end{cases} \end{aligned}$$

where $\eta \in \Omega_{K(x)}$ is as in Theorem 1.78. Call $\delta(x)$ the Weil differential associated with x.

Theorem 5.40 (Theorem 4.3.2)**.** *Let x be a separating element of F/K.*

(i) The mapping $\delta : F \to \Omega_F$ is a derivation of F/K.

(ii) Let $y \in F$. Then
$$\delta(y) = \frac{dy}{dx} \cdot \delta(x).$$

(iii) The mapping
$$\mu : \Delta_F \to \Omega_F;$$
$$\mu(z\,dx) = z \cdot \delta(x)$$
is an isomorphism of Δ_F onto Ω_F. This isomorphism satisfies $\mu \circ d = \delta$ where the derivation d is as in Definition 5.11 and the derivation δ is as in Definition 5.39.

(iv) Let $P \in \mathbb{P}_F^{(1)}$. Let $\omega = z \cdot \delta(x) \in \Omega_F$ where $z \in F$. Then for every $u \in F$
$$\omega_P(u) = \big(z \cdot \delta(x)\big)_P(u) = \mathrm{res}_P\,\big((uz)\,dx\big).$$

(v) If $\omega = z \cdot \delta(t)$ where t is a uniformizing element of a place $P \in \mathbb{P}_F^{(1)}$ then $W_P = \upsilon_P(z)$ where $(\omega) = W$.

5.4 How to calculate the residue

In this section the residue, the P-adic power series expansion and the relationship between Ω_F and Δ_F are used to develop a tool computing ω_P where ω is as in Theorem 3.4 and $e\big(P \mid P \cap K(x)\big) = 1$.

Firstly, define the derivation of higher order for an arbitrary algebraic function field. Show how to use it for a calculation of a residue. Next, consider an elliptic function field E/K. The main results of this section are Algorithms 5.60 and 5.63 and Theorem 5.66. They provides the calculation of $\mathrm{res}_{P,x-\alpha}$ and ω_P where $P = P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ such that $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 1$.

Convention 5.41. *Throughout this section consider that F/K is an algebraic function field, $P \in \mathbb{P}_F^{(1)}$ and $t \in F$ is a uniformizing element of P.*

Definition 5.42. Let $\epsilon \in \mathrm{Der}_F$. Define the derivation of order $n \in \mathbb{N}$ $\epsilon^{(n)} := \underbrace{\epsilon \circ \ldots \circ \epsilon}_{n-times}$.

Lemma 5.43. *If $z = \sum_{i=n}^{\infty} a_i \cdot t^i$ is the P-adic power series expansion of $z \in F$ with respect t and $m \in \mathbb{N}$ then*
$$\big(\delta_t(z)\big)^{(m)} = \sum_{i=n}^{\infty} \Big(\prod_{j=0}^{m-1} (i-j)\Big) \cdot a_i \cdot t^{i-m};$$
$$\sum_{i=0}^{m-1} \Big(\prod_{j=0}^{m-1} (i-j)\Big) \cdot a_i \cdot t^{i-m} = 0.$$

Proof. Prove the first statement by an induction.

(i) Let $m = 1$. It is proved by Theorem 5.33.

(ii) Let the statement be satisfied for $m \in \mathbb{N}$. Prove it for $m + 1$. By Theorem 5.33

$$\big(\delta_t(z)\big)^{(m+1)}(z) = \delta_t \circ \big(\delta_t(z)\big)^{(m)}(z) = \delta_t\Big(\sum_{i=n}^{\infty}\Big(\prod_{j=0}^{m-1}(i-j)\Big)\cdot a_i \cdot t^{i-m}\Big)$$
$$= \sum_{i=n}^{\infty}\Big(\prod_{j=0}^{m}(i-j)\Big)\cdot a_i \cdot t^{i-m-1}.$$

If $i = j$ then $\prod_{j=0}^{m-1}(i-j) = 0$. Thus the other statement is proved. ☐

Lemma 5.44. *Let $z \in F$ be such that $\upsilon_P(z) \geq 0$. Then $\upsilon_P\Big(\delta_t^{(m)}(z)\Big) \geq 0$ for every $m \in \mathbb{N}$.*

Proof. Let $z = \sum_{i=0}^{\infty} a_i \cdot t^i$ be the P-adic power series expansion of $z \in F$ with respect t. By Lemma 5.43

$$\delta_t^{(m)}(z) = \sum_{i=0}^{\infty}\Big(\prod_{j=0}^{m-1}(i-j)\Big)\cdot a_i \cdot t^{i-m} = \sum_{i=m}^{\infty}\Big(\prod_{j=0}^{m-1}(i-j)\Big)\cdot a_i \cdot t^{i-m}.$$

By Theorem 5.32 $\upsilon_P\big(\delta_t^{(m)}(z)\big) \geq 0$. ☐

Lemma 5.45. *Let F/K be determined by an irreducible polynomial $f(x,y) = 0$. Let $P = P_{(\alpha,\beta)} \in \mathbb{P}_F^{(1)}$. Let $z \in F$ be such that $\upsilon_{(\alpha,\beta)}(z) \geq 0$. Expand z to the P-adic power series expansion with respect t $z = \sum_{i=0}^{\infty} a_i \cdot t^i$. Consider $z(x,y)$ as a function $z : K \times K \to K$. Then $z(\alpha,\beta) = a_0$.*

Proof. Define $\tilde{z} := \sum_{i=1}^{\infty} a_i \cdot t^i$. By Theorem 5.32 $\upsilon_{(\alpha,\beta)}(\tilde{z}) > 0$. By Theorem 1.103 $\tilde{z}(\alpha,\beta) = 0$. Thus $z(\alpha,\beta) = a_0 + \tilde{z}(\alpha,\beta) = a_0$. ☐

Theorem 5.46. *Let F/K be determined by an irreducible polynomial $f(x,y) = 0$. Let $P = P_{(\alpha,\beta)} \in \mathbb{P}_F^{(1)}$. Let $z \in F$ be such that $n := \upsilon_{(\alpha,\beta)}(z) < 0$. Let $m \leq \upsilon_{(\alpha,\beta)}(z) < 0$. Consider every $u(x,y) \in F$ as a function $u : K \times K \to K$. If* $\operatorname{char} K = 0$ *or* $\operatorname{char} K \geq -m$ *then*

$$\operatorname{res}_{P,t}(z) = \frac{\Big(\delta_t^{(-m-1)}\big(z \cdot t^{-m}\big)\Big)(\alpha,\beta)}{(-m-1)!}.$$

Proof. If $z = \sum_{i=n}^{\infty} a_i \cdot t^i$ is the P-adic power series expansion of $z \in F$ with respect t then

$$z \cdot t^{-m} = \sum_{i=n}^{\infty} a_i \cdot t^{i-m} = \sum_{i=n-m}^{\infty} a_{i+m} \cdot t^i.$$

By Lemma 5.43

$$\delta_t^{(-m-1)}(z \cdot t^{-m}) = \sum_{i=n-m}^{\infty}\Big(\prod_{j=0}^{-m-2}(i-j)\Big)\cdot a_{i+m} \cdot t^{i+m+1}.$$

Recall that $n-m \geq 0$. By Lemma 5.43 $\sum_{i=n-m}^{-m-2} \Big(\prod_{j=0}^{-m-2} (i-j) \Big) \cdot a_{i+m} \cdot t^{i+m+1} = 0$. Put

$$v := \delta_t^{(-m-1)}(z \cdot t^{-m}) = \sum_{i=-m-1}^{\infty} \Big(\prod_{j=0}^{-m-2} (i-j) \Big) \cdot a_{i+m} \cdot t^{i+m+1}$$
$$= \sum_{i=0}^{\infty} \Big(\prod_{j=0}^{-m-2} (i-m-1-j) \Big) \cdot a_{i-1} \cdot t^{i}.$$

The valuation $\upsilon_{(\alpha,\beta)}\big(z \cdot t^{-m}\big) = \upsilon_{(\alpha,\beta)}(z) - m \geq 0$. Thus $\upsilon_{(\alpha,\beta)}\big(\delta_t^{(-m-1)}(z \cdot t^{-m})\big) \geq 0$ by Lemma 5.44. Lemma 5.45 can be used

$$v(\alpha, \beta) = \Big(\prod_{j=0}^{-m-2} (-m-1-j) \Big) \cdot a_{-1} = (-m-1)! \cdot a_{-1}.$$

Thus

$$\mathrm{res}_{P,t}(z) = \frac{\Big(\delta_t^{(-m-1)}\big(z \cdot t^{-m}\big)\Big)(\alpha, \beta)}{(-m-1)!}.$$

□

Remark 5.47. In Theorem 5.46 and Lemma 5.45 there is an evaluation of an element $u(x, y) \in F$ by (α, β). Because $\upsilon_{(\alpha,\beta)}(u) \geq 0$, there exists a representation $u(x, y) = \frac{a(x,y)}{b(x,y)}$ such that $b(\alpha, \beta) \neq 0$. Recall $a(x, y), b(x, y) \in K[x, y]$.

Convention 5.48. *Throughout this section consider an elliptic function field E/K determined by a short smooth Weierstrass equation $w(x, y) = 0$.*

Remark 5.49. (i) Let $Q \in \mathbb{P}_{K(x)}$ and $P \in \mathbb{P}_E$ such that $P \mid Q$. Recall that $e(P \mid Q) = 1$ or 2.

(ii) Section 4.2 describes how to compute ω_P if $e(P \mid Q) = 2$. Recall that ω is as in Theorem 3.4.

(iii) Focus to develop an algorithm computing ω_P if $P = P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ and $e(P \mid Q) = 1$. In the context of Definition 2.15, this places are of *Type 1*, *Type 4*, *Type 5* and *Type 6*. Start with help algorithms.

Lemma 5.50. *Let $P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ and $P_\alpha \in \mathbb{P}_{K(x)}^{(1)}$ such that $e\big(P_{(\alpha,\beta)} \mid P_\alpha\big) = 1$. Then $x - \alpha$ is a uniformizing element of $P_{(\alpha,\beta)}$.*

Proof. The valuation $\upsilon_{(\alpha,\beta)}(x - \alpha) = e\big(P_{(\alpha,\beta)} \mid P_\alpha\big) \cdot \upsilon_\alpha(x - \alpha) = 1$. □

Lemma 5.51. *If $\alpha \in K$ and $\delta_x, \delta_{x-\alpha} \in \mathrm{Der}_E$ then $\delta_x = \delta_{x-\alpha}$.*

Proof. By Lemma 5.9 $\delta_{x-\alpha} = \delta_{x-\alpha}(x) \cdot \delta_x$. Derive $\delta_{x-\alpha}(x) = \delta_{x-\alpha}(x - \alpha) + \delta_{x-\alpha}(\alpha) = 1 + 0 = 1$. □

Corollary 5.52. *If $m \in \mathbb{N}$ then $\delta_x^{(m)} = \delta_{x-\alpha}^{(m)}$.*

Algorithm 5.53. ***The inputs*** *of the algorithm are* *(i)* $P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ *such that* $e(P_{(\alpha,\beta)} \mid P_\alpha) = 1$;

(ii) $u(x,y) = \frac{a(x)+y\cdot b(x)}{c(x)} \in E$ *where* $a(x), b(x), c(x) \in K[x]$ *(cf. Theorem 1.116); The element* $u \in E$ *satisfies* $\upsilon_{(\alpha,\beta)}(u) \geq 0$.

The output *of the algorithm is the evaluation* $u(\alpha,\beta) \in K$.
The algorithm:

1. *If* $c(\alpha) \neq 0$ *then evaluate* $\frac{a(\alpha)+\beta\cdot b(\alpha)}{c(\alpha)}$ *and finish the algorithm.*

2. *Assume that* $c(\alpha) = 0$. *Decompose* $a(x) = (x-\alpha)^A \cdot \tilde{a}(x)$, $b(x) = (x-\alpha)^B \cdot \tilde{b}(x)$, $c(x) = (x-\alpha)^C \cdot \tilde{c}(x)$ *such that* $\tilde{a}(\alpha) \neq 0$, $\tilde{b}(\alpha) \neq 0$, $\tilde{c}(\alpha) \neq 0$.

3. *Define* $M := \min(A,B)$. *If* $M \geq C$ *then evaluate*

$$u(\alpha,\beta) = \frac{(\alpha-\alpha)^{M-C} \cdot \big(\tilde{a}(\alpha)\cdot(\alpha-\alpha)^{A-M} + \beta\cdot(\alpha-\alpha)^{B-M}\cdot\tilde{a}(\alpha)\big)}{\tilde{c}(\alpha)}$$

   and finish the algorithm.

4. *Let* $\beta \neq 0$.

   (i) Let $w(x,y) = y^2 - f(x)$ *and* $\operatorname{char} K \neq 2$. *Express*

$$\frac{\tilde{a}^2(x) - f(x)\cdot\tilde{b}^2(x)}{\tilde{c}(x)\cdot(x-\alpha)^{C-M}} = (x-\alpha)^i \cdot \frac{p(x)}{q(x)}$$

   such that $p(\alpha) \neq 0$ *and* $q(\alpha) \neq 0$. *If* $i \geq 1$ *then* $u(\alpha,\beta) = 0$. *If* $i = 0$ *then evaluate*

$$u(\alpha,\beta) = \frac{p(\alpha)}{q(\alpha)} \cdot \frac{1}{\tilde{a}(\alpha) - \beta\cdot\tilde{b}(\alpha)}$$

   and finish the algorithm.

   (ii) Let $w(x,y) = y^2 + x\cdot y - f_1(x)$ *and* $\operatorname{char} K = 2$. *Express*

$$\frac{\tilde{a}^2(x) + x\cdot\tilde{a}(x)\cdot\tilde{b}(x) + f_1(x)\cdot\tilde{b}^2(x)}{\tilde{c}(x)\cdot(x-\alpha)^{C-M}} = (x-\alpha)^i \cdot \frac{p(x)}{q(x)}$$

   such that $p(\alpha) \neq 0$ *and* $q(\alpha) \neq 0$. *If* $i \geq 1$ *then* $u(\alpha,\beta) = 0$. *If* $i = 0$ *then evaluate*

$$u(\alpha,\beta) = \frac{p(\alpha)}{q(\alpha)} \cdot \frac{1}{\alpha\cdot\tilde{b}(\alpha)}$$

   and finish the algorithm.

   (iii) Let $w(x,y) = y^2 + a_3\cdot y - f_2(x)$ *and* $\operatorname{char} K = 2$. *Express*

$$\frac{\tilde{a}^2(x) + a_3\cdot\tilde{a}(x)\cdot\tilde{b}(x) + f_2(x)\cdot\tilde{b}^2(x)}{\tilde{c}(x)\cdot(x-\alpha)^{C-M}} = (x-\alpha)^i \cdot \frac{p(x)}{q(x)}$$

   such that $p(\alpha) \neq 0$ *and* $q(\alpha) \neq 0$. *If* $i \geq 1$ *then* $u(\alpha,\beta) = 0$. *If* $i = 0$ *then evaluate*

$$u(\alpha,\beta) = \frac{p(\alpha)}{q(\alpha)} \cdot \frac{1}{a_3\cdot\tilde{b}(\alpha)}$$

   and finish the algorithm.

5. If $\beta = 0$ then $\operatorname{char} K = 2$*.*

(ii) Let $w(x,y) = y^2 + x \cdot y - f_1(x)$. Express

$$\frac{\tilde{a}^2(x) \cdot (x-\alpha)^{2A-2M} + x \cdot \tilde{a}(x) \cdot (x-\alpha)^{A-M} \cdot \tilde{b}(x) + f_1(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} = (x-\alpha)^i \cdot \frac{p(x)}{q(x)}$$

such that $p(\alpha) \neq 0$ and $q(\alpha) \neq 0$. If $i \geq 1$ then $u(\alpha,\beta) = 0$. If $i = 0$ then evaluate

$$u(\alpha,\beta) = \frac{p(\alpha)}{q(\alpha)} \cdot \frac{1}{\alpha \cdot \tilde{b}(\alpha)}$$

and finish the algorithm.

(iii) Let $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$. Express

$$\frac{\tilde{a}^2(x) \cdot (x-\alpha)^{2A-2M} + a_3 \cdot \tilde{a}(x) \cdot (x-\alpha)^{A-M} \cdot \tilde{b}(x) + f_2(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} = (x-\alpha)^i \cdot \frac{p(x)}{q(x)}$$

such that $p(\alpha) \neq 0$ and $q(\alpha) \neq 0$. If $i \geq 1$ then $u(\alpha,\beta) = 0$. If $i = 0$ then evaluate

$$u(\alpha,\beta) = \frac{p(\alpha)}{q(\alpha)} \cdot \frac{1}{a_3 \cdot \tilde{b}(\alpha)}$$

and finish the algorithm.

Remark 5.54. Keep the assumption from Algorithm 5.53. Before proving that Algorithm 5.53 is correct there is a description of the main thought of the proof and an explanation why $e(P_{(\alpha,\beta)} \mid P_\alpha)$ has to be 1.

A critical situation is if the numerator and the denominator are evaluated to zero. Recall that if the numerator is evaluated to non-zero and the denominator is evaluated to zero then $v_{(\alpha,\beta)}(u) < 0$. It is a contradiction with the assumption. First, try to shorten

$$u = \frac{(x-\alpha)^M \cdot \left(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-M}\right)}{\tilde{c}(x) \cdot (x-\alpha)^C}.$$

If $M < C$ then the denominator is evaluated to zero. Thus the numerator has to be evaluated to zero too. Thus it is still the situation $\frac{0}{0}$. Our goal is to transfer the problem to another problem in $K(x)$ that is known. Extend the fraction by

$$p(x,y) := \delta_2\left(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-M}\right)$$

where δ_2 is the non-identical Galois mapping (cf. Lemma 2.3). Prove that $p(\alpha,\beta) \neq 0$. If $e(P_{(\alpha,\beta)} \mid P_\alpha) = 2$ then $p(\alpha,\beta) = 0$. Thus the algorithm would not be correct. Recall that $N_{E/K(x)} = \delta_1 \cdot \delta_2 = \mathrm{id} \cdot \delta_2$. Thus $u(x,y) = v(x) \cdot \frac{1}{p(x,y)}$ where $v(x) \in K(x)$. A fraction in $K(x)$ can be shortened easily.

Theorem 5.55. *Algorithm 5.53 is correct.*

Proof. Keep the assumption from Algorithm 5.53. Because $\upsilon_{(\alpha,\beta)}(u) \geq 0$, $u(\alpha,\beta)$ is defined. If $c(\alpha) \neq 0$ then the evaluation is without any problem. Assume that $c(\alpha) = 0$. We have

$$\begin{aligned}\frac{a(x) + y \cdot b(x)}{c(x)} &= \frac{\tilde{a}(x) \cdot (x-\alpha)^A + y \cdot \tilde{b}(x) \cdot (x-\alpha)^B}{\tilde{c}(x) \cdot (x-\alpha)^C} \\ &= \frac{(x-\alpha)^M \cdot \big(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-M}\big)}{\tilde{c}(x) \cdot (x-\alpha)^C}.\end{aligned}$$

If $M \geq C$ then the fraction can be shortened and it can be evaluated without any problem.

Assume that $M < C$ and $\beta \neq 0$. Then

$$u(x,y) = \frac{\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-M}}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}.$$

If $A > B$ then the numerator is $\tilde{a}(x) \cdot (x-\alpha)^{A-B} + y \cdot \tilde{b}(x)$. Evaluate $\tilde{a}(\alpha) \cdot (\alpha-\alpha)^{A-B} + \beta \cdot \tilde{b}(\alpha) = \beta \cdot \tilde{b}(\alpha) \neq 0$. Thus $\upsilon_{(\alpha,\beta)}\big(\tilde{a}(x) \cdot (x-\alpha)^{A-B} + y \cdot \tilde{b}(x)\big) = 0$. If $A < B$ then then the numerator $\tilde{a}(x) + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-A}$. Evaluate $\tilde{a}(\alpha) + \beta \cdot \tilde{b}(\alpha) \cdot (\alpha-\alpha)^{B-A} = \tilde{a}(\alpha) \neq 0$. Thus $\upsilon_{(\alpha,\beta)}\big(\tilde{a}(x) + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-A}\big) = 0$. The valuation $\upsilon_{(\alpha,\beta)}\big(\tilde{c}(x) \cdot (x-\alpha)^{C-M}\big) = \upsilon_{(\alpha,\beta)}\big(\tilde{c}(x)\big) + \upsilon_{(\alpha,\beta)}\big((x-\alpha)^{C-M}\big) = 0 + e(P_{(\alpha,\beta)} \mid P_\alpha) \cdot (C-M) = C - M > 0$. Thus if $A \neq B$ then

$$\upsilon_{(\alpha,\beta)}\Big(\frac{\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-M}}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}\Big) = 0 + M - C < 0.$$

It is a contradiction. Thus $A = B = M$ and

$$u(x,y) = \frac{\tilde{a}(x) + y \cdot \tilde{b}(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}.$$

If $\tilde{a}(\alpha) + \beta \cdot \tilde{b}(\alpha) \neq 0$ then $\upsilon_{(\alpha,\beta)}\big(\tilde{a}(x) + y \cdot \tilde{b}(x)\big) = 0$. Thus

$$\upsilon_{(\alpha,\beta)}\Big(\frac{\tilde{a}(x) + y \cdot \tilde{b}(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}\Big) < 0.$$

It is a contradiction. Thus $\tilde{a}(\alpha) + \beta \cdot \tilde{b}(\alpha) = 0$. Lemma 2.3 describes the Galois mapping $\delta_2 : E \to E$ such that $\delta_2 \neq \mathrm{id}$. Expand

$$u(x,y) = \frac{\tilde{a}(x) + y \cdot \tilde{b}(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} \cdot \frac{\delta_2\big(\tilde{a}(x) + y \cdot \tilde{b}(x)\big)}{\delta_2\big(\tilde{a}(x) + y \cdot \tilde{b}(x)\big)}.$$

(i) Let $w(x,y) = y^2 - f(x)$ and $\operatorname{char} K \neq 2$. The mapping $\delta_2\big(\tilde{a}(x) + y \cdot \tilde{b}(x)\big) = \tilde{a}(x) - y \cdot \tilde{b}(x)$. If $\tilde{a}(\alpha) - \beta \cdot \tilde{b}(\alpha) = 0$ then

$$\begin{aligned}\tilde{a}(\alpha) + \beta \cdot \tilde{b}(\alpha) &= \tilde{a}(\alpha) - \beta \cdot \tilde{b}(\alpha); \\ 2 \cdot \beta \cdot \tilde{b}(\alpha) &= 0.\end{aligned}$$

It is a contradiction. Thus $\tilde{a}(\alpha) - \beta \cdot \tilde{b}(\alpha) \neq 0$ and $\upsilon_{(\alpha,\beta)}\big(\tilde{a}(x) - y \cdot \tilde{b}(x)\big) = 0$. Put

$$r(x,y) := \frac{1}{\tilde{a}(x) - y \cdot \tilde{b}(x)} \in E.$$

The evaluation $r(\alpha,\beta) \in K$ is well-defined and $v_{(\alpha,\beta)}\big(r(x,y)\big) = 0$. We have

$$u(x,y) = \frac{\tilde{a}^2(x) - f(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} \cdot r(x,y).$$

Express

$$\frac{\tilde{a}^2(x) - f(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} = (x-\alpha)^i \cdot \frac{p(x)}{q(x)}$$

such that $p(\alpha) \neq 0$ and $q(\alpha) \neq 0$. We have

$$u(x,y) = \frac{p(x)}{q(x)} \cdot (x-\alpha)^i \cdot r(x,y).$$

(ii) Let $w(x,y) = y^2 + x \cdot y - f_1(x)$ and $\operatorname{char} K = 2$. The mapping $\delta_2\big(\tilde{a}(x) + y \cdot \tilde{b}(x)\big) = \tilde{a}(x) + (y+x) \cdot \tilde{b}(x)$. Evaluate

$$\tilde{a}(\alpha) + (\beta+\alpha) \cdot \tilde{b}(\alpha) = \tilde{a}(\alpha) + \beta \cdot \tilde{b}(\alpha) + \alpha \cdot \tilde{b}(\alpha) = \alpha \cdot \tilde{b}(\alpha).$$

If $\alpha = 0$ then $0 = w(0,\beta) = y^2 + 0 \cdot y - f_1(0) = y^2 - a_6$. Thus $\beta = \sqrt{a_6}$. But $P_{(0,\sqrt{a_6})}$ is of *Type 3* (cf. Definition 2.15). Thus $e\big(P_{(0,\sqrt{a_6})} \mid P_0\big) = 2$ where $P_0 \in \mathbb{P}_{K(x)}$. It is a contradiction. Thus $\alpha \neq 0$ and $\tilde{a}(\alpha) + (\beta+\alpha) \cdot \tilde{b}(\alpha) \neq 0$. Thus $v_{(\alpha,\beta)}\big(\tilde{a}(x) + (y+x) \cdot \tilde{b}(x)\big) = 0$. Put

$$r(x,y) := \frac{1}{\tilde{a}(x) + (y+x) \cdot \tilde{b}(x)} \in E.$$

The evaluation $r(\alpha,\beta) \in K$ is well-defined and $v_{(\alpha,\beta)}\big(r(x,y)\big) = 0$. We have

$$u(x,y) = \frac{\tilde{a}^2(x) + x \cdot \tilde{a}(x) \cdot \tilde{b}(x) + f_1(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} \cdot r(x,y).$$

Express

$$\frac{\tilde{a}^2(x) + x \cdot \tilde{a}(x) \cdot \tilde{b}(x) + f_1(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} = (x-\alpha)^i \cdot \frac{p(x)}{q(x)}$$

such that $p(\alpha) \neq 0$ and $q(\alpha) \neq 0$. We have

$$u(x,y) = \frac{p(x)}{q(x)} \cdot (x-\alpha)^i \cdot r(x,y).$$

(iii) Let $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$ and $\operatorname{char} K = 2$. The mapping $\delta_2\big(\tilde{a}(x) + y \cdot \tilde{b}(x)\big) = \tilde{a}(x) + (y+a_3) \cdot \tilde{b}(x)$. Evaluate

$$\tilde{a}(\alpha) + (\beta+a_3) \cdot \tilde{b}(\alpha) = \tilde{a}(\alpha) + \beta \cdot \tilde{b}(\alpha) + a_3 \cdot \tilde{b}(\alpha) = a_3 \cdot \tilde{b}(\alpha) \neq 0.$$

Thus $v_{(\alpha,\beta)}\big(\tilde{a}(x) + (y+a_3) \cdot \tilde{b}(x)\big) = 0$. Put

$$r(x,y) := \frac{1}{\tilde{a}(x) + (y+a_3) \cdot \tilde{b}(x)} \in E.$$

The evaluation $r(\alpha,\beta) \in K$ is well-defined and $\upsilon_{(\alpha,\beta)}\big(r(x,y)\big) = 0$. We have

$$u(x,y) = \frac{\tilde{a}^2(x) + a_3 \cdot \tilde{a}(x) \cdot \tilde{b}(x) + f_2(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} \cdot r(x,y).$$

Express

$$\frac{\tilde{a}^2(x) + a_3 \cdot \tilde{a}(x) \cdot \tilde{b}(x) + f_2(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} = (x-\alpha)^i \cdot \frac{p(x)}{q(x)}$$

such that $p(\alpha) \neq 0$ and $q(\alpha) \neq 0$. We have

$$u(x,y) = \frac{p(x)}{q(x)} \cdot (x-\alpha)^i \cdot r(x,y).$$

Assume that $M < C$ and $\beta = 0$. If $\beta = 0$ and $e(P_{(\alpha,\beta)} \mid P_\alpha) = 1$ then $\operatorname{char} K$ has to be 2 (cf. Definition 2.15). Then

$$u(x,y) = \frac{\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-M}}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}.$$

If $A = M$ then $\tilde{a}(\alpha) + 0 \cdot \tilde{b}(\alpha) \cdot (\alpha - \alpha)^{B-M} = \tilde{a}(\alpha) \neq 0$. Thus

$$\upsilon_{(\alpha,\beta)}\Big(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-M}\Big) = 0.$$

Recall that $\upsilon_{(\alpha,\beta)}\big(\tilde{c}(x) \cdot (x-\alpha)^{C-M}\big) > 0$. Thus

$$\upsilon_{(\alpha,\beta)}\Big(\frac{\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x) \cdot (x-\alpha)^{B-M}}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}\Big) < 0.$$

It is a contradiction. Thus $A > M$ and $B = M$. Then

$$u(x,y) = \frac{\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}.$$

Lemma 2.3 describes the Galois mapping $\delta_2 : E \to E$ such that $\delta_2 \neq \mathrm{id}$. Expand

$$u(x,y) = \frac{\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} \cdot \frac{\delta_2\big(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x)\big)}{\delta_2\big(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x)\big)}.$$

(ii) Let $w(x,y) = y^2 + x \cdot y - f_1(x)$ and $\operatorname{char} K = 2$. The mapping $\delta_2\big(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x)\big) = \tilde{a}(x) \cdot (x-\alpha)^{A-M} + (y+x) \cdot \tilde{b}(x)$. Evaluate

$$\tilde{a}(\alpha) \cdot (\alpha-\alpha)^{A-M} + (0+\alpha) \cdot \tilde{b}(\alpha) = \alpha \cdot \tilde{b}(\alpha).$$

If $\alpha = 0$ then $w(0,0) = 0$. it means that $w(x,y) = 0$ is a singular curve. It is a contradiction. Thus $\alpha \neq 0$ and $\tilde{a}(\alpha) \cdot (\alpha-\alpha)^{A-M} + (0+\alpha) \cdot \tilde{b}(\alpha) \neq 0$. Thus $\upsilon_{(\alpha,\beta)}\big(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + (y+x) \cdot \tilde{b}(x)\big) = 0$. Put

$$r(x,y) := \frac{1}{\tilde{a}(x) \cdot (x-\alpha)^{A-M} + (y+x) \cdot \tilde{b}(x)}.$$

The evaluation $r(\alpha, \beta)$ is well-defined and $\upsilon_{(\alpha,\beta)}\big(r(x, y)\big) = 0$. We have

$$\begin{aligned}&u(x, y)\\=&\frac{\tilde{a}^2(x) \cdot (x-\alpha)^{2A-2M} + x \cdot \tilde{a}(x) \cdot (x-\alpha)^{A-M} \cdot \tilde{b}(x) + f_1(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} \cdot r(x, y).\end{aligned}$$

Express

$$\begin{aligned}&\frac{\tilde{a}^2(x) \cdot (x-\alpha)^{2A-2M} + x \cdot \tilde{a}(x) \cdot (x-\alpha)^{A-M} \cdot \tilde{b}(x) + f_1(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}\\=&(x-\alpha)^i \cdot \frac{p(x)}{q(x)}\end{aligned}$$

such that $p(\alpha) \neq 0$ and $q(\alpha) \neq 0$. We have

$$u(x, y) = \frac{p(x)}{q(x)} \cdot (x-\alpha)^i \cdot r(x, y).$$

(iii) Let $w(x, y) = y^2 + a_3 \cdot y - f_2(x)$ and $\operatorname{char} K = 2$. The mapping $\delta_2\big(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + y \cdot \tilde{b}(x)\big) = \tilde{a}(x) \cdot (x-\alpha)^{A-M} + (y + a_3) \cdot \tilde{b}(x)$. Evaluate

$$\tilde{a}(\alpha) \cdot (\alpha-\alpha)^{A-M} + (0 + a_3) \cdot \tilde{b}(\alpha) = a_3 \cdot \tilde{b}(\alpha) \neq 0.$$

Thus $\upsilon_{(\alpha,\beta)}\big(\tilde{a}(x) \cdot (x-\alpha)^{A-M} + (y + a_3) \cdot \tilde{b}(x)\big) = 0$. Put

$$r(x, y) := \frac{1}{\tilde{a}(x) \cdot (x-\alpha)^{A-M} + (y + a_3) \cdot \tilde{b}(x)}.$$

The evaluation $r(\alpha, \beta)$ is well-defined and $\upsilon_{(\alpha,\beta)}\big(r(x, y)\big) = 0$. We have

$$\begin{aligned}&u(x, y)\\=&\frac{\tilde{a}^2(x) \cdot (x-\alpha)^{2A-2M} + a_3 \cdot \tilde{a}(x) \cdot (x-\alpha)^{A-M} \cdot \tilde{b}(x) + f_2(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}} \cdot r(x, y).\end{aligned}$$

Express

$$\begin{aligned}&\frac{\tilde{a}^2(x) \cdot (x-\alpha)^{2A-2M} + a_3 \cdot \tilde{a}(x) \cdot (x-\alpha)^{A-M} \cdot \tilde{b}(x) + f_2(x) \cdot \tilde{b}^2(x)}{\tilde{c}(x) \cdot (x-\alpha)^{C-M}}\\=&(x-\alpha)^i \cdot \frac{p(x)}{q(x)}\end{aligned}$$

such that $p(\alpha) \neq 0$ and $q(\alpha) \neq 0$. It gives

$$u(x, y) = \frac{p(x)}{q(x)} \cdot (x-\alpha)^i \cdot r(x, y).$$

Calculate

$$\begin{aligned}&0 \leq \upsilon_{(\alpha,\beta)}\big(u(x, y)\big) = \upsilon_{(\alpha,\beta)}\Big(\frac{p(x)}{q(x)} \cdot (x-\alpha)^i \cdot r(x, y)\Big)\\&= \upsilon_{(\alpha,\beta)}\big(p(x)\big) - \upsilon_{(\alpha,\beta)}\big(q(x)\big) + \upsilon_{(\alpha,\beta)}\big(r(x, y)\big) + i \cdot e\big(P_{(\alpha,\beta)} \mid P_\alpha\big) \cdot \upsilon_\alpha\big((x-\alpha)\big)\\&= 0 + 0 + 0 + i = i;\\&0 \leq i.\end{aligned}$$

Thus the evaluation $u(\alpha, \beta) = \frac{p(\alpha)}{q(\alpha)} \cdot (\alpha-\alpha)^i \cdot r(\alpha, \beta)$ is well-defined. □

Algorithm 5.56. ***The input*** *of the algorithm is* $\frac{p(x,y)}{q(x,y)} \in E$ *such that*

$$p(x,y) = \sum_{i\geq 0, j\geq 0} p_{i,j} \cdot x^i \cdot y^j \in K[x,y] \ ; \ q(x,y) = \sum_{i\geq 0, j\geq 0} q_{i,j} \cdot x^i \cdot y^j \in K[x,y].$$

The output *of the algorithm is a representation* $\frac{p(x,y)}{q(x,y)} = \frac{a(x)+y\cdot b(x)}{c(x)}$ *such that* $a(x), b(x), c(x) \in K[x]$.

The algorithm: *Firstly, convert* $p(x,y)$ *(respectively* $q(x,y)$*) to* $p(x,y) = p_1(x) + y \cdot p_2(x)$ *(respectively* $q(x,y) = q_1(x) + y \cdot q_2(x)$*). Repeat the following process until* $\deg_y\big(p(x,y)\big) < 2$.

1. Split

$$p(x,y) = \sum_{i,j \text{ even}} p_{i,j} \cdot x^i \cdot (y^2)^{\frac{j}{2}} + y \cdot \sum_{i,j \text{ odd}} p_{i,j} \cdot x^i \cdot (y^2)^{\frac{j-1}{2}}.$$

2. Change the term y^2 *by*

(i) $y^2 = f(x)$ *if* $w(x,y) = y^2 - f(x)$ *and* $\operatorname{char} K \neq 2$*;*

(ii) $y^2 = f_1(x) - x \cdot y$ *if* $w(x,y) = y^2 + x \cdot y - f_1(x)$ *and* $\operatorname{char} K = 2$*;*

(iii) $y^2 = f_2(x) - a_3 \cdot y$ *if* $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$ *and* $\operatorname{char} K = 2$.

Consider that $p(x,y) = p_1(x) + y \cdot p_2(x)$ *and* $q(x,y) = q_1(x) + y \cdot q_2(x)$*. If* $w(x,y) = y^2 - f(x)$ *and* $\operatorname{char} K \neq 2$ *then*

- $a(x) := p_1(x) \cdot q_1(x) - f(x) \cdot p_2(x) \cdot q_2(x);$
- $b(x) := p_2(x) \cdot q_1(x) - p_1(x) \cdot q_2(x);$
- $c(x) := q_1^2(x) - f(x) \cdot q_2^2(x).$

If $w(x,y) = y^2 + x \cdot y - f_1(x)$ *and* $\operatorname{char} K = 2$ *then*

- $a(x) := p_1(x) \cdot q_1(x) + x \cdot p_1(x) \cdot q_2(x) + f_1(x) \cdot p_2(x) \cdot q_2(x);$
- $b(x) := p_2(x) \cdot q_1(x) + p_1(x) \cdot q_2(x);$
- $c(x) := q_1^2(x) + x \cdot q_1(x) \cdot q_2(x) + f_1(x) \cdot q_2^2(x).$

If $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$ *and* $\operatorname{char} K = 2$ *then*

- $a(x) := p_1(x) \cdot q_1(x) + a_3 \cdot p_1(x) \cdot q_2(x) + f_2(x) \cdot p_2(x) \cdot q_2(x);$
- $b(x) := p_2(x) \cdot q_1(x) + p_1(x) \cdot q_2(x);$
- $c(x) := q_1^2(x) + a_3 \cdot q_1(x) \cdot q_2(x) + f_2(x) \cdot q_2^2(x).$

Theorem 5.57. *Algorithm 5.56 is correct.*

Proof. Keep the assumption from Algorithm 5.56. In the first part, the changing of the term y^2 follows from the equation $w(x,y) = 0$. Put $d_y := \deg_y\big(p(x,y)\big)$. Every step decreases d_y to $\lceil \frac{d_y}{2} \rceil$ at least. The decreasing stops if $d_y \leq 1$. Thus $p(x,y)$ can be represented as $p(x,y) = p_1(x) + y \cdot p_2(x)$.

In the other part, Lemma 2.3 describes the Galois mapping $\delta_2 : E \to E$ such that $\delta_2 \neq \mathrm{id}$. Expand

$$\frac{p_1(x) + y \cdot p_2(x)}{q_1(x) + y \cdot q_2(x)} \cdot \frac{\delta_2\big(q_1(x) + y \cdot q_2(x)\big)}{\delta_2\big(q_1(x) + y \cdot q_2(x)\big)}.$$

(i) If $w(x,y) = y^2 - f(x)$ and $\operatorname{char} K \neq 2$ then $\delta_2\big(q_1(x) + y \cdot q_2(x)\big) = q_1(x) - y \cdot q_2(x)$. Calculate

$$\begin{aligned}
&\frac{p_1(x) + y \cdot p_2(x)}{q_1(x) + y \cdot q_2(x)} \cdot \frac{q_1(x) - y \cdot q_2(x)}{q_1(x) - y \cdot q_2(x)} \\
=&\frac{p_1(x) \cdot q_1(x) - f(x) \cdot p_2(x) \cdot q_2(x) + y \cdot \big(p_2(x) \cdot q_1(x) - p_1(x) \cdot q_2(x)\big)}{q_1^2(x) - f(x) \cdot q_2^2(x)}.
\end{aligned}$$

(ii) If $w(x,y) = y^2 + x \cdot y - f_1(x)$ and $\operatorname{char} K \neq 2$ then $\delta_2\big(q_1(x) + y \cdot q_2(x)\big) = q_1(x) + (y + x) \cdot q_2(x)$. Calculate

$$\begin{aligned}
&\frac{p_1(x) + y \cdot p_2(x)}{q_1(x) + y \cdot q_2(x)} \cdot \frac{q_1(x) + (y+x) \cdot q_2(x)}{q_1(x) + (y+x) \cdot q_2(x)} \\
=&\frac{p_1(x) \cdot q_1(x) + y \cdot \big(p_2(x) \cdot q_1(x)\big)}{q_1^2(x) + y \cdot q_1(x) \cdot q_2(x) + (y+x) \cdot q_1(x) \cdot q_2(x) + f_1(x) \cdot q_2^2(x)} \\
+&\frac{(y+x) \cdot \big(p_1(x) \cdot q_2(x)\big) + f_1(x) \cdot p_2(x) \cdot q_2(x)}{q_1^2(x) + y \cdot q_1(x) \cdot q_2(x) + (y+x) \cdot q_1(x) \cdot q_2(x) + f_1(x) \cdot q_2^2(x)} \\
=&\frac{p_1(x) \cdot q_1(x) + x \cdot p_1(x) \cdot q_2(x) + f_1(x) \cdot p_2(x) \cdot q_2(x)}{q_1^2(x) + x \cdot q_1(x) \cdot q_2(x) + f_1(x) \cdot q_2^2(x)} \\
+&\frac{y \cdot \big(p_2(x) \cdot q_1(x) + p_1(x) \cdot q_2(x)\big)}{q_1^2(x) + x \cdot q_1(x) \cdot q_2(x) + f_1(x) \cdot q_2^2(x)}.
\end{aligned}$$

(iii) If $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$ and $\operatorname{char} K \neq 2$ then $\delta_2\big(q_1(x) + y \cdot q_2(x)\big) = q_1(x) + (y + a_3) \cdot q_2(x)$. Calculate

$$\begin{aligned}
&\frac{p_1(x) + y \cdot p_2(x)}{q_1(x) + y \cdot q_2(x)} \cdot \frac{q_1(x) + (y+a_3) \cdot q_2(x)}{q_1(x) + (y+a_3) \cdot q_2(x)} \\
=&\frac{p_1(x) \cdot q_1(x) + y \cdot \big(p_2(x) \cdot q_1(x)\big)}{q_1^2(x) + y \cdot q_1(x) \cdot q_2(x) + (y+a_3) \cdot q_1(x) \cdot q_2(x) + f_2(x) \cdot q_2^2(x)} \\
+&\frac{(y+a_3) \cdot \big(p_1(x) \cdot q_2(x)\big) + f_2(x) \cdot p_2(x) \cdot q_2(x)}{q_1^2(x) + y \cdot q_1(x) \cdot q_2(x) + (y+a_3) \cdot q_1(x) \cdot q_2(x) + f_2(x) \cdot q_2^2(x)} \\
=&\frac{p_1(x) \cdot q_1(x) + a_3 \cdot p_1(x) \cdot q_2(x) + f_2(x) \cdot p_2(x) \cdot q_2(x)}{q_1^2(x) + a_3 \cdot q_1(x) \cdot q_2(x) + f_2(x) \cdot q_2^2(x)} \\
+&\frac{y \cdot \big(p_2(x) \cdot q_1(x) + p_1(x) \cdot q_2(x)\big)}{q_1^2(x) + a_3 \cdot q_1(x) \cdot q_2(x) + f_2(x) \cdot q_2^2(x)}.
\end{aligned}$$

□

Algorithm 5.58. ***The inputs*** *of the algorithm are*

(i) $m \in \mathbb{N}$*;*

(ii) $u = \frac{a_1(x)}{c_1(x)} + y \cdot \frac{b_2(x)}{c_2(x)} \in E$ *where* $a_1(x), c_1(x), b_2(x), c_2(x) \in K[x]$*.*

The outputs *of the algorithm are* $r_1(x), t_1(x), s_2(x), t_2(x) \in K[x]$ *such that* $\delta_x^{(m)}(u) = \frac{r_1(x)}{t_1(x)} + y \cdot \frac{s_2(x)}{t_2(x)}$*.*

The algorithm: *Initialize* $r_1(x) := a_1(x)$*,* $t_1(x) := c_1(x)$ *,* $s_2(x) := b_2(x)$ *and* $t_2(x) := c_2(x)$*. Repeat the following process until* $m = 0$*.*

1. Assign the polynomials.

(i) If $w(x,y) = y^2 - f(x)$ *and* $\operatorname{char} K \neq 2$ *then assign*

- $r_1(x) := r_1'(x) \cdot t_1(x) - r_1(x) \cdot t_1'(x)$;
- $t_1(x) := t_1^2(x)$;
- $s_2(x) := f'(x) \cdot s_2(x) \cdot t_2(x) + 2 \cdot f(x) \cdot \big(s_2'(x) \cdot t_2(x) - s_2(x) \cdot t_2'(x)\big)$;
- $t_2(x) := 2 \cdot f(x) \cdot t_2^2(x)$;

(ii) If $w(x,y) = y^2 + x \cdot y - f_1(x)$ *and* $\operatorname{char} K = 2$ *then assign*

- $r_1(x) := x \cdot t_2(x) \cdot \big(r_1'(x) \cdot t_1(x) - r_1(x) \cdot t_1'(x)\big) + t_1^2(x) \cdot s_2(x) \cdot f_1'(x)$;
- $t_1(x) := t_1^2(x) \cdot t_2(x) \cdot x$;
- $s_2(x) := s_2(x) \cdot t_2(x) + x \cdot \big(s_2'(x) \cdot t_2(x) - s_2(x) \cdot t_2'(x)\big)$;
- $t_2(x) := x \cdot t_2^2(x)$;

(iii) If $w(x,y) = y^2 + a_3 \cdot y - f_2(x)$ *and* $\operatorname{char} K = 2$ *then assign*

- $r_1(x) := a_3 \cdot t_2(x) \cdot \big(r_1'(x) \cdot t_1(x) - r_1(x) \cdot t_1'(x)\big) + t_1^2(x) \cdot s_2(x) \cdot f_2'(x)$;
- $t_1(x) := t_1^2(x) \cdot t_2(x) \cdot a_3$;
- $s_2(x) := s_2'(x) \cdot t_2(x) - s_2(x) \cdot t_2'(x)$;
- $t_2(x) := t_2^2(x)$;

2. Put the polynomials in the form such that

$$\gcd\big(r_1(x), t_1(x)\big) = 1 \text{ and } \gcd\big(s_2(x), t_2(x)\big) = 1.$$

3. $m := m - 1$.

Return $r_1(x)$, $t_1(x)$, $s_2(x)$ *and* $t_2(x)$.

Theorem 5.59. *Algorithm 5.58 is correct.*

Proof. Keep the assumption from Algorithm 5.58. Show that one iteration is $\delta_x\big(\frac{r_1(x)}{t_1(x)} + y \cdot \frac{s_2(x)}{t_2(x)}\big)$. Recall that by Remark 5.8 $\delta_x\big(f(x)\big) = f'(x)$ for every $f(x) \in K[x]$.

(i) Let $w(x,y) = y^2 - f(x)$ and $\operatorname{char} K \neq 2$. The derivation $\delta_x(y) = \frac{f'(x)}{2y} = \frac{y \cdot f'(x)}{2 \cdot f(x)}$ by Lemma 5.10. Derive

$$\begin{aligned}
&\delta_x\Big(\frac{r_1(x)}{t_1(x)} + y \cdot \frac{s_2(x)}{t_2(x)}\Big) \\
=&\frac{r_1'(x) \cdot t_1(x) - r_1(x) \cdot t_1'(x)}{t_1^2(x)} + \delta_x(y) \cdot \frac{s_2(x)}{t_2(x)} + y \cdot \frac{s_2'(x) \cdot t_2(x) - s_2(x) \cdot t_2'(x)}{t_2^2(x)} \\
=&\frac{r_1'(x) \cdot t_1(x) - r_1(x) \cdot t_1'(x)}{t_1^2(x)} \\
+&y \cdot \frac{f'(x) \cdot s_2(x) \cdot t_2(x)}{2 \cdot f(x) \cdot t_2^2(x)} + y \cdot \frac{2 \cdot f(x) \cdot \big(s_2'(x) \cdot t_2(x) - s_2(x) \cdot t_2'(x)\big)}{2 \cdot f(x) \cdot t_2^2(x)}.
\end{aligned}$$

(ii) Let $w(x,y) = y^2+x\cdot y-f_1(x)$ and $\operatorname{char} K = 2$. The derivation $\delta_x(y) = \frac{f_1'(x)+y}{x}$ by Lemma 5.10. Derive

$$\begin{aligned}
&\delta_x\Big(\frac{r_1(x)}{t_1(x)} + y\cdot\frac{s_2(x)}{t_2(x)}\Big)\\
=&\frac{r_1'(x)\cdot t_1(x) - r_1(x)\cdot t_1'(x)}{t_1^2(x)} + \delta_x(y)\cdot\frac{s_2(x)}{t_2(x)} + y\cdot\frac{s_2'(x)\cdot t_2(x) - s_2(x)\cdot t_2'(x)}{t_2^2(x)}\\
=&\frac{t_2(x)\cdot x\cdot\big(r_1'(x)\cdot t_1(x) - r_1(x)\cdot t_1'(x)\big)}{t_1^2(x)\cdot t_2(x)\cdot x} + \frac{t_1^2(x)\cdot s_2(x)\cdot f_1'(x)}{t_1^2(x)\cdot t_2(x)\cdot x}\\
+&y\cdot\frac{s_2(x)\cdot t_2(x)}{x\cdot t_2^2(x)} + y\cdot\frac{x\cdot\big(s_2'(x)\cdot t_2(x) - s_2(x)\cdot t_2'(x)\big)}{x\cdot t_2^2(x)}.
\end{aligned}$$

(iii) Let $w(x,y) = y^2+a_3\cdot y-f_2(x)$ and $\operatorname{char} K = 2$. The derivation $\delta_x(y) = \frac{f_2'(x)}{a_3}$ by Lemma 5.10. Derive

$$\begin{aligned}
&\delta_x\Big(\frac{r_1(x)}{t_1(x)} + y\cdot\frac{s_2(x)}{t_2(x)}\Big)\\
=&\frac{r_1'(x)\cdot t_1(x) - r_1(x)\cdot t_1'(x)}{t_1^2(x)} + \delta_x(y)\cdot\frac{s_2(x)}{t_2(x)} + y\cdot\frac{s_2'(x)\cdot t_2(x) - s_2(x)\cdot t_2'(x)}{t_2^2(x)}\\
=&\frac{t_2(x)\cdot a_3\cdot\big(r_1'(x)\cdot t_1(x) - r_1(x)\cdot t_1'(x)\big)}{t_1^2(x)\cdot t_2(x)\cdot a_3} + \frac{t_1^2(x)\cdot s_2(x)\cdot f_2'(x)}{t_1^2(x)\cdot t_2(x)\cdot a_3}\\
+&y\cdot\frac{s_2'(x)\cdot t_2(x) - s_2(x)\cdot t_2'(x)}{t_2^2(x)}.
\end{aligned}$$

□

Algorithm 5.60. ***The inputs*** *of the algorithm are $u \in E$ and $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ such that $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 1$.*

The output *of the algorithm is the value of $\operatorname{res}_{P,x-\alpha}(u)$ or a fault.*

The algorithm

1. *Use Algorithm 5.56 to express*

$$u = (x-\alpha)^i\cdot\frac{\tilde{a}_1(x)}{\tilde{c}_1(x)} + (x-\alpha)^j\cdot\frac{\tilde{b}_2(x)\cdot y}{\tilde{c}_2(x)}$$

where $\tilde{a}_1(x), \tilde{b}_2(x), \tilde{c}_1(x), \tilde{c}_2(x) \in K[x]$ such that $\tilde{a}_1(\alpha) \neq 0$, $\tilde{b}_2(\alpha) \neq 0$, $\tilde{c}_1(\alpha) \neq 0$, $\tilde{c}_2(\alpha) \neq 0$, $\gcd\big(\tilde{a}_1(x), \tilde{c}_1(x)\big) = 1$ and $\gcd\big(\tilde{b}_2(x), \tilde{c}_2(x)\big) = 1$.

Define $C \in \{0,\dots,3\}$ according to the type of $P_{(\alpha,\beta)}$ (cf. Definition 2.15).

(i) If $P_{(\alpha,\beta)}$ is of Type 1 then $C := 0$.

(ii) If $P_{(\alpha,\beta)}$ is of Type 4 then $C := 1$.

(iii) If $P_{(\alpha,\beta)}$ is of Type 5 then $C := 2$.

(iv) If $P_{(\alpha,\beta)}$ is of Type 6 then $C := 3$.

Define $I := \min(i, j+C)$.

2. *If* $I \geq 0$ *then* $\operatorname{res}_{P,x-\alpha}(u) = 0$ *and finish the algorithm. If* $\operatorname{char} K > 0$ *and* $-I > \operatorname{char} K$ *then the algorithm finish with the fault. Assume that if* $\operatorname{char} K > 0$ *then* $0 > I \geq -\operatorname{char} K$ *or if* $\operatorname{char} K = 0$ *then* $0 > I$.

3. *Put*

$$u_1 := (x-\alpha)^{-I} \cdot u = (x-\alpha)^{i-I} \cdot \frac{\tilde{a_1}(x)}{\tilde{c_1}(x)} + (x-\alpha)^{j-I} \cdot \frac{\tilde{b_2}(x) \cdot y}{\tilde{c_2}(x)}.$$

4. *Use Algorithm 5.58 to derive*

$$u_2 := \delta_x^{(-I-1)}(u_1).$$

5. *Use Algorithm 5.53 to evaluate* $v := u_2(\alpha, \beta) \in K$. *The residue* $\operatorname{res}_{P,x-\alpha}(u) = \frac{v}{(-I-1)!}$.

Theorem 5.61. *Algorithm 5.60 is correct.*

Proof. Keep the assumption from Algorithm 5.60. Clearly, the assumption of Algorithm 5.56 is satisfied. By Lemma 5.50 $x - \alpha$ is a uniformizing element of $P_{(\alpha,\beta)}$. In other words, $\upsilon_{(\alpha,\beta)}(x-\alpha) = 1$. Prove that $C = \upsilon_{(\alpha,\beta)}(y)$. If $P_{(\alpha,\beta)}$ is of *Type 1* then $\beta \neq 0$ and $\upsilon_{(\alpha,\beta)}(y) = 0 = C$. If $P_{(\alpha,\beta)}$ is of *Type 4*, *Type 5* or *Type 6* then $\beta = 0$ and $\upsilon_{(\alpha,\beta)}(y) = e(P_{(\alpha,\beta)} \mid P_\beta) \cdot \upsilon_\beta(y) = e(P_{(\alpha,\beta)} \mid P_\beta)$ where $P_\beta \in \mathbb{P}^{(1)}_{K(y)}$. Then $\upsilon_{(\alpha,\beta)}(y) = C$ by Theorem 2.16. Recall that if $P_{(\alpha,\beta)}$ is of *Type 2* or *Type 3* then $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 2$ and it is a contradiction with the assumption of Algorithm 5.60. The valuation

$$\begin{aligned}\upsilon_{(\alpha,\beta)}(u) &\geq \min\left(\upsilon_{(\alpha,\beta)}\left((x-\alpha)^i \cdot \frac{\tilde{a_1}(x)}{\tilde{c_1}(x)}\right), \upsilon_{(\alpha,\beta)}\left((x-\alpha)^j \cdot \frac{\tilde{b_2}(x)}{\tilde{c_1}(x)}\right) + \upsilon_{(\alpha,\beta)}(y)\right) \\ &= \min(i, j+C) = I.\end{aligned}$$

If $I \geq 0$ then $\operatorname{res}_{P,x-\alpha}(u) = 0$ by Remark 5.35. Assume that $I < 0$. If $\operatorname{char} K > 0$ and $-I > \operatorname{char} K$ then the assumption of Theorem 5.46 is not satisfied. Thus the algorithm finishes with the fault. Now, the assumption of Theorem 5.46 is satisfied. By Theorem 5.46

$$\operatorname{res}_{P,x-\alpha}(u) = \frac{\left(\delta_{x-\alpha}^{(-I-1)}\big(u \cdot (x-\alpha)^{-I}\big)\right)(\alpha,\beta)}{(-I-1)!} = \frac{\left(\delta_{x-\alpha}^{(-I-1)}(u_1)\right)(\alpha,\beta)}{(-I-1)!}.$$

By Corollary 5.52 $\delta_{x-\alpha}^{(-I-1)} = \delta_x^{(-I-1)}$. Continue

$$\operatorname{res}_{P,x-\alpha}(u) = \frac{\left(\delta_x^{(-I-1)}(u_1)\right)(\alpha,\beta)}{(-I-1)!} = \frac{u_2(\alpha,\beta)}{(-I-1)!} = \frac{v}{(-I-1)!}.$$

Clearly the assumption of Algorithm 5.58 is satisfied. Show that $\upsilon_{(\alpha,\beta)}(u_2) \geq 0$. The valuation $\upsilon_{(\alpha,\beta)}\left(u \cdot (x-\alpha)^{-I}\right) = \upsilon_{(\alpha,\beta)}(u) - I \geq 0$ by the definition of I. By Lemma 5.44 $\upsilon_{(\alpha,\beta)}(u_2) \geq 0$. Thus the assumption of Algorithm 5.53 is satisfied. □

Remark 5.62. During Algorithm 5.60 u is kept in the representation

$$u = (x-\alpha)^i \cdot \frac{\tilde{a_1}(x)}{\tilde{c_1}(x)} + (x-\alpha)^j \cdot \frac{\tilde{b_2}(x) \cdot y}{\tilde{c_2}(x)}$$

because $v_{(\alpha,\beta)}(u)$ can be approximated from this representation and Algorithms 5.58 and 5.53 need this representation at the input.

Algorithm 5.63. ***The inputs*** *of the algorithm are* $u \in E$ *and* $P := P_{(\alpha,\beta)} \in \mathbb{P}_E^{(1)}$ *such that* $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 1$.

The output *of the algorithm is the value of* $\mathrm{res}_{P,x-\alpha}(u)$.

The algorithm

1. *Use Algorithm 5.56 to express*

$$u = (x-\alpha)^i \cdot \frac{\tilde{a_1}(x)}{\tilde{c_1}(x)} + (x-\alpha)^j \cdot \frac{\tilde{b_2}(x) \cdot y}{\tilde{c_2}(x)}$$

where $\tilde{a_1}(x), \tilde{b_2}(x), \tilde{c_1}(x), \tilde{c_2}(x) \in K[x]$ *such that* $\tilde{a_1}(\alpha) \neq 0$, $\tilde{b_2}(\alpha) \neq 0$, $\tilde{c_1}(\alpha) \neq 0$, $\tilde{c_2}(\alpha) \neq 0$, $\gcd\big(\tilde{a_1}(x), \tilde{c_1}(x)\big) = 1$ *and* $\gcd\big(\tilde{b_2}(x), \tilde{c_2}(x)\big) = 1$.

Define $C \in \{0, \ldots, 3\}$ *according to the type of* $P_{(\alpha,\beta)}$ *(cf. Definition 2.15).*

   (i) If $P_{(\alpha,\beta)}$ *is of Type 1 then* $C := 0$.

   (ii) If $P_{(\alpha,\beta)}$ *is of Type 4 then* $C := 1$.

   (iii) If $P_{(\alpha,\beta)}$ *is of Type 5 then* $C := 2$.

   (iv) If $P_{(\alpha,\beta)}$ *is of Type 6 then* $C := 3$.

Define $I := \min(i, j + C)$.

2. *If* $I \geq 0$ *then* $\mathrm{res}_{P,x-\alpha}(u) = 0$ *and finish the algorithm. Assume that* $I < 0$.

3. *Initialize* $r := u \in E$ *and* $o := 0 \in K$. *Repeat the following process until* $I = 0$.

   (a) Put $\tilde{r} := r \cdot (x-\alpha)^{-I}$.

   (b) Use Algorithm 5.53 to evaluate $o := \tilde{r}(\alpha, \beta)$.

   (c) Put $r := r - o \cdot (x-\alpha)^{-I}$

   (d) Use Algorithm 5.56 to express in the form $r = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x) \cdot y}{c_2(x)}$.

   (e) Put $I := I + 1$.

4. *The residue* $\mathrm{res}_{P,x-\alpha}(u) = o$ *and finish the algorithm.*

Remark 5.64. (i) The idea of Algorithm 5.63 is to find a_I and to shorten $r = \sum_{i=I}^{\infty} a_i \cdot (x-\alpha)^i$ by $r - a_I \cdot (x-\alpha)^I$ in each step of the cycle. It is inspired by the proof of Theorem 4.2.6 from [1].

(ii) The element r has to be kept in the representation $r = \frac{a_1(x)}{c_1(x)} + \frac{b_2(x) \cdot y}{c_2(x)}$ because Algorithm 5.53 needs this representation at the input. It is the reason why Step 3d is included in the algorithm.

Theorem 5.65. *Algorithm 5.63 is correct.*

Proof. Keep the assumption from Algorithm 5.63. Clearly, the assumption of Algorithm 5.56 is satisfied. By Lemma 5.50 $x-\alpha$ is a uniformizing element of $P_{(\alpha,\beta)}$. In other words, $\upsilon_{(\alpha,\beta)}(x-\alpha)=1$. Prove that $C=\upsilon_{(\alpha,\beta)}(y)$. If $P_{(\alpha,\beta)}$ is of *Type 1* then $\beta\neq 0$ and $\upsilon_{(\alpha,\beta)}(y)=0=C$. If $P_{(\alpha,\beta)}$ is of *Type 4*, *Type 5* or *Type 6* then $\beta=0$ and $\upsilon_{(\alpha,\beta)}(y)=e(P_{(\alpha,\beta)}\mid P_\beta)\cdot\upsilon_\beta(y)=e(P_{(\alpha,\beta)}\mid P_\beta)$ where $P_\beta\in\mathbb{P}^{(1)}_{K(y)}$. Then $\upsilon_{(\alpha,\beta)}(y)=C$ by Theorem 2.16. Recall that if $P_{(\alpha,\beta)}$ is of *Type 2* or *Type 3* then $e\big(P_{(\alpha,\beta)}\mid P_{(\alpha,\beta)}\cap K(x)\big)=2$ and it is a contradiction with the assumption of Algorithm 5.63. The valuation

$$\begin{aligned}\upsilon_{(\alpha,\beta)}(u)\geq&\min\left(\upsilon_{(\alpha,\beta)}\Big((x-\alpha)^i\cdot\frac{\tilde{a_1}(x)}{\tilde{c_1}(x)}\Big),\upsilon_{(\alpha,\beta)}\Big((x-\alpha)^j\cdot\frac{\tilde{b_2}(x)}{\tilde{c_1}(x)}\Big)+\upsilon_{(\alpha,\beta)}(y)\right)\\=&\min(i,j+C)=I.\end{aligned}$$

If $I\geq 0$ then $\operatorname{res}_{P,x-\alpha}(u)=0$ by Remark 5.35. Assume that $I<0$.

Let $u=\sum_{i=I}^\infty a_i\cdot(x-\alpha)^i$ be the P-adic power series expansion of $u\in F$ with respect $x-\alpha$. In case $\upsilon_{(\alpha,\beta)}(u)>I$, fill the series by zero coefficients. Prove that after every round of the cycle $o=a_{I-1}$. Thus if the cycle stop when $I=0$ then $o=a_{-1}=\operatorname{res}_{P,x-\alpha}(u)$.

Assume that $\upsilon_{(\alpha,\beta)}(r)\geq I$ at the start of every round of the cycle. After the initialization $\upsilon_{(\alpha,\beta)}(r)=\upsilon_{(\alpha,\beta)}(u)\geq I$. Thus the assumption is satisfied after the initialization. The valuation $\upsilon_{(\alpha,\beta)}(\tilde{r})=\upsilon_{(\alpha,\beta)}(r)-I\geq 0$. Thus Algorithm 5.53 can be used. Express $\tilde{r}=\sum_{i=I}^\infty a_i\cdot(x-\alpha)^{i-I}$ The value $o=\tilde{r}(\alpha,\beta)=a_I$ by Lemma 5.45.

The element $r-o\cdot(x-\alpha)^I=\sum_{i=I+1}^\infty a_i\cdot(x-\alpha)^i$. If $o=0$ then $\upsilon_{(\alpha,\beta)}(r)>I$ by Theorem 5.32. Thus $\upsilon_{(\alpha,\beta)}(r)\geq I+1$. If $o\neq 0$ then $\upsilon_{(\alpha,\beta)}(r)=I$ by Theorem 5.32. Then $\upsilon_{(\alpha,\beta)}\big(r-o\cdot(x-\alpha)^I\big)\geq I+1$. Put $r:=r-o\cdot(x-\alpha)^I$ and $I:=I+1$. Then $o=a_{I-1}$ and $\upsilon_{(\alpha,\beta)}(r)\geq I$. □

Theorem 5.66. *Consider $\omega\in\Omega_E$ as in Theorem 3.4. Let $P:=P_{(\alpha,\beta)}\in\mathbb{P}^{(1)}_E$ be such that $e\big(P_{(\alpha,\beta)}\mid P_{(\alpha,\beta)}\cap K(x)\big)=1$. Define $C\in E$ by the following way.*

(i) If $\operatorname{char}K\neq 2$ and $w(x,y)=y^2-f(x)$ then $C:=\frac{1}{2\cdot y}$.

(ii) If $\operatorname{char}K=2$ and $w(x,y)=y^2+x\cdot y-f_1(x)$ then $C:=\frac{1}{x}$.

(iii) If $\operatorname{char}K=2$ and $w(x,y)=y^2+a_3\cdot y-f_2(x)$ then $C:=\frac{1}{a_3}$.

Then $\omega_{(\alpha,\beta)}(u)=\operatorname{res}_{P,x-\alpha}(C\cdot u)$ for every $u\in E$.

Proof. Recall that $x-\alpha$ is a uniformizing element of $P_{(\alpha,\beta)}$ by Lemma 5.50. By Definition 5.39 $\delta(x)=\operatorname{Cotr}_{F/K(x)}(\eta)$ where $\eta\in\Omega_{K(x)}$ is as in Theorem 1.78. By Theorem 4.16 $\omega=C\cdot\delta(x)$. By Theorem 5.40

$$\omega_{(\alpha,\beta)}(u)=\big(C\cdot\delta(x)\big)_P(u)=\operatorname{res}_P\big((u\cdot C)\,dx\big).$$

By Definition 5.11 and Lemma 5.51

$$(u\cdot C)\,dx=\big(u\cdot C\cdot\delta_{x-\alpha}(x)\big)\,d(x-\alpha)=\big(u\cdot C\big)\,d(x-\alpha).$$

Continue

$$\omega_{(\alpha,\beta)}(u)=\operatorname{res}_P\Big(\big(u\cdot C\big)\,d(x-\alpha)\Big)=\operatorname{res}_{P,x-\alpha}(u\cdot C).$$

□

Remark 5.67.

(i) Theorem 5.66 converts the calculation of $\omega_{(\alpha,\beta)}$ to the calculation of $\mathrm{res}_{P,x-\alpha}$. Algorithms 5.60 and 5.63 calculate $\mathrm{res}_{P,x-\alpha}$. Thus there are two ways how to calculate $\omega_{(\alpha,\beta)}$ if $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 1$.

(ii) Recall that if $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 2$ then Section 4.2 provides a useful tool calculating $\omega_{(\alpha,\beta)}$. The tool works for P_∞.

(iii) Now, describe why the result of this chapter does not work if $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 2$. Recall that $P_{(\alpha,\beta)}$ is of *Type 2* or *Type 3*. Let $t \in E$ be a uniformizing element of $P_{(\alpha,\beta)}$. If $P_{(\alpha,\beta)}$ is of *Type 2* then $t = y$ by Lemma 2.11. If $P_{(\alpha,\beta)}$ is of *Type 3* then $t = y - \sqrt{a_6}$ by Lemma 2.8.

Both Algorithms 5.60 and 5.63 use the evaluation from Algorithm 5.53 that does not work if $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 2$.

Algorithm 5.60 uses Theorem 5.46. Thus it is needed to calculate $\delta_y^{(m)}$. By Lemma 5.9 $\delta_y = \delta_x \cdot \delta_y(x)$. By Lemma 5.10 $\delta_y(x)$ is known. Thus Algorithm 5.58 can be modified to calculate $\delta_y^{(m)}$. The modification is that after each step of the cycle a result is multiple by $\delta_y(x)$.

Recall that at the beginning of Algorithms 5.60 and 5.63 it is needed to know $I \in \mathbb{Z}$ such that $\upsilon_{(\alpha,\beta)}(u) \geq I$. Recall that $\upsilon_{(\alpha,\beta)}(x-\alpha) = 2$. Thus the approximation can be done.

All this issues could be solved. But it does not have any meaning because Section 4.2 solves the calculation of $\omega_{(\alpha,\beta)}$ if $e\big(P_{(\alpha,\beta)} \mid P_{(\alpha,\beta)} \cap K(x)\big) = 2$.

Conclusion

The algorithms calculating local components of Weil differentials have been developed in this thesis. They are based on different approaches. The following remarks serve to the purpose of explaining how the three approaches exposed in this thesis might be extended and generalized.

The algorithm from Chapter 3 is simplified if $\operatorname{char} K \neq 2$. There is a possibility to be interested whether a similar simplification is possible for $\operatorname{char} K = 2$.

The algorithms from Section 5.4 work for a place P such that $e\big(P \mid P \cap K(x)\big) = 1$. Remark 5.67 explains where a problem is if $e\big(P \mid P \cap K(x)\big) = 2$. Shortly, Algorithm 5.53 does not work if $e\big(P \mid P \cap K(x)\big) = 2$. There is a possibility to develop a new algorithm for the evaluation if $e\big(P \mid P \cap K(x)\big) = 2$.

The algorithm from Section 4.2 is based on converting the calculation form an elliptic function field E/K to a rational function field $K(x)/K$. The algorithm is only one which can calculate a local component of a place of a higher degree. It is caused by that Algorithm 1.87 for $K(x)/K$ can do it. But there is an obstacle that the converting works only for a place P such that $e\big(P \mid P \cap K(x)\big) = 2$. The exact problem is described in Remark 4.18. If the problem of the converting was solved then there would be a tool calculating local components of every place. By my opinion it could be solved two ways. The first one is to distinguish anyway between local components of $P_1, P_2 \in \mathbb{P}_E$ such that $P_1 \cap K(x) = P_2 \cap K(x)$. The other one is to build an algorithm at the thought of the proof of Theorem 3.4.6 from [1, Chapter 3.4]. It uses Theorem 1.22. Thus it is needed to make an algorithm calculating Theorem 1.22.

There is another possibility how to progress calculating a local component for a place of a higher degree. Let P be a place of an elliptic function field F/K. Let $\deg(P) = n \geq 2$. Then there exists a constant field extension F'/K' of F/K such that is finite and P breaks down to $P_1, \ldots, P_n \in \mathbb{P}_{F'}^{(1)}$. The local components of $P_1, \ldots, P_n$ can be calculated by an arbitrary algorithm. The finalization could be done by the theory of $\operatorname{Cotr}_{F'/F}$. The first idea could be to use Remark 4.9.

All algorithms in this thesis have been designed to be able to program them. They employ implementations of arithmetic K, polynomial arithmetic $K[x]$, $K[x, y]$ and other usual algorithms like the extended Euclidean algorithm. Their implementations could be made as another progress.

Bibliography

[1] Stichtenoth, Henning. *Algebraic Function Fields and Codes*. Second Edition Berlin Heidelberg: Springer-Verlag, 2009. ISBN 978-3-540-76877-7.